

40. I3C Bus Interface (I3C)

Specific HDR Modes are not covered by conformance test suite (CTS).

When CTS for HDR Modes is available, Renesas will confirm CTS tests.

40.1 Overview

40.1.1 Functional Overview

The I3C bus interface (I3C) has 1 channel. The I3C module conform with and provide a subset of the NXP I²C (Inter-Integrated Circuit) bus interface functions and a subset of the MIPI I3C.

In this section, PCLK refers to PCLKA, TCLK refers to I3CCLK.

[Table 40.1](#) lists the I²C specifications, and [Table 40.2](#) lists the I3C specifications.

Table 40.1 I²C specifications (1 of 2)

Parameter	Specifications
Operation mode	Master mode and slave mode selectable
Data handler	Single buffer transfer
Communication protocol	- I²C bus format - Standard-mode (Sm) : 0 to 100 kbps - Fast-mode (Fm) : 0 to 400 kbps - Fast-mode Plus (Fm+) : 0 to 1 Mbps - High-speed mode (Hs-mode) : 0 to 3.4 Mbps - SMBus format : 10 to 100 kbps
Address format	7-bit address 10-bit address
Address detection	- Slave address (static address) (max 3 address) - General call address - Hs-mode master code - Device ID - Host address 10-bit slave addressing
Clock stretching	Clock stretching capability
Noise-filter	- Analog noise-filter - Digital noise-filter
Interrupt source	- Normal Rx data buffer full - Normal Tx data buffer empty - START condition detection - STOP condition detection - Transmit end - NACK detection - Arbitration lost - Timeout detection - Wake-up condition detection
Error detection	- Non-recoverable internal error - NACK received - Receive overflow or transfer underflow error - Arbitration lost error - Timeout error
Event link output	- Normal Rx data buffer full event - Normal Tx data buffer empty event - START condition event - STOP condition event - Transmit end event - NACK event - Arbitration lost event - Timeout event
Wake-up source	Address detection of slave address

Table 40.1 I²C specifications (2 of 2)

Parameter	Specifications
Module-stop function	Module-stop state can be set to reduce power consumption
Trust Zone Filter	Security and Privilege attribution can be set

Table 40.2 I3C specifications (1 of 2)

Parameter	Specifications
Operation mode	Master (main master/secondary master) mode and slave mode selectable
Data handler	- Master : - High priority FIFO buffer transfer - Normal FIFO buffer transfer - Slave : - Normal FIFO buffer transfer
Communication protocol	- SDR (I3C Single Data Rate) mode : up to 12.5 Mbps - Private message - Broadcast message (common command code) - Direct message (common command code) - HDR (I3C High Data Rate) mode - HDR-DDR (Dual Data Rate) mode : up to 25 Mbps - HDR-TSL (Ternary Symbol Legacy) mode : up to 27.5 Mbps - HDR-TSP (Ternary Symbol Pure-bus) mode : up to 39.5 Mbps - Legacy I²C message - Fast-mode (Fm) : 0 to 400 kbps - Fast-mode Plus (Fm+) : 0 to 1 Mbps
In-band interrupt	- Slave interrupt request - Master ship request (secondary master only)
Address format	7-bit address
Address detection	- Slave address (static address or dynamic address) - Broadcast address (0x7E)
Clock stalling	Clock stalling capability
Timing control	- Synchronous timing control - Sync mode : Synchronous basic mode - Asynchronous timing control - Async mode 0 : Asynchronous basic mode - Async mode 1 : Asynchronous advanced mode
Interrupt source	- Non-recoverable internal error - Normal Transfer error - Normal Transfer abort - Normal Response queue full - Normal Command queue empty - Normal IBI Queue Empty/Full - Normal Rx Data buffer full - Normal Tx Data buffer empty - Normal Receive status queue full - High Priority Response queue full - High Priority Command queue empty - High Priority Rx Data buffer full - High Priority Tx Data buffer empty - High Priority Transfer Error - High Priority Transfer Abort - START condition detection - STOP condition detection - HDR exit pattern detection - Timeout detection - Synchronous Timing - MREF Counter Overflow - MREF Capture - Additional Master-initiated bus Event - Wake-up condition detection

Table 40.2 I3C specifications (2 of 2)

Parameter	Specifications
Error detection	• Non-recoverable internal error • CRC error • Parity error • Frame error • Address header error • Address NACKed or dynamic address assignment NACKed • Receive overflow or transfer underflow error • Aborted • NACK received for the I²C write data transfer • Timeout error
Event link output	• Normal Response queue full event • Normal Command queue empty event • Normal IBI Queue Empty/Full • Normal Rx data buffer full event • Normal Tx data buffer empty event • Normal Receive Status queue full event • High Priority Response queue full event • High Priority Command queue empty event • High Priority Rx Data buffer full event • High Priority Tx Data buffer empty event • Non-recoverable internal error event • Normal Transfer Error event • Normal Transfer Abort event • High Priority Transfer Error event • High Priority Transfer Abort event • START condition event • STOP condition event • HDR exit pattern event • Timeout event • Synchronous timing event • MREF counter overflow event • MREF capture event • Additional master-initiated bus Event
Wake-up source	• Master : SDA assert of IBI (START condition detection) • Slave : Address detection of broadcast address (0x7E) and slave address
Module-stop function	Module-stop state can be set to reduce power consumption
Trust Zone Filter	Security and Privilege attribution can be set

Table 40.3 I3C I/O pins

Channel	Pin name	I/O	Function
I3C	I3C_SCL0	I/O	I3C serial clock I/O pin
	I3C_SDA0	I/O	I3C serial data I/O pin

Note: In this section, I3C_SCL refers to I3C_SCL0, I3C_SDA refers to I3C_SDA0.

40.1.2 Block Diagram

Figure 40.1 shows the main components of this I3C.

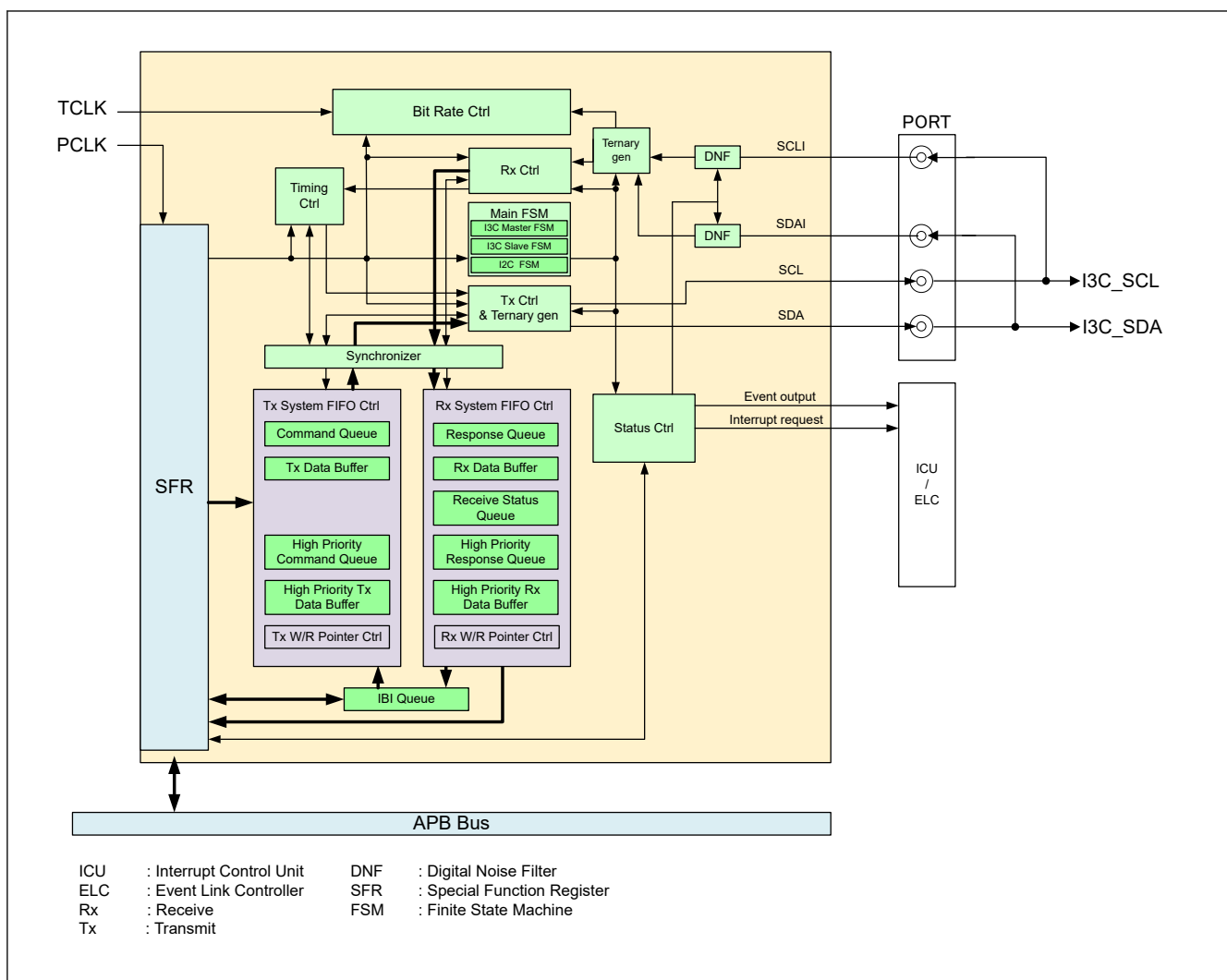


Figure 40.1 I3C block diagram

40.2 Registers

40.2.1 PRTS : Protocol Selection Register

Base address: I3C = 0x4035_F000
 I3C_NS = 0x5035_F000

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PRTM D
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	PRTMD	Protocol Mode 0: I3C protocol mode 1: I ² C protocol mode	R/W

Bit	Symbol	Function	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

PRTMD bit (Protocol Mode)

PRTMD = 0 : I3C FIFO buffer transfer (Equivalent to HCI)

PRTMD = 1 : I²C single buffer transfer

There are notes about rewriting the PRTS.PRTMD bit, see [section 40.7.4. Notes on Rewriting PRTS.PRTMD Bit](#).

40.2.2 BCTL : Bus Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	BUSE	RSM	ABT	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	BMDS	—	—	—	—	—	—	INCBA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	INCBA	Include I3C Broadcast Address* ¹ 0: Do not include I3C broadcast address for private transfers 1: Include I3C broadcast address for private transfers	R/W
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	BMDS	Bus Mode Selection* ¹ 0: Legacy inclusive Bus mode disabled 1: Legacy inclusive (mix) Bus mode enabled	R/W
28:8	—	These bits are read as 0. The write value should be 0.	R/W
29	ABT	Abort* ¹ 0: I3C is running. 1: I3C has aborted a transfer.	R/W
30	RSM	Resume* ² Values when read: 0: I3C is running. 1: I3C is suspended.	R/W
31	BUSE	Bus Enable 0: I3C bus operation is disabled. 1: I3C bus operation is enabled.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports I3C master mode and I3C secondary master mode.

Note 2. This bit supports all I3C mode.

INCBA bit (Include I3C Broadcast Address)

This bit controls whether the I3C broadcast address (0x7E) is included for private transfers.

If the I3C broadcast address is not included for private transfers, then IBIs driven from Slaves might not win the arbitration, potentially delaying acceptance of the IBIs.

BMDS bit (Bus Mode Selection)

Indicates whether or not any Legacy I²C Devices are present on the I3C Bus. Whenever this bit is set, I3C shall use the HDR-TSL protocol (Ternary Symbol Legacy Inclusive Bus) for HDR-TS Transfers.

The specific mode of the transfer (SDR, DDR, HDRTS) is controlled on a per-Command basis.

I3C shall use this field to select either TSL (1) or TSP (0) for HDR-TS transfers.

ABT bit (Abort)

When set to 1, this bit allows I3C to relinquish control of the I3C Bus before completing the currently issued transfer.

In response to an ABORT request, I3C issues the STOP condition on the I3C Bus after the complete data byte is transferred or received.

The Driver shall clear the ABT bit to allow operation on the Bus.

If BCTL.ABT is set and ABORT processing is performed, please ignore ERR_STATUS of Response Descriptor.

RSM bit (Resume)

This bit is used to resume I3C operation following the Halt state.

I3C enters the Halt state (as indicated in register PRSTDBG) as a result of any type of error occurring in a transfer.

The error type is indicated by the field ERR_STATUS in register NRSPQP, HRSPQP, NRSQP and NIBIQP.

After I3C has entered the Halt state, the application must write the value 1 to the RSM bit to resume I3C operation. I3C shall auto-clear the RSM bit once it has resumed making transfers (it has initiated the next Command).

BUSE bit (Bus Enable)

Enables or disables the operation on the I3C Bus by I3C.

Set the BUSE bit to 1 when using I3C. The I3C_SCL and I3C_SDA pins are placed in the active state when the BUSE bit is set to 1. Set the BUSE bit to 0 when I3C is not to be used. The I3C_SCL and I3C_SDA pins are placed in the inactive state when the BUSE bit is set to 0.

If the software sets this bit, then it also confirms that initialization is done, and that I3C can use the programmed register values (For example, generation of SCL on IBI detection, etc.). If this bit is not set, then I3C shall not generate SCL for incoming IBI.

Software may disable I3C bus operation while it is active, However:

- If a disable request occurs while receiving IBI, the actual disabling will not occur until reception of the IBI is complete.
- When the software reads the value 0 from this field, this indicates that I3C bus operation disable operation has completed.

If commands remain in the command queue, do not set BUSE = 0.

40.2.3 MSDVAD : Master Device Address Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x018

Bit position:	31	30	29	28	27	26	25	24	23	22						16
Bit field:	MDYA DV	—	—	—	—	—	—	—	—	—	MDYAD[6:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
22:16	MDYAD[6:0]	Master Dynamic Address	R/W
30:23	—	These bits are read as 0. The write value should be 0.	R/W
31	MDYADV	Master Dynamic Address Valid 0: The master dynamic address field is not valid. 1: The master dynamic address field is valid.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode.

MDYAD[6:0] bits (Master Dynamic Address)

This field is used to program I3C master dynamic address. I3C uses this address to respond to master transactions in I3C interface mode (slave or secondary master role).

In I3C main master mode, the software shall program the dynamic address as it self-assigns its dynamic address.

MDYADV bit (Master Dynamic Address Valid)

This bit indicates whether or not the value in the MDYAD field is valid.

In I3C main master mode, the user sets this bit to 1 as it self-assigns its dynamic address.

Note: After setting MSDVAD, and setting BCTL.BUSE = 1, the device will act as main master.

Without setting MSDVAD, setting SVDCT.TBCR76[1:0] = 00b (Device Role Slave), and setting BCTL.BUSE = 1, the device will act as slave.

Without setting MSDVAD, setting MSDCTm.RBCR76[1:0] = 01b (Device Role Master), and setting BCTL.BUSE = 1, the device will act as slave.

40.2.4 RSTCTL : Reset Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	INTLR ST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	HRDB RST	HTDB RST	HRSP QRST	HCMD QRST	—	—	RSQR ST	IBIQR ST	RDBR ST	TDBR ST	RSPQ RST	CMDQ RST	RI3CR ST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RI3CRST	I3C Software Reset 0: Release I3C reset. 1: Initiate I3C reset.	R/W
1	CMDQRST	Normal Command Queue Software Reset* ¹ 0: The Normal Command Queues in I3C is not flushed. 1: The Normal Command Queues in I3C is flushed.	R/W
2	RSPQRST	Normal Response Queue Software Reset* ¹ 0: The Normal Response Queues in I3C is not flushed. 1: The Normal Response Queues in I3C is flushed.	R/W
3	TDBRST	Normal Tx Data Buffer Software Reset* ¹ 0: The Normal Tx Data buffers in I3C is not flushed. 1: The Normal Tx Data buffers in I3C is flushed.	R/W

Bit	Symbol	Function	R/W
4	RDBRST	Normal Rx Data Buffer Software Reset* ¹ 0: The Normal Rx Data buffers in I3C is not flushed. 1: The Normal Rx Data buffers in I3C is flushed.	R/W
5	IBIQRST	Normal IBI Queue Software Reset* ¹ 0: The Normal IBI Queues in I3C is not flushed. 1: The Normal IBI Queues in I3C is flushed.	R/W
6	RSQRST	Normal Receive Status Queue Software Reset* ² 0: The Normal Receive Status Queue in I3C is not flushed. 1: The Normal Receive Status Queue in I3C is flushed.	R/W
8:7	—	These bits are read as 0. The write value should be 0.	R/W
9	HCMDQRST	High Priority Command Queue Software Reset* ³ 0: The High Priority Command Queues in I3C is not flushed. 1: The High Priority Command Queues in I3C is flushed.	R/W
10	HRSPQRST	High Priority Response Queue Software Reset* ³ 0: The High Priority Response Queues in I3C is not flushed. 1: The High Priority Response Queues in I3C is flushed.	R/W
11	HTDBRST	High Priority Tx Data Buffer Software Reset* ³ 0: The High Priority Tx Data buffers in I3C is not flushed. 1: The High Priority Tx Data buffers in I3C is flushed.	R/W
12	HRDBRST	High Priority Rx Data Buffer Software Reset* ³ 0: The High Priority Rx Data buffers in I3C is not flushed. 1: The High Priority Rx Data buffers in I3C is flushed.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
16	INTLRST	Internal Software Reset 0: Releases of some registers and internal state. 1: Resets of some registers and internal state.	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports all I3C mode.

Note 2. This bit supports I3C secondary master mode and I3C slave mode.

Note 3. This bit supports I3C master mode and I3C secondary master mode.

For details on reset for each register, see [section 40.6. Reset Descriptions](#).

RI3CRST bit (I3C Software Reset)

On Driver setting this bit to 1, I3C shall be reset and disabled.

All registers shall return to their reset values, and the software shall re-initialize I3C.

This field is cleared automatically upon I3C reset completion. This field also resets all Queues in I3C.

Note: Programming this field while it contains a value of 1 may result in undefined behavior.

CMDQRST bit (Normal Command Queue Software Reset)

On software setting this bit to 1, the Normal Command Queues in I3C shall be flushed.

This field shall be cleared automatically upon Normal Command Queue reset completion.

RSPQRST bit (Normal Response Queue Software Reset)

On software setting this bit to 1, the Normal Response Queues in I3C shall be flushed.

This field shall be cleared automatically upon Normal Response Queue reset completion.

TDBRST bit (Normal Tx Data Buffer Software Reset)

On software setting this bit to 1, the Normal Tx Data Buffers in I3C shall be flushed.

This field shall be cleared automatically upon Normal Tx Data Buffer reset completion.

RDBRST bit (Normal Rx Data Buffer Software Reset)

On software setting this bit to 1, the Normal Rx Data Buffers in I3C shall be flushed.

This field shall be cleared automatically upon completion of Normal Rx Data Buffer reset.

IBIQRST bit (Normal IBI Queue Software Reset)

On software setting this bit to 1, the Normal IBI Queues in I3C shall be flushed.

This field shall be cleared automatically upon completion of Normal IBI Queue reset.

RSQRST bit (Normal Receive Status Queue Software Reset)

On software setting this bit to 1, the Normal Receive Status Queues in I3C shall be flushed.

This field shall be cleared automatically upon Normal Receive Status Queue reset completion.

HCMDQRST bit (High Priority Command Queue Software Reset)

On software setting this bit to 1, the High Priority Command Queues in I3C shall be flushed.

This field shall be cleared automatically upon High Priority Command Queue reset completion.

HRSPQRST bit (High Priority Response Queue Software Reset)

On software setting this bit to 1, the High Priority Response Queues in I3C shall be flushed.

This field shall be cleared automatically upon High Priority Response Queue reset completion.

HTDBRST bit (High Priority Tx Data Buffer Software Reset)

On software setting this bit to 1, the High Priority Tx Data Buffers in I3C shall be flushed.

This field shall be cleared automatically upon High Priority Tx Data Buffer reset completion.

HRDBRST bit (High Priority Rx Data Buffer Software Reset)

On software setting this bit to 1, the High Priority Rx Data Buffers in I3C shall be flushed.

This field shall be cleared automatically upon completion of High Priority Rx Data Buffer reset.

INTLRST bit (Internal Software Reset)

When set to 1, some of the registers are reset. For details on the registers to be reset, see [section 40.6. Reset Descriptions](#).

Note: When set internal software reset during bus operation enable, use DISEC CCC in advance to disable IBI transmission to I3C Slave in order to avoid conflict with IBI from I3C Slave connected to I3C Bus.

40.2.5 PRSST : Present State Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x024

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	PRSS TWP	—	—	TRMD	—	CRMS	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
2	CRMS	Current Master* ² 0: The Master is not the Current Master, and must request and acquire bus ownership before initiating any transfer. 1: The Master is the Current Master, and as a result can initiate transfers.	R/W* ¹
3	—	This bit is read as 0. The write value should be 0.	R/W
4	TRMD	Transmit/Receive Mode* ³ 0: Receive mode 1: Transmit mode	R
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	PRSSTWP	Present State Write Protect* ² 0: CRMS bit is protected. 1: CRMS bit can be written when writing simultaneously with the value of the target bit.	W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. When the PRSSTWP bit is set to 1, the CRMS bit can be written to.

Note 2. This bit supports I²C, I3C master, and I3C secondary master mode.

Note 3. This bit supports I²C mode.

CRMS bit (Current Master)

Indicates the set condition and reset condition of each operation mode.

Operation Mode [I²C/I3C common]

[Clearing conditions]

- When 0 written to the PRSST.CRMS by the software.

[Setting conditions]

- When 1 written to the PRSST.CRMS by the software.

Operation Mode [I²C]

[Clearing conditions]

- When STOP is issued.
- When Master Arbitration-Lost.

[Setting conditions]

- When START is issued.

Operation Mode [I3C Main Master]

[Clearing conditions]

- When 0 written to the MSDVAD.MDYADV by the software.
- When GETACCMST transmission is successfully completed by issuing STOP, after responding ACK to the Mastership-Request received from the Secondary Master.

[Setting conditions]

- When 1 written to the MSDVAD.MDYADV by the software.
- When GETACCMST reception is successfully completed by issuing STOP, after the ACK is responded to the Mastership-Request transmitted to the Secondary Master.

Operation Mode [I3C Secondary Master]

[Clearing condition]

- When GETACCMST transmission is successfully completed by issuing STOP, after responding ACK to the Mastership-Request received from the Non-Current Master.

[Setting condition]

- When GETACCMST reception is successfully completed by issuing STOP, after the ACK is responded to the Mastership-Request transmitted to the Current Master.

The PRSST register returns I3C current state.

State has two parts: this register which is mandatory, and an additional optional PRSST_DEBUG register intended for debug purposes (see the Debug Capability registers in the Extended Capabilities list).

TRMD bit (Transmit/Receive Mode)

This bit indicates transmit or receive mode.

I3C is in receive mode when the TRMD bit is set to 0 and is in transmit mode when the bit is set to 1. Combination of this bit and the CRMS bit indicates the operating mode of I3C.

The value of TRMD bit is automatically changed to 1 for transmit mode or 0 for receive mode by issuing or detection of a START condition and setting of the R/W# bit.

[Setting conditions]

- When a START condition is issued normally according to the START condition issuance request (when a START condition is detected with the CNDCTL.STCND bit set to 1).
- When a Repeated START condition is issued normally according to the Repeated START condition issuance request (when a Repeated START condition is detected with the CNDCTL.SRCND bit set to 1).
- When the R/W# bit added to the slave address is set to 0 in master mode.
- When the address received in slave mode matches the address enabled in SVCTL, with the R/W# bit set to 1.

[Clearing conditions]

- When a STOP condition is detected.
- The ALF (arbitration-lost) flag in BST being set to 1.
- In master mode, reception of a slave address to which an R/W# bit with the value 1 is appended.
- In slave mode, a match between the received address and the address enabled in SVCTL when the value of the received R/W# bit is 0 (including cases where the received address is the general call address).
- In slave mode, a Repeated START condition is detected (a Repeated START condition is detected with BCST.BFREF = 0 and CRMS = 0).

PRSSTWP bit (Present State Write Protect)

PRSSTWP is always 0 when reading.

When writing to PRSST, writing 1 to this bit at the same time enables writing to CRMS bit.

40.2.6 INST : Internal Status Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x030

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	INEF	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	—	These bits are read as 0. The write value should be 0.	R/W
10	INEF	Internal Error Flag 0: I3C Internal Error has not detected. 1: I3C Internal Error has detected.	R/W ¹
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports all I3C mode.

Note 1. Clearing (to 0) condition : Writing 0 after the 1 state is read.

The Interrupt Status register reflects the status of outstanding interrupts.

The status fields are either write 0 to clear, or else are cleared based on queue operations.

INEF bit (Internal Error Flag)

When this bit is 1, it indicates that I3C Internal Error has detected.

When this bit is 0, it indicates that I3C Internal Error has not detected.

[Setting conditions]

- The following 1 is satisfied and any of the following 2 to 9 are satisfied.
 1. The INSTE.INEE bit = 1
 2. When transmit data is written to the Tx Data Buffer that is completely full.
 3. When received data is read from the Rx Data Buffer that is completely empty.
 4. When Command Descriptor is written to the Command Queue that is completely full.
 5. When Response Descriptor is read from the Response Queue that is completely empty.
 6. When Receive Status Descriptor is read from the Receive Status Queue that is completely empty.
 7. When IBI Status Descriptor is read from the IBI Queue under the condition that the IBI Queue is completely empty and PRSST.CRMS = 1.
 8. When IBI Data is written to the IBI Queue under the condition that the IBI Queue is completely full and PRSST.CRMS = 0.
 9. When the Response Queue, IBI Queue or Receive Status Queue overflows.

[Clearing condition]

- When 0 is written to the INEF bit after reading INEF bit = 1.

40.2.7 INSTE : Internal Status Enable Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x034

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	INEE	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
10	INEE	Internal Error Enable 0: Disable INST.INEF 1: Enable INST.INEF	R/W
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports all I3C mode.

INEE bit (Internal Error Enable)

When this bit set to 1, it enables detection of I3C Internal Error.

When this bit set to 0, it disables detection of I3C Internal Error.

40.2.8 INIE : Internal Interrupt Enable Register

Base address: I3C = 0x4035_F000

I3C_NS = 0x5035_F000

Offset address: 0x038

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	INEIE	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	—	These bits are read as 0. The write value should be 0.	R/W
10	INEIE	Internal Error Interrupt Enable 0: Disables Non-recoverable Internal Error Interrupt Signal. 1: Enables Non-recoverable Internal Error Interrupt Signal.	R/W
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports all I3C mode.

INEIE bit (Internal Error Interrupt Enable)

When set to 1 and register INEF is set, the hardware Controller asserts an interrupt to the Host.

40.2.9 INSTFC : Internal Status Force Register

Base address: I3C = 0x4035_F000

I3C_NS = 0x5035_F000

Offset address: 0x03C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	INEFC	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	—	The write value should be 0.	W
10	INEFC	Internal Error Force 0: Not force a specific interrupt 1: Force a specific interrupt	W
31:11	—	The write value should be 0.	W

Note: S-TYPE3, P-TYPE3

Note: This register supports all I3C mode.

INEFC bit (Internal Error Force)

For debug, helps to force this interrupt.

40.2.10 DVCT : Device Characteristic Table Register

Base address: I3C = 0x4035_F000

I3C_NS = 0x5035_F000

Offset address: 0x044

Bit position:	31	30	29	28	27	26	25	24	23				19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	IDX[4:0]				—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
18:0	—	These bits are read as 0.	R
23:19	IDX[4:0]	DCT Table Index Current index of the DCT, which is used as the starting index for the I3C ENTDAACCCC.	R
31:24	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

IDX[4:0] bits (DCT Table Index)

Once the complete characteristics of device that won the arbitration are written to the DCT (during ENTDAAC using Address Assignment Command) this index is incremented by 1.

Note: How to check the progress of ENTDAAC using this bit:

1. Read the value of this bit before setting the Command Descriptor for issuing the ENTDAAC command.
2. After starting the ENTDAAC command, until the value of this bit is updated (that is, it changes from the value read in advance), it indicates that the Dynamic Address is being assigned to the device specified by the first index value (value set in DEV_INDEX[4:0] of Command Descriptor).
3. After the value of this bit is updated, it indicates that Dynamic Address is being assigned according to the value set in DEV_INDEX[4:0] and DEV_COUNT[3:0] of Command Descriptor to the device of the first index value or later.

40.2.11 IBINCTL : IBI Notify Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x058

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	NRSIR CTL	—	NRMR CTL	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	NRMRCTL	Notify Rejected Master Request Control 0: Do not pass rejected IBI Status to the Normal IBI Queue, if the incoming Master Request is NACKed and is auto-disabled based on DVMRRJ field in relevant DAT entry. 1: Pass rejected IBI Status to the Normal IBI Queue if the incoming Master Request is NACKed and is auto-disabled based on DVMRRJ field in relevant DAT entry.	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	NRSIRCTL	Notify Rejected Slave Interrupt Request Control 0: Do not pass rejected IBI Status to the Normal IBI Queue, if the incoming SIR is NACKed and is auto-disabled based on DVSIRRJ field in relevant DAT entry. 1: Pass rejected IBI Status to the Normal IBI Queue, if the incoming SIR is NACKed and is auto-disabled based on DVSIRRJ field in relevant DAT entry.	R/W
31:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

NRMRCTL bit (Notify Rejected Master Request Control)

Enables or disables reporting rejection of individual Master Requests.

NRSIRCTL bit (Notify Rejected Slave Interrupt Request Control)

Enables or disables reporting rejection of individual Slave Interrupt Requests (SIR).

40.2.12 BFCTL : Bus Function Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x060

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	HSME	FMPE	—	SMBS	—	—	—	SCSY NE	—	—	—	—	—	SALE	NALE	MALE
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	MALE	Master Arbitration-Lost Detection Enable 0: Master arbitration-lost detection disables. Disables the arbitration-lost detection function and does not clear the CRMS and TRMD bits in PRSST automatically when arbitration is lost. 1: Master arbitration-lost detection enables. Enables the arbitration-lost detection function and clears the CRMS and TRMD bits in PRSST automatically when arbitration is lost.	R/W
1	NALE	NACK Transmission Arbitration-Lost Detection Enable 0: NACK transmission arbitration-lost detection disables. 1: NACK transmission arbitration-lost detection enables.	R/W
2	SALE	Slave Arbitration-Lost Detection Enable 0: Slave arbitration-lost detection disables. 1: Slave arbitration-lost detection enables.	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
8	SCSYNE	SCL Synchronous Circuit Enable 0: No SCL synchronous circuit uses. 1: An SCL synchronous circuit uses.	R/W
11:9	—	These bits are read as 0. The write value should be 0.	R/W
12	SMBS	SMBus/I ² C Bus Selection 0: The I ² C bus select. 1: The SMBus select.	R/W
13	—	This bit is read as 0. The write value should be 0.	R/W
14	FMPE	Fast-mode Plus Enable 0: No Fm+ slope control circuit uses for the I3C_SCL pin and I3C_SDA pin. (n = 0) 1: An Fm+ slope control circuit uses for the I3C_SCL pin and I3C_SDA pin. (n = 0)	R/W
15	HSME	High Speed Mode Enable 0: Disable High Speed Mode. 1: Enable High Speed Mode.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports for I2C mode.

MALE bit (Master Arbitration-Lost Detection Enable)

This bit is used to specify whether to use the arbitration-lost detection function in master mode. Normally, set this bit to 1.

NALE bit (NACK Transmission Arbitration-Lost Detection Enable)

This bit is used to specify whether to cause arbitration to be lost when ACK is detected during transmission of NACK in receive mode (such as when slaves with the same address exist on the bus or when two or more masters select the same slave device simultaneously with different number of receive bytes).

SALE bit (Slave Arbitration-Lost Detection Enable)

This bit is used to specify whether to cause arbitration to be lost when a value different from the value being transmitted is detected on the bus in slave transmit mode (such as when slaves with the same address exist on the bus or when a mismatch with the transmit data occurs due to noise).

SCSYNE bit (SCL Synchronous Circuit Enable)

This bit is used to specify whether to synchronize the SCL clock with the SCL input clock. Normally, set this bit to 1.

When the SCSYNE bit set to 0 (no SCL synchronous circuit used), I3C does not synchronize the SCL clock with the SCL input clock. In this setting, I3C outputs the SCL clock with the transfer rate set in STDBR and EXTBR regardless of the I3C_SCL line state. For this reason, if the bus load of the I²C bus line is much larger than the specification value or if the SCL clock output overlaps in multiple masters, the short-cycle SCL clock that does not meet the specification may be output. When no SCL synchronous circuit uses, it also affects the issuance of a START condition, Repeated START condition, and STOP condition, and the continuous output of extra SCL clock cycles.

This bit must not be set to 0 except for checking the output of the set transfer rate.

FMPE bit (Fast-mode Plus Enable)

This bit is used to specify whether to use a slope control circuit for Fast-mode Plus [Fm+].

When this bit is set to 1, a slope control circuit conforming to the Fast-mode Plus [Fm+] slope control specification (tof) of the I3C-bus is selected. When this bit is set to 0, a slope control circuit conforming to the Standard-mode [Sm] and Fast-mode [fm] slope control specification (tof) of the I3C-bus is selected.

Set this bit to 1 when using the transmission rate within a range up to 1 Mbps (Fast-mode Plus [Fm+]) of the I3C-bus specification. Set this bit to 0 when using the transmission rate at other rates (up to 100 kbps [Sm], up to 400 kbps [Fm]) or for SMBus (10 to 100 kbps).

Note: When communicating in Hs-mode, set as follows.

- Set FMPE to 0 when sending Hs-mode master code (0000 1XXXb) with Fast-mode.
- Set FMPE to 1 when sending Hs-mode master code (0000 1XXXb) with Fast-mode Plus.

HSME bit (High Speed Mode Enable)

This bit is used for communicating in Hs-mode.

When this bit is set to 1, the Hs-mode master code is recognized and Hs-mode communication is possible.

After the START condition is detected, if Hs-mode master code (0000 1XXXb) transmission is recognized, Hs-mode communication starts from Repeated START after receiving the NACK response.

It communicates at the bit rate set in STDBR until the NACK response, and automatically switches from Repeated START condition issuance after receiving the NACK response to the bit rate set in EXTBR.

Hs-mode continues until a STOP condition is detected.

When the STOP condition is detected, the bit rate is automatically switched to the bit rate set in STDBR.

Note: When this bit is set to 1, the BST.NACKDF bit will not be set even if a NACK response is received after sending the Hs-mode master code.

40.2.13 SVCTL : Slave Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x064

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	16	
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SVAE[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	HOAE	—	—	—	—	—	—	—	—	DVIDE	HSMC E	—	—	—	—	GCAE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	GCAE	General Call Address Enable*1 0: General call address detection disables. 1: General call address detection enables.	R/W
4:1	—	These bits are read as 0. The write value should be 0.	R/W
5	HSMCE	Hs-mode Master Code Enable*1 0: Hs-mode Master Code Detection disables. 1: Hs-mode Master Code Detection enables.	R/W
6	DVIDE	Device-ID Address Enable*1 0: Device-ID address detection disables. 1: Device-ID address detection enables.	R/W
14:7	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
15	HOAE	Host Address Enable* ¹ 0: Host address detection disables. 1: Host address detection enables.	R/W
18:16	SVAE[2:0]	Slave Address Enable n (n = 0 to 2)* ² 0: Slave n disables 1: Slave n enables	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports I²C mode.

Note 2. These bits support I²C, I3C secondary master, and I3C slave mode.

GCAE bit (General Call Address Enable)

This bit is used to specify whether to ignore the general call address (0000 000 + 0 (write): All 0) when it is received. When this bit is set to 1, if the received slave address matches the general call address, I3C recognizes the received slave address as the general call address independently of the slave addresses set in the SVDVADn.SVAD[9:0] bits (n = 0 to 2) and performs data receive operation.

When this bit is set to 0, the received slave address is ignored even if it matches the general call address.

HSMCE bit (Hs-mode Master Code Enable)

This bit is used to specify whether to recognize and execute the Hs-mode master code (00001xxxb) is received in the first byte after a START condition is detected.

When this bit is set to 1, if the received first byte matches the Hs-mode master code, I3C recognizes that the Hs-mode master code has been received.

The first byte after Repeated START after NACK response to Hs-mode master code is recognized as a slave address and compared with the slave address set by SVDVADn.SVAD[9:0]bits (n = 0 to 2).

If the addresses match, the transmission / reception operation continues according to the R/W# bit value.

Hs-mode continues until a STOP condition is detected.

When this bit is set to 0, I3C will ignore the pattern until a STOP condition is detected, even if it matches the Hs-mode master code.

Note: When this bit is set to 1, SCSTRCTL.ACKTWE bit must be set to 0 and SCSTRCTL.RWE bit must be set to 1.

DVIDE bit (Device-ID Address Enable)

This bit is used to specify whether to recognize and execute the Device-ID address when a device ID (1111 100) is received in the first byte after a START condition or Repeated START condition is detected.

When this bit is set to 1, if the received first byte matches the Device-ID, I3C recognizes that the Device-ID address has been received. When the following R/W# bit is 0 (write), I3C recognizes the second and the following bytes as slave addresses and continues the receive operation.

When this bit is set to 0, I3C ignores the received first byte even if it matches the Device ID address and recognizes the first byte as a normal slave address.

For details on the Device-ID address detection, see [\(3\) Device-ID Address Detection \[I²C mode\]](#).

HOAE bit (Host Address Enable)

This bit is used to specify whether to ignore received host address (0001 000) when the BFCTL.SMBS bit = 1.

When this bit is set to 1 while the SMBS bit = 1, if the received slave address matches the host address, I3C recognizes the received slave address as the host address independently of the slave addresses set in the SVDVADn.SVAD[9:0] bits (n = 0 to 2) and performs the receive operation.

When the SMBS bit or the HOAE bit is set to 0, the received slave address is ignored even if it matches the host address.

SVAE[2:0] bits (Slave Address Enable n (n = 0 to 2))

This bit is used to enable or disable the slave address set in the SVDVADn.SVAD[9:0] bits.

When this bit is set to 1, the slave address set in the SVAD[9:0] bits is enabled and is compared with the received slave address.

When this bit is set to 0, the slave address set in the SVAD[9:0] bits is disabled and is ignored even if it matches the received slave address.

40.2.14 REFCKCTL : Reference Clock Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x070

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2		0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	IREFCKS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	IREFCKS[2:0] ^{*1}	Internal Reference Clock Selection Selects the internal reference clock source (I3Cφ) for I3C. 0 0 0: TCLK/1 clock 0 0 1: TCLK/2 clock 0 1 0: TCLK/4 clock 0 1 1: TCLK/8 clock 1 0 0: TCLK/16 clock 1 0 1: TCLK/32 clock 1 1 0: TCLK/64 clock 1 1 1: TCLK/128 clock	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. Set the IREFCKS[2:0] bit to 000 in I3C mode.

40.2.15 STDBR : Standard Bit Rate Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x074

Bit position:	31	30	29					24	23	22	21					16
Bit field:	DSBR PO	—						SBRHP[5:0]	—	—					SBRLP[5:0]	
Value after reset:	0	0	1	1	1	1	1	1	0	0	1	1	1	1	1	1
Bit position:	15							8	7							0
Bit field:								SBRHO[7:0]							SBRLO[7:0]	
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
7:0	SBRLO[7:0]	Standard Bit Rate Low-Level Period Open-Drain Count value of the low-level period of SCL clock ^{*1}	R/W
15:8	SBRHO[7:0]	Standard Bit Rate High-Level Period Open-Drain Count value of the high-level period of SCL clock ^{*1}	R/W

Bit	Symbol	Function	R/W
21:16	SBRLP[5:0]	Standard Bit Rate Low-level Period Push-Pull* ² Count value of the low-level period of SCL clock	R/W
23:22	—	These bits are read as 0. The write value should be 0.	R/W
29:24	SBRHP[5:0]	Standard Bit Rate High-Level Period Push-Pull* ³ Count value of the high-level period of SCL clock	R/W
30	—	This bit is read as 0. The write value should be 0.	R/W
31	DSBRPO	Double the Standard Bit Rate Period for Open-Drain* ⁴ 0: The time period set for SBRHO[7:0] and SBRLO[7:0] is not doubled. 1: The time period set for SBRHO[7:0] and SBRLO[7:0] is doubled.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. These bits support I²C, I3C master, and I3C secondary master mode.

Note 2. These bits support I3C master mode and I3C secondary master mode.

Note 3. These bits support all I3C mode.

Note 4. This bit supports I²C, I3C master, and I3C secondary master mode.

The STDBR register sets the bit rate according to the operating speed.

- I²C mode: Bit rate setting when communicating with Standard-mode / Fast-mode / Fast-mode plus
- I3C master mode: Bit rate setting selected by MODE bit of command descriptor
- I3C slave mode: I3C bit rate setting

The I²C transfer rate and the SCL clock duty are calculated using the following expression.

$$\text{Transfer rate} = 1 / \{[(\text{High-Level Period} + \alpha^*1) + (\text{Low-Level Period} + \alpha)] / \text{I3C}\phi^2 + \text{I3C_SCL line rising time [tr]}^3 + \text{I3C_SCL line falling time [tf]}^3\}$$

$$\text{Duty cycle} = \{\text{I3C_SCL line rising time [tr]} + (\text{High-Level Period} + \alpha) / \text{I3C}\phi\} / \{\text{I3C_SCL line falling time [tf]} + (\text{Low-Level Period} + \alpha) / \text{I3C}\phi\}$$

Note 1. α depend on the number of stages in the noise filter.

Note 2. $\text{I3C}\phi = \text{TCLK} \times \text{Division ratio}$

Note 3. The I3C_SCL line rising time [tr] and I3C_SCL line falling time [tf] depend on the total bus line capacitance [Cb] and the pull-up resistor [Rp]. For details, see the I²C-bus specification from NXP Semiconductors.

The I3C transfer rate and the SCL clock duty are calculated using the following expression.

$$\text{Transfer rate} = 1 / [(\text{High-Level Period} + \text{Low-Level Period}) / \text{I3C}\phi + \text{I3C_SCL line rising time [tr]} + \text{I3C_SCL line falling time [tf]}]$$

$$\text{Duty cycle} = [\text{I3C_SCL line rising time [tr]} + \text{High-Level Period} / \text{I3C}\phi] / [\text{I3C_SCL line falling time [tf]} + \text{Low-Level Period} / \text{I3C}\phi]$$

SBRLO[7:0] bits (Standard Bit Rate Low-Level Period Open-Drain)

The SBRLO[7:0] bits are used to set the low-level period of SCL clock in Open-Drain mode.

I3C counts the low-level period with the internal reference clock source (I3C ϕ) specified by the REFCKCTL.IREFCKS[2:0] bits. It also works to generate the data setup time for automatic SCL low-hold operation (see [section 40.3.2.3.6. Clock Stretching \[I²C Mode\]](#)); when I3C is used in I²C slave mode, these bits need to be set to a value longer than the data setup time*¹.

If the digital noise filter is enabled (INCTL.DNFE = 1), set the SBRLO[7:0] bits to a value at least one greater than the number of stages in the noise filter. Regarding the number of stages in the noise filter, see the description of the INCTL.DNFS[3:0] bits.

Note 1. Data setup time (t_{SU}: DAT)

250 ns (up to 100 kbps: Standard-mode [Sm])

100 ns (up to 400 kbps: Fast-mode [Fm])

50 ns (up to 1 Mbps: Fast-mode plus [Fm+])

10 ns (up to 3.4 Mbps: Hs-mode [HS])

SBRHO[7:0] bits (Standard Bit Rate High-Level Period Open-Drain)

The SBRHO[7:0] bits are used to set the high-level period of SCL clock in Open-Drain mode. SBRHO[7:0] bits are valid in master mode. If I3C is used only in I²C slave mode, these bits need not set the high-level period.

I3C counts the high-level period with the internal reference clock source (I3C ϕ) specified by the REFCKCTL.IREFCKS[2:0] bits.

If the digital noise filter is enabled (the INCTL.DNFE bit = 1), set the SBRHO[7:0] bits to a value at least one greater than the number of stages in the noise filter. Regarding the number of stages in the noise filter, see the description of the INCTL.DNFS[3:0] bits.

SBRLP[5:0] bits (Standard Bit Rate Low-level Period Push-Pull)

SBRLP[5:0] bits are used to set the low-level period of SCL clock in Push-Pull.

I3C counts the low-level period with the internal reference clock source (I3C ϕ) specified by the REFCKCTL.IREFCKS[2:0] bits.

If the digital noise filter is enabled (the INCTL.DNFE bit = 1), set the SBRLP[5:0] bits to a value at least one greater than the number of stages in the noise filter. Regarding the number of stages in the noise filter, see the description of the INCTL.DNFS[3:0] bits.

SBRHP[5:0] bits (Standard Bit Rate High-Level Period Push-Pull)

SBRHP[5:0] bits is used to set the high-level period of SCL clock in Push-Pull mode.

SBRHP[5:0] bits are valid in master mode. If I3C is used only in I²C slave mode, these bits need not set the high-level period.

I3C counts the high-level period with the internal reference clock source (I3C ϕ) specified by the REFCKCTL.IREFCKS[2:0] bits.

If the digital noise filter is enabled (the INCTL.DNFE bit = 1), set the SBRHP[5:0] bits to a value at least one greater than the number of stages in the noise filter. Regarding the number of stages in the noise filter, see the description of the INCTL.DNFS[3:0] bits.

DSBRPO bit (Double the Standard Bit Rate Period for Open-Drain)

When DSBRPO = 1, double the high-level period that is set in SBRHO[7:0] and double the low-level period that is set in SBRLO[7:0].

Table 40.4 Requirement and usage of setting in each mode

Bit name	Device mode				
	I ² C master	I ² C slave	I3C Master	I3C Secondary Master	I3C Slave
SBRHP[5:0]	Do not use	Do not use	Setting required*3	Setting required*4	Setting required*6
SBRLP[5:0]	Do not use	Do not use	Setting required*3	Setting required*5	Do not use
SBRHO[7:0]	Setting required*1	Do not use	Setting required*3	Setting required*5	Do not use
SBRLO[7:0]	Setting required*1	Setting required*2	Setting required*3	Setting required*5	Do not use

Note 1. The setting value is used for the data rate of ST, FM, and FM+ mode.

Note 2. The setting value is used for the data setup time of automatic SCL low-hold operation.

Note 3. The setting value is used for the data rate of each communication.

Note 4. When operating with I3C Master, the setting value is used for the data rate of each communication.
When operating with I3C Slave, the setting value is used for the data rate of HDR-TSP/TSL mode.

Note 5. When operating with I3C Master, the setting value is used for the data rate of each communication.
When operating with I3C Slave, do not use.

Note 6. The setting value is used for the data rate of HDR-TSP/TSL mode.

40.2.16 EXTBR : Extended Bit Rate Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x078

Bit position:	31	29	24	21	16	15	8	7	0
Bit field:	—	—	EBRHP[5:0]	—	—	EBRLP[5:0]	EBRHO[7:0]	EBRLO[7:0]	
Value after reset:	0	0	1	1	1	1	1	0	0

Bit	Symbol	Function	R/W
7:0	EBRLO[7:0]	Extended Bit Rate Low-Level Period Open-Drain* ¹ Count value of the low-level period of SCL clock	R/W
15:8	EBRHO[7:0]	Extended Bit Rate High-Level Period Open-Drain* ¹ Count value of the high-level period of SCL clock	R/W
21:16	EBRLP[5:0]	Extended Bit Rate Low-Level Period Push-Pull* ² Count value of the low-level period of SCL clock	R/W
23:22	—	These bits are read as 0. The write value should be 0.	R/W
29:24	EBRHP[5:0]	Extended Bit Rate High-Level Period Push-Pull* ² Count value of the high-level period of SCL clock	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. These bits support I²C, I3C master, and I3C secondary master mode.

Note 2. These bits support I3C master mode and I3C secondary master mode.

The EXTBR register sets the bit rate according to the operating speed.

- I²C mode: Bit rate setting for communicating in high-speed mode
- I3C master mode: Bit rate setting selected by MODE bit of command descriptor
- I3C slave mode: unused

EBRLO[7:0] bits (Extended Bit Rate Low-Level Period Open-Drain)

See SBRLO[7:0] bits of [section 40.2.15. STDBR : Standard Bit Rate Register](#) for details. Watch SBRHO, SBRLO as EBRHO[7:0], EBRLO[7:0].

EBRHO[7:0] bits (Extended Bit Rate High-Level Period Open-Drain)

See SBRHO[7:0] bits of [section 40.2.15. STDBR : Standard Bit Rate Register](#) for details. Watch SBRHO, SBRLO as EBRHO[7:0], EBRLO[7:0].

EBRLP[5:0] bits (Extended Bit Rate Low-Level Period Push-Pull)

See SBRLP[5:0] bits of [section 40.2.15. STDBR : Standard Bit Rate Register](#) for details. Watch SBRHP, SBRLP as EBRHP[5:0], EBRLP[5:0].

EBRHP[5:0] bits (Extended Bit Rate High-Level Period Push-Pull)

See SBRHP[5:0] bits of [section 40.2.15. STDBR : Standard Bit Rate Register](#) for details. Watch SBRHP, SBRLP as EBRHP[5:0], EBRLP[5:0].

Table 40.5 Requirement and usage of setting in each mode (1 of 2)

Bit name	Device mode				
	I ² C master	I ² C slave	I3C Master	I3C Secondary Master	I3C Slave
EBRHP[5:0]	do not use	do not use	Setting required* ³	Setting required* ⁴	do not use
EBRLP[5:0]	do not use	do not use	Setting required* ³	Setting required* ⁴	do not use

Bit name	Device mode				
	I ² C master	I ² C slave	I3C Master	I3C Secondary Master	I3C Slave
EBRHO[7:0]	Setting required*1	do not use	Setting required*3	Setting required*4	do not use
EBRLO[7:0]	Setting required*1	Setting required*2	Setting required*3	Setting required*4	do not use

Note 4. When operating with I3C Master, the setting value is used for the data rate of each communication.
When operating with I3C Slave, do not use.

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These bits set the Bus Available period. This Bus Available period is counted by the internal reference clock (I3C ϕ) selected by the REFCKCTL.IREFCKS[2:0] bits. See the BCST.BAVLF flag for Bus Available detection behavior.

40.2.19 BIDLCDT : Bus Idle Condition Detection Time Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x084

Bit position:	31	18	17	0
Bit field:	—	—	—	IDLCYC[17:0]
Value after reset:	0	0	0	0

Bit	Symbol	Function	R/W
17:0	IDLCYC[17:0]	Bus Idle Condition Detection Cycle The count value is a period for detecting the Bus idle condition.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports all I3C mode.

IDLCYC[17:0] bits (Bus Idle Condition Detection Cycle)

I3C counts the period for detecting the Bus idle condition with the I3C ϕ .

These bits set the Bus Idle period. This Bus Idle period is counted by the internal reference clock (I3C ϕ) selected by the REFCKCTL.IREFCKS[2:0] bits. See the BCST.BIDLf flag for Bus Available detection behavior.

40.2.20 OUTCTL : Output Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x088

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	8	7	6	5	4	3	2	1	0
Bit field:	SDOD CS	—	—	—	—	SDOD[2:0]	—	—	—	EXCY C	—	SOCW P	SCOC	SDOC	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	1	1	

Bit	Symbol	Function	R/W
0	SDOC	SDA Output Control*1 0: I3C drives the I3C_SDA pin low. 1: I3C releases the I3C_SDA pin.	R/W
1	SCOC	SCL Output Control*1 High level output is achieved through an external pull-up resistor. 0: I3C drives the I3C_SCL pin low. 1: I3C releases the I3C_SCL pin.	R/W
2	SOCWP	SCL/SDA Output Control Write Protect*1 0: Bits SCOC and SDOC are protected. 1: Bits SCOC and SDOC can be written (When writing simultaneously with the value of the target bit). This bit is read as 0.	W
3	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
4	EXCYC	Extra SCL Clock Cycle Output ^{*3} The EXCYC bit is cleared automatically after one clock cycle is output. 0: Does not output an extra SCL clock cycle (default). 1: Outputs an extra SCL clock cycle.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
10:8	SDOD[2:0]	SDA Output Delay ^{*2} 0 0 0: No output delay 0 0 1: 1 I3C ϕ cycle (When OUTCTL.SDODCS = 0 (I3C ϕ)) 1 or 2 I3C ϕ cycles (When OUTCTL.SDODCS = 1 (I3C ϕ /2)) 0 1 0: 2 I3C ϕ cycles (When OUTCTL.SDODCS = 0 (I3C ϕ)) 3 or 4 I3C ϕ cycles (When OUTCTL.SDODCS = 1 (I3C ϕ /2)) 0 1 1: 3 I3C ϕ cycles (When OUTCTL.SDODCS = 0 (I3C ϕ)) 5 or 6 I3C ϕ cycles (When OUTCTL.SDODCS = 1 (I3C ϕ /2)) 1 0 0: 4 I3C ϕ cycles (When OUTCTL.SDODCS = 0 (I3C ϕ)) 7 or 8 I3C ϕ cycles (When OUTCTL.SDODCS = 1 (I3C ϕ /2)) 1 0 1: 5 I3C ϕ cycles (When OUTCTL.SDODCS = 0 (I3C ϕ)) 9 or 10 I3C ϕ cycles (When OUTCTL.SDODCS = 1 (I3C ϕ /2)) 1 1 0: 6 I3C ϕ cycles (When OUTCTL.SDODCS = 0 (I3C ϕ)) 11 or 12 I3C ϕ cycles (When OUTCTL.SDODCS = 1 (I3C ϕ /2)) 1 1 1: 7 I3C ϕ cycles (When OUTCTL.SDODCS = 0 (I3C ϕ)) 13 or 14 I3C ϕ cycles (When OUTCTL.SDODCS = 1 (I3C ϕ /2))	R/W
14:11	—	These bits are read as 0. The write value should be 0.	R/W
15	SDODCS	SDA Output Delay Clock Source Selection ^{*3} 0: The internal reference clock (I3C ϕ) is selected as the clock source of the SDA output delay counter. 1: The internal reference clock divided by 2 (I3C ϕ /2) is selected as the clock source of the SDA output delay counter. ^{*4}	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports I²C, I3C master, and I3C secondary master mode.

Note 2. These bits support I²C mode.

Note 3. This bit supports I²C mode.

Note 4. The setting SDODCS = 1 (I3C ϕ /2) only becomes valid when SCL is at the low level. When SCL is at the high level, the setting SDODCS = 1 becomes invalid and the clock source becomes the internal reference clock (I3C ϕ).

SDOC bit (SDA Output Control) and SCOC bit (SCL Output Control)

These bits are used to directly control the I3C_SDA and I3C_SCL signals output from this I3C.

When writing to these bits, also write 1 to the SOCWP bit at the same time.

The result of setting these bits is input to I3C via the input buffer. When slave mode is selected, a START condition may be detected and the bus may be released depending on the bit settings.

Do not rewrite these bits during a START condition, STOP condition, Repeated START condition, or during transmission or reception. Operation after rewriting under the above conditions is not guaranteed.

EXCYC bit (Extra SCL Clock Cycle Output)

This bit is used to output an extra SCL clock cycle for debugging or error processing.

Normally, set the bit to 0. Setting the bit to 1 in a normal communication state causes a communication error.

For details on this function, see [section 40.3.2.3.11. Port Control](#), (1) Extra SCL Clock Cycle Output Function.

40.2.21 INCTL : Input Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x08C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3			0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	DNFE	DNFS[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	1	1	0	1	0	0	0	0

Bit	Symbol	Function	R/W
3:0	DNFS[3:0]	Digital Noise Filter Stage Selection 0x0: Noise of up to one I3C ϕ cycle is filtered out (single-stage filter). 0x1: Noise of up to two I3C ϕ cycles is filtered out (2-stage filter). 0x2: Noise of up to three I3C ϕ cycles is filtered out (3-stage filter). 0x3: Noise of up to four I3C ϕ cycles is filtered out (4-stage filter). 0x4: Noise of up to five I3C ϕ cycles is filtered out (5-stage filter). ⋮ 0xF: Noise of up to sixteen I3C ϕ cycles is filtered out (16-stage filter).	R/W
4	DNFE	Digital Noise Filter Circuit Enable 0: No digital noise filter circuit is used. 1: A digital noise filter circuit is used.	R/W
5	—	This bit is read as 0. The write value should be 0.	R/W
7:6	—	These bits are read as 1. The write value should be 1.	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I²C mode.

DNFS[3:0] bits (Digital Noise Filter Stage Selection)

These bits are used to select the number of stages in the digital noise filter.

For details on the digital noise filter function, see [section 40.3.2.6.3. Digital Noise-Filter Circuits \[I²C Mode\]](#).

In I²C High Speed mode, I3C changes the number of noise filter stage to a quarter of the number of noise filter stage automatically.

- Note:
- Set the noise range to be filtered out by the noise filter within a range less than the I3C_SCL line high-level period or low-level period. If the noise range is set to a value of (SCL clock width: high-level period or low-level period, whichever is shorter) - [1.5 internal reference clock (I3C ϕ) cycles] or more, the SCL clock is regarded as noise by the noise filter function of I3C, which may prevent I3C from operating normally.
 - In I²C High Speed mode, the lower 2 bits of the DNFS [3:0] bits are ignored, and the number of filter stages for 1 to 4 stages is selected by the upper 2 bits.

40.2.22 TMOCTL : Timeout Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x090

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	TOMDS[1:0]		TOHC TL	TOLC TL	—	—	TODTS[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0

Bit	Symbol	Function	R/W
1:0	TODTS[1:0]	Timeout Detection Time Selection 0 0: 16bit-timeout 0 1: 14bit-timeout 1 0: 8bit-timeout 1 1: 6bit-timeout	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	TOLCTL	Timeout L Count Control 0: Count is disabled while the I3C_SCL line is at a low level. 1: Count is enabled while the I3C_SCL line is at a low level.	R/W
5	TOHCTL	Timeout H Count Control 0: Count is disabled while the I3C_SCL line is at a high level. 1: Count is enabled while the I3C_SCL line is at a high level.	R/W
7:6	TOMDS[1:0]	Timeout Operation Mode Selection 0 0: Timeout is detected during the following conditions: - The bus is busy (BCST.BFREF = 0) in master mode. - I3C's own slave address is detected and the bus is busy in slave mode. - The bus is free (BCST.BFREF = 1) while generation of a START condition is requested (CNDCTL.STCND = 1). 0 1: Timeout is detected while the bus is busy. 1 0: Timeout is detected while the bus is free. 1 1: Setting prohibited	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

TODTS[1:0] bits (Timeout Detection Time Selection)

These bits are used to select for the timeout detection time when the timeout function is enabled (BSTE.TODE bit = 1).

When these bits are set to 00b, the timeout detection internal counter functions as a 16-bit counter.

When these bits are set to 01b, the counter functions as a 14-bit counter.

When these bits are set to 10b, the counter functions as an 8-bit counter.

When these bits are set to 11b, the counter functions as a 6-bit counter.

While the I3C_SCL line is in the state that enables this counter as specified by bits TOHCTL and TOLCTL, the counter counts up in synchronization with the internal reference clock (I3Cφ) as a count source.

For details on the timeout function, see [section 40.3.2.4.5. Timeout Error Detection](#).

TOLCTL bit (Timeout L Count Control)

This bit is used to enable or disable the internal counter of the timeout function to count up while the I3C_SCL line is held low when the timeout function is enabled (BSTE.TODE = 1).

TOHCTL bit (Timeout H Count Control)

This bit is used to enable or disable the internal counter of the timeout function to count up while the I3C_SCL line is held high when the timeout function is enabled (BSTE.TODE = 1).

TOMDS[1:0] bits (Timeout Operation Mode Selection)

These bits are used to select the detection condition for timeout when the timeout function is enabled.

Note: When working with I²C Slave, during 10-bit address communication, the timeout count starts when the upper address match is detected.

40.2.23 WUCTL : Wake Up Unit Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x098

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	WUFE	WUFS YNE	—	WUAN FS	—	—	—	WUAC KS
Value after reset:	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	WUACKS	Wake-Up Acknowledge Selection* ¹ Choice of four response mode with a combination of RSTCTL.INTLRST bit and WUACKS bit. Shown in Table 40.6 .	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	WUANFS	Wake-Up Analog Noise Filter Selection* ¹ 0: Do not add the Wake-up analog filter. 1: Add the Wake-up analog filter.	R/W
5	—	This bit is read as 0. The write value should be 0.	R/W
6	WUFSYNE	Wake-Up function PCLK Synchronous Enable 0: I3C asynchronous circuit enable 1: I3C synchronous circuit enable	R/W
7	WUFE	Wake-up function Enable Do not set WUFE = 0 during Wake-up operation. 0: Wake-up function disables 1: Wake-up function enables	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports I²C mode.

Table 40.6 Wake-Up Mode

INTLRST	WUACKS	Operation mode	Description
0	0	Normal Wake-Up mode 1	ACK response at 9th SCL and SCL low hold after at 9th SCL.
0	1	Normal Wake-Up mode 2	No ACK response immediately and SCL low hold between 8th and 9th SCL. Release SCL low hold and ACK response at 9th SCL.
1	0	Command recovery mode	ACK response at 9th SCL and not SCL low hold.
1	1	EEP response mode	NACK response at 9th SCL and not SCL low hold.

Note: In Wake-Up mode 2, HS mode cannot be used.

WUFSYNE bit (Wake-Up function PCLK Synchronous Enable)

This bit is used to switch between the PCLK synchronous operation and the PCLK asynchronous operation.

The bit is used in combination with the WUASYNF flag at Wake-Up effective function (WUCTL.WUFE bit = 1).

[When switching from the PCLK synchronous operation to the PCLK asynchronous operation]

I3C operation changes into the PCLK asynchronous operation during BCST.BFREF flag = 1, when the WUASYNF flag set to 1 during WUFSYNE = 0.

The reception can operate without depending on the state of operation of PCLK (With PCLK stopped) after it switches to the PCLK asynchronous operation (Wake-Up event detection operation).

[When switching from the PCLK asynchronous operation to the PCLK synchronous operation]

I3C operation changes into the PCLK synchronous operation at the following conditions. (At the same timing when WUFSYNE flag becomes 0)

In the case Wake-Up event detects : right after WUFSYNE bit is set to 1.

In the case Wake-Up event does not detect : when STOP condition is detected after WUFSYNE bit is set to 1.

[Setting condition]

- When 1 is written to the WUFSYNE bit.
- WUCTL.WUFE = 0

[Clearing conditions]

- When 0 is written to the WUFSYNE bit.

40.2.24 ACKCTL : Acknowledge Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x0A0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	ACKT WP	ACKT	ACKR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ACKR	Acknowledge Reception 0: A 0 is received as the acknowledge bit (ACK reception). 1: A 1 is received as the acknowledge bit (NACK reception).	R
1	ACKT	Acknowledge Transmission 0: A 0 is sent as the acknowledge bit (ACK transmission). 1: A 1 is sent as the acknowledge bit (NACK transmission).	R/W
2	ACKTWP	ACKT Write Protect 0: The ACKT bit is protected. 1: The ACKT bit can be written (when writing simultaneously with the value of the target bit). This bit is read as 0.	W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I²C mode.

ACKR bit (Acknowledge Reception)

This bit is used to store the acknowledge bit information received from the receive device in transmit mode.

[Setting condition]

- When 1 is received as the acknowledge bit with the PRSST.TRMD bit set to 1.

[Clearing condition]

- When 0 is received as the acknowledge bit with the PRSST.TRMD bit set to 1.

ACKT bit (Acknowledge Transmission)

[Setting condition]

- When 1 is written to the ACKT bit and 1 is written to the ACKTWP bit at the same time.

[Clearing conditions]

- When 0 is written to the ACKT bit and 1 is written to the ACKTWP bit at the same time.
- When a STOP condition is detected. (when a STOP condition is detected with the CNDCTL.SPCND bit set to 1.)

Note: Set the ACKT bit to 0 in I²C Slave mode.

ACKTWP bit (ACKT Write Protect)

This bit is used to control the modification of the ACKT bit.

When changing the ACKT bit, setting this bit to 1 at the same time can change the ACKT bit.

When this bit is read, 0 is always read.

40.2.25 SCSTRCTL : SCL Stretch Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x0A4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RWE	ACKTWE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ACKTWE	Acknowledge Transmission Wait Enable 0: NTST.RDBFF0 is set at the rising edge of the ninth SCL clock cycle. (The I3C_SCL line is not held low at the falling edge of the eighth clock cycle.) 1: NTST.RDBFF0 is set at the rising edge of the eighth SCL clock cycle. (The I3C_SCL line is held low at the falling edge of the eighth clock cycle.) Low-hold is released by writing a value to the ACKCTL.ACKT bit.	R/W
1	RWE	Receive Wait Enable 0: No WAIT (The period between ninth clock cycle and first clock cycle is not held low.) 1: WAIT (The period between ninth clock cycle and first clock cycle is held low.) Low-hold is released by reading NTDTBP0.	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I²C mode.

ACKTWE bit (Acknowledge Transmission Wait Enable)

This bit is used to select the NTST.RDBFF0 flag set timing in receive mode and also to select whether to hold the I3C_SCL line low at the falling edge of the eighth SCL clock cycle.

When ACKTWE = 0, the I3C_SCL line is not held low at the falling edge of the eighth SCL clock cycle, and the NTST.RDBFF0 flag is set to 1 at the rising edge of the ninth SCL clock cycle.

When ACKTWE = 1, the NTST.RDBFF0 flag is set to 1 at the rising edge of the eighth SCL clock cycle and the I3C_SCL line is held low at the falling edge of the eighth SCL clock cycle. The low-hold of the I3C_SCL line is released by writing a value to the ACKCTL.ACKT bit.

After data is received with this setting, the I3C_SCL line is automatically held low before the acknowledge bit is sent. This enables processing to send ACK (ACKCTL.ACKT = 0) or NACK (ACKCTL.ACKT = 1) according to receive data.

RWE bit (Receive Wait Enable)

This bit is used to control whether to hold the period between the ninth SCL clock cycle and the first SCL clock cycle low until the receive data buffer (NTDTBP0) is completely read each time single-byte data is received in receive mode.

When RWE = 0, the receive operation is continued without holding the period between the ninth and the first SCL clock cycle low. When both the ACKTWE and RWE bits = 0, continuous receive operation is enabled with the double buffer.

When RWE = 1, the I3C_SCL line is held low from the falling edge of the ninth clock cycle until the NTDTBP0 value is read each time single-byte data is received.

This enables receive operation in byte units.

Note: When the value of the RWE bit is to be read, be sure to read the NTDTBP0 beforehand.

40.2.26 SCSTLCTL : SCL Stalling Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x0B0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ACKP E	PARP E	—	AAPE	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15															0
Bit field:	STLCYC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	STLCYC[15:0]	Stalling Cycle Counter setting of stall period (I3Cφ cycle). Common use for each phase.	R/W
27:16	—	These bits are read as 0. The write value should be 0.	R/W
28	AAPE	Assigned Address Phase Enable Enable bit that allows stall by the first bit at address assignment 0: Does not stall the SCL clock during the address assignment phase. 1: Stall the SCL clock during address assignment phase.	R/W
29	—	This bit is read as 0. The write value should be 0.	R/W
30	PARPE	Parity Phase Enable Stall enable bit in parity bit period 0: Does not stall the SCL clock during the parity bit period. 1: Stall the SCL clock during the parity bit period.	R/W
31	ACKPE	ACK phase Enable Stall enable bit during ACK/NACK phase 0: Does not stall the SCL clock during the ACK/NACK phase. 1: Stall the SCL clock during the ACK/NACK phase.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

When setting this register, follow Chapter 5.1.2.5 Master Clock Stalling of MIPI I3C Spec V1.0, and use it only when necessary because of its negative impacts on bus performance.

STLCYC[15:0] bits (Stalling Cycle)

These bits set the SCL stall period. The SCL stall period is counted by the internal reference clock (I3C ϕ). This is a counter common to the enable bits of each phase.

AAPE bit (Assigned Address Phase Enable)

The master can stall SCL during the low period of the first bit of the assigned address phase of the Enter Dynamic Address Assignment CCC command. It can gain time in assigning dynamic address to the device based on the BCR and DCR of the slave. However, because the Dynamic Address Assignment procedure sends the dynamic address set in the DATBASm (m = 0 to 7) register in sequence, it is not necessary to set this bit and it is prohibited.

PARPE bit (Parity Phase Enable)

The parity bit of the transmission data of I3C write transfer can be used for SCL stalling to avoid underrun of the transmission data FIFO. However, when the transmission data FIFO of the I3C master becomes empty, SCL stalling is performed regardless of the setting of this bit, it is not necessary to set this bit and it is prohibited. It is necessary to set this bit when the I3C slave requires preparation time to receive data.

ACKPE bit (ACK phase Enable)

Determine the need to perform SCL stalling in the ACK/NACK phase based on the following criteria:

- It is necessary to set this bit when the I3C and I2C slaves connected to the bus require preparation time to receive or transmit data.
- In legacy I²C communication, if there is a possibility that the data FIFO of the I3C master might underrun or overflow, it is not necessary to set this bit because SCL Stalling is performed by FIFO Empty or Full regardless of the setting of this bit.
- Other than legacy I²C communication, the data FIFO of I3C master might underrun or overflow, and if SCL stalling is required in ACK phase, this bit can be set. However, it is necessary to build the software so that the FIFO does not underrun or overflow due to the interrupt generated according to the FIFO threshold setting (NQTHCTL, NTBTHCTL0, NRQTHCTL, HQTHCTL, HTBTHCTL).
- When I3C master responds ACK/NACK to IBI, it is not necessary to set this bit because ACK/NACK response can be set in advance by BCTL.HJACK, DATBASm.DVMRRJ and DATBASm.DVS IRRJ (m = 0 to 7).
- It is necessary to set this bit when the I3C slave connected to the bus requires preparation time to transmit data for Direct GET CCC.

40.2.27 SVTDLG0 : Slave Transfer Data Length Register 0

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x0C0

Bit position: 31

16 15

0

Bit field:

STDLG[15:0]

Value after reset: 0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
31:16	STDLG[15:0]	Slave Transfer Data Length Indicates the number of bytes to be transferred.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

40.2.28 STCTL : Synchronous Timing Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x120

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STOE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STOE	Synchronous Timing output Enable 0: Disable 1: Enable	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports all I3C mode.

40.2.29 ATCTL : Asynchronous Timing Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x124

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CDIV[7:0]								—	—	—	—	—	AMEOE	MREFOE	ATTRGS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ATTRGS	Asynchronous Timing Trigger Select* ¹ 0: Software trigger 1: Hardware trigger	R/W
1	MREFOE	MREF Output Enable (Capture Event / Counter Overflow)* ² 0: Disable 1: Enable	R/W
2	AMEOE	Additional Master-initiated bus Event Output Enable* ² 0: Disable 1: Enable	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
15:8	CDIV[7:0]	TCLK Counter Divide Setting* ³	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports I3C secondary master mode and I3C slave mode.

Note 2. This bit supports I3C master mode and I3C secondary master mode.

Note 3. These bits support all I3C mode.

40.2.30 ATTRG : Asynchronous Timing Trigger Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x128

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ATST RG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ATSTRG	Asynchronous Timing Software Trigger 0: Do nothing 1: Software trigger (one-shot pulse) output This bit is always read as 0.	W
31:1	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

40.2.31 ATCCNTE : Asynchronous Timing Control Counter Enable Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x12C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ATCE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ATCE	Asynchronous Timing Counter Enable for MREF, MC2, SC1, SC2. 0: Disable 1: Enable	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports all I3C mode.

40.2.32 CNDCTL : Condition Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x140

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SPCND	SRCND	STCND
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STCND	START (S) Condition Issuance 0: Does not request to issue a START condition. 1: Requests to issue a START condition.	R/W
1	SRCND	Repeated START (Sr) Condition Issuance 0: Does not request to issue a Repeated START condition. 1: Requests to issue a Repeated START condition.	R/W
2	SPCND	STOP (P) Condition Issuance 0: Does not request to issue a STOP condition. 1: Requests to issue a STOP condition.	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I²C mode.

STCND bit (START (S) Condition Issuance)

This bit is used to request transition to master mode and issuance of a START condition.

For details on the START condition issuance, see [section 40.3.2.3.3. START Condition / Repeated START Condition / STOP Condition Issuing Function](#).

[Setting condition]

- When 1 is written to the STCND bit

[Clearing conditions]

- When 0 is written to the STCND bit
- When a START condition has been issued (A START condition is detected)
- When the BST.ALF (arbitration-lost) flag is set to 1

Note: Set the STCND bit to 1 (START condition issuance request) when the BCST.BFREF flag is set to 1 (bus free state).

Note that arbitration may be lost due to a START condition issuance error if the STCND bit is set to 1 (START condition issuance request) when the BFREF flag is set to 0 (bus busy state).

SRCND bit (Repeated START (Sr) Condition Issuance)

This bit is used to request that a Repeated START condition be issued in master mode.

When this bit is set to 1 to request to issue a Repeated START condition, a Repeated START condition is issued when the BFREF flag is set to 0 (bus busy state) and the PRSST.CRMS bit is set to 1 (master mode).

For details on the Repeated START condition issuance, see [section 40.3.2.3.3. START Condition / Repeated START Condition / STOP Condition Issuing Function](#).

[Setting condition]

- When 1 is written to the SRCND bit with the BCST.BFREF flag set to 0

[Clearing conditions]

- When 0 is written to the SRCND bit
- When a Repeated START condition has been issued (A Repeated START condition is detected)
- When the BST.ALF (arbitration-lost) flag is set to 1

Note: Do not set the SRCND bit to 1 while issuing a STOP condition.

Note: If 1 (requests to issue a Repeated START condition) is written to the SRCND bit in slave mode, the Repeated START condition is not issued but the SRCND bit remains set to 1.

If the operating mode changes to master mode with the bit not being cleared, note that the Repeated START condition may be issued.

SPCND bit (STOP (P) Condition Issuance)

This bit is used to request that a STOP condition be issued in master mode.

When this bit is set to 1 to request to issue a STOP condition, a STOP condition is issued when the BCST.BFREF flag is set to 0 (bus busy state) and the PRSST.CRMS bit is set to 1 (master mode).

For details on the STOP condition issuance, see [section 40.3.2.3.3. START Condition / Repeated START Condition / STOP Condition Issuing Function](#).

[Setting condition]

- When 1 is written to the SPCND bit with the BCST.BFREF flag set to 0 and the PRSST.CRMS bit set to 1

[Clearing conditions]

- When 0 is written to the SPCND bit
- When a STOP condition has been issued (A STOP condition is detected)
- When the BST.ALF (arbitration-lost) flag is set to 1
- When a START condition and a Repeated START condition are detected

Note: Writing to the SPCND bit is not possible while the setting of the BCST.BFREF flag = 1 (bus free state).

Note: Do not set the SPCND bit to 1 while a Repeated START condition is being issued.

40.2.33 NCMDQP : Normal Command Queue Port Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x150

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Normal Command Queue Port	W

Note: S-TYPE3, P-TYPE3

Note: This register supports all I3C mode.

32-bit mailbox register NCMDQP contains a command descriptor structure that depends on the requested transfer type:

1. Address Assignment Command (see [section 40.3.1.1.1. Address Assign Command](#))
2. Immediate Data Transfer (see [section 40.3.1.1.2. Immediate Transfer Command](#))
3. Regular Data Transfer (see [section 40.3.1.1.3. Regular Transfer Command](#))
4. Write + Write/Read Combo Transfer (see [section 40.3.1.1.4. Combo Transfer Command](#))
5. Internal Control Command (see [section 40.3.1.1.5. Internal Control Command](#))

Within the command descriptor, DWORDs appear starting with the Least Significant DWORD, in order until the Most Significant DWORD.

40.2.34 NRSPQP : Normal Response Queue Port Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x154

Bit position: 31 0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Normal Response Queue Port	R

Note: S-TYPE3, P-TYPE3

Note: This register supports all I3C mode.

32-bit mailbox register NRSPQP contains a response structure (see [section 40.3.1.4. Receive Status Descriptor](#)).

40.2.35 NTDTBP0/NTDTBP0_BY : Normal Transfer Data Buffer Port Register 0

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x158

Bit position: 31 0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Normal Transfer Data Buffer Port NTDTBP0 is a 32-bit read/write register. NTDTBP0_BY (NTDTBP0[7:0]) is an 8-bit read/write register.	R/W

Note: S-TYPE3, P-TYPE3

Note: NTDTBP0 is 32-bit access in I3C mode.

NTDTBP0_BY is 8-bit access in I²C mode.

32-bit mailbox register NTDTBP0 is a 32-bit bidirectional data transfer register which is used both to read from the Normal Rx Data Buffer, and to write to the Normal Tx Data Buffer.

In other words, the Normal Rx Data Buffer and the Normal Tx Data Buffer have the same offset, forming a single bidirectional port for transmitting or receiving I3C data.

Read Operations:

[I3C protocol mode]

Data Read from the Normal Rx Data Buffer. It should be read based on Normal Queue Status Level indications. The Rx data is always aligned to a 4-byte boundary, and stored in the Normal Rx Data Buffer. If the length of the data transfer is not aligned to a 4-byte boundary, then there will be extra (unused) bytes at the end of the transferred data. The valid data must be identified using the DATA_LENGTH field in the Response Descriptor.

[I2C protocol mode]

When 1 byte of data has been received, the received data is transferred from the internal shift register to NTDTBP0 to enable the next data to be received. The double-buffer structure of the internal shift register and NTDTBP0 allows continuous receive operation if the received data has been read from NTDTBP0 while the internal shift register is receiving

data. Read data from NTDTBP0 once when a Normal Rx Data buffer full interrupt (I3C_RX) request is generated. If NTDTBP0 receives the next Rx data before the current data is read from NTDTBP0 (while the RDBFF0 flag in NTST is 1), this module automatically holds the SCL clock low one cycle before the RDBFF0 flag is set to 1 next. The lower 8 bits of the read 32-bit data are valid as received data.

Write Operations:

[I3C protocol mode]

Data Written to the Normal Tx Data Buffer. Data DWORDs written to the Normal Tx Data Buffer are placed onto the I3C bus one byte at a time, with the DWORD LSB first. Within each byte, bits are placed onto the I3C bus in big-endian order, with bit 7 going out first on the bus. The Tx data should always start aligned to a 4-byte boundary, and written to the NTDTBP0 register. If the length of the transfer is not aligned to a 4-byte boundary, then there will be extra (unused) bytes at the end of the transferred data. I3C shall only send the valid number of bytes indicated in the DATA_LENGTH field of the Command Descriptor.

[I2C protocol mode]

When NTDTBP0 detects a space in the internal shift register, it transfers the Tx data that has been written to NTDTBP0 to the internal shift register and starts transmitting data in transmit mode. The double-buffer structure of NTDTBP0 and the internal shift register allows continuous transmit operation if the next Tx data has been written to NTDTBP0 while the internal shift register data is being transmitted. Write Tx data to NTDTBP0 once when a Normal Tx Data buffer empty interrupt (I3C_TX) request is generated. The lower 8 bits of the written 32-bit data are valid as transmission data.

40.2.36 NIBIQP : Normal IBI Queue Port Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x17C

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Normal IBI Queue Port	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports all I3C mode.

When receiving an IBI, 32-bit mailbox register NIBIQP is used for both:

- Read the IBI status descriptor (see [section 40.3.1.3. IBI Status Descriptor](#))
- Read the IBI data (which is raw/opaque data).

The IBI status descriptor is a read-only structure describing an IBI event received from a Slave device on the I3C bus.

Note: If the I3C HCI auto-read feature is used, then the IBI data includes the data received from the auto-generated private read operation.

Even if LAST_STATUS is set to 0, the driver software still evaluates the data payload length by examining the CHUNKS field.

40.2.37 NRSQP : Normal Receive Status Queue Port Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x180

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Normal Receive Status Queue Port	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

32-bit mailbox register NRSQP contains a receive status structure (see [section 40.3.1.4. Receive Status Descriptor](#)).

40.2.38 HCMDQP : High Priority Command Queue Port Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x184

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	High Priority Command Queue Port	W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

32-bit mailbox register HCMDQP contains a command descriptor structure that depends on the requested transfer type:

1. Address Assignment Command (see [section 40.3.1.1.1. Address Assign Command](#))
2. Immediate Data Transfer (see [section 40.3.1.1.2. Immediate Transfer Command](#))
3. Regular Data Transfer (see [section 40.3.1.1.3. Regular Transfer Command](#))
4. Write + Write/Read Combo Transfer (see [section 40.3.1.1.4. Combo Transfer Command](#))
5. Internal Control Command (see [section 40.3.1.1.5. Internal Control Command](#))

Within the command descriptor, DWORDs appear starting with the least significant DWORD, in order until the most significant DWORD.

40.2.39 HRSPQP : High Priority Response Queue Port Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x188

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	High Priority Response Queue Port	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

32-bit mailbox register HRSPQP contains a response structure. (see [section 40.3.1.2. Response Descriptor](#))

40.2.40 HTDTBP : High Priority Transfer Data Buffer Port Register

Base address: I3C = 0x4035_F000

I3C_NS = 0x5035_F000

Offset address: 0x18C

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	High Priority Transfer Data Buffer Port	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

The HTDTBP register is a 32-bit bidirectional data transfer register which is used both to read from the high priority Rx data, and to write to the high priority Tx data.

For Read Operation:

To receive data from the High Priority Rx Data Buffer, read from the HTDTBP register. It should be read based on queue status indication.

The Rx data is always aligned to a 4-byte boundary, and stored in the High Priority Rx Data Buffer.

If the length of the data transfer is not aligned to a 4-byte boundary, then there will be extra (unused) bytes at the end of the transferred data.

The valid data must be identified using the DATA_LENGTH field in the response descriptor.

For Write Operation:

To send data to the High Priority Tx Data Buffer, write to the HTDTBP register. Data DWORDs written to the Data port are placed onto the I3C bus one byte at a time, with the DWORD's LSB first. Within each byte, bits are placed onto the I3C bus in big-endian order, with bit 7 going out first on the bus.

The High Priority Tx Data Buffer Port is mapped to the High Priority Tx Data Buffer.

The Tx data should always start aligned to a 4byte boundary, and written to the Tx data port register.

If the length of the transfer is not aligned to a 4-byte boundary, then there will be extra (unused) bytes at the end of the transferred data.

I3C shall only send the valid number of bytes indicated in the DATA_LENGTH field of the command descriptor.

40.2.41 NQTHCTL : Normal Queue Threshold Control Register

Base address: I3C = 0x4035_F000

I3C_NS = 0x5035_F000

Offset address: 0x190

Bit position: 31

24 23

16 15

8 7

0

Bit field:

IBIQTH[7:0]

IBIDSSZ[7:0]

RSPQTH[7:0]

CMDQTH[7:0]

Value after reset: 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1

Bit	Symbol	Function	R/W
7:0	CMDQTH[7:0]	Normal Command Queue Threshold* ¹ 0x00: Interrupt is issued when Normal Command Queue is completely empty. Others: Interrupt is issued when Normal Command Queue contains N empties. (N = CMDQTH[7:0])	R/W
15:8	RSPQTH[7:0]	Normal Response Queue Threshold* ¹ 0x00: Interrupt is issued when Normal Response Queue contains 1 entry (DWORD). Others: Interrupt is triggered when Normal Response Queue contains N+1 entries (DWORD). (N = CMDQTH[7:0])	R/W
23:16	IBIDSSZ[7:0]	Normal IBI Data Segment Size* ² Supported Values: Minimum: 1 (4 bytes) Maximum: 63 (252 bytes), provided that the configured IBI Queue depth is 64 or more. When ATCCNTE.ATCE = 1, restrict to the number of slices ≥ 2 .	R/W
31:24	IBIQTH[7:0]	Normal IBI Queue Threshold* ¹ 0x00: I3C Protocol mode (Master): Interrupt is generated when the Outstanding IBI Status count is 1 or more. I3C Protocol mode (Slave): Interrupt is issued when IBI Data Buffer is completely empty. Others: I3C Protocol mode (Master): Interrupt is generated when the Outstanding IBI Status count is N + 1 or more. (N = CMDQTH[7:0]) I3C Protocol mode (Slave): Interrupt is issued when IBI Data Buffer contains N empties.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. These bits support all I3C mode.

Note 2. These bits support I3C master mode and I3C secondary master mode.

The Queue Threshold Control register controls the interrupt trigger thresholds for the Command Queue, the Response Queue, and the IBI Queue.

The specific reset values are indicative, and could be hardware implementation specific.

CMDQTH[7:0] bits (Normal Command Queue Threshold)

Controls the minimum number of Normal Command Queue empties needed to trigger the I3C_CMD interrupt.

If this field is greater than (Normal Command Queue size*¹ – 1), then only the number of bits required to address the full buffer depth will be considered.

RSPQTH[7:0] bits (Normal Response Queue Threshold)

Controls the minimum number of Normal Response Queue entries needed to trigger the I3C_RESP interrupt.

If this field is greater than (Normal Response Queue size*² – 1), then only the number of bits required to address the full buffer depth will be considered.

IBIDSSZ[7:0] bits (Normal IBI Data Segment Size)

This is the IBI data segment size, in DWORDs (4 bytes).

In PIO mode, this field allows the incoming Normal IBI data to be sliced into multiple segments generating status individually, to support cut-through readout of a long IBI payload data.

When Asynchronous Timing Control mode is supported, this field should be set to a value other than 1 or 3 to allow the single data segment to contain the entire Master time-stamp value (for example, both MREF and MC2).

IBIQTH[7:0] bits (Normal IBI Queue Threshold)

For I3C protocol mode (Master): PRTS.PRTMD = 0 and PRSST.CRMS = 1.

Controls generation of the I3C_IBI interrupt, based on the value of the Normal IBI Queue's Outstanding IBI status count.

Each IBI status entry can represent either the complete IBI payload (if the IBI payload byte size is $4 \times \text{IBIDSSZ}$ or less), or a segment of the IBI payload (if the IBI payload byte size is more than $4 \times \text{IBIDSSZ}$).

For I3C protocol mode (Slave): PRTS.PRTMD bit = 0, PRSST.CRMS bit = 0.

Controls the minimum number of IBI Data Buffer empties needed to trigger the I3C_IBI interrupt.

If this field is greater than (IBI Data Buffer size^{*3} - 1), then only the number of bits required to address the full buffer depth will be considered.

Note 1. Normal Command Queue size is 4.

Note 2. Normal Response Queue size is 4.

Note 3. IBI Data Buffer size is 8.

Note: It is assumed that I3C has exactly one Command Queue, exactly one Response Queue, and exactly one IBI Queue.

40.2.42 NTBTHCTL0 : Normal Transfer Data Buffer Threshold Control Register 0

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x194

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	RXSTTH[2:0]			—	—	—	—	—	TXSTTH[2:0]		
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	RXDBTH[2:0]			—	—	—	—	—	TXDBTH[2:0]		
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
2:0	TXDBTH[2:0]	Normal Tx Data Buffer Threshold ^{*1} 0 0 0: Interrupt triggers at 2 Tx Buffer empties, DWORDs 0 0 1: Interrupt triggers at 4 Tx Buffer empties, DWORDs 0 1 0: Interrupt triggers at 8 Tx Buffer empties, DWORDs 0 1 1: Interrupt triggers at 16 Tx Buffer empties, DWORDs Others: Setting prohibited	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
10:8	RXDBTH[2:0]	Normal Rx Data Buffer Threshold ^{*1} 0 0 0: Interrupt triggers at 2 Rx Buffer entries, DWORDs 0 0 1: Interrupt triggers at 4 Rx Buffer entries, DWORDs 0 1 0: Interrupt triggers at 8 Rx Buffer entries, DWORDs 0 1 1: Interrupt triggers at 16 Rx Buffer entries, DWORDs Others: Setting prohibited	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
18:16	TXSTTH[2:0]	Normal Tx Start Threshold ^{*2} 0 0 0: Wait for 2 entry DWORDs 0 0 1: Wait for 4 entry DWORDs 0 1 0: Wait for 8 entry DWORDs 0 1 1: Wait for 16 entry DWORDs Others: Setting prohibited	R/W
23:19	—	These bits are read as 0. The write value should be 0.	R/W
26:24	RXSTTH[2:0]	Normal Rx Start Threshold ^{*2} 0 0 0: Wait for 2 empty DWORDs 0 0 1: Wait for 4 empty DWORDs 0 1 0: Wait for 8 empty DWORDs 0 1 1: Wait for 16 empty DWORDs Others: Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. These bits support all I3C mode.

Note 2. These bits support I3C master mode and I3C secondary master mode.

The Data Buffer Control register controls the interrupt trigger thresholds for the Rx Data Buffer Queue and the Tx Data Buffer Queue.

TXDBTH[2:0] bits (Normal Tx Data Buffer Threshold)

Minimum number of Tx Data Buffer empties, in DWORDs, that will trigger the I3C_TX interrupt.

The software must program a value less than Tx Data Buffer size in this register.

RXDBTH[2:0] bits (Normal Rx Data Buffer Threshold)

Minimum number of Rx Data Buffer entries in DWORDs that will trigger the I3C_RX interrupt.

The software must program a value less than Rx Data Buffer size in this register.

TXSTTH[2:0] bits (Normal Tx Start Threshold)

When preparing to initiate a Write Transfer on the I3C Bus, I3C shall wait until the Transmit Buffer has at least the indicated number of locations available.

Two optional configurable Modes are available:

- ### 1. Store and Forward Mode

If the TXSTTH[2:0] field is set to the Transmit Buffer size, then I3C shall delay initiation of the Write Command as follows:

- If the data length to be transferred is more than the Transmit Buffer size, then this module shall wait until the Tx Data Buffer is completely full.
- If the data length to be transferred is less than the Transmit Buffer size, then I3C shall wait until enough Tx Data Buffer locations are available to store the data to be transferred.

- ## 2. Threshold Mode

If the TXSTTH[2:0] field value is less than the Transmit Buffer size, then I3C shall initiate the Write Command as soon as the indicated number of Tx Data Buffer locations are entries.

RXSTTH[2:0] bits (Normal Rx Start Threshold)

When preparing to initiate a Read Transfer on the I3C bus, I3C shall wait until the Receive Buffer has at least the indicated number of empty locations in DWORDs.

Two optional configurable Modes are available:

- ### 1. Store and Forward Mode

If the RXSTTH[2:0] field is set to the Receive Buffer size, then I3C shall delay initiation of the Read Command as follows:

- If the data length to be transferred is more than the Receive Buffer size, then this module shall wait until the Rx Data Buffer is completely empty.
- If the data length to be transferred is less than the Receive Buffer size, then I3C shall wait until enough Rx Data Buffer locations are available to store the data to be transferred.

- ## 2. Threshold Mode

If the RXSTTH[2:0] field value is less than the Receive Buffer size, then I3C shall initiate the Read Command as soon as the indicated number of Rx Data Buffer locations are empty.

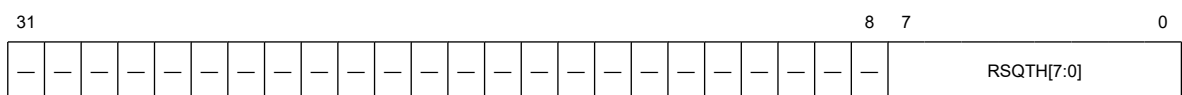
40.2.43 NRQTHCTL : Normal Receive Status Queue Threshold Control Register

Base address: I3C = 0x4035_F000
I3C NS = 0x5035_F000

Offset address: 0x1C0

Bit position: 31

Bit field:

[illegible]

Bit	Symbol	Function	R/W
7:0	RSQTH[7:0]	Normal Receive Status Queue Threshold 0x00: Interrupt is issued when Normal Receive Status Queue contains 1 entry (DWORD). Others: Interrupt is triggered when Normal Receive Status Queue contains N+1 entries (DWORD). (N = RSQTH[7:0])	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports for I3C secondary master and I3C slave mode.

RSQTH[7:0] bits (Normal Receive Status Queue Threshold)

Controls the minimum number of normal receive status queue entries needed to trigger the I3C_RCV interrupt.

If this field is greater than (Normal Receive Status Queue size^{*1} - 1), then only the number of bits required to address the full buffer depth will be considered.

Note 1. Normal Receive Status Queue size is 2.

40.2.44 HQTCTL : High Priority Queue Threshold Control Register

Base address: I3C = 0x4035_F000
I3C NS = 0x5035_F000

Offset address: 0x1C4

Bit position: 31 15 8 7 0

Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RSPQTH[7:0]	CMDQTH[7:0]
------------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-------------	-------------

Value after reset: 0 1 0 0 0 0 0 0 0 0 1

Bit	Symbol	Function	R/W
7:0	CMDQTH[7:0]	High Priority Command Queue Threshold 0x00: Interrupt is issued when High Priority Command Queue is completely empty. Others: Interrupt is issued when High Priority Command Queue contains N entries. (N = CMDQTH[7:0])	R/W
15:8	RSPQTH[7:0]	High Priority Response Queue Threshold 0x00: Interrupt is issued when High Priority Response Queue contains 1 entry (DWORD). Others: Interrupt is triggered when High Priority Response Queue contains N+1 entries (DWORD). (N = RSPQTH[7:0])	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

The High Priority Queue Threshold Control register controls the interrupt trigger thresholds for the High Priority Command Queue, the High Priority Response Queue, and the IBI Queue.

The specific reset values are indicative, and could be hardware implementation specific.

CMDQTH[7:0] bits (High Priority Command Queue Threshold)

Controls the minimum number of empty High Priority Command Queue entries needed to trigger the I3C HCMD interrupt.

If this field is greater than (High Priority Command Queue size*1 - 1), then only the number of bits required to address the full buffer depth will be considered.

RSPQTH[7:0] bits (High Priority Response Queue Threshold)

Controls the minimum number of High Priority Response Queue entries needed to trigger the I3C HRESP interrupt.

If this field is greater than (High Priority Response Queue size^{*2} - 1), then only the number of bits required to address the full buffer depth will be considered.

Note 1. High Priority Command Queue size is 2.

Note 2. High Priority Response Queue size is 2.

Note: It is assumed that I3C has exactly one High Priority Command Queue, exactly one High Priority Response Queue, and exactly one IBI Queue.

40.2.45 HTBTHCTL : High Priority Transfer Data Buffer Threshold Control Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x1C8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	RXSTTH[2:0]			—	—	—	—	—	TXSTTH[2:0]		
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	RXDBTH[2:0]			—	—	—	—	—	TXDBTH[2:0]		
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
2:0	TXDBTH[2:0]	High Priority Tx Data Buffer Threshold 0 0 0: Interrupt triggers at 2 High Priority Tx Buffer empties, DWORDs 0 0 1: Reserved Others Setting prohibited	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
10:8	RXDBTH[2:0]	High Priority Rx Data Buffer Threshold 0 0 0: Interrupt triggers at 2 High Priority Rx Buffer entries, DWORDs 0 0 1: Reserved Others Setting prohibited	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
18:16	TXSTTH[2:0]	High Priority Tx Start Threshold 0 0 0: Wait for 2 entry DWORDs 0 0 1: Reserved Others Setting prohibited	R/W
23:19	—	These bits are read as 0. The write value should be 0.	R/W
26:24	RXSTTH[2:0]	High Priority Rx Start Threshold 0 0 0: Wait for 2 empty DWORDs 0 0 1: Reserved Others Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

TXDBTH[2:0] bits (High Priority Tx Data Buffer Threshold)

Minimum number of High Priority Tx Data Buffer empties, in DWORDs, that will trigger the I3C_HTX interrupt. The software must program a value less than High Priority Tx Data Buffer size in this register.

RXDBTH[2:0] bits (High Priority Rx Data Buffer Threshold)

Minimum number of High Priority Rx Data Buffer entries in DWORDs that will trigger the I3C_HRX interrupt. The software must program a value less than High Priority Rx Data Buffer size in this register.

TXSTTH[2:0] bits (High Priority Tx Start Threshold)

When preparing to initiate a Write Transfer on the I3C bus, I3C shall wait until the High Priority Transmit Buffer has at least the indicated number of locations available.

Two optional configurable modes are available:

1. Store and Forward Mode

If the TXSTTH[2:0] field is set to the High Priority Transmit Buffer size, then I3C shall delay initiation of the write command as follows:

- If the data length to be transferred is more than the High Priority Transmit Buffer size, then I3C shall wait until the High Priority Tx Data Buffer is completely full.
- If the data length to be transferred is less than the High Priority Transmit Buffer size, then I3C shall wait until enough High Priority Tx Data Buffer locations are available to store the data to be transferred.

2. Threshold mode

If the TXSTTH[2:0] field value is less than the High Priority Transmit Buffer size, then I3C shall initiate the write command as soon as the indicated number of High Priority Tx Data Buffer locations are empty.

RXSTTH[2:0] bits (High Priority Rx Start Threshold)

When preparing to initiate a Read Transfer on the I3C bus, I3C shall wait until the High Priority Receive Buffer has at least the indicated number of empty locations in DWORDs.

Two optional configurable modes are available:

1. Store and forward mode

If the RXSTTH[2:0] field is set to the High Priority Receive Buffer size, then I3C shall delay initiation of the read command as follows:

- If the data length to be transferred is more than the High Priority Receive Buffer size, then I3C shall wait until the High Priority Rx Data Buffer is completely empty.
- If the data length to be transferred is less than the High Priority Receive Buffer size, then I3C shall wait until enough High Priority Rx Data Buffer locations are available to store the data to be transferred.

2. Threshold mode

If the RXSTTH[2:0] field value is less than the High Priority Receive Buffer size, then I3C shall initiate the read command as soon as the indicated number of High Priority Rx Data Buffer locations are empty.

40.2.46 BST : Bus Status Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x1D0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	WUCN DDF	—	—	—	TODF	—	—	—	ALF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	TEND F	—	—	—	NACK DF	—	HDRE XDF	SPCN DDF	STCN DDF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STCNDDF	START Condition Detection Flag 0: START condition is not detected. 1: START condition is detected.	R/W ³
1	SPCNDDF	STOP Condition Detection Flag 0: STOP condition is not detected. 1: STOP condition is detected.	R/W ³
2	HDREXDF	HDR Exit Pattern Detection Flag ^{*1} 0: HDR Exit Pattern is not detected 1: HDR Exit Pattern is detected.	R/W ³
3	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
4	NACKDF	NACK Detection Flag ^{*2} 0: NACK is not detected. 1: NACK is detected.	R/W ^{*3}
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	TENDF	Transmit End Flag ^{*2} 0: Data is being transmitted. 1: Data has been transmitted.	R/W ^{*3}
15:9	—	These bits are read as 0. The write value should be 0.	R/W
16	ALF	Arbitration Lost Flag ^{*2} 0: Arbitration is not lost 1: Arbitration is lost.	R/W ^{*3}
19:17	—	These bits are read as 0. The write value should be 0.	R/W
20	TODF	Timeout Detection Flag 0: Timeout is not detected. 1: Timeout is detected.	R/W ^{*3}
23:21	—	These bits are read as 0. The write value should be 0.	R/W
24	WUCNDDF	Wake-Up Condition Detection Flag 0: Wake-Up is not detected. 1: Wake-Up is detected.	R/W ^{*3}
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports all I3C mode.

Note 2. This bit supports I²C mode.

Note 3. Clearing (to 0) condition : Writing 0 after 1 is read.

STCNDDF bit (START Condition Detection Flag)

[Setting conditions]

- All of the followings are satisfied:
 1. The BSTE.STCNDDE bit = 1.
 2. When a START condition (or a Repeated START condition) is detected.

[Clearing conditions]

- When 0 is written to the STCNDDF flag after reading STCNDDF flag = 1.
- When a STOP condition is detected.

SPCNDDF bit (STOP Condition Detection Flag)

[Setting conditions]

- All of the followings are satisfied:
 1. The BSTE.SPCNDDE bit = 1.
 2. When a STOP condition is detected.

[Clearing condition]

- When 0 is written to the SPCNDDF flag after reading SPCNDDF flag = 1.

HDREXDF bit (HDR Exit Pattern Detection Flag)

[Setting conditions]

- All of the followings are satisfied:
 1. The BSTE.HDREXDE bit = 1.
 2. When an HDR EXIT pattern is detected.

[Clearing condition]

- When 0 is written to the HDREXDF flag after reading HDREXDF flag = 1.

NACKDF bit (NACK Detection Flag)

[Setting conditions]

- All of the followings are satisfied:
 1. The PRTS.PRTMD bit = 1 (I²C protocol mode).
 2. The BSTE.NACKDE bit = 1 (Enables NACK detection interrupt status logging).
 3. When acknowledge is not received (NACK is received) from the receive device in transmit mode.

[Clearing condition]

- When 0 is written to the NACKDF flag after reading NACKDF flag = 1.

TENDF bit (Transmit End Flag)

[Setting conditions]

- All of the followings are satisfied:
 1. The PRTS.PRTMD bit = 1 (I²C protocol mode).
 2. The BSTE.TENDE bit = 1 (Enables Transmit End Interrupt Status logging).
 3. At the rising edge of the ninth SCL clock cycle while the NTST.TDBEF0 flag = 1.
Excluding when sending an address.

[Clearing conditions]

- When 0 is written to the TENDF flag after reading TENDF flag = 1.
- When data is written to the NTDTBP0 register.
- When a STOP condition is detected.

ALF bit (Arbitration Lost Flag)

[Setting conditions]

When master arbitration-lost detection is enabled: BSTE.ALE bit = 1, BFCTL.MALE = 1.

- When the internal SDA output state does not match the I3C_SDA line level at the rising edge of SCL clock except for the ACK period during data (including slave address) transmission in master transmit mode (when the I3C_SDA line is driven low while the internal SDA output is at a high level (the I3C_SDA pin is in the high-impedance state)).
- All of the followings are satisfied.
 1. When the START condition is detected while the CNDCTL.STCND bit = 1.
 2. When the internal SDA output state does not match the I3C_SDA line level.
- When the CNDCTL.STCND bit is set to 1 (START condition issuance request) while the BCST.BFREF flag = 0.

When NACK arbitration-lost detection is enabled: BSTE.ALE bit = 1, BFCTL.NALE = 1.

- When the internal SDA output state does not match the I3C_SDA line level at the rising edge of SCL clock in the ACK period during NACK transmission in receive mode.

When slave arbitration-lost detection is enabled: BSTE.ALE bit = 1, BFCTL.SALE = 1.

- When the internal SDA output state does not match the I3C_SDA line level at the rising edge of SCL clock except for the ACK period during data transmission in slave transmit mode.

[Clearing condition]

- When 0 is written to the ALF flag after reading ALF flag = 1.

TODF bit (Timeout Detection Flag)

[Setting conditions]

- All of the followings are satisfied.
 1. The BSTE.TODE bit = 1 (Enables Timeout Detection Interrupt Status logging).
 2. When the master mode or the received slave address matches the slave address n (n = 0 to 2) in Slave mode.
 3. When the I3C_SCL line state remains unchanged for the period specified by TMOCTL register.

[Clearing condition]

- When 0 is written to the TODF flag after reading TODF flag = 1.

WUCNDDF bit (Wake-Up Condition Detection Flag)

[Setting condition]

For I²C protocol mode: PRTS.PRTMD bit = 1

- When PCLK and TCLK are supplied after all of the followings are satisfied.
 1. The WUCTL.WUFE bit = 1 (Wake-up function is enabled).
 2. The BSTE.WUCNDDE bit = 1 (Enables Wake-up Condition Detection Status logging).
 3. The WUST.WUASYNF flag = 1.
 4. When the address received in slave mode matches the address of slave enabled in the SVCTL.SVAE[2:0] bit (except for the Device-ID address).

For I3C Protocol mode (Master): PRTS.PRTMD bit = 0, PRSST.CRMS bit = 1.

- When PCLK and TCLK are supplied after all of the followings are satisfied.
 1. The WUCTL.WUFE bit = 1 (Wake-up function is enabled).
 2. The BSTE.WUCNDDE bit = 1 (Enables Wake-up Condition Detection Status logging).
 3. The WUST.WUASYNF flag = 1.
 4. When low level of the I3C_SDA line is detected (When the START condition is detected).

For I3C Protocol mode (Slave) : PRTS.PRTMD bit = 0, PRSST.CRMS bit = 0.

- When PCLK and TCLK are supplied after all of the followings are satisfied.
 1. The WUCTL.WUFE bit = 1 (Wake-up function is enabled).
 2. The BSTE.WUCNDDE bit = 1 (Enables Wake-up Condition Detection Status logging).
 3. The WUST.WUASYNF flag = 1.
 4. When the broadcast address (0x7E) is detected after a START (or Repeated START) condition and the own dynamic address is detected after the Repeated START condition following the broadcast address.

[Clearing condition]

- When 0 is written to the WUCNDDF flag after reading WUCNDDF flag = 1 while the WUST.WUASYNF flag = 0.

40.2.47 BSTE : Bus Status Enable Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x1D4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	WUCN DDE	—	—	—	TODE	—	—	—	ALE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	TEND E	—	—	—	NACK DE	—	HDRE XDE	SPCN DDE	STCN DDE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STCNDDE	START Condition Detection Enable 0: Disables START condition Detection Interrupt Status logging. 1: Enables START condition Detection Interrupt Status logging.	R/W
1	SPCNDDE	STOP Condition Detection Enable 0: Disables STOP condition Detection Interrupt Status logging. 1: Enables STOP condition Detection Interrupt Status logging.	R/W
2	HDREXDE	HDR Exit Pattern Detection Enable*1 0: Disables HDR Exit Pattern Detection Interrupt Status logging. 1: Enables HDR Exit Pattern Detection Interrupt Status logging.	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	NACKDE	NACK Detection Enable*2 0: Disables NACK Detection Interrupt Status logging. 1: Enables NACK Detection Interrupt Status logging.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	TENDE	Transmit End Enable*2 0: Disables Transmit End Interrupt Status logging. 1: Enables Transmit End Interrupt Status logging.	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
16	ALE	Arbitration Lost Enable*2 0: Disables Arbitration Lost Interrupt Status logging. 1: Enables Arbitration Lost Interrupt Status logging.	R/W
19:17	—	These bits are read as 0. The write value should be 0.	R/W
20	TODE	Timeout Detection Enable 0: Disables Timeout Detection Interrupt Status logging. 1: Enables Timeout Detection Interrupt Status logging.	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
24	WUCNDDE	Wake-up Condition Detection Enable 0: Disables Wake-up Condition Detection Status logging. 1: Enables Wake-up Condition Detection Status logging.	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports all I3C mode.

Note 2. This bit supports I²C mode.

STCNDDE bit (START Condition Detection Enable)

When this bit is 1, operation of BST.STCNDDE is enabled. For the setting conditions and clearing conditions of the BST.STCNDDE flag, see the details of BST.STCNDDE.

SPCNDDE bit (STOP Condition Detection Enable)

When this bit is 1, operation of BST.SPCNDDF is enabled. For the setting conditions and clearing conditions of the BST.SPCNDDF flag, see the details of BST.SPCNDDF.

HDREXDE bit (HDR Exit Pattern Detection Enable)

When this bit is 1, the operation of BST.HDREXDF is enabled. For the setting conditions and clearing conditions of the BST.HDREXDF flag, see the details of BST.HDREXDF.

NACKDE bit (NACK Detection Enable)

When this bit is 1, the operation of BST.NACKDF is enabled. This bit is used to specify whether to continue or discontinue the transfer operation when NACK is received from the slave device in transmit mode. Normally, set this bit to 1. For the setting conditions and clearing conditions of the BST.NACKDF flag, see the details of BST.NACKDF.

TENDE bit (Transmit End Enable)

When this bit is 1, the operation of BST.TENDF is enabled. For the setting conditions and clearing conditions of the BST.TENDF flag, see the details of BST.TENDF.

ALE bit (Arbitration Lost Enable)

When this bit is 1, the operation of BST.ALF is enabled. For the setting conditions and clearing conditions of the BST.ALF flag, see the details of BST.ALF.

TODE bit (Timeout Detection Enable)

When this bit is 1, the operation of BST.TODF is enabled. For the setting conditions and clearing conditions of the BST.TODF flag, see the details of BST.TODF.

WUCNDDE bit (Wake-up Condition Detection Enable)

When this bit is 1, the operation of BST.WUCNDDF is enabled. For the setting conditions and clearing conditions of the BST.WUCNDDF flag, see the details of BST.WUCNDDF.

40.2.48 BIE : Bus Interrupt Enable Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x1D8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	WUCNDDIE	—	—	—	TODIE	—	—	—	ALIE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	TENDIE	—	—	—	NACKDIE	—	HDREXDIE	SPCNDDIE	STCNDDIE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STCNDDIE	START Condition Detection Interrupt Enable 0: Disables START condition Detection Interrupt Signal. 1: Enables START condition Detection Interrupt Signal.	R/W
1	SPCNDDIE	STOP Condition Detection Interrupt Enable 0: Disables STOP condition Detection Interrupt Signal. 1: Enables STOP condition Detection Interrupt Signal.	R/W
2	HDREXDIE	HDR Exit Pattern Detection Interrupt Enable*1 0: Disables HDR Exit Pattern Detection Interrupt Signal. 1: Enables HDR Exit Pattern Detection Interrupt Signal.	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
4	NACKDIE	NACK Detection Interrupt Enable ^{*2} 0: Disables NACK Detection Interrupt Signal. 1: Enables NACK Detection Interrupt Signal.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	TENDIE	Transmit End Interrupt Enable ^{*2} 0: Disables Transmit End Interrupt Signal. 1: Enables Transmit End Interrupt Signal.	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
16	ALIE	Arbitration Lost Interrupt Enable ^{*2} 0: Disables Arbitration Lost Interrupt Signal. 1: Enables Arbitration Lost Interrupt Signal.	R/W
19:17	—	These bits are read as 0. The write value should be 0.	R/W
20	TODIE	Timeout Detection Interrupt Enable 0: Disables Timeout Detection Interrupt Signal. 1: Enables Timeout Detection Interrupt Signal.	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
24	WUCNDDIE	Wake-Up Condition Detection Interrupt Enable 0: Disables Wake-Up Condition Detection Interrupt Signal. 1: Enables Wake-Up Condition Detection Interrupt Signal.	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports all I3C mode.

Note 2. This bit supports I²C mode.

The BIE register enables signaling of outstanding bus interrupts received by I3C.

STCNDDIE bit (START Condition Detection Interrupt Enable)

This bit enables or disables the START Condition Detection interrupt requests when the BST.STCNDDF flag is set to 1.

SPCNDDIE bit (STOP Condition Detection Interrupt Enable)

This bit enables or disables the STOP Condition Detection interrupt requests when the BST.SPCNDDF flag is set to 1.

HDREXDIE bit (HDR Exit Pattern Detection Interrupt Enable)

This bit enables or disables the HDR Exit Pattern Detection interrupt requests when the BST.HDREXDF flag is set to 1.

NACKDIE bit (NACK Detection Interrupt Enable)

This bit enables or disables the NACK Detection interrupt requests when the BST.NACKDF flag is set to 1.

TENDIE bit (Transmit End Interrupt Enable)

This bit enables or disables the Transmit End interrupt (I3C_TEND) requests when the BST.TENDF flag is set to 1.

ALIE bit (Arbitration Lost Interrupt Enable)

This bit enables or disables the Arbitration lost interrupt requests when the BST.ALF flag is set to 1.

TODIE bit (Timeout Detection Interrupt Enable)

This bit enables or disables the Timeout Detection interrupt requests when the BST.TODF flag is set to 1.

WUCNDDIE bit (Wake-Up Condition Detection Interrupt Enable)

This bit enables or disables the Wake-up Condition Detection interrupt (I3C_WU) requests when the BST.WUCNDDF flag is set to 1.

40.2.49 BSTFC : Bus Status Force Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x1DC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	WUCNDDFC	—	—	—	TODFC	—	—	—	ALFC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	TENDFC	—	—	—	NACKDFC	—	HDREXDFC	SPCNDDFC	STCNDDFC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STCNDDFC	START condition Detection Force 0: Not Force START condition Detection Interrupt for software testing. 1: Force START condition Detection Interrupt for software testing.	W
1	SPCNDDFC	STOP condition Detection Force 0: Not Force STOP condition Detection Interrupt for software testing. 1: Force STOP condition Detection Interrupt for software testing.	W
2	HDREXDFC	HDR Exit Pattern Detection Force*1 0: Not Force HDR Exit Pattern Detection Interrupt for software testing. 1: Force HDR Exit Pattern Detection Interrupt for software testing.	W
3	—	This bit is read as 0.	R
4	NACKDFC	NACK Detection Force*2 0: Not Force NACK Detection Interrupt for software testing. 1: Force NACK Detection Interrupt for software testing.	W
7:5	—	These bits are read as 0.	R
8	TENDFC*3	Transmit End Force*2 0: Not Force Transmit End Interrupt for software testing. 1: Force Transmit End Interrupt for software testing.	W
15:9	—	These bits are read as 0.	R
16	ALFC	Arbitration Lost Force*2 0: Not Force Arbitration Lost Interrupt for software testing. 1: Force Arbitration Lost Interrupt for software testing.	W
19:17	—	These bits are read as 0.	R
20	TODFC	Timeout Detection Force 0: Not Force Timeout Detection Interrupt for software testing. 1: Force Timeout Detection Interrupt for software testing.	W
23:21	—	These bits are read as 0.	R
24	WUCNDDFC	Wake-Up Condition Detection Force 0: Not Force Wake-Up Condition Detection Interrupt for software testing. 1: Force Wake-Up Condition Detection Interrupt for software testing.	W
31:25	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports all I3C mode.

Note 2. This bit supports I²C mode.

Note 3. TENDFC does not work unless TDBEF0 = 1.

40.2.50 NTST : Normal Transfer Status Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x1E0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	RSQF F	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TEF	—	—	—	TABTF	RSPQ FF	CMDQ EF	IBIQE FF	RDBF F0	TDBE F0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TDBEF0	Normal Tx Data Buffer Empty Flag 0 0: For I2C protocol mode: PRTS.PRTMD bit = 1. Normal Tx Data Buffer contains transmit data. For I3C protocol mode: PRTS.PRTMD bit = 0. The number of empties in the Normal Tx Data Buffer is less than the NTBTHCTL0.TXDBTH[2:0] threshold. 1: For I2C protocol mode: PRTS.PRTMD bit = 1. Normal Tx Data Buffer contains no transmit data. For I3C protocol mode: PRTS.PRTMD bit = 0. The number of empties in the Normal Tx Data Buffer is the NTBTHCTL0.TXDBTH [2:0] threshold or more.	R/W ^{*3}
1	RDBFF0	Normal Rx Data Buffer Full Flag 0 0: For I2C protocol mode: PRTS.PRTMD bit = 1. Normal Rx Data Buffer contains no receive data. For I3C Protocol mode: PRTS.PRTMD bit = 0. The number of entries in the Normal Rx Data Buffer is less than the NTBTHCTL0.RXDBTH[2:0] threshold. 1: For I2C protocol mode: PRTS.PRTMD bit = 1. Normal Rx Data Buffer contains receive data. For I3C Protocol mode: PRTS.PRTMD bit = 0. The number of entries in the Normal Rx Data Buffer is the NTBTHCTL0.RXDBTH [2:0] threshold or more.	R/W ^{*3}
2	IBIQEFF	Normal IBI Queue Empty/Full Flag ^{*1} 0: For I3C protocol mode (Master): PRTS.PRTMD bit = 0, PRSST.CRMS bit = 1. The number of Normal IBI Queue entries is the NQTHCTL.IBIQTH threshold or less. For I3C protocol mode (Slave) : PRTS.PRTMD bit = 0, PRSST.CRMS bit = 0. If the NQTHCTL.IBIQTH = 0: The number of IBI Data Buffer empties is less than the IBI Data Buffer size. If the NQTHCTL.IBIQTH is other than 0: The number of IBI Data Buffer empties is less than the NQTHCTL.IBIQTH threshold. 1: For I3C protocol mode (Master): PRTS.PRTMD bit = 0, PRSST.CRMS bit = 1. The number of Normal IBI Queue entries is more than the NQTHCTL.IBIQTH threshold. For I3C protocol mode (Slave) : PRTS.PRTMD bit = 0, PRSST.CRMS bit = 0. If the NQTHCTL.IBIQTH = 0: The number of IBI Data Buffer empties is the IBI Data Buffer size. If the NQTHCTL.IBIQTH is other than 0: The number of IBI Data Buffer empties is the NQTHCTL.IBIQTH threshold or more.	R/W ^{*3}

Bit	Symbol	Function	R/W
3	CMDQEF	Normal Command Queue Empty Flag ^{*1} 0: If the NQTHCTL.CMDQTH = 0: The number of Normal Command Queue empties is less than the Normal Command Queue size. If the NQTHCTL.CMDQTH is other than 0: The number of Normal Command Queue empties is less than the NQTHCTL.CMDQTH threshold. 1: If the NQTHCTL.CMDQTH = 0: The number of Normal Command Queue empties is the Normal Command Queue size. If the NQTHCTL.CMDQTH is other than 0: 1: The number of Normal Command Queue empties is the NQTHCTL.CMDQTH threshold or more.	R/W ^{*3}
4	RSPQFF	Normal Response Queue Full Flag ^{*1} 0: The number of Normal Response Queue entries is the NQTHCTL.RSPQTH threshold or less. 1: The number of Normal Response Queue entries is more than the NQTHCTL.RSPQTH threshold.	R/W ^{*3}
5	TABTF	Normal Transfer Abort Flag ^{*1} 0: Normal Transfer Abort does not occur. 1: Normal Transfer Abort occur. To clear, write 0 to this bit after 1 state is read.	R/W ^{*3}
8:6	—	These bits are read as 0. The write value should be 0.	R/W
9	TEF	Normal Transfer Error Flag ^{*1} 0: Normal Transfer Error does not occur. 1: Normal Transfer Error occurs. To clear, write 0 to this bit after 1 state is read.	R/W ^{*3}
19:10	—	These bits are read as 0. The write value should be 0.	R/W
20	RSQFF	Normal Receive Status Queue Full Flag ^{*2} 0: The number of Normal Receive Status Queue entries is the NRQTHCTL.RSQTH threshold or less. 1: The number of Normal Receive Status Queue entries is more than the NRQTHCTL.RSQTH threshold.	R/W ^{*3}
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports all I3C mode.

Note 2. This bit supports I3C secondary master mode and I3C slave mode.

Note 3. Clearing (to 0) condition : Writing 0 after the 1 state is read.

TDBEF0 bit (Normal Tx Data Buffer Empty Flag 0)

[Setting conditions]

For I²C Protocol mode: PRTS.PRTMD bit = 1.

The following condition 1 is satisfied and any of the following conditions 2 to 4 are satisfied:

1. The NTSTE.TDBEE0 bit = 1 (enables Normal Tx Data Buffer Empty Interrupt Status logging).
2. When data has been transferred from the Normal Tx Data Buffer to the Shift Register and the Normal Tx Data Buffer becomes empty^{*1}.
3. When the PRSST.TRMD bit is set to 1.
4. When the received slave address matches while the TRMD bit = 1.

For I3C Protocol mode: PRTS.PRTMD bit = 0.

The following conditions 1 and 2 are satisfied:

1. The NTSTE.TDBEE0 bit = 1 (enables Normal Tx Data Buffer Empty Interrupt Status logging).
2. When the number of empties in the Normal Tx Data Buffer is the NTBTHCTL0.TXDBTH[2:0] threshold or more (see NTBTHCTL0 register).

[Clearing conditions]

For I²C Protocol mode: PRTS.PRTMD bit = 1.

- When data is written to NTDTBP0.
- When the TRMD bit in PRSST is set to 0.

For I3C protocol mode: PRTS.PRTMD bit = 0.

- Write 0 to this bit after 1 is read.
- On completion of the last write access to Normal Tx Data by DMAC/DTC.

Note 1. When the BST.NACKDF flag is set to 1 while the BSTE.NACKDE bit = 1, I3C aborts data transmission/reception. If the TDBEF0 flag = 0 (next transmit data has been written), data is transferred to the Shift Register and the Normal Tx Data Buffer register becomes empty at the rising edge of the 9th clock cycle, but the TDBEF0 flag is not set to 1.

RDBFF0 bit (Normal Rx Data Buffer Full Flag 0)

[Setting conditions]

For I²C Protocol mode: PRTS.PRTMD bit = 1.

The following condition 1 is satisfied and any of the following condition 2 or 3 is satisfied:

1. The NTSTE.RDBFE0 bit = 1 (enables Normal Rx Data Buffer Full Interrupt Status logging).
2. When Rx data is transferred from Shift Register to Normal Rx Data Buffer.
The RDBFF0 flag is set to 1 on the rising edge of the 8th or 9th SCL clock cycle (selected in the ACKTWE bit in SCSTRCTL).
3. When the received slave address matches after a START (or Repeated START) condition is detected with the TRMD bit in PRSST set to 0.

For I3C Protocol mode: PRTS.PRTMD bit = 0.

The following conditions 1 and 2 are satisfied:

1. The NTSTE.RDBFE0 bit = 1 (enables Normal Rx Data Buffer Full Interrupt Status logging).
2. When the number of Normal Rx Data Buffer entries is the NTBTHCTL0.RXDBTH[2:0] threshold or more (see NTBTHCTL0 register).

[Clearing conditions]

For I²C Protocol mode: PRTS.PRTMD bit = 1.

- When data is read from NTDTBP0.

For I3C Protocol mode: PRTS.PRTMD bit = 0.

- Write 0 to this bit after 1 is read.
- On completion of the last read access to Normal Rx Data by DMAC/DTC.

IBIQEFF bit (Normal IBI Queue Empty/Full Flag)

[Setting conditions]

The following 2 conditions are satisfied:

1. The NTSTE.IBIQFE bit = 1 (enables Normal IBI Queue Empty/Full Interrupt Status logging)
2. For I3C protocol mode (master): PRTS.PRTMD bit = 0, PRSST.CRMS bit = 1.
 - When the number of Normal IBI Queue entries is more than the NQTHCTL.IBIQTH threshold (see NQTHCTL register).

For I3C protocol mode (slave) : PRTS.PRTMD bit = 0, PRSST.CRMS bit = 0.

If the NQTHCTL.IBIQTH = 0:

- When IBI Data Buffer is completely empty.

If the NQTHCTL.IBIQTH is other than 0:

- When the number of IBI Data Buffer empties is the NQTHCTL.IBIQTH threshold or more (see NQTHCTL register).

[Clearing conditions]

For I3C protocol mode (master): PRTS.PRTMD bit = 0, PRSST.CRMS bit = 1.

- Write 0 to this bit after 1 is read.
- On completion of the last read access to IBI Status by DMAC/DTC.

For I3C protocol mode (slave): PRTS.PRTMD bit = 0, PRSST.CRMS bit = 0.

- Write 0 to this bit after 1 is read.
- On completion of the last write access to IBI Status by DMAC/DTC.

CMDQEF bit (Normal Command Queue Empty Flag)

[Setting conditions]

The following 2 conditions are satisfied:

1. The NTSTE.CMDQEE bit = 1 (enables Normal Command Queue Empty Interrupt Status logging).
2. If the NQTHCTL.CMDQTH = 0:
 - When Normal Command Queue is completely empty.

If the NQTHCTL.CMDQTH is other than 0:

- When the number of Normal Command Queue empties is the NQTHCTL.CMDQTH threshold or more (see NQTHCTL register).

[Clearing conditions]

- Write 0 to this bit after 1 is read.
- On completion of the last write access to Normal Command by DMAC/DTC.

RSPQFF bit (Normal Response Queue Full Flag)

[Setting conditions]

The following 2 conditions are satisfied:

1. The NTSTE.RSPQFE bit = 1 (enables Normal Response Queue Full Interrupt Status logging).
2. When the number of Normal Response Queue entries is more than the NQTHCTL.RSPQTH threshold (see NQTHCTL register).

[Clearing conditions]

- Write 0 to this bit after 1 is read.
- On completion of the last read access to Normal Receive Status by DMAC/DTC.

TABTF bit (Normal Transfer Abort Flag)

[Setting conditions]

The following 2 conditions are satisfied:

1. The NTSTE.TABTE bit = 1 (enables Normal Transfer Abort Interrupt Status logging).
2. When any transfer is aborted.

[Clearing condition]

- Write 0 to this bit after 1 is read.

TEF bit (Normal Transfer Error Flag)

[Setting conditions]

The following 2 conditions are satisfied:

1. The NTSTE.TEE bit = 1 (enables Normal Transfer Error Interrupt Status logging).
2. When any transfer error occurs on the I3C bus. The Error type for this error is available in the Response or Receive Status structure corresponding to the Transfer command.

[Clearing condition]

- Write 0 to this bit after 1 is read.

RSQFF bit (Normal Receive Status Queue Full Flag)

[Setting conditions]

The following 2 conditions are satisfied:

1. The NTSTE.RSQFE bit = 1 (Normal Receive Status Queue Full Enable).
2. When the number of Normal Receive Status Queue entries is more than the NRQTHCTL.RSQTH threshold (see NRQTHCTL register).

[Clearing conditions]

- Write 0 to this bit after 1 is read.
- On completion of the last read access to Normal Receive Status by DMAC/DTC.

40.2.51 NTSTE : Normal Transfer Status Enable Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x1E4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	RSQF E	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TEE	—	—	—	TABTE	RSPQ FE	CMDQ EE	IBIQE FE	RDBF E0	TDBE E0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TDBEE0	Normal Tx Data Buffer Empty Enable 0 0: Disables Normal Tx Data Buffer Empty Interrupt Status logging. 1: Enables Normal Tx Data Buffer Empty Interrupt Status logging.	R/W
1	RDBFE0	Normal Rx Data Buffer Full Enable 0 0: Disables Normal Rx Data Buffer Full Interrupt Status logging. 1: Enables Normal Rx Data Buffer Full Interrupt Status logging.	R/W
2	IBIQEFE	Normal IBI Queue Empty/Full Enable*1 0: Disables Normal IBI Queue Empty/Full Interrupt Status logging. 1: Enables Normal IBI Queue Empty/Full Interrupt Status logging.	R/W
3	CMDQEE	Normal Command Queue Empty Enable*1 0: Disables Normal Command Queue Empty Interrupt Status logging. 1: Enables Normal Command Queue Empty Interrupt Status logging.	R/W
4	RSPQFE	Normal Response Queue Full Enable*1 0: Disables Normal Response queue Full Interrupt Status logging. 1: Enables Normal Response queue Full Interrupt Status logging.	R/W

Bit	Symbol	Function	R/W
5	TABTE	Normal Transfer Abort Enable* ¹ 0: Disables Normal Transfer Abort Interrupt Status logging. 1: Enables Normal Transfer Abort Interrupt Status logging.	R/W
8:6	—	These bits are read as 0. The write value should be 0.	R/W
9	TEE	Normal Transfer Error Enable* ¹ 0: Disables Normal Transfer Error Interrupt Status logging. 1: Enables Normal Transfer Error Interrupt Status logging.	R/W
19:10	—	These bits are read as 0. The write value should be 0.	R/W
20	RSQFE	Normal Receive Status Queue Full Enable* ² 0: Disables Normal Receive Status Queue Full Interrupt Status logging. 1: Enables Normal Receive Status Queue Full Interrupt Status logging.	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports all I3C mode.

Note 2. This bit supports I3C secondary master mode and I3C slave mode.

TDBEE0 bit (Normal Tx Data Buffer Empty Enable 0)

When this bit is 1, the operation of NTST.TDBEF0 is enabled.

For the setting conditions and clearing conditions of the NTST.TDBEF0 flag, see the details of NTST.TDBEF0.

RDBFE0 bit (Normal Rx Data Buffer Full Enable 0)

When this bit is 1, the operation of NTST.RDBFF0 is enabled.

For the setting conditions and clearing conditions of the NTST.RDBFF0 flag, see the details of NTST.RDBFF0.

IBIQEFE bit (Normal IBI Queue Empty/Full Enable)

When this bit is 1, the operation of NTST.IBIQEFF is enabled.

For the setting conditions and clearing conditions of the NTST.IBIQEFF flag, see the details of NTST.IBIQEFF.

CMDQEE bit (Normal Command Queue Empty Enable)

When this bit is 1, the operation of NTST.CMDQEF is enabled.

For the setting conditions and clearing conditions of the NTST.CMDQEF flag, see the details of NTST.CMDQEF.

RSPQFE bit (Normal Response Queue Full Enable)

When this bit is 1, the operation of NTST.RSPQFF is enabled.

For the setting conditions and clearing conditions of the NTST.RSPQFF flag, see the details of NTST.RSPQFF.

TABTE bit (Normal Transfer Abort Enable)

When this bit is 1, the operation of NTST.TABTF is enabled.

For the setting conditions and clearing conditions of the NTST.TABTF flag, see the details of NTST.TABTF.

TEE bit (Normal Transfer Error Enable)

When this bit is 1, the operation of NTST.TEF is enabled.

For the setting conditions and clearing conditions of the NTST.TEF flag, see the details of NTST.TEF.

RSQFE bit (Normal Receive Status Queue Full Enable)

When this bit is 1, the operation of NTST.RSQFF is enabled.

For the setting conditions and clearing conditions of the NTST.RSQFF flag, see the details of NTST.RSQFF.

40.2.52 NTIE : Normal Transfer Interrupt Enable Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x1E8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	RSQFI E	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TEIE	—	—	—	TABTI E	RSPQ FIE	CMDQ EIE	IBIQE FIE	RDBFI E0	TDBEI E0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TDBEIE0	Normal Tx Data Buffer Empty Interrupt Enable 0 0: Disables Normal Tx Data Buffer Empty Interrupt Signal. 1: Enables Normal Tx Data Buffer Empty Interrupt Signal.	R/W
1	RDBFIE0	Normal Rx Data Buffer Full Interrupt Enable 0 0: Disables Normal Rx Data Buffer Full Interrupt Signal. 1: Enables Normal Rx Data Buffer Full Interrupt Signal.	R/W
2	IBIQEFIE	Normal IBI Queue Empty/Full Interrupt Enable* ¹ 0: Disables Normal IBI Queue Empty/Full Interrupt Signal. 1: Enables Normal IBI Queue Empty/Full Interrupt Signal.	R/W
3	CMDQEIE	Normal Command Queue Empty Interrupt Enable* ¹ 0: Disables Normal Command Queue Empty Interrupt Signal. 1: Enables Normal Command Queue Empty Interrupt Signal.	R/W
4	RSPQFIE	Normal Response Queue Full Interrupt Enable* ¹ 0: Disables Normal Response Queue Full Interrupt Signal. 1: Enables Normal Response Queue Full Interrupt Signal.	R/W
5	TABTIE	Normal Transfer Abort Interrupt Enable* ¹ 0: Disables Normal Transfer Abort Interrupt Signal. 1: Enables Normal Transfer Abort Interrupt Signal.	R/W
8:6	—	These bits are read as 0. The write value should be 0.	R/W
9	TEIE	Normal Transfer Error Interrupt Enable* ¹ 0: Disables Normal Transfer Error Interrupt Signal. 1: Enables Normal Transfer Error Interrupt Signal.	R/W
19:10	—	These bits are read as 0. The write value should be 0.	R/W
20	RSQFIE	Normal Receive Status Queue Full Interrupt Enable* ² 0: Disables Normal Receive Status Queue Full Interrupt Signal. 1: Enables Normal Receive Status Queue Full Interrupt Signal.	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports all I3C mode.

Note 2. This bit supports I3C secondary master mode and I3C slave mode.

The PIO Interrupt Signal Enable register enables signaling of outstanding interrupts received by I3C.

TDBEIE0 bit (Normal Tx Data Buffer Empty Interrupt Enable 0)

This bit is used to enable or disable the Normal Tx Data buffer empty interrupt (I3C_TX) requests when the NTST.TDBEF0 flag is set to 1.

RDBFIE0 bit (Normal Rx Data Buffer Full Interrupt Enable 0)

This bit is used to enable or disable the Normal Rx Data buffer full interrupt (I3C_RX) requests when the NTST.RDBFF0 flag is set to 1.

IBIQEFIE bit (Normal IBI Queue Empty/Full Interrupt Enable)

This bit is used to enable or disable the Normal IBI Queue Empty/Full interrupt (I3C_IBI) requests when the NTST.IBIQEFC flag is set to 1.

CMDQEIE bit (Normal Command Queue Empty Interrupt Enable)

This bit is used to enable or disable the Normal Command queue empty interrupt (I3C_CMD) requests when the NTST.CMDQEFC flag is set to 1.

RSPQFIE bit (Normal Response Queue Full Interrupt Enable)

This bit is used to enable or disable the Normal Response queue full interrupt (I3C_RESP) requests when the NTST.RSPQFFC flag is set to 1.

TABTIE bit (Normal Transfer Abort Interrupt Enable)

This bit is used to enable or disable the Normal Transfer Abort interrupt (I3C_EEI) requests when the NTST.TABTF flag is set to 1.

TEIE bit (Normal Transfer Error Interrupt Enable)

This bit is used to enable or disable the Normal Transfer Error interrupt (I3C_EEI) requests when the NTST.TEF flag is set to 1.

RSQFIE bit (Normal Receive Status Queue Full Interrupt Enable)

This bit is used to enable or disable the Normal Receive Status queue full interrupt (I3C_RCV) requests when the NTST.RSQFFC flag is set to 1.

40.2.53 NTSTFC : Normal Transfer Status Force Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x1EC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	RSQF FC	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TEFC	—	—	—	TABTF C	RSPQ FFC	CMDQ EFC	IBIQE FFC	RDBF FC0	TDBE FC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TDBEFC0	Normal Tx Data Buffer Empty Force 0 0: Not Force Normal Tx Data Buffer Empty Interrupt for software testing. 1: Force Normal Tx Data Buffer Empty Interrupt for software testing.	W
1	RDBFFC0	Normal Rx Data Buffer Full Force 0 0: Not Force Normal Rx Data Buffer Full Interrupt for software testing. 1: Force Normal Rx Data Buffer Full Interrupt for software testing.	W
2	IBIQEFC	Normal IBI Queue Empty/Full Force*1 0: Not Force Normal IBI Queue Empty/Full Interrupt for software testing. 1: Force Normal IBI Queue Empty/Full Interrupt for software testing.	W
3	CMDQEFC	Normal Command Queue Empty Force*1 0: Not Force Normal Command Queue Empty Interrupt for software testing. 1: Force Normal Command Queue Empty Interrupt for software testing.	W
4	RSPQFFC	Normal Response Queue Full Force*1 0: Not Force Normal Response Queue Full Interrupt for software testing. 1: Force Normal Response Queue Full Interrupt for software testing.	W

Bit	Symbol	Function	R/W
5	TABTFC	Normal Transfer Abort Force* ¹ 0: Not Force Normal Transfer Abort Interrupt for software testing. 1: Force Normal Transfer Abort Interrupt for software testing.	W
8:6	—	These bits are read as 0.	R
9	TEFC	Normal Transfer Error Force* ¹ 0: Not Force Normal Transfer Error Interrupt for software testing. 1: Force Normal Transfer Error Interrupt for software testing.	W
19:10	—	The write value should be 0.	W
20	RSQFFC	Normal Receive Status Queue Full Force* ² 0: Not Force Normal Receive Status Queue Full Interrupt for software testing. 1: Force Normal Receive Status Queue Full Interrupt for software testing.	W
31:21	—	The write value should be 0.	W

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports all I3C mode.

Note 2. This bit supports I3C secondary master mode and I3C slave mode.

The PIO Interrupt Force register is used to force specific interrupt. It can be used for debug purposes.

TDBEFC0 bit (Normal Tx Data Buffer Empty Force 0)

For software testing, when set to 1, forces the corresponding interrupt, subject to TDBEE0 and TDBEIE0 configuration.

RDBFFC0 bit (Normal Rx Data Buffer Full Force 0)

For software testing, when set to 1, forces the corresponding interrupt, subject to RDBFE0 and RDBFIE0 configuration.

IBIQEFC bit (Normal IBI Queue Empty/Full Force)

For software testing, when set to 1, forces the corresponding interrupt, subject to IBIQEF0 and IBIQEFIE configuration.

CMDQEFC bit (Normal Command Queue Empty Force)

For software testing, when set to 1, forces the corresponding interrupt, subject to CMDQEE and CMDQEIE configuration.

RSPQFFC bit (Normal Response Queue Full Force)

For software testing, when set to 1, forces the corresponding interrupt, subject to RSPQFE and RSPQFIE configuration.

TABTFC bit (Normal Transfer Abort Force)

For software testing, forces the corresponding interrupt, subject to TABTE and TABTIE configuration.

TEFC bit (Normal Transfer Error Force)

For software testing, when set to 1, forces the corresponding interrupt, subject to TEE and TEIE configuration.

RSQFFC bit (Normal Receive Status Queue Full Force)

For software testing, when set to 1, forces the corresponding interrupt, subject to RSQFE and RSQFIE configuration.

40.2.54 HTST : High Priority Transfer Status Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x200

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TEF	—	—	—	TABTF	RSPQ FF	CMDQ EF	—	RDBF F	TDBE F
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TDBEF	High Priority Tx Data Buffer Empty Flag 0: The number of empties in the High Priority Tx Data Buffer is less than the HTBTHCTL.TXDBTH[2:0] threshold. 1: The number of empties in the High Priority Tx Data Buffer is or more than the HTBTHCTL.TXDBTH[2:0] threshold.	R/W ^{*1}
1	RDBFF	High Priority Rx Data Buffer Full Flag 0: The number of entries in the High Priority Rx Data Buffer is less than the HTBTHCTL.RXDBTH[2:0] threshold. 1: The number of entries in the High Priority Rx Data Buffer is or more than the HTBTHCTL.RXDBTH[2:0] threshold.	R/W ^{*1}
2	—	This bit is read as 0. The write value should be 0.	R/W
3	CMDQEF	High Priority Command Queue Empty Flag 0: If HQTHTCTL.CMDQTH is 0, the number of High Priority Command Queue empties is less than the Command Queue size. If HQTHTCTL.CMDQTH is other than 0, the number of High Priority Command Queue empties is less than the HQTHTCTL.CMDQTH threshold. 1: If HQTHTCTL.CMDQTH is 0, the number of High Priority Command Queue empties is the Command Queue size. If HQTHTCTL.CMDQTH is other than 0, the number of High Priority Command Queue empties is or more than the HQTHTCTL.CMDQTH threshold.	R/W ^{*1}
4	RSPQFF	High Priority Response Queue Full Flag 0: The number of High Priority Response Queue entries is less than the HQTHTCTL.RSPQTH threshold. 1: The number of High Priority Response Queue entries is or more than the HQTHTCTL.RSPQTH threshold.	R/W ^{*1}
5	TABTF	High Priority Transfer Abort Flag 0: High Priority Transfer Abort does not occur. 1: High Priority Transfer Abort occurs. To clear, write 0 to this bit after 1 is read.	R/W ^{*1}
8:6	—	These bits are read as 0. The write value should be 0.	R/W
9	TEF	High Priority Transfer Error Flag 0: High Priority Transfer Error does not occur. 1: High Priority Transfer Error occurs. To clear, write 0 to this bit after 1 is read.	R/W ^{*1}
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

Note 1. Clearing (to 0) condition : Writing 0 after the 1 state is read.

TDBEF bit (High Priority Tx Data Buffer Empty Flag)

[Setting condition]

The following 2 conditions are satisfied:

1. The HTSTE.TDBEE bit = 1 (Enables High Priority Tx Data Buffer Empty Interrupt Status Logging).
2. When the number of empties in the High Priority Tx Data Buffer is the HTBTHCTL.TXDBTH[2:0] threshold (see register HTBTHCTL) or more.

[Clearing condition]

- Write 0 to this bit after 1 is read.
- On completion of the last write access to “High Priority Tx Data” by DMAC/DTC.

RDBFF bit (High Priority Rx Data Buffer Full Flag)

[Setting condition]

The following 2 conditions are satisfied:

1. The HTSTE.RDBFE bit = 1 (Enables High Priority Rx Data Buffer Full Interrupt Status Logging).
2. When the number of High Priority Rx Data Buffer entries is $\geq$ the HTBTHCTL.RXDBTH[2:0] threshold (see register HTBTHCTL).

[Clearing condition]

- Write 0 to this bit after 1 is read.
- On completion of the last read access to “High Priority Rx Data” by DMAC/DTC.

CMDQEF bit (High Priority Command Queue Empty Flag)

[Setting condition]

The following 2 conditions are satisfied:

1. The HTSTE.CMDQEE bit = 1 (Enables High Priority Command Queue Empty Interrupt Status Logging).
2. If the HQTHTCTL.CMDQTH = 0:
 - When High Priority Command Queue is completely empty

If the HQTHTCTL.CMDQTH is other than 0:

- When the number of High Priority Command Queue empties is the HQTHTCTL.CMDQTH threshold (see register HQTHTCTL) or more.

[Clearing condition]

- Write 0 to this bit after 1 is read.
- On completion of the last write access to “High Priority Command” by DMAC/DTC.

RSPQFF bit (High Priority Response Queue Full Flag)

[Setting condition]

The following 2 conditions are satisfied:

1. The HTSTE.RSPQFE bit = 1 (Enables High Priority Response Queue Full Interrupt Status Logging).
2. When the number of High Priority Response Queue entries is $>$ the HQTHTCTL.RSPQTH threshold (see register HQTHTCTL).

[Clearing condition]

- Write 0 to this bit after 1 is read.
- On completion of the last read access to “High Priority Receive Status” by DMAC/DTC.

See the description of the RSPQFF bit for the elements used.

TABTF bit (High Priority Transfer Abort Flag)

[Setting condition]

The following 2 conditions are satisfied:

1. The HTSTE.TABTE bit = 1 (Enables High Priority Transfer Abort Interrupt Status Logging).
2. When any transfer is aborted.

[Clearing condition]

- Write 0 to this bit after 1 is read.

TEF bit (High Priority Transfer Error Flag)

[Setting condition]

The following 2 conditions are satisfied:

1. The HTSTE.TEE bit = 1 (Enables High Priority Transfer Error Interrupt Status Logging).
2. When any transfer error occurs on the I3C Bus. The error type for this error is available in the Response structure corresponding to this transfer/command.

[Clearing condition]

- Write 0 to this bit after 1 is read.

40.2.55 HTSTE : High Priority Transfer Status Enable Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x204

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TEE	—	—	—	TABTE	RSPQFE	CMDQEE	—	RDBFE	TDBEE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TDBEE	High Priority Tx Data Buffer Empty Enable 0: Disables High Priority Tx Data Buffer Empty Interrupt Status logging. 1: Enables High Priority Tx Data Buffer Empty Interrupt Status logging.	R/W
1	RDBFE	High Priority Rx Data Buffer Full Enable 0: Disables High Priority Rx Data Buffer Full Interrupt Status logging. 1: Enables High Priority Rx Data Buffer Full Interrupt Status logging.	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	CMDQEE	High Priority Command Queue Empty Enable 0: Disables High Priority Command Queue Empty Interrupt Status logging. 1: Enables High Priority Command Queue Empty Interrupt Status logging.	R/W
4	RSPQFE	High Priority Response Queue Full Enable 0: Disables High Priority Response Queue Full Interrupt Status logging. 1: Enables High Priority Response Queue Full Interrupt Status logging.	R/W
5	TABTE	High Priority Transfer Abort Enable 0: Disables High Priority Transfer Abort Interrupt Status logging. 1: Enables High Priority Transfer Abort Interrupt Status logging.	R/W
8:6	—	These bits are read as 0. The write value should be 0.	R/W
9	TEE	High Priority Transfer Error Enable 0: Disables High Priority Transfer Error interrupt Stats logging. 1: Enables High Priority Transfer Error interrupt Stats logging.	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

TDBEE bit (High Priority Tx Data Buffer Empty Enable)

When TDBEE is 1, the operation of HTST.TDBEF is enabled.

For the setting conditions and clearing conditions of the HTST.TDBEF flag, see the details of HTST.TDBEF.

RDBFE bit (High Priority Rx Data Buffer Full Enable)

When RDBFE is 1, the operation of HTST.RDBFF is enabled.

For the setting conditions and clearing conditions of the HTST.RDBFF flag, see the details of HTST.RDBFF.

CMDQEE bit (High Priority Command Queue Empty Enable)

When CMDQEE is 1, the operation of HTST.CMDQEF is enabled.

For the setting conditions and clearing conditions of the HTST.CMDQEF flag, see the details of HTST.CMDQEF.

RSPQFE bit (High Priority Response Queue Full Enable)

When RSPQFE is 1, the operation of HTST.RSPQFF is enabled.

For the setting conditions and clearing conditions of the HTST.RSPQFF flag, see the details of HTST.RSPQFF.

TABTE bit (High Priority Transfer Abort Enable)

When TABTE is 1, the operation of HTST.TABTF is enabled.

For the setting conditions and clearing conditions of the HTST.TABTF flag, see the details of HTST.TABTF.

TEE bit (High Priority Transfer Error Enable)

When TEE is 1, the operation of HTST.TEF is enabled.

For the setting conditions and clearing conditions of the HTST.TEF flag, see the details of HTST.TEF.

40.2.56 HTIE : High Priority Transfer Interrupt Enable Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x208

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TEIE	—	—	—	TABTIE	RSPQFIE	CMDQEIE	—	RDBFIE	TDBEIE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TDBEIE	High Priority Tx Data Buffer Empty Interrupt Enable 0: Disables High Priority Tx Data Buffer Empty Interrupt Signal. 1: Enables High Priority Tx Data Buffer Empty Interrupt Signal.	R/W
1	RDBFIE	High Priority Rx Data Buffer Full Interrupt Enable 0: Disables High Priority Rx Data Buffer Full Interrupt Signal. 1: Enables High Priority Rx Data Buffer Full Interrupt Signal.	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	CMDQEIE	High Priority Command Queue Empty Interrupt Enable 0: Disables High Priority Command Queue Empty Interrupt Signal. 1: Enables High Priority Command Queue Empty Interrupt Signal.	R/W

Bit	Symbol	Function	R/W
4	RSPQFIE	High Priority Response Queue Full Interrupt Enable 0: Disables High Priority Response Queue Full Interrupt Signal. 1: Enables High Priority Response Queue Full Interrupt Signal.	R/W
5	TABTIE	High Priority Transfer Abort Interrupt Enable 0: Disables High Priority Transfer Abort interrupt Signal. 1: Enables High Priority Transfer Abort interrupt Signal.	R/W
8:6	—	These bits are read as 0. The write value should be 0.	R/W
9	TEIE	High Priority Transfer Error Interrupt Enable 0: Disables High Priority Transfer Error Interrupt Signal. 1: Enables High Priority Transfer Error Interrupt Signal.	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

The high priority interrupt signal enable register enables signaling of outstanding high priority interrupts received by I3C.

TDBEIE bit (High Priority Tx Data Buffer Empty Interrupt Enable)

TDBEIE is used to enable or disable the High Priority Tx Data buffer empty interrupt (I3C_HTX) requests when the HTST.TDBEF flag is set to 1.

RDBFIE bit (High Priority Rx Data Buffer Full Interrupt Enable)

RDBFIE is used to enable or disable the High Priority Rx Data buffer full interrupt (I3C_HRX) requests when the HTST.RDBFF flag is set to 1.

CMDQEIE bit (High Priority Command Queue Empty Interrupt Enable)

CMDQEIE is used to enable or disable the High Priority Command Queue empty interrupt (I3C_HCMD) requests when the HTST.CMDQEF flag is set to 1.

RSPQFIE bit (High Priority Response Queue Full Interrupt Enable)

RSPQFIE is used to enable or disable the High Priority Response Queue full interrupt (I3C_HRESP) requests when the HTST.RSPQFF flag is set to 1.

TABTIE bit (High Priority Transfer Abort Interrupt Enable)

TABTIE is used to enable or disable the High Priority Transfer Abort interrupt (I3C_EEI) requests when the HTST.TABTF flag is set to 1.

TEIE bit (High Priority Transfer Error Interrupt Enable)

TEIE is used to enable or disable the High Priority Transfer Error interrupt (I3C_EEI) requests when the HTST.TEF flag is set to 1.

40.2.57 HTSTFC : High Priority Transfer Status Force Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x20C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TEFC	—	—	—	TABTF C	RSPQ FFC	CMDQ EFC	—	RDBF FC	TDBE FC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TDBEFC	High Priority Tx Data Buffer Empty Force 0: Not Force High Priority Tx Data Buffer Empty Interrupt for software testing. 1: Force High Priority Tx Data Buffer Empty Interrupt for software testing.	W
1	RDBFFC	High Priority Rx Data Buffer Full Force 0: Not Force High Priority Rx Data Buffer Full Interrupt for software testing. 1: Force High Priority Rx Data Buffer Full Interrupt for software testing.	W
2	—	This bit is read as 0.	R
3	CMDQEFC	High Priority Command Queue Empty Force 0: Not Force High Priority Command Queue Empty Interrupt for software testing. 1: Force High Priority Command Queue Empty Interrupt for software testing.	W
4	RSPQFFC	High Priority Response Queue Full Force 0: Not Force High Priority Response Queue Full Interrupt for software testing. 1: Force High Priority Response Queue Full Interrupt for software testing.	W
5	TABTFC	High Priority Transfer Abort Force 0: Not Force High Priority Transfer Abort Interrupt for software testing. 1: Force High Priority Transfer Abort Interrupt for software testing.	W
8:6	—	These bits are read as 0.	R
9	TEFC	High Priority Transfer Error Force 0: Not Force High Priority Transfer Error Interrupt for software testing. 1: Force High Priority Transfer Error Interrupt for software testing.	W
31:10	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

40.2.58 BCST : Bus Condition Status Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x210

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	BIDLF	BAVLF	BFRE F
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BFREF	Bus Free Detection Flag 0: Have not Detected Bus Free 1: Have Detected Bus Free	R
1	BAVLF	Bus Available Detection Flag*1 0: Have not Detected Bus Available 1: Have Detected Bus Available	R
2	BIDLF	Bus Idle Detection Flag*1 0: Have not Detected Bus Idle 1: Have Detected Bus Idle	R
31:3	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

Note 1. This bit supports all I3C mode.

BFREF bit (Bus Free Detection Flag)

The Bus Free Condition is a period occurring after a STOP and before a START, and with the following duration:

- For Pure Bus: A duration of at least t_{CAS} (see [section 60, Electrical Characteristics](#))
- For Mixed Bus (at least one Legacy I²C is present on the I3C Bus): A duration of at least t_{BUF} (see [section 60, Electrical Characteristics](#))

[Setting conditions]

- After a STOP condition is detected, when the number of cycles ($I3C\phi$) that are set by `BFRECDT.FRECYC[8:0]` has passed in the state of `SCL = SDA = 1`.
- After setting `BCTL.BUSE` to 1, when the number of cycles ($I3C\phi$) that are set by `BFRECDT.FRECYC[8:0]` has passed in the state of `SCL = SDA = 1`.

[Clearing conditions]

- When `SCL` and `SDA` are other than high.
- When the `BCTL.BUSE` bit is set to 0.

BAVLF bit (Bus Available Detection Flag)

The Bus Available Condition is a period during which the Bus Free Condition is sustained continuously for a duration of at least t_{AVAL} (see [section 60, Electrical Characteristics](#)). A Slave can only issue a START Request (for an In-Band Interrupt, or for a Master Handoff Request) after a Bus Available Condition.

[Setting conditions]

- After a STOP condition is detected, when the number of cycles ($I3C\phi$) that are set by `BAVLCDT.AVLCYC[8:0]` has passed in the state of `SCL = SDA = 1`.
- After setting `BCTL.BUSE` to 1, when the number of cycles ($I3C\phi$) that are set by `BAVLCDT.AVLCYC[8:0]` has passed in the state of `SCL = SDA = 1`.

[Clearing conditions]

- When `SCL` and `SDA` are other than high.
- When the `BCTL.BUSE` bit is set to 0.

BIDLF bit (Bus Idle Detection Flag)

The Bus Idle Condition is a period during which the Bus Available Condition is sustained continuously for a duration of at least t_{IDLE} (see [section 60, Electrical Characteristics](#)).

[Setting conditions]

- After a STOP condition is detected, when the number of cycles ($I3C\phi$) that are set by `BIDLCDT.IDLCYC[17:0]` has passed in the state of `SCL = SDA = 1`.
- After setting `BCTL.BUSE` to 1, when the number of cycles ($I3C\phi$) that are set by `BIDLCDT.IDLCYC[17:0]` has passed in the state of `SCL = SDA = 1`.

[Clearing conditions]

- When `SCL` and `SDA` are other than high.
- When the `BCTL.BUSE` bit is set to 0.

40.2.59 SVST : Slave Status Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x214

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SVA[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	HOAF	—	—	—	—	—	—	—	—	DVIDF	HSMCF	—	—	—	—	GCAF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	GCAF	General Call Address Detection Flag 0: General call address does not detect. 1: General call address detects.	R/W ^{*1}
4:1	—	These bits are read as 0. The write value should be 0.	R/W
5	HSMCF	Hs-mode Master Code Detection Flag 0: Hs-mode Master Code does not detect. 1: Hs-mode Master Code detects.	R/W ^{*1}
6	DVIDF	Device-ID Address Detection Flag 0: Device-ID command does not detect. 1: Device-ID command detects. • This bit set to 1 when the first frame received immediately after a START condition is detected matches a value of (device ID (1111 100) + 0[W]).	R/W ^{*1}
14:7	—	These bits are read as 0. The write value should be 0.	R/W
15	HOAF	Host Address Detection Flag 0: Host address does not detect. 1: Host address detects. • This bit set to 1 when the received slave address matches the host address (0001 000).	R/W ^{*1}
18:16	SVA[2:0]	Slave Address Detection Flag n (n = 0 to 2) 0: Slave n does not detect 1: Slave n detect	R/W ^{*1}
31:19	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I²C mode.

Note 1. Clearing (to 0) condition : Writing 0 after the 1 state is read.

GCAF flag (General Call Address Detection Flag)

I²C Normal Wake-Up Mode1 / 2 sets GCAF to 1 when switching from asynchronous operation to synchronous unit.

[Setting conditions]

- This flag is set to 1 at the rising edge of the ninth SCL clock cycle in the first byte when all of the followings are satisfied.
 - The SVCTL.GCAE bit = 1 (General call address detection is enabled).
 - When the received slave address matches the general call address (0000 000 + 0 (write)).

[Clearing conditions]

- When 0 is written to the GCAF flag after reading GCAF flag to be 1.
- When a STOP condition is detected.
- When a Repeated START condition is detected.

HSMCF flag (Hs-mode Master Code Detection Flag)

The I²C Normal Wake-Up Mode 1/2 sets 1 to HSMCF when switching from asynchronous operation to synchronous unit.

[Setting conditions]

- This flag is set to 1 at the rising edge of the ninth SCL clock cycle in the first byte when all of the followings are satisfied.
 1. The SVCTL.HSMCE bit = 1 (Hs-mode master code detection is enabled).
 2. When the first byte received immediately after a START condition is detected matches a value of Hs-mode master code (0000 1XXX) + 1 (NACK).

[Clearing conditions]

- When 0 is written to the HSMCF flag after reading HSMCF flag to be 1.
- When a STOP condition is detected.

DVIDF flag (Device-ID Address Detection Flag)**[Setting conditions]**

- This flag is set to 1 at the rising edge of the ninth SCL clock cycle in the first byte when all of the followings are satisfied.
 1. The SVCTL.DVIDE bit = 1 (Device-ID address detection is enabled).
 2. When the first byte received immediately after a START condition or Repeated START condition is detected matches a value of Device ID (1111 100) + 0 (write).

[Clearing conditions]

- When 0 is written to the DVIDF flag after reading DVIDF flag to be 1.
- When a STOP condition is detected.
- This flag is set to 0 at the rising edge of the ninth SCL clock cycle in the first byte when the following 1 and 2 or 1 and 3 are satisfied.
 1. The SVCTL.DVIDE bit = 1 (Device-ID address detection is enabled).
 2. When the first byte received immediately after a START condition or Repeated START condition is detected does not match a value of Device ID (1111 100).
 3. When the first byte received immediately after a START condition or Repeated START condition is detected matches a value of (device ID (1111 100) + 0 [W]) and the second byte does not match any of slave addresses 0 to 2.

HOAF flag (Host Address Detection Flag)

I²C Normal Wake-Up Mode1 / 2 sets HOAF to 1 at the time of switching from asynchronous operation to synchronous unit.

[Setting conditions]

- This flag is set to 1 at the rising edge of the ninth SCL clock cycle in the first byte when all of the followings are satisfied.
 1. The SVCTL.HOAE bit = 1 (Host address detection is enabled).
 2. When the received slave address matches the host address (0001 000).

[Clearing conditions]

- When 0 is written to the HOAF flag after reading HOAF flag to be 1.
- When a STOP condition is detected.
- When a Repeated START condition is detected.

SVAF[2:0] flags (Slave Address Detection Flag n (n = 0 to 2))

I²C Normal Wake-Up Mode1 / 2 sets 1 to SVAF[2:0] when switching from asynchronous operation to synchronous unit.

[Setting conditions]

For 7-bit address format: SVDVADn.SADLG bit = 0.

- This flag is set to 1 at the rising edge of the ninth SCL clock cycle in the first byte when all of the followings are satisfied.
 1. The SVCTL.SVAEn bit = 1 (Slave n enabled).
 2. When the received slave address matches the SVDVADn.SVAD[6:0] bits value.

For 10-bit address format: SVDVADn.SADLG bit = 1.

- This flag is set to 1 at the rising edge of the ninth SCL clock cycle in the second byte when all of the followings are satisfied.
 1. The SVCTL.SVAEn bit = 1 (Slave n enabled).
 2. When the received slave address matches a value of 11110 + SVDVADn.SVAD[9:8] bits and the following address matches the SVDVADn.SVAD[7:0] value.

[Clearing conditions]

- When 0 is written to the SVAF[2:0] flag after reading SVAF[2:0] flag to be 1.
- When a STOP condition is detected.

For 7-bit address format: SVDVADn.SADLG bit = 0.

- This flag is set to 0 at the rising edge of the ninth SCL clock cycle in the first byte when all of the followings are satisfied.
 1. The SVCTL.SVAEn bit = 1 (Slave n enabled).
 2. When the received slave address does not match SVDVADn.SVAD[6:0] bits value.

For 10-bit address format: SVDVADn.SADLG bit = 1.

- This flag is set to 0 at the rising edge of the ninth SCL clock cycle in the first byte when all of the followings are satisfied.
 1. The SVCTL.SVAEn bit = 1 (Slave n enabled).
 2. When the received slave address does not match a value of 11110 + SVDVADn.SVAD[9:8] bits.
- This flag is set to 0 at the rising edge of the ninth SCL clock cycle in the second byte when all of the followings are satisfied.
 1. The SVCTL.SVAEn bit = 1 (Slave n enabled).
 2. When the received slave address matches a value of 11110 + SVDVADn.SVAD[9:8] bits and the following address does not match the SVDVADn.SVAD[7:0] value.

40.2.60 WUST : Wake Up Unit Operating Status Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x218

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	WUAS YNF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	WUASYNF	Wake-up function asynchronous operation status flag 0: I3C synchronous circuit enable condition. 1: I3C asynchronous circuit enable condition.	R
31:1	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

WUASYNF flag (Wake-up function asynchronous operation status flag)

This bit shows whether I3C is in the TCLK asynchronous operation (WUCTL.WUFE bit = 1).

[Setting condition]

- All of the followings are satisfied.
 1. The WUCTL.WUFE bit = 1 (Wake-up function is enabled)
 2. When the BCST.BFREF flag = 1 after 0 is written to the WUCTL.WUFSYNE bit

[Clearing condition : I²C slave]

- The WUCTL.WUFE bit = 0 (Wake-up function is disabled)
- All of the followings are satisfied.
 1. The WUCTL.WUFE bit = 1 (Wake-up function is enabled)
 2. Wake-up event is detected
 3. When 1 is written to the WUCTL.WUFSYNE bit during WUASYNF flag = 1

[Clearing condition : I3C slave]

- The WUCTL.WUFE bit = 0 (Wake-up function is disabled)
- All of the followings are satisfied.
 1. The WUCTL.WUFE bit = 1 (Wake-up function is enabled)
 2. Wake-up event is detected
 3. When 1 is written to the WUCTL.WUFSYNE bit during WUASYNF flag = 1
 4. When a STOP condition is detected.

[Clearing condition : I²C/I3C slave]

- All of the followings are satisfied.
 1. The WUCTL.WUFE bit = 1 (Wake-up function is enabled)
 2. Wake-Up event is not detected
 3. The WUASYNF flag = 1
 4. The WUCTL.WUFSYNE bit = 1
 5. When a STOP condition is detected.

[Clearing condition : I3C master]

- The WUCTL.WUFE bit = 0 (Wake-up function is disabled)
- All of the followings are satisfied.
 1. The WUCTL.WUFE bit = 1 (Wake-up function is enabled)
 2. Wake-up event is detected.
 3. The WUASYNF flag = 1
 4. The WUCTL.WUFSYNE bit = 1

40.2.61 MRCCPT : MSyncCNT Counter Capture Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x21C

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	MSyncCNT Counter Capture Used in Async Mode 1, not used in Async Mode0.	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

MRCCPT[31:0] Bits

- Async Mode 1 (Asynchronous Advanced Mode)

When ATCCNTE.ATCE is enabled, it starts counting. It captures as MSyncCNT for each aME (SDA falling edge of START condition), and store it in the capture register.

40.2.62 DATBASm : Device Address Table Basic Register m (m = 0 to 7)

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x224 + 0x08 × m

Bit position:

31

30

29

28

27

26

25

24

23

16

Bit field:

DVTY P	DVNACK[1:0]	—	—	—	—	—	—	DVDYAD[7:0]
--------	-------------	---	---	---	---	---	---	-------------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:

15

14

13

12

11

10

9

8

7

6

0

Bit field:

DVIBITS	DVMRRJ	DVSIR RJ	DVIBI PL	—	—	—	—	—	DVSTAD[6:0]
---------	--------	----------	----------	---	---	---	---	---	-------------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
6:0	DVSTAD[6:0]	Device Static Address I3C/I ² C Static Address	R/W
11:7	—	These bits are read as 0. The write value should be 0.	R/W
12	DVIBIPL	Device IBI Payload 0: IBIs from this Device do not carry a Data Payload. 1: IBIs from this Device do carry a Data Payload.	R/W
13	DVSIR RJ	Device In-Band Slave Interrupt Request Reject 0: This Device shall ACK the SIR. 1: This Device shall NACK the SIR and send the auto-disable CCC.	R/W
14	DVMRRJ	Device In-Band Master Request Reject 0: This Device shall ACK Master Requests. 1: This Device shall NACK Master Requests and send the auto-disable command.	R/W
15	DVIBITS	Device IBI Time-stamp 0: The Master shall not time-stamp IBIs from this Device with Master Time-stamps. 1: The Master shall time-stamp IBIs for this Device with Master Time-stamps.	R/W
23:16	DVDYAD[7:0]	Device I3C Dynamic Address Bit 23 is the parity bit, per the I3C specification, computed and updated by the software driver.	R/W

Bit	Symbol	Function	R/W
28:24	—	These bits are read as 0. The write value should be 0.	R/W
30:29	DVNACK[1:0]	Device NACK Retry Count Device-specific retry count	R/W
31	DVTYP	Device Type 0: I3C Device 1: I ² C Device	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

DVIBIPL bit (Device IBI Payload)

Indicates whether IBIs from this Device have a Data Payload. This field reflects the IBI Payload bit in the Device's Bus Characteristics Register (BCR).

During IBI handling for this Device, the Master shall use this field to determine whether or not to drive reception of the IBI Data Payload. Data continuation is indicated by the T-Bit.

DVSIRRJ bit (Device In-Band Slave Interrupt Request Reject)

Controls whether this Device, when operating in the Master role, will accept vs. reject Slave Interrupt Requests from other Devices.

DVMRRJ bit (Device In-Band Master Request Reject)

Controls whether this Device, when operating in the Master role, will accept vs. reject Master requests from other Devices.

This bit is only valid if I3C declares Non-Current Master Capability.

DVIBITS bit (Device IBI Time-stamp)

Enables or disables IBI time-stamping for a specific Device.

Note: The IBI Status Descriptor for each IBI event indicates whether or not the individual IBI event was actually time-stamped. Set to 0 except for Async mode 0 and Async mode 1 of timing control.

DVNACK[1:0] bits (Device NACK Retry Count)

These bits set the number of retries when a NACK response is received from the slave for the transaction set in the Command Descriptor.

Note: When ENTDA is executed by Address Assign Command, the setting of this bit is ignored and the transaction ends when NACK is received once.

Note: I3C will retry according to the setting of DVNACK[1:0] bit, even if it receives a NACK for the broadcast address.

Note: If DVNACK[1:0] bits are 0x0, I3C will not retry even for Direct CCCs.

40.2.63 EXDATBAS : Extended Device Address Table Basic Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x2A0

Bit position:	31	30	29	28	27	26	25	24	23								16
Bit field:	EDTY P	EDNACK[1:0]		—	—	—	—	—	EDDYAD[7:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit position:	15	14	13	12	11	10	9	8	7	6							0
Bit field:	—	—	—	—	—	—	—	—	—	EDSTAD[6:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Bit	Symbol	Function	R/W
6:0	EDSTAD[6:0]	Extended Device Static Address I3C/I ² C static address	R/W
15:7	—	These bits are read as 0. The write value should be 0.	R/W
23:16	EDDYAD[7:0]	Extended Device I3C Dynamic Address Bit 23 is the parity bit, per the I3C specification, computed and updated by the software driver.	R/W
28:24	—	These bits are read as 0. The write value should be 0.	R/W
30:29	EDNACK[1:0]	Extended Device NACK Retry Count Device-specific retry count	R/W
31	EDTYP	Extended Device Type 0: I3C Device 1: I ² C Device	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

40.2.64 SDATBASn : Slave Device Address Table Basic Register n (n = 0 to 2)

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x2B0 + 0x04 × n

Bit position:	31	30	29	28	27	26	25	24	23	22							16
Bit field:	—	—	—	—	—	—	—	—	—	SDDYAD[6:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit position:	15	14	13	12	11	10	9									0	
Bit field:	—	—	—	SDIBI PL	—	SDAD LS	SDSTAD[9:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Bit	Symbol	Function	R/W
9:0	SDSTAD[9:0]	Slave Device Static Address* ² I3C Static Address	R/W
10	SDADLS	Slave Device Address Length Selection* ³ 0: Slave device address length 7 bits selected. 1: Slave device address length 10 bits selected. (I ² C device only)	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
12	SDIBIPL* ¹	Slave Device IBI Payload* ⁴ This bit is the mirror bit of the SVDCT.TBCR2. 0: IBIs from this device do not carry a data payload. 1: IBIs from this device carry a data payload.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
22:16	SDDYAD[6:0]* ¹	Slave Device I3C Dynamic Address* ⁵	R/W
31:23	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: SW write to the SDATBAS register of the main master is prohibited.

Note 1. This bit is valid only in SDATBAS0 register.

Note 2. These bits support I²C, I3C secondary master, and I3C slave mode.

Note 3. This bit supports I²C mode.

Note 4. This bit supports I3C secondary master mode and I3C slave mode.

Note 5. These bits support I3C secondary master mode and I3C slave mode.

SDSTAD[9:0] bits (Slave Device Static Address)

When the 7-bit address format is selected (SDADLS bit is 0), the lower 7 bits of SDSTAD[9:0] function as the 7-bit address.

When the 10-bit address format is selected (SDADLS bit is 1), the SDSTAD[9:0] function as the 10-bit address. While the SVCTL.SVAEn bit is 0, the setting of this bit is ignored.

SDIBIPL bit (Slave Device IBI Payload)

Indicates whether IBIs from this Device have a Data Payload. This field reflects the IBI Payload bit in the Device's Bus Characteristics Register (BCR).

During IBI handling for this Device, the Master shall use this field to determine whether or not to drive reception of the IBI Data Payload. Data continuation is indicated by the T-Bit.

SDDYAD[6:0] bits (Slave Device I3C Dynamic Address)

[Update conditions]

- When writing Dynamic Address value.
- When Slave Address value is its own Static Address in receiving SETDASA CCC (Direct), these bits are updated to Dynamic Address value.^{*1}
- When Dynamic Address Assignment procedure that starts by receiving ENTDAACCC (Broadcast) is established.^{*1}
- When receiving RSTDAA CCC (Broadcast), all bits are cleared to 0.^{*1}
- When Slave Address value is its own Dynamic Address in receiving RSTDAA CCC (Direct), all bits are cleared to 0.^{*1}
- When Slave Address value is its own Dynamic Address in receiving SETNEWDA CCC (Direct), these bits are updated to the Dynamic Address value.^{*1}
- When receiving SETAASA CCC (Broadcast), these bits are updated to the value of SDSTAD[6:0] bits^{*2}.

Note 1. See the MIPI I3C Specification v1.0.

Note 2. See the MIPI I3C Basic Specification v1.0.

40.2.65 MSDCTm : Master Device Characteristic Table Register m (m = 0 to 7)

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x2D0 + 0x04 × m

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RBCR76[1:0]	RBCR 5	RBCR 4	RBCR 3	RBCR 2	RBCR 1	RBCR 0	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	RBCR0	Max Data Speed Limitation ^{*1} 0: No Limitation 1: Limitation	R/W
9	RBCR1	IBI Request Capable 0: Not Capable 1: Capable	R/W

Bit	Symbol	Function	R/W
10	RBCR2	IBI Payload 0: No data byte follows the accepted IBI. 1: Mandatory one or more data bytes follow the accepted IBI. Data byte continuation is indicated by T-Bit.	R/W
11	RBCR3	Offline Capable* ² 0: Device will always respond to I3C bus commands. 1: Device will not always respond to I3C bus commands.	R/W
12	RBCR4	Bridge Identifier* ³ 0: Not a Bridge Device 1: A Bridge Device	R/W
13	RBCR5	SDR Only / SDR and HDR Capable 0: SDR only 1: HDR Capable	R/W
15:14	RBCR76[1:0]	Device Role 0 0: I3C Slave 0 1: I3C Master* ⁴ Others: Setting prohibited	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

Note 1. Master shall use the GETMXDS CCC to interrogate the Slave for specific limitation.

Note 2. Offline Capable Devices retain the Dynamic Address.

Note 3. Bridge Devices are required to comply with this MIPI Specification for I3C.

Note 4. For an I3C Device acting as I3C Main Master, the BCR Device Role bits will contain the value 01.

The DCT table captures the Device characteristics (PID, BCR, DCR) and assigned Dynamic Address of each Device on the I3C Bus that participates in the Dynamic Address Allocation (ENTDAA) procedure.

RBCRn bits (Received Bus Characteristic Register)

Each I3C Device that is connected to the I3C Bus shall have an associated read-only Bus Characteristics Register (BCR). This read-only register describes the I3C compliant Device's role and capabilities for use in Dynamic Address assignment and Common Command Codes.

Note: When RBCR2 is 0 and when ACK response to Slave Interrupt Request from I3C Slave by DATBASm.DVSIRRJ = 0 (m = 0 to 7), STOP Condition is issued after ACK response. When RBCR2 is 1 and when ACK response to Slave Interrupt Request from I3C Slave by DATBASm.DVSIRRJ = 0 (m = 0 to 7), IBI Payload is received after ACK response. STOP Condition is issued after end of IBI Payload.

[Update condition]

- When receiving of Bus Characteristics Register (BCR) from Device in the Dynamic Address Assignment procedure starting by receiving ENTDAACCC (Broadcast).^{*1}

Note 1. See the MIPI I3C Specification v1.0

40.2.66 SVDCT : Slave Device Characteristic Table Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x320

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7							0
Bit field:	TBCR76[1:0]		TBCR5	TBCR4	TBCR3	TBCR2	TBCR1	TBCR0	TDCR[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	TDCR[7:0]	Transfer Device Characteristic Register 255 available codes for describing the type of sensor, or device. Examples: Accelerometer, gyroscope, composite devices Default value is 0x0: Generic Device	R/W
8	TBCR0	Max Data Speed Limitation*1 0: No Limitation 1: Limitation	R/W
9	TBCR1	IBI Request Capable 0: Not Capable 1: Capable	R/W
10	TBCR2	IBI Payload 0: No data byte follows the accepted IBI. 1: Mandatory one or more data bytes follow the accepted IBI. Data byte continuation is indicated by T-Bit.	R/W
11	TBCR3	Offline Capable*2 0: Device will always respond to I3C bus commands. 1: Device will not always respond to I3C bus commands.	R/W
12	TBCR4	Bridge Identifier*3 0: Not a Bridge Device 1: A Bridge Device	R/W
13	TBCR5	SDR Only / SDR and HDR Capable 0: SDR only 1: HDR Capable	R/W
15:14	TBCR76[1:0]	Device Role 0 0: I3C Slave 0 1: I3C Master*4 Others: Setting prohibited	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

Note 1. Master shall use the GETMXDS CCC to interrogate the Slave for specific limitation.

Note 2. Offline Capable Devices retain the Dynamic Address.

Note 3. Bridge Devices are required to comply with this MIPI Specification for I3C.

Note 4. For an I3C Device acting as I3C Main Master, the BCR Device Role bits will contain the value 01b.

The DCT table captures the Device characteristics (PID, BCR, DCR) and assigned Dynamic Address of each device on the I3C bus that participates in the Dynamic Address Allocation (ENTDAA) procedure.

TDCR[7:0] bits (Transfer Device Characteristic Register)

Each I3C device that is connected to the I3C bus has an associated Device Characteristics Register (DCR). This register describes the I3C compliant device type such as accelerometer and gyroscope, for use in Dynamic Address assignment and Common Command Codes.

TBCRn bits (Transfer Bus Characteristic Register)

Each I3C device that is connected to the I3C bus has an associated Bus Characteristics Register (BCR). This register describes the role and capabilities of the I3C compliant device for use in Dynamic Address assignment and Common Command Codes.

When I3C Slave issues IBI by Command Descriptor, the condition of TBCRn is described as follows:

[Slave Interrupt Request : No IBI Payload follow the accepted IBI]

- TBCR1 = 1
- TBCR2 = 0

Note: Set DATA_LENGTH[15:0] of Command Descriptor to 0.

[Slave Interrupt Request : IBI Payload follow the accepted IBI]

- TBCR1 = 1
- TBCR2 = 1

Note: Set DATA_LENGTH[15:0] of Command Descriptor to any value.

[Mastership Request]

- TBCR1 = 1
- TBCR76[1:0] = 01b

When I3C Slave receives CCC from I3C Master, it performs the following operations according to the setting of TBCRn:

- When TBCR2 = 1, CMRLG.IBIPSZ[7:0] is sent as the 3rd byte data to GETMRL CCC from I3C Master
- When TBCR0 = 0, NACK responses to GETMXDS CCC from I3C Master
- When TBCR0 = 1, ACK responses to GETMXDS CCC from I3C Master and sends data from CMDSPW, CMDSPR, and CMDSPS registers

40.2.67 SDCTPIDL : Slave Device Characteristic Table Provisional ID Low Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x324

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Transfer Device Provisional ID Low Bits 31 to 16 are read as 0. Bits 15 to 0 are bits [15:0] of device's I3C PID.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

40.2.68 SDCTPIDH : Slave Device Characteristic Table Provisional ID High Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x328

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Transfer Device Provisional ID High Bits [47:16] of device's I3C PID.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

40.2.69 SVDVADn : Slave Device Address Register n (n = 0 to 2)

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x330 + 0x04 × n

Bit position:

31

30

29

28

27

26

25

16

Bit field:

SDYA DV	SSTA DV	—	—	SADL G	—	SVAD[9:0]									
------------	------------	---	---	-----------	---	-----------	--	--	--	--	--	--	--	--	--

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:

15

14

13

12

11

10

9

8

7

6

5

4

3

2

1

0

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0.	R
25:16	SVAD[9:0]	Slave Address* ¹ A slave address is set. When rewriting SVAD, change to SVAE = 0 and rewrite.	R
26	—	This bit is read as 0.	R
27	SADLG	Slave Address Length* ² 0: The 7-bit address format is selected. 1: The 10-bit address format is selected.	R
29:28	—	These bits are read as 0.	R
30	SSTADV	Slave Static Address Valid* ¹ 0: Slave address is disabled. 1: Slave address is enabled.	R
31	SDYADV* ⁴	Slave Dynamic Address Valid* ³ 0: Dynamic Address is disabled. 1: Dynamic Address is enabled.	R

Note: S-TYPE3, P-TYPE3

Note 1. These bits support I²C, I3C secondary master, and I3C slave mode.

Note 2. This bit supports I²C mode.

Note 3. This bit supports I3C secondary master mode and I3C slave mode.

Note 4. This bit is valid only in SVDVAD0 register.

SVAD[9:0] bits (Slave Address)

The SVAD[9:0] bits indicate a valid slave address.

[The SVDVAD0.SDYADV bit = 1]

Note: This condition is only for SVDVAD0.SVAD[9:0].

- The SVAD[9:7] bits = 0
- The SVAD[6:0] bits = the SDATBAS0.SDDYAD[6:0] bits

[The SVDVADn.SSTADV bit = 1 and the SVDVADn.SADLG bit = 0]

- The SVAD[9:7] bits = 0
- The SVAD[6:0] bits = the SDATBASn.SDSTAD[6:0] bits

[The SVDVADn.SSTADV bit = 1 and the SVDVADn.SADLG bit = 1]

- The SVAD[9:0] bits = the SDATBASn.SDSTAD[9:0] bits

SADLG bit (Slave Address Length)

[Setting conditions]

- All of the followings are satisfied:
 1. The PRTS.PRTMD bit = 1 (I²C Protocol mode)
 2. The SVCTL.SVAEn bit = 1 (Slave n is enabled)
 3. The SDATBASn.SDADLS bit = 1 (The address length is 10 bits)

[Clearing condition]

- [Setting condition] is not satisfied.

SSTADV bit (Slave Static Address Valid)

[Setting conditions]

- All of the followings are satisfied:
 1. The SVCTL.SVAEn bit = 1 (Slave n is enabled)
 2. The SVDVAD0.SDYADV bit = 0 (Dynamic Address is disabled)

Note: This condition is only for SVDVAD0.SSTADV.

3. If the SVDVADn.SADLG bit = 0, the SDATBASn.SDSTAD[6:0] bits are not all 0
If the SVDVADn.SADLG bit = 1, the SDATBASn.SDSTAD[9:0] bits are not all 0

[Clearing condition]

- [Setting condition] is not satisfied.

SDYADV bit (Slave Dynamic Address Valid)

[Setting conditions]

- All of the followings are satisfied:
 1. The PRTS.PRTMD bit = 0 (I3C Protocol mode)
 2. The SVCTL.SVAEn bit = 1 (Slave n is enabled)
 3. The SDATBAS0.SDDYAD[6:0] bits are not all 0

Note: This condition is only for SVDVAD0.SDYADV.

[Clearing condition]

- [Setting condition] is not satisfied.

40.2.70 CSECMD : CCC Slave Events Command Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x350

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MSRQ E	SVIRQ E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SVIRQE	Slave Interrupt Requests Enable 0: DISABLED: Slave-initiated Interrupts is Disabled by the Master to control. 1: ENABLED: Slave-initiated Interrupts is Enabled by the Master to control.	R/W
1	MSRQE	Mastership Requests Enable 0: DISABLED: Mastership requests from Secondary Masters is Disabled by the Current Master to control. 1: ENABLED: Mastership requests from Secondary Masters is Enabled by the Current Master to control.	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

SVIRQE bit (Slave Interrupt Requests Enable)

This bit allows the Master to control when Slave-initiated Interrupts are allowed on the I3C Bus.

These four Direct (ENEC/DISEC Format 1) or Broadcast (ENEC/DISEC Format 2) CCCs allows the Master to control when Slave-initiated traffic is (Enable) vs. is not (Disable) allowed on the I3C Bus. This control governs a Slave's attempts to request an Interrupt (ENI), or to request Mastership (ENMR).

[Setting conditions]

- When writing 1
- When receiving ENEC CCC (Broadcast) with ENINT bit = 1.^{*1}
- When ENINT bit = 1 with own Slave Address in receiving ENEC CCC (Direct).^{*1}

[Clearing conditions]

- When writing 0.
- When receiving DISEC CCC (Broadcast) with DISINT bit = 1.^{*1}
- When DISINT bit = 1 with own Slave Address in receiving DISEC CCC (Direct).^{*1}

MSRQE bit (Mastership Requests Enable)

This bit allows the Current Master to control when Mastership requests from Secondary Masters are allowed on the I3C Bus.

[Setting conditions]

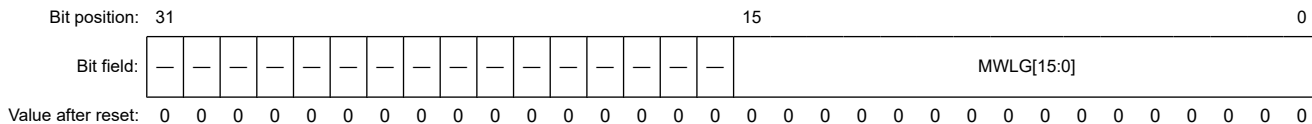
- When writing 1.
- When receiving ENEC CCC (Broadcast) with ENMR bit = 1.^{*1}
- When ENMR bit = 1 with own Slave Address in receiving ENEC CCC (Direct).^{*1}

[Clearing conditions]

40.2.72 CMWLGL : CCC Max Write Length Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x358



Bit	Symbol	Function	R/W
15:0	MWLG[15:0]	Max Write Length	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

MWLG[15:0] bits (Max Write Length)

These bits use for the I3C Master to set or get a maximum data write length in bytes for one Slave Device.

This Max Write Length does not affect data write lengths for Broadcast CCCs. The Set/Get Max Write Length value is transmitted over two bytes, with the most significant byte (MSB) transmitted first. The minimum value that Max Write Length can be set to is 8.

[Update conditions]

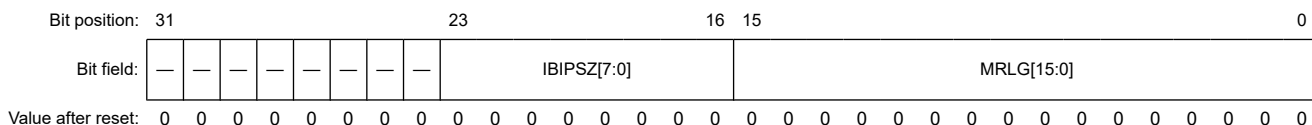
- When writing Max Write Length value.
- When receiving SETMWL CCC (Broadcast), these bits are updated to MWL value.*¹
- When Slave Address value is its own Slave Address in receiving SETMWL CCC (Direct), these bits are updated to MWL value.*¹

Note 1. See the MIPI I3C Specification v1.0

40.2.73 CMRLGL : CCC Max Read Length Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x35C



Bit	Symbol	Function	R/W
15:0	MRLG[15:0]	Max Read Length	R/W
23:16	IBIPSZ[7:0]	IBI Payload Size	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

MRLG[15:0] bits (Max Read Length)

These bits use for the I3C Master to set or get a maximum data read length for one Slave Device.

The Set/Get Max Read Length value is transmitted over the first two bytes, with most significant byte (MSB) transmitted first. The minimum value to which Max Read Length can be set is 16.

[Update conditions]

The Broadcast CCC informs all I3C Devices that the Master is entering a specified Test Mode during manufacturing or Device test. The Enter Test Mode command Frame format includes a byte that specifies which Test Mode to enter. Supporting I3C Devices shall enter the indicated Test Mode upon receipt of the Enter Test Mode CCC.

[Update condition]

- When receiving ENT TM CCC (Broadcast), these bits are updated to Test Mode Byte value.*1

Note 1. See the MIPI I3C Specification v1.0.

40.2.75 CGDVST : CCC Get Device Status Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x364

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15							8	7	6	5	4	3			0
Bit field:	VDRSV[7:0]							ACTMD[1:0]		PRTE	—	PNDINT[3:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	PNDINT[3:0]	Pending Interrupt Contains the interrupt number of any pending interrupt, or 0 if no interrupts are pending. This encoding allows for up to 15 numbered interrupts. If more than one interrupt is set, then the highest priority interrupt shall be returned.	R/W
4	—	This bit is read as 0. The write value should be 0.	R/W
5	PRTE	Protocol Error 0: The Slave has not detected a protocol error since the last Status read. 1: The Slave has detected a protocol error since the last Status read.	R/W
7:6	ACTMD[1:0]	Slave Device's current Activity Mode 0 0: Activity Mode 0 0 1: Activity Mode 1 1 0: Activity Mode 2 1 1: Activity Mode 3	R/W
15:8	VDRSV[7:0]	Vendor Reserved Reserved for vendor-specific meaning	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

PRTE bit (Protocol Error)

If this bit set to 1, then the Slave detects a protocol error since the last Status read.

The Slave checks for such errors. Note that this value self-clears by the hardware upon every successful completion of a Master read of the Slave's Status.

The Direct CCC is a Get request for one I3C Slave Device to return its current Status, in the two-byte format detailed. Note that byte 0 is the LSB, and byte 1 is the MSB.

[Setting condition]

- When the Slave detected a protocol error.*1

[Clearing condition]

- When transmission by own Slave Address is completed without error after receiving GETSTATUS CCC (Direct).*1

ACTMD[1:0] bits (Slave Device's current Activity Mode)

Contains the two-bit ID of the Slave Device's current Activity Mode (readiness to support data read of sensor or related information).

Note 1. See the MIPI I3C Specification v1.0.

40.2.76 CMDSPW : CCC Max Data Speed W (Write) Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x368

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2		0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MSWDR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	MSWDR[2:0]	Maximum Sustained Write Data Rate 0 0 0: fsci Max (default value) 0 0 1: 8 MHz 0 1 0: 6 MHz 0 1 1: 4 MHz 1 0 0: 2 MHz Others: Setting prohibited	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

40.2.77 CMDSPR : CCC Max Data Speed R (Read) Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x36C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5		3	2		0
Bit field:	—	—	—	—	—	—	—	—	—	—		CDTTIM[2:0]		MSRDR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	MSRDR[2:0]	Maximum Sustained Read Data Rate 0 0 0: fsci Max (default value) 0 0 1: 8 MHz 0 1 0: 6 MHz 0 1 1: 4 MHz 1 0 0: 2 MHz Others: Setting prohibited	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

[illegible]

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

Base address: I3C = 0x4035_F000
I3C NS = 0x5035_F000

Value after reset: 0

Bit	Symbol	Function	R/W
0	SPTSYN	Supports Sync Mode 0: Sync Mode is not supported. 1: Sync Mode is supported.	R/W
1	SPTASYN0	Support Async Mode 0 0: Async Mode 0 is not supported. 1: Async Mode 0 is supported.	R/W
2	SPTASYN1	Support Async Mode 1 0: Async Mode 1 is not supported. 1: Async Mode 1 is supported.	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
15:8	FREQ[7:0]	Frequency Byte This byte represents the Slave's internal oscillator frequency in increments of 0.5 MHz (500 kHz), up to 127.5 MHz. 0x00: 32.0 KHz 0x01: 0.5 MHz 0x02: 1.0 MHz ⋮ 0xFD: 126.5 MHz 0xFE: 127.0 MHz 0xFF: 127.5 MHz	R/W
23:16	INAC[7:0]	Inaccuracy Byte This byte represents the maximum variation of the Slave's internal oscillator in 1/10th percent (0.1%) increments, up to 25.5%. 0x00: 0.0% 0x01: 0.1% 0x02: 0.2% ⋮ 0xFD: 25.3% 0xFE: 25.4% 0xFF: 25.5%	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

SPTSYN bit (Supports Sync Mode)

Bit mask indicating which Supports Sync Mode of Timing Control Mode (s) the target Slave supports.

If this bit set (has value 1), then that Slave supports the corresponding Supports Sync Mode of Timing Control Mode.

SPTASYN0 bit (Support Async Mode 0)

Bit mask indicating which Supports Async Mode 0 of Timing Control Mode (s) the target Slave supports.

If this bit set (has value 1), then that Slave supports the corresponding Supports Async Mode 0 of Timing Control Mode.

SPTASYN1 bit (Support Async Mode 1)

Bit mask indicating which Supports Async Mode 1 of Timing Control Mode (s) the target Slave supports.

If this bit set (has value 1), then that Slave supports the corresponding Supports Async Mode 1 of Timing Control Mode.

The Directed CCC provides the framework for the Master to query the Exchange Timing capabilities supported by the I3C Slaves. The Get Exchange Timing Support Information CCC causes the addressed Slave to return four data bytes containing key information on supported Timing Control modes, current state, and internal oscillator/clock frequency and inaccuracy.

40.2.80 CETSS : CCC Exchange Timing Support Information S (State) Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x378

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	ICOVF	—	—	—	—	ASYNE[1:0]	SYNE	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SYNE	Sync Mode Enabled 0: Sync Mode Disabled 1: Sync Mode Enabled	R/W
2:1	ASYNE[1:0]	Async Mode Enabled Async Mode 3, 2 are unsupported and set to 0. 0 0: All Mode Disable 0 1: Async Mode 0 Enabled 1 0: Async Mode 1 Enabled Others: Setting prohibited	R/W
6:3	—	These bits are read as 0. The write value should be 0.	R/W
7	ICOVF	Internal Counter Overflow 0: Slave has not experienced a counter overflow since the most recent previous check. 1: Slave experienced a counter overflow since the most recent previous check.	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

Bit mask indicating which Timing Control Mode (if any) is currently enabled for the target Slave, and whether any counter overflows have occurred since the most recent previous check. If a Timing Control Mode bit is set (has value 1), then that Slave has currently enabled the corresponding Timing Control Mode. If the Overflow bit is set (has value 1), then that Slave experienced a counter overflow since the most recent previous check.

ASYNE[0] Bit (Async Mode 0 Enabled)

Slave Timing Control Async Mode 0 is enabled.

[Setting condition]

- When writing 1.
- When CETSM.SPTASYN[0] bit = 1 and either of the following 1 or 2 are satisfied.

1. When receiving SETXTIME CCC (Broadcast) with Defining byte value 0xDF.
2. When Slave Address value is its own Slave Address in receiving SETXTIME CCC (Direct) with Defining byte value 0xDF.

[Clearing condition]

- When writing 0.
- When CETSM.SPTASYN[0] bit = 1 and either of the following 1 or 2 are satisfied.

1. When receiving SETXTIME CCC (Broadcast) with Defining byte value 0xEF.

- When Slave Address value is its own Slave Address in receiving SETXTIME CCC (Direct) with Defining byte value 0xEF.

ASYNE[1] Bit (Async Mode 1 Enabled)

Slave Timing Control Async Mode 1 is enabled.

[Setting condition]

- When writing 1.
- When CETSM.SPTASYN[1] bit = 1 and either of the following 1 or 2 are satisfied.

- When receiving SETXTIME CCC (Broadcast) with Defining byte value 0xEF.
- When Slave Address value is its own Slave Address in receiving SETXTIME CCC (Direct) with Defining byte value 0xEF.

[Clearing condition]

- When writing 0.
- When CETSM.SPTASYN[1] bit = 1 and either of the following 1 or 2 are satisfied.

- When receiving SETXTIME CCC (Broadcast) with Defining byte value 0xDF.
- When Slave Address value is its own Slave Address in receiving SETXTIME CCC (Direct) with Defining byte value 0xDF.

40.2.81 CGHDRCAP : CCC Get HDR Capability Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x37C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TSLEN	TSPEN	DDREN

Value after reset: 0

Bit	Symbol	Function	R/W
0	DDREN	HDR-DDR Operation Enable 0: HDR-DDR Operation is Disabled. 1: HDR-DDR Operation is Enabled.	R/W
1	TSPEN	HDR-TSP Operation Enable 0: HDR-TSP Operation is Disabled. 1: HDR-TSP Operation is Enabled.	R/W
2	TSLEN	HDR-TSL Operation Enable 0: HDR-TSL Operation is Disabled. 1: HDR-TSL Operation is Enabled.	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R

Note: S-TYPE3, P-TYPE3

Note: This register supports for I3C secondary master mode and I3C slave mode.

DDREN bit (HDR-DDR Operation Enable)

This bit is used to enable or disable the HDR-DDR communication when used with I3C Slave.

HDR-DDR communication is permitted only when SVDCT.TBCR5 = 1 and DDREN = 1.

[Setting condition]

- When writing 1.

[Clearing condition]

- When writing 0.

TSPEN bit (HDR-TSP Operation Enable)

This bit is used to enable or disable the HDR-TSP communication when used with I3C Slave.

HDR-TSP communication is permitted only when SVDCT.TBCR5 = 1 and TSPEN = 1.

[Setting condition]

- When writing 1.

[Clearing condition]

- When writing 0.

TSLEN bit (HDR-TSL Operation Enable)

This bit is used to enable or disable the HDR-TSL communication when used with I3C Slave.

HDR-TSL communication is permitted only when SVDCT.TBCR5 = 1 and TSLEN = 1.

[Setting condition]

- When writing 1.

[Clearing condition]

- When writing 0.

Note: Refer to the MIPI I3C Specification v1.0.

The lower 8 bits of this register are used as the response data of GETHDRCAP CCC.

40.2.82 BITCNT : Bit Count Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x380

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BCNT[4:0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	BCNT[4:0]	Bit Counter Indicates the number of bits remaining to be transferred. For details on the values, see Table 40.7 and Table 40.8 .	R
31:5	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

BCNT[4:0] bits (Bit Counter)

These bits function as a counter that indicates the number of bits remaining to be transferred at the detection of a sampling edge on the I3C_SCL line.

Table 40.7 I²C / Legacy I²C transfer (1 of 2)

BCNT[4:0]	Master		Slave	
	Address phase	Data phase	Address phase	Data phase
0x00	2 to 1 bits	2 to 1 bits	3 to 1 bits	2 to 1 bits
0x01	3 bits	3 bits	4 bits	3 bits

Table 40.7 I²C / Legacy I²C transfer (2 of 2)

BCNT[4:0]	Master		Slave	
	Address phase	Data phase	Address phase	Data phase
0x02	4 bits	4 bits	5 bits	4 bits
0x03	5 bits	5 bits	6 bits	5 bits
0x04	6 bits	6 bits	7 bits	6 bits
0x05	7 bits	7 bits	8 bits	7 bits
0x06	8 bits	8 bits	9 bits	8 bits
0x07	9 bits	9 bits	—	9 bits

Table 40.8 I3C transfer

BCNT[4:0]	SDR*1		HDR-DDR		HDR-TS
	Transmission	Reception	Command/Data	CRC	
0x00	1 bit	2 to 1 bits	19, 1 bits	11, 1 bits	1 Symbol
0x01	2 bits	3 bits	20, 2 bits	12, 2 bits	2 Symbols
0x02	3 bits	4 bits	3 bits	3 bits	3 Symbols
0x03	4 bits	5 bits	4 bits	4 bits	4 Symbols
0x04	5 bits	6 bits	5 bits	5 bits	5 Symbols
0x05	6 bits	7 bits	6 bits	6 bits	6 Symbols
0x06	7 bits	8 bits	7 bits	7 bits	7 Symbols
0x07	8 bits	9 bits	8 bits	8 bits	8 Symbols
0x08	9 bits	—	9 bits	9 bits	9 Symbols
0x09	—	—	10 bits	10 bits	10 Symbols
0x0A	—	—	11 bits	—	11 Symbols
0x0B	—	—	12 bits	—	12 Symbols
0x0C	—	—	13 bits	—	—
0x0D	—	—	14 bits	—	—
0x0E	—	—	15 bits	—	—
0x0F	—	—	16 bits	—	—
0x10	—	—	17 bits	—	—
0x11	—	—	18 bits	—	—

Note 1. The address phase is the same as in Table 40.7.

40.2.83 NQSTLV : Normal Queue Status Level Register

Base address: I3C = 0x4035_F000
I3C NS = 0x5035_F000

Offset address: 0x394

Bit position: 31 30 29 28 24 23 16 15 8 7 0

Bit field:	—	—	—	IBISCNT[4:0]	IBIQLV[7:0]	RSPQLV[7:0]	CMDQFLV[7:0]
------------	---	---	---	--------------	-------------	-------------	--------------

[illegible]

Bit	Symbol	Function	R/W
7:0	CMDQFLV[7:0]	Normal Command Queue Free Level* ¹ Number of free buffer entries currently in the Normal Command Queue. Reset value is the depth of the Normal Command Queue.	R

Note: S-TYPE3, P-TYPE3
 Note 1. These bits support all I3C mode.
 Note 2. These bits support I3C master mode and I3C secondary master mode.

Base address: I3C = 0x4035_F000
I3C NS = 0x5035_F000

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 8 7 0

[illegible]

Note: S-TYPE3, P-TYPE3
Note: This register supports all I3C mode.

Base address: I3C = 0x4035_F000
I3C NS = 0x5035_F000

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 0

Value after reset: 0

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

40.2.86 HQSTLV : High Priority Queue Status Level Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x3C4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RSPQLV[7:0]								CMDQLV[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
7:0	CMDQLV[7:0]	High Priority Command Queue Level Number of free buffer entries currently in the High Priority Command Queue. Reset value is the depth of the High Priority Command Queue.	R
15:8	RSPQLV[7:0]	High Priority Response Queue Level Number of buffer entries currently in the High Priority Response Queue.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

40.2.87 HDBSTLV : High Priority Data Buffer Status Level Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x3C8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15							8	7							0
Bit field:	RDBLV[7:0]								TDBFLV[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
7:0	TDBFLV[7:0]	High Priority Tx Data Buffer Free Level Indicates the number of free High Priority Tx Data Buffer entries in the High Priority Tx Data Queue. Reset value is the depth of the High Priority Tx Data Queue.	R
15:8	RDBLV[7:0]	High Priority Rx Data Buffer Level Indicates the number of High Priority Rx Data Buffer entries in the High Priority Rx Data Queue.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x3CC

[illegible]

Bit	Symbol	Function	R/W
0	SCILV	I3C_SCL Line Signal Level This bit is used to check the I3C_SCL Line level, in order to recover from errors and for debugging.	R
1	SDILV	I3C_SDA Line Signal Level This bit is used to check the I3C_SDA Line level, in order to recover from errors and for debugging.	R
2	SCOLV	SCL Output Level 0: I3C has driven the I3C_SCL pin low. 1: I3C has released the I3C_SCL pin.	R
3	SDOLV	SDA Output Level 0: I3C has driven the I3C_SDA pin low. 1: I3C has released the I3C_SDA pin.	R
31:4	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

This bit is used to check the I3C SCL Line level, in order to recover from errors and for debugging.

This bit is used to check the I3C SDA Line level, in order to recover from errors and for debugging.

This bit is used to select the output level of I3C SCL pin.

This bit is used to select the output level of I3C SDA pin.

```
Base address: I3C = 0x4035_F000
              I3C_NS = 0x5035_F000
Offset address: 0x3D0
```

[illegible]

Bit	Symbol	Function	R/W
7:0	M2ECNT[7:0]	M2 Error Counter Counts I3C Type M2 errors on the I3C Bus. Cleared upon read out.	R
31:8	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C master mode and I3C secondary master mode.

40.2.90 SC1CPT : SC1 Capture Monitor Register

Base address: I3C = 0x4035_F000

I3C_NS = 0x5035_F000

Offset address: 0x3E0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SC1C[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	SC1C[15:0]	SC1 Capture	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

SC1C[15:0] bit (SC1 Capture)

- Async Mode 0 (Asynchronous Basic Mode)

After enabling ATCCNTE.ATCE, SC1C[15:0] Counter counts up from SC1C[15:0] count trigger^{*1} to SCL rise edge next to ACK for the IBI, and capture it as SC1C[15:0].

- Async Mode 1 (Asynchronous Advanced Mode)

After enabling ATCCNTE.ATCE, SC1C[15:0] Counter counts up from SC1C[15:0] count trigger^{*1} to the first aME, and capture it as SC1C[15:0].

Note: As the timing control specification, the SC1C[15:0] counter value is included in the IBI frame as IBI data and is sent to I3C Master, therefore it is not necessary for the I3C Slave to read this register. If the I3C Slave needs to read this register, read it after completing the IBI frame.

Note 1. SW or external trigger can be selected by selection bits.

40.2.91 SC2CPT : SC2 Capture Monitor Register

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x3E4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SC2C[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	SC2C[15:0]	SC2 Capture	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

Note: This register supports I3C secondary master mode and I3C slave mode.

SC2C[15:0] bits (SC2 Capture)

- Async Mode 0 (Asynchronous Basic Mode)
After enabling ATCCNTE.ATCE, SC2C[15:0] Counter counts up from SCL rise edge next to ACK for the IBI transmitted from I3C Slave to SCL rise edge next to Tbit after Mandatory Byte, and capture it as SC2C[15:0].
- Async Mode 1 (Asynchronous Advanced Mode)
After enabling ATCCNTE.ATCE, SC2C[15:0] Counter counts up from SCL rise edge next to ACK for the IBI transmitted from I3C Slave to SCL rise edge next to Tbit after Mandatory Byte, and capture it as SC2C[15:0].

Note: As the timing control specification, the SC2C[15:0] counter value is included in the IBI frame as IBI data and is sent to I3C Master, therefore it is not necessary for the I3C Slave to read this register. If the I3C Slave needs to read this register, read it after completing the IBI frame.

40.2.92 CECTL : Clock Enable Control Resister

Base address: I3C = 0x4035_F000
I3C_NS = 0x5035_F000

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLKE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLKE	Clock Enable 0: Clock disable 1: Clock enable	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

CLKE bit (Clock Enable)

- This bit controls enabling / disabling of clock supply of the communication function.

40.3 Operation**40.3.1 Data Structures****40.3.1.1 Command Descriptor**

The write-only Command Descriptor structure is 64 bits in length. The Command Descriptor is put to the Command Queue with writes to the Command Queue Port (High Priority or Normal).

Write to the Command Queue Port (High Priority or Normal) in the following order:

1. First write : The least significant DWORD (Command Descriptor Structure Low).
2. Second write : The most significant DWORD (Command Descriptor Structure High).

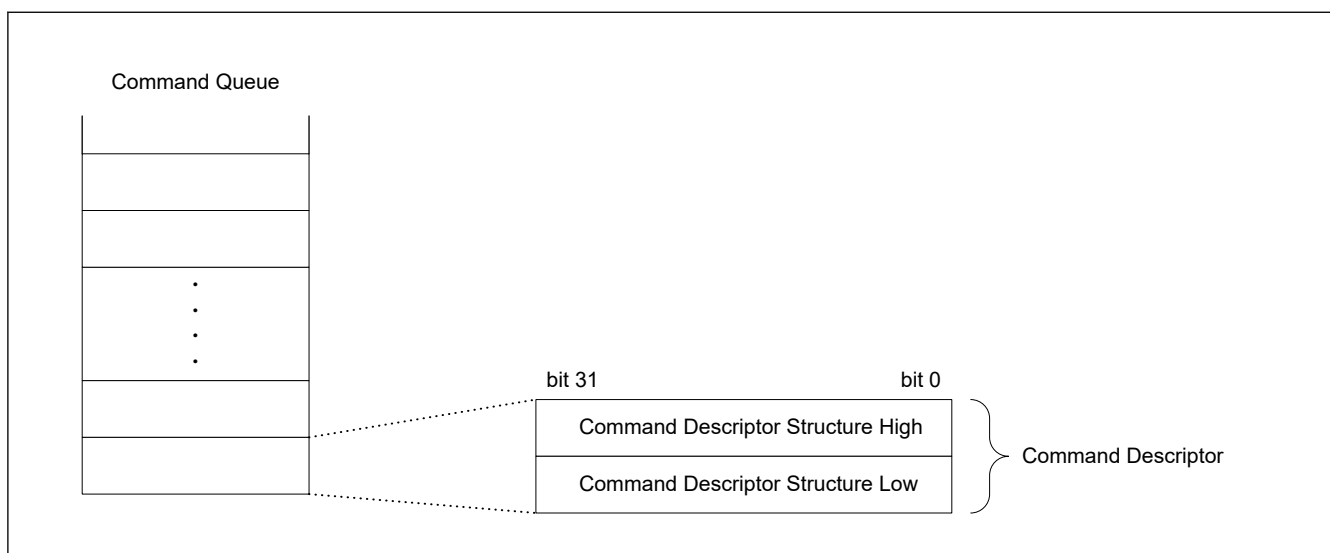


Figure 40.2 Command descriptor data structure

I3C provides a Command Descriptor structure for each command type as follows:

- Address Assign Command
- Immediate Transfer Command
- Regular Transfer Command
- Combo Transfer Command
- Internal Control Command

Details are explained in the following sections.

40.3.1.1.1 Address Assign Command

This command is used for address assignment (ENTDAA, SETDASA).

Note: When issuing SETAASA CCC, use the Immediate Transfer command.

The I3C provides an address assign command for the following mode:

- I3C Master mode

Details of the Address Assign command structure are as follows.

Bit position:	63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48	
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit position:	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32	
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Bit field:	TOC	ROC	DEV_COUNT[3:0]				—	—	—	—	EXT_DEVICE	DEV_INDEX[4:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Bit field:	—	CMD[7:0]								TID[3:0]				CMD_ATTR[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Bit	Symbol	Function	R/W
2:0	CMD_ATTR[2:0]	Command Attributes 0x0: XFER: Regular Transfer 0x1: IMMED_DATA_XFER: Immediate Data Transfer 0x2: ADDR_ASSGN_CMD: Address Assignment Command 0x3: WWR_COMBO_XFER: Write + Write/Read Combo Transfer 0x7: INTERNAL_CONTROL: Internal Control command Others: Setting prohibited	W
6:3	TID[3:0]	Transaction ID	W
14:7	CMD[7:0]	Transfer Command CCC Value	W
15	—	The write value should be 0.	W
20:16	DEV_INDEX[4:0]	Device Index	W
21	EXT_DEVICE	Extended Device Index 0: Use the DATBASm table indicated by DEV_INDEX[4:0]. 1: Use the EXDATBAS table.	W
25:22	—	The write value should be 0.	W
29:26	DEV_COUNT[3:0]	Device Count	W
30	ROC	Response on Completion 0: NOT_REQUIRED: Response Status is not required. 1: REQUIRED: Response Status is required.	W
31	TOC	Terminate on Completion 0: RESTART: Issue Repeated START (Sr) at end of transfer 1: STOP: Issue STOP (P) at end of transfer	W
63:32	—	The write value should be 0.	W

CMD_ATTR[2:0] bits (Command Attributes)

Command Type, defining the format of the other fields.

TID[3:0] bits (Transaction ID)

Used as an identification tag for this command. This field shall be populated by the software Driver, and the same value shall be reflected in the Response Descriptor.

CMD[7:0] bits (Transfer Command CCC Value)

Specifies CCC code indicating whether Address Assignment uses ENTDAAs or SETDASAs commands. The field comprises the entire command code (ENTDAAs or SETDASAs).

DEV_INDEX[4:0] bits (Device Index)

Indicates the DATBASm table index for the Slave device being addressed with the transfer. Static and device addressing related information are stored to this index in the DATBASm.

DEV_COUNT[3:0] bits (Device Count)

Indicates the number of devices that a dynamic address is assigned to.

ROC bit (Response on Completion)

Controls whether Response Status is sent after successful completion of the Transfer command. The successful completion is read from register NRSPQP. Upon unsuccessful transfer the Response Status is sent.

TOC bit (Terminate on Completion)

Controls what bus condition to issue after the Transfer command completes.

For ENTDAAs, a STOP condition is issued regardless of the setting value of TOC. It is meaningful for SETDASAs transfers.

When sending SETDASAs CCC by TOC = 0 (RESTART), the next command must be set to SETDASAs CCC with the Address Assign Command.

When the next command is not the same SETDASAs CCC flame, it must be set to TOC = 1 (STOP).

40.3.1.1.2 Immediate Transfer Command

This structure directly contains data (max 4 bytes) to be transferred, and as a result is only useful for Transfers/CCCs that write data. This structure shall not be used for Read operations (for example, to Rx data).

When transmitting data of 4 bytes or less, use this Immediate Transfer Command to communicate.

When transmitting data of 5 bytes or more, use the Regular Transfer Command to communicate.

For the Regular Transfer Command, see [section 40.3.1.1.3. Regular Transfer Command](#).

I3C provides an Immediate Transfer Command for the following mode:

- I3C Master Mode

Details of the Immediate Transfer Command Structure of each mode are shown in this section.

Bit position:	63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48
Bit field:	DATA_BYTE_4[7:0]								DATA_BYTE_3[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32
Bit field:	DATA_BYTE_2[7:0]								DATA_BYTE_1[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	TOC	ROC	RNW	MODE[2:0]			BYTE_CNT[2:0]			—	EXT_DEVICE	DEV_INDEX[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CP	CMD[7:0]							TID[3:0]			CMD_ATTR[2:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CMD_ATTR[2:0]	Immediate Data Transfer Command Attribute 0x0: XFER: Regular Transfer 0x1: IMMED_DATA_XFER: Immediate Data Transfer 0x2: ADDR_ASSGN_CMD: Address Assignment Command 0x3: WWR_COMBO_XFER: Write + Write/Read Combo Transfer 0x7: INTERNAL_CONTROL: Internal Control command Others: Setting prohibited	W
6:3	TID[3:0]	Immediate Data Transfer Transaction ID	W
14:7	CMD[7:0]	Immediate Data Transfer CCC / HDR Command Code Value For CCC: 8 bits For HDR: 7 bits	W
15	CP	Immediate Data Transfer Command Present 0: TRANSFER: This structure describes an SDR transfer, so the CMD field is not valid. 1: CCC_HDR: This structure describes a CCC or HDR transfer, so the CMD field is valid.	W
20:16	DEV_INDEX[4:0]	Immediate Data Transfer Device Index	W
21	EXT_DEVICE	Immediate Data Transfer Extended Device Index 0: Use the DATBASm table indicated by DEV_INDEX[4:0]. 1: Use the EXDATBAS table.	W
22	—	The write value should be 0.	W
25:23	BYTE_CNT[2:0]	Immediate Data Transfer Byte Count 0x0: No payload 0x1 to N bytes are valid. 0x4: Others: Setting prohibited	W
28:26	MODE[2:0]	Immediate Data Transfer Mode and Speed Values 0x0: I3C SDR0 / Data rate : STDBR (I3C mode) Legacy I ² C Message 0 / Data rate : STDBR (I ² C mode) 0x1: I3C SDR1 / Data rate : EXTBR (I3C mode) Legacy I ² C Message 0 / Data rate : EXTBR (I ² C mode) 0x2: I3C SDR2 / Data rate : STDBR × 2 (I3C mode) Reserved (I ² C mode) 0x3: I3C SDR3 / Data rate : EXTBR × 2 (I3C mode) Reserved (I ² C mode) 0x4: I3C SDR4 / Data rate : EXTBR × 4 (I3C mode) Reserved (I ² C mode) 0x5: I3C HDR-TS / Data rate : STDBR (I3C mode) HDR-Ternary Mode. Selection of HDR-TSP vs. HDR-TSL depends on the value of register BCTL, field BMDS Reserved (I ² C mode) 0x6: I3C HDR-DDR / Data rate : STDBR (I3C mode) Reserved (I ² C mode) Others: Setting prohibited	W
29	RNW	Immediate Data Transfer R/W 0: WRITE: Write transfer 1: READ: Read transfer	W

Bit	Symbol	Function	R/W
30	ROC	Immediate Data Transfer Response on Completion 0: NOT_REQUIRED: Response Status is not required. 1: REQUIRED: Response Status is required.	W
31	TOC	Immediate Data Transfer Terminate on Completion 0: RESTART: Issue Repeated START (Sr) at end of data transfer 1: STOP: Issue STOP (P) at end of data transfer	W
39:32	DATA_BYTE_1[7:0]	Immediate Data Transfer Data Byte 1 Direct argument	W
47:40	DATA_BYTE_2[7:0]	Immediate Data Transfer Data Byte 2 Direct argument	W
55:48	DATA_BYTE_3[7:0]	Immediate Data Transfer Data Byte 3 Direct argument	W
63:56	DATA_BYTE_4[7:0]	Immediate Data Transfer Data Byte 4 Direct argument	W

CMD_ATTR[2:0] bits (Immediate Data Transfer Command Attribute)

Command Type, defining the format of the other fields.

TID[3:0] bits (Immediate Data Transfer Transaction ID)

Used as an identification tag for this command. This field shall be populated by the software Driver, and the same value shall be reflected in the Response Descriptor.

CP bit (Immediate Data Transfer Command Present)

Indicates whether CMD field is valid for CCC or HDR Transfer.

DEV_INDEX[4:0] bits (Immediate Data Transfer Device Index)

Indicates the DATBASm Table index for the Slave Device being addressed with the transfer. Static and Device addressing related information will be stored to this index in the DATBASm.

BYTE_CNT[2:0] bits (Immediate Data Transfer Byte Count)

Number of valid data bytes to use in this Immediate Data Transfer Descriptor.

This field must be set to non-zero value, except for CCCs that does not have payload defined.

This field must be set to even value, when specifying HDR mode (MODE[2:0] = 0x5 or 0x6).

MODE[2:0] bits (Immediate Data Transfer Mode and Speed Values)

Sets the mode and speed for the I3C or I²C transfer.

Interpretation of this field depends on whether the Device is in I3C Mode vs. I²C Mode (see the DEVICE field in the DATBASm Table entry indexed by field DEV_INDEX).

RNW bit (Immediate Data Transfer R/W)

Identifies direction of the transfer.

This field shall always be set to 0, because Immediate transfers are valid for Write transactions only.

ROC bit (Immediate Data Transfer Response on Completion)

Controls whether Response Status is required after successful completion of the data transfer command. The successful completion shall be read from NRSPQP register. Upon unsuccessful transfer the Response Status shall always be sent.

TOC bit (Immediate Data Transfer Terminate on Completion)

Controls what Bus condition is issued after completion of the data transfer.

When sending Direct CCC by TOC = 0 (RESTART), next command must be set to same Direct CCC.

When the next command is not the same Direct CCC, must be set to TOC = 1 (STOP).

40.3.1.1.3 Regular Transfer Command

This structure does not contain data to be transferred.

For Master Mode, the data buffer is available through Transfer Data Queue Port (Rx Data Queue Port and Tx Data Queue Port).

When transmitting data of 5 bytes or more, use this Regular Transfer Command to communicate.

When transmitting data of 4 bytes or less, use the Immediate Transfer Command to communicate.

For the Immediate Transfer Command, see [section 40.3.1.1.2. Immediate Transfer Command](#).

For I3C Slave Mode, the IBI Payload buffer is available through IBI Queue Port.

I3C provides a Regular Transfer Command for the following modes:

- I3C Master Mode
- I3C Slave Mode

Details of the regular transfer command structure of each mode are as follows.

(1) I3C Master Mode

Bit position:	63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48
Bit field:	DATA_LENGTH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	TOC	ROC	RNW	MODE[2:0]			—	—	—	—	EXT_DEVICE	DEV_INDEX[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CP	CMD[7:0]							TID[3:0]				CMD_ATTR[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CMD_ATTR[2:0]	Data Transfer Command Attribute Command Type, defining the format of the other fields. Values: 0x0: XFER: Regular Transfer 0x1: IMMED_DATA_XFER: Immediate Data Transfer 0x2: ADDR_ASSGN_CMD: Address Assignment Command 0x3: WWR_COMBO_XFER: Write + Write/Read Combo Transfer 0x7: INTERNAL_CONTROL: Internal Control command Others: Setting prohibited	W
6:3	TID[3:0]	Data Transfer Transaction ID Identification tag for this command	W
14:7	CMD[7:0]	Data Transfer CCC / HDR Command Code Value Specifies the I3C Command code For CCC: 8 bits For HDR: 7 bits	W
15	CP	Data Transfer Command Present 0: TRANSFER: This structure describes an SDR transfer, so the CMD field is not valid. 1: CCC_HDR: This structure describes a CCC or HDR transfer, so the CMD field is valid.	W

Bit	Symbol	Function	R/W
20:16	DEV_INDEX[4:0]	Data Transfer Device Index	W
21	EXT_DEVICE	Data Transfer Extended Device Index 0: Use the DATBASm Table indicated by DEV_INDEX[4:0]. 1: Use the EXDATBAS table.	W
25:22	—	The write value should be 0.	W
28:26	MODE[2:0]	Data Transfer Speed and Mode 0x0: I3C SDR0 / Data rate : STDBR (I3C mode) Legacy I ² C Message 0 / Data rate : STDBR (I ² C mode) 0x1: I3C SDR1 / Data rate : EXTBR (I3C mode) Legacy I ² C Message 0 / Data rate : EXTBR (I2C mode) 0x2: I3C SDR2 / Data rate : STDBR × 2 (I3C mode) Reserved (I ² C mode) 0x3: I3C SDR3 / Data rate : EXTBR × 2 (I3C mode) Reserved (I ² C mode) 0x4: I3C SDR4 / Data rate : EXTBR × 4 (I3C mode) Reserved (I ² C mode) 0x5: I3C HDR-TS / Data rate : STDBR × 4 (I3C mode) HDR-Ternary Mode. Selection of HDR-TSP vs. HDR-TSL depends on the value of register BCTL, field BMDS Reserved (I ² C mode) 0x6: I3C HDR-DDR / Data rate : STDBR (I3C mode) Reserved (I ² C mode) Others: Setting prohibited	W
29	RNW	Data Transfer R/W 0: WRITE: Write transfer 1: READ: Read transfer	W
30	ROC	Data Transfer Response on Completion 0: NOT_REQUIRED: Response Status is not required. 1: REQUIRED: Response Status is required.	W
31	TOC	Data Transfer Terminate on Completion 0: RESTART: Issue Repeated START (Sr) at end of transfer 1: STOP: Issue STOP (P) at end of transfer	W
47:32	—	The write value should be 0.	W
63:48	DATA_LENGTH [15:0]	Data Transfer Data Length Indicates the number of bytes to be transferred. This field must be set to non-zero value, except for CCCs that does not have payload defined.	W

CMD_ATTR[2:0] bits (Data Transfer Command Attribute)

Command Type, defining the format of the other fields.

TID[3:0] bits (Data Transfer Transaction ID)

Used as an identification tag for this command. This field shall be populated by the software Driver, and the same value shall be reflected in the Response Descriptor.

CP bit (Data Transfer Command Present)

Indicates whether the contents of the CMD field is valid for a CCC or HDR Transfer.

DEV_INDEX[4:0] bits (Data Transfer Device Index)

Indicates the DATBASm Table index for the Slave Device being addressed with the transfer. Static and Device addressing related information will be stored to this index in the DATBASm.

MODE[2:0] bits (Data Transfer Speed and Mode)

Sets the mode and speed for the I3C or I²C transfer.

Interpretation of this field depends on whether the Device is in I3C Mode vs. I²C Mode (see the DEVICE field in the DATBASm Table entry indexed by field DEV_INDEX).

RNW bit (Data Transfer R/W)

Identifies direction of the transfer.

ROC bit (Data Transfer Response on Completion)

Controls whether Response Status is required after successful completion of the transfer command. The successful completion shall be read from NRSPQP register. Upon unsuccessful transfer the Response Status shall always be sent.

TOC bit (Data Transfer Terminate on Completion)

Controls what Bus condition will be issued after completion of the transfer.

When sending Direct CCC by TOC = 0 (RESTART), next command must be set to same Direct CCC.

When the next command is not the same Direct CCC, must be set to TOC = 1 (STOP).

DATA_LENGTH[15:0] bits (Data Transfer Data Length)

Number of valid data bytes to use in this Regular Transfer Descriptor.

This field must be set to non-zero value, except for CCCs that does not have payload defined.

This field must be set to even value, when specifying HDR mode (MODE[2:0] = 0x5 or 0x6).

Length setting of GETMXDS command should be fixed to 5.

(2) I3C Slave Mode

Bit position:	63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48
Bit field:	DATA_LENGTH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	ROC	RNW	—	—	—	—	—	—	ITS	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	TID[3:0]			CMD_ATTR[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CMD_ATTR[2:0]	Data Transfer Command Attribute Command Type, defining the format of the other fields. Values: 0x0: XFER: Regular Transfer 0x1: IMMED_DATA_XFER: Immediate Data Transfer 0x2: ADDR_ASSGN_CMD: Address Assignment Command 0x3: WWR_COMBO_XFER: Write + Write/Read Combo Transfer 0x7: INTERNAL_CONTROL: Internal Control command Others: Setting prohibited	W
6:3	TID[3:0]	Data Transfer Transaction ID Identification tag for this command	W

Bit	Symbol	Function	R/W
21:7	—	The write value should be 0.	W
22	ITS	Include timestamp for Async Mode 0: Do not include timestamp. 1: Include timestamp.	W
28:23	—	The write value should be 0.	W
29	RNW	Data Transfer R/W 0: WRITE: Write transfer (Mastership Request) 1: READ: Read transfer (Slave Interrupt Request)	W
30	ROC	Data Transfer Response on Completion 0: NOT_REQUIRED: Response Status is not required. 1: REQUIRED: Response Status is required.	W
47:31	—	The write value should be 0.	W
63:48	DATA_LENGTH [15:0]	Data Transfer Data Length Indicates the number of bytes to be transferred. This field must be set to non-zero value, except for CCCs that does not have payload defined.	W

CMD_ATTR[2:0] bits (Data Transfer Command Attribute)

Command Type, defining the format of the other fields.

TID[3:0] bits (Data Transfer Transaction ID)

Used as an identification tag for this command. This field shall be populated by the software Driver, and the same value shall be reflected in the Response Descriptor.

RNW bit (Data Transfer R/W)

Identifies direction of the transfer.

ROC bit (Data Transfer Response on Completion)

Controls whether Response Status is required after successful completion of the transfer command. The successful completion shall be read from NRSPQP register. Upon unsuccessful transfer the Response Status shall always be sent.

40.3.1.1.4 Combo Transfer Command

This structure contains a combined Write + Read/Write operation.

The data buffer is available through Transfer Data Queue Port (Rx Data Queue Port and Tx Data Queue Port).

I3C provides a Combo Transfer Command for the following mode:

- I3C Master mode

Details of the Combo Transfer Command Structure of each mode are as follows.

Bit position:	63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48
Bit field:	DATA_LENGTH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32
Bit field:	OFFSET[15:0]/SUBOFFSET[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Bit field:	TOC	ROC	RNW	MODE[2:0]			16_BIT_SUBOFFSET	FIRST_PHASE_MODE	DATA_LENGTH_POSITION[1:0]		EXT_DEVICE	DEV_INDEX[4:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Bit field:	CP	CMD[7:0]								TID[3:0]				CMD_ATTR[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Bit	Symbol	Function	R/W
2:0	CMD_ATTR[2:0]	Combo Transfer Command Attribute Command Type, defining the format of the other fields. 0x0: XFER: Regular Transfer 0x1: IMMED_DATA_XFER: Immediate Data Transfer 0x2: ADDR_ASSGN_CMD: Address Assignment Command 0x3: WWR_COMBO_XFER: Write + Write/Read Combo Transfer 0x7: INTERNAL_CONTROL: Internal Control command Others: Setting prohibited	W
6:3	TID[3:0]	Combo Transfer Transaction ID Identification tag for the command	W
14:7	CMD[7:0]	Combo Transfer HDR Command Code Value Specifies the I3C Command code (7 bits).	W
15	CP	Combo Transfer Command Present Indicates whether the CMD field is valid for an HDR Transfer 0: TRANSFER: This structure describes an SDR transfer, so the CMD field is not valid. 1: CCC_HDR: This structure describes an HDR transfer, so the CMD field is valid.	W
20:16	DEV_INDEX[4:0]	Combo Transfer Device Index	W
21	EXT_DEVICE	Combo Transfer Extended Device Index 0: Use the DATBASm table indicated by DEV_INDEX[4:0]. 1: Use the EXDATBAS table.	W
23:22	DATA_LENGTH_POSITION[1:0]	Data Length Field Position 0 0: NO: Do not put length field. 0 1: FIRST: Put length as first field. 1 0: SECOND: Put length as second field. Others: Setting prohibited	W
24	FIRST_PHASE_MODE	Combo Transfer First Phase Mode 0: SDR: First phase is executed in SDR mode. 1: MODE: First phase is executed in the mode indicated by the MODE field.	W
25	16_BIT_SUBOFFSET	Combo Transfer Sub Offset Size 0: 8_BIT_SUBOFFSET: Sub-offset is 8-bits long. Value is encoded in Lower Byte of OFFSET/SUBOFFSET field. 1: 16_BIT_SUBOFFSET: Sub-offset is 16-bits long.	W
28:26	MODE[2:0]	Combo Transfer Speed and Mode Values for I3C Mode 0x0: I3C SDR0/Data rate: STDBR 0x1: I3C SDR1/Data rate: EXTBR 0x2: I3C SDR2/Data rate: STDBR × 2 0x3: I3C SDR3/Data rate: EXTBR × 2 0x4: I3C SDR4/Data rate: EXTBR × 4 0x5: I3C HDR-TS/Data rate: STDBR × 4 HDR-Ternary Mode. Selection of HDR-TSP vs. HDR-TSL depends on the value of register BCTL, field BMDS 0x6: I3C HDR-DDR/Data rate: STDBR Others: Setting prohibited	W

Bit	Symbol	Function	R/W
29	RNW	Combo Transfer R/W Identifies direction of the transfer 0: WRITE: Write transfer 1: READ: Read transfer	W
30	ROC	Combo Transfer Response on Completion 0: NOT_REQUIRED: Response Status is not required. 1: REQUIRED: Response Status is required.	W
31	TOC	Combo Transfer Terminate on Completion 0: RESTART: Issue Repeated START (Sr) at end of transfer 1: STOP: Issue STOP (P) at end of transfer	W
47:32	OFFSET[15:0]/ SUBOFFSET[15:0]	Combo Transfer Offset / Sub-Offset Offset of the target operation	W
63:48	DATA_LENGTH [15:0]	Combo Transfer Data Length Number of bytes to be transferred. This field must be set to non-zero value.	W

CMD_ATTR[2:0] bits (Combo Transfer Command Attribute)

Command Type, defining the format of the other fields.

TID[3:0] bits (Combo Transfer Transaction ID)

Used as an identification tag for this command. This field shall be populated by the software Driver, and the same value shall be reflected in the Response Descriptor.

CP bit (Combo Transfer Command Present)

Indicates whether the contents of the CMD field is valid for an HDR Transfer.

DEV_INDEX[4:0] bits (Combo Transfer Device Index)

Indicates the DATBASm table index for the Slave Device being addressed with the transfer. Static and Device addressing related information will be stored to this index in the DATBASm.

DATA_LENGTH_POSITION[1:0] bits (Data Length Field Position)

Indicates whether and where to put Data Length (DATA_LENGTH) in the first phase of the transfer. This field is only applicable if First phase of the transfer is executed in HDR mode.

Whether 8-bit or 16-bit of Data Length field is used is indicated with 16_BIT_SUBOFFSET field. In case of 8-bit value, it is encoded in Lower Byte of DATA_LENGTH field.

FIRST_PHASE_MODE bits (Combo Transfer First Phase Mode)

Indicates whether the first phase of the Combo Transfer is executed in SDR Mode, vs. the Mode indicated by the MODE field.

MODE[2:0] bits (Combo Transfer Speed and Mode Values for I3C Mode)

Sets the mode and speed for the I3C or I²C transfer.

Interpretation of this field depends on whether the Device is in I3C Mode vs. I²C Mode (see the DEVICE field in the DATBASm Table entry indexed by field DEV_INDEX).

RNW bit (Combo Transfer R/W Identifies direction of the transfer)

Identifies direction of the transfer.

ROC bit (Combo Transfer Response on Completion)

Controls whether Response Status is required after successful completion of the data transfer command. The successful completion shall be read from NRSPQP register. Upon unsuccessful transfer the Response Status shall always be sent.

TOC bit (Combo Transfer Terminate on Completion)

Controls what Bus condition is issued after completion of the data transfer.

When sending 2nd flame by TOC = 0 (RESTART) and HDR mode (MODE[2:0] = 0x5 or 0x6), next command must be set to that 1st flame will be the same as HDR mode (MODE[2:0] = 0x5 or 0x6).

When the next command is not the same HDR mode (MODE[2:0] = 0x5 or 0x6), must be set to TOC = 1 (STOP).

DATA_LENGTH[15:0] bit (Combo Transfer Data Length)

Number of valid data bytes to use in this Combo Transfer Descriptor.

This field must be set to non-zero value.

This field must be set to even value, when specifying HDR mode (MODE[2:0] = 0x5 or 0x6).

40.3.1.1.5 Internal Control Command

This structure is used for controlling I3C itself (not for transfer commands).

I3C provides an Internal Control Command for the following mode:

- I3C Master mode

Details of the Internal Control Command Structure are as follows:

Bit position:	63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	ON OFF	MIPI_CMD[3:0]				—	TID[3:0]			CMD_ATTR[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CMD_ATTR[2:0]	Command Attribute*2 Command Type, defining the format of the other fields. 0x0: XFER: Regular Transfer 0x1: IMMED_DATA_XFER: Immediate Data Transfer 0x2: ADDR_ASSGN_CMD: Address Assignment Command 0x3: WWR_COMBO_XFER: Write + Write/Read Combo Transfer 0x7: INTERNAL_CONTROL: Internal Control command Others: Setting prohibited	W
6:3	TID[3:0]	Transaction ID Identification tag for the command	W
7	—	The write value should be 0.	W
11:8	MIPI_CMD[3:0]	MIPI Alliance Command 0x00: NoOp, so the ON_OFF field is not valid. 0x02: Include 7E (IBA), so the ON_OFF field is valid. Others: Setting prohibited	W
12	ON_OFF	Bus Instance 7E On / Off*1 Enables or disables automatic transmission of the I3C Broadcast Header after every START condition on this I3C Bus instance. 0: IBA_INCLUDE off 1: IBA_INCLUDE on	W

Bit	Symbol	Function	R/W
63:13	—	The write value should be 0.	W

Note 1. The IBA_INCLUDE on state set by MIPI_CMD [3:0] = 0x2 and ON_OFF = 1 is cleared by setting RSTCTL.INTLRST to 1.

Note 2. The Response descriptor is not stored when the Internal Control Command is executed.

40.3.1.2 Response Descriptor

The Response Descriptor is a read-only structure describing the success or failure of a command, and the amount of data transferred.

The Response Descriptor is read from Response Queue with reads from Response Queue Port.

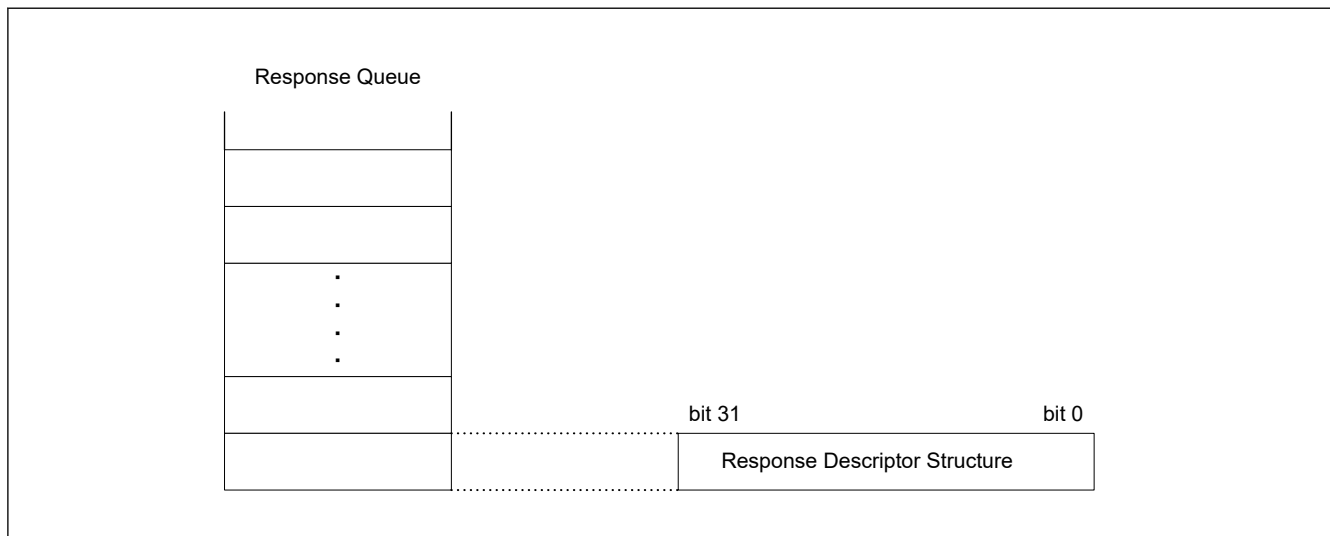


Figure 40.3 Response descriptor data structure

I3C provides a Response Descriptor for the following modes:

- I3C Master mode
- I3C Slave mode

Details of the Response Descriptor structure of each mode are shown in the following sections.

(1) I3C Master Mode

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ERR_STATUS[3:0]				TID[3:0]				—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA_LENGTH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DATA_LENGTH [15:0]	Data Length / Device Count The meaning of this field depends on the context: For Write Transfer: Remaining data length (in bytes) For Read Transfer: Received data length (in bytes) For Address Assignment: Remaining Device count	R
23:16	—	These bits are read as 0.	R

Bit	Symbol	Function	R/W
27:24	TID[3:0]	Command/Response Transaction ID Identification tag for the command. This value shall match one of commands sent on the Bus. 0x0-0x7: Valid Transaction IDs Others: Reserved	R
31:28	ERR_STATUS[3:0]	Response Error Status 0x0: SUCCESS: Transfer successful, no error 0x1: CRC: CRC Error 0x2: PARITY: Parity Error 0x3: FRAME: Frame Error 0x4: ADDR_HEADER: Address Header Error 0x5: NACK: Address NACKed or Dynamic Address Assignment NACKed 0x6: OVL: Receive Overflow or Transfer Underflow Error 0x8: ABORTED: Aborted 0x9: I ² C_WR_DATA_NACK: NACK received for the I ² C Write Data transfer 0xA: NOT_SUPPORTED: Command with specific parameters not supported by I3C implementation (for example, specific Internal Control codes may not be supported) Others: Reserved	R

Note: In I3C Master mode, when an abnormal command with a specific parameter that is not supported is stored in Command Descriptor, it is indicated as NOT_SUPPORTED (0xA) in ERR_STATUS [3:0].

(2) I3C Slave Mode

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ERR_STATUS[3:0]				TID[3:0]				—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA_LENGTH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DATA_LENGTH [15:0]	Data Length Remaining data length (in bytes) for Slave Interrupt Request	R
23:16	—	These bits are read as 0.	R
27:24	TID[3:0]	Command/Response Transaction ID Identification tag for the command. This value matches one of commands sent on the bus. 0x0-0x7: Valid Transaction IDs Others: Reserved	R
31:28	ERR_STATUS[3:0]	Response Error Status 0x0: SUCCESS: Transfer successful, no error. 0x3: FRAME: Frame Error 0x4: ADDR_HEADER: Address Header Error 0x5: NACK: Address NACK'ed or Dynamic Address Assignment NACK'ed 0x6: OVL: Receive Overflow or Transfer Underflow Error 0x8: ABORTED: Aborted 0xA: NOT_SUPPORTED: Command with specific parameters not supported by I3C implementation (for example, specific Internal Control codes may not be supported) Others: Reserved	R

Note: In I3C Slave mode, it is indicated as NOT_SUPPORTED (0xA) in ERR_STATUS[3:0] in the following cases:

- When an abnormal command with a specific parameter that is not supported is stored in the Command Descriptor.
- When the IBI to be transmitted is disabled in the CSECMD register.
- After the normal command for IBI transmission is prepared in the Command Queue, when that IBI is disabled in the CSECMD register by the DISEC CCC frame from the I3C Master.

40.3.1.3 IBI Status Descriptor

The IBI Status Descriptor is a read-only structure describing an IBI event received from a Slave device on the I3C Bus.

The IBI Status Descriptor is read from IBI Queue with reads from IBI Queue Port.

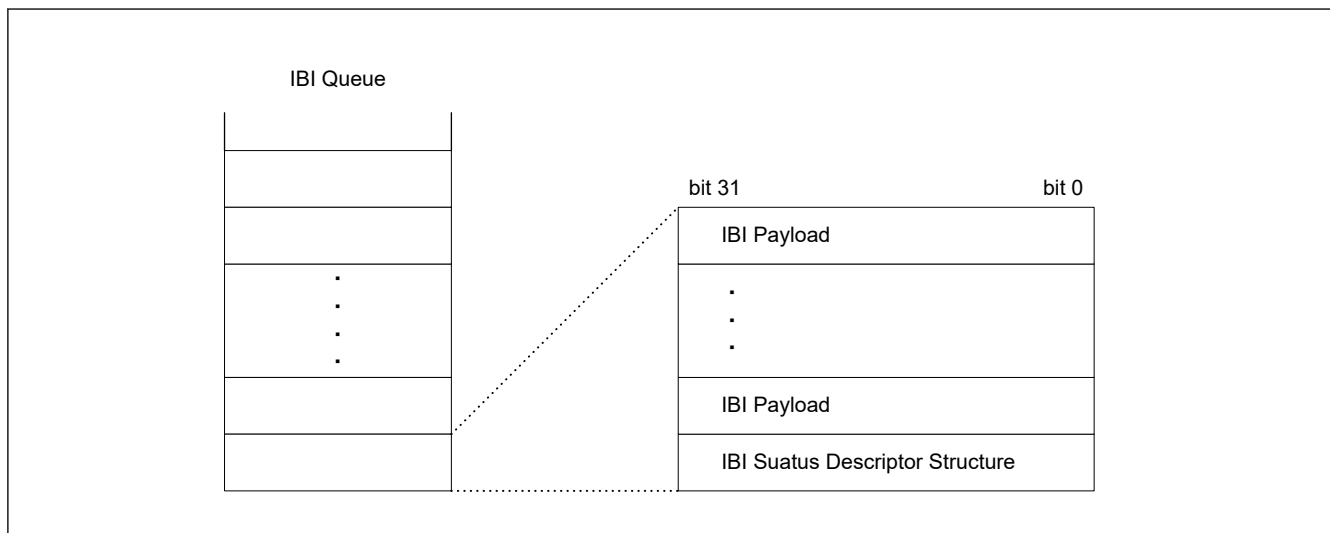


Figure 40.4 IBI status descriptor data structure

I3C provides a IBI Status Descriptor for the following mode:

- I3C Master mode

Details of the IBI Status Descriptor Structure are as follows.

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	IBI_ST	—	—	ERR_STATUS[2:0]		TS	LAST_STATUS	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	IBI_ID[7:0]								DATA_LENGTH[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	DATA_LENGTH[7:0]	IBI Data Length Number of data bytes in IBI Data.	R
15:8	IBI_ID[7:0]	IBI Received ID The meaning of this field depends on the context: For Slave Interrupt or Master Request: Bits 15:9 contain the Slave's Device Address, and bit 8 contains the R/W bit.	R
23:16	—	These bits are read as 0. The write value should be 0.	R
24	LAST_STATUS	Last IBI Status Last IBI status for the IBI transaction.	R

Bit	Symbol	Function	R/W
25	TS	IBI Time-stamp Present Indicates whether a time-stamp is available for the IBI. 0: OFF: IBI is not time-stamped. 1: ON: IBI is time-stamped.	R
28:26	ERR_STATUS[2:0]	IBI Error Status 0x0: SUCCESS 0x3: ERROR: FRAME (Frame Error) 0x4: ERROR: ADDR_HEADER (Address Header Error) 0x5: NACK: Address NACKed 0x7: ERROR: ABORT (Aborted to Master) Others: Reserved	R
30:29	—	These bits are read as 0.	R
31	IBI_ST	IBI Received Status Indicates how the received IBI was handled. 0: The IBI was handled with ACK. 1: NACK: The IBI was handled with NACK, and then Auto-Disabled.	R

LAST_STATUS bits (Last IBI Status)

Even if LAST_STATUS is set to 0, the software driver still evaluates the data payload length by examining the DATA_LENGTH field.

40.3.1.4 Receive Status Descriptor

The Receive Status Descriptor is a read-only structure describing the success or failure of read/write operation from the master, and the amount of data transferred.

The Receive Status Descriptor is read from Receive Status Queue with reads from Receive Status Queue Port.

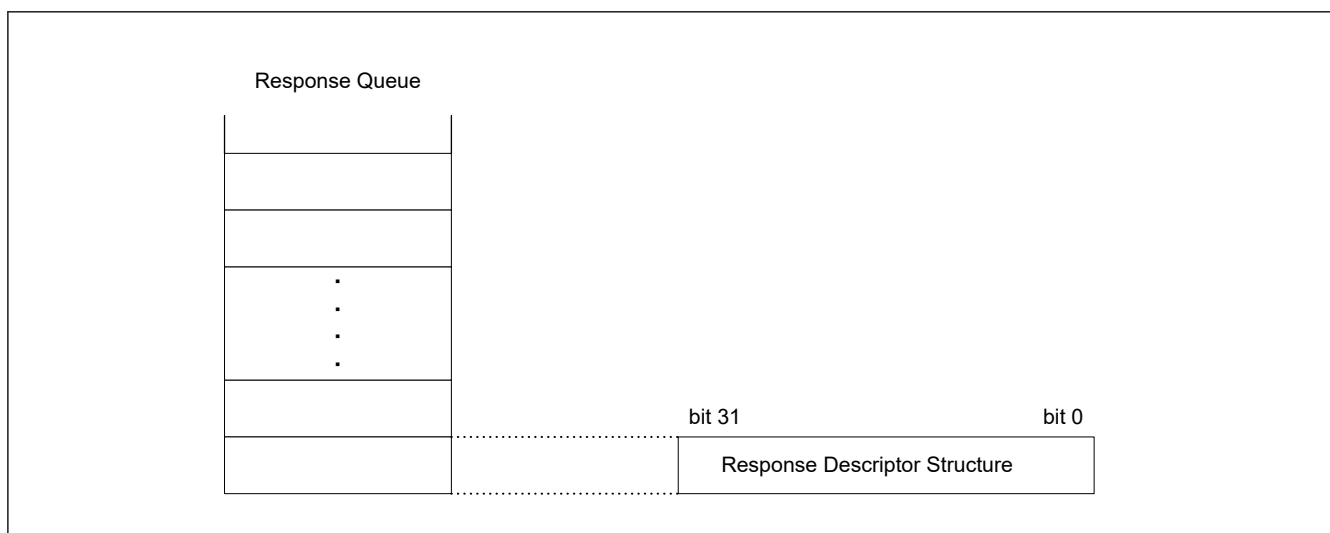


Figure 40.5 Receive status descriptor data structure

I3C provides a Receive Status Descriptor for the following mode:

- I3C Slave mode

Details of the Receive Status Descriptor structure of each mode are as follows.

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DEV_INDEX[2:0]			TRANSFER_TY PE[1:0]		ERR_STATUS[2:0]			CMD[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA_LENGTH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DATA_LENGTH [15:0]	Data Length The meaning of this field depends on the context. For Write Transfer: Received data length (in bytes) For Read Transfer: Transmitted data length (in bytes)	R
23:16	CMD[7:0]	The contents are different depending on the operation mode. Details are follows: [SDR Private Message Mode] CMD[7]: R/W Type CMD[6:4]: Reserved CMD[3]: I3C_I ² C Type CMD[2:0]: Reserved [SDR CCC Mode] CCC code[7:0] [HDR-DDR mode] HDR CMD[7:0] 0x00 – 0x7F: Write Command 0x80 – 0xFF: Read Command [HDR-TSL/TSP mode] HDR CMD[7:0]	R
26:24	ERR_STATUS[2:0]	Error Status 0x0: SUCCESS 0x1: ERROR: CRC (CRC Error) 0x2: ERROR: PARITY (Parity Error) 0x3: ERROR: FRAME (Frame Error) 0x4: ERROR: ADDR_HEADER (Address Header Error) 0x5: ERROR: NACK (Slave NACKed) 0x6: ERROR: OVL (FIFO Overflow/Underflow) 0x7: ERROR: ABORT (Aborted to Master)	R
28:27	TRANSFER_TYPE [1:0]	Transfer Type 0 0: I3C SDR/I ² C Message 0 1: I3C CCC 1 0: I3C HDR-DDR 1 1: I3C HDR-TS	R
31:29	DEV_INDEX[2:0]	Device Index Indicates the SVDVADn index for the response with the transfer.	R

40.3.2 Details of Function

40.3.2.1 Operation Mode

The support relationship between the mode select (I3C mode / I²C mode) and operation mode (Master / Slave) on the I3C bus or the I²C bus is shown in [Table 40.9](#).

Table 40.9 Support of operating mode

I3C/I ² C Bus	I3C mode		I ² C mode	
	Master	Slave	Master	Slave
I3C Bus	✓	✓	—	✓
I ² C Bus	—	—	✓	✓

Note: ✓: Supported

—: Unsupported

40.3.2.1.1 Master Mode Operation

(1) I²C Master Operation

(a) Data Write Transfer (Single Buffer transfer)

In master transmit operation, I3C outputs the SCL clock and transmitted data signals as the master device, and the slave device returns acknowledgments. [Figure 40.143](#) shows an example of usage of master transmission and [Figure 40.6](#) to [Figure 40.8](#) show the timing of operations in master transmission.

The following describes the procedure and operations for master transmission.

1. Initial settings. For details, see [section 40.3.3.1. Initial Setting Flow](#).
2. Read the BCST.BFREF flag to check that the bus is open, and then set the CNDCTL.STCND bit to 1 (START condition issuance request). Upon receiving the request, I3C issues a START condition. At the same time, the BFREF flag bit is automatically set to 0, the BST.STCNDDF flag is automatically set to 1 and the STCND bit is automatically set to 0. At this time, if the START condition is detected and the internal levels for the SDA output state and the levels on the I3C_SDA line have matched while the STCND bit = 1, I3C recognizes that issuing of the START condition as requested by the STCND bit has been successfully completed, and bits CRMS and TRMD in the PRSST register are automatically set to 1, placing I3C in master transmit mode. The NTST.TDBEF0 flag is also automatically set to 1 in response to setting of the TRMD bit to 1.
3. Check that the NTST.TDBEF0 flag = 1, and then write the value for transmission (the slave address and the R/W# bit) to the NTDTBP0 register. Once the data for transmission are written to the NTDTBP0 register, the TDBEF0 flag is automatically set to 0, the data are transferred from the Normal Tx Data Buffer to the Shift Register, and the TDBEF0 flag is again set to 1. After the byte containing the slave address and R/W# bit has been transmitted, the value of the TRMD bit is automatically updated to select master transmit or master receive mode in accord with the value of the transmitted R/W# bit. If the value of the R/W# bit was 0, I3C continues in master transmit mode. Because the BST.NACKDF flag being 1 at this time indicates that no slave device recognized the address or there was an error in communications, write 1 to the CNDCTL.SPCND bit to issue a STOP condition. For data transmission with an address in the 10-bit format, start by writing 1111 0, the 2 higher-order bits of the slave address, and W to the NTDTBP0 register as the first address transmission. Then, as the second address transmission, write the 8 lower-order bits of the slave address to the NTDTBP0 register.
4. After confirming that the NTST.TDBEF0 flag = 1, write the data for transmission to the NTDTBP0 register. I3C automatically holds the I3C_SCL line low until the data for transmission are ready or a STOP condition is issued.
5. After all bytes of data for transmission have been written to the NTDTBP0 register, wait until the value of the BST.TENDF flag returns to 1, and then set the CNDCTL.SPCND bit to 1 (STOP condition issuance request). Upon receiving a STOP condition issuance request, I3C issues the STOP condition.
6. Upon detecting the STOP condition, I3C automatically sets bits CRMS and TRMD in the PRSST register to 00 and enters slave receive mode. Furthermore, it automatically sets the TDBEF0 and TENDF flags to 0, and sets the BST.SPCNDDF flag to 1.
7. After checking that the BST.SPCNDDF flag = 1, set the BST.NACKDF and SPCNDDF flags to 0 for the next transfer operation.

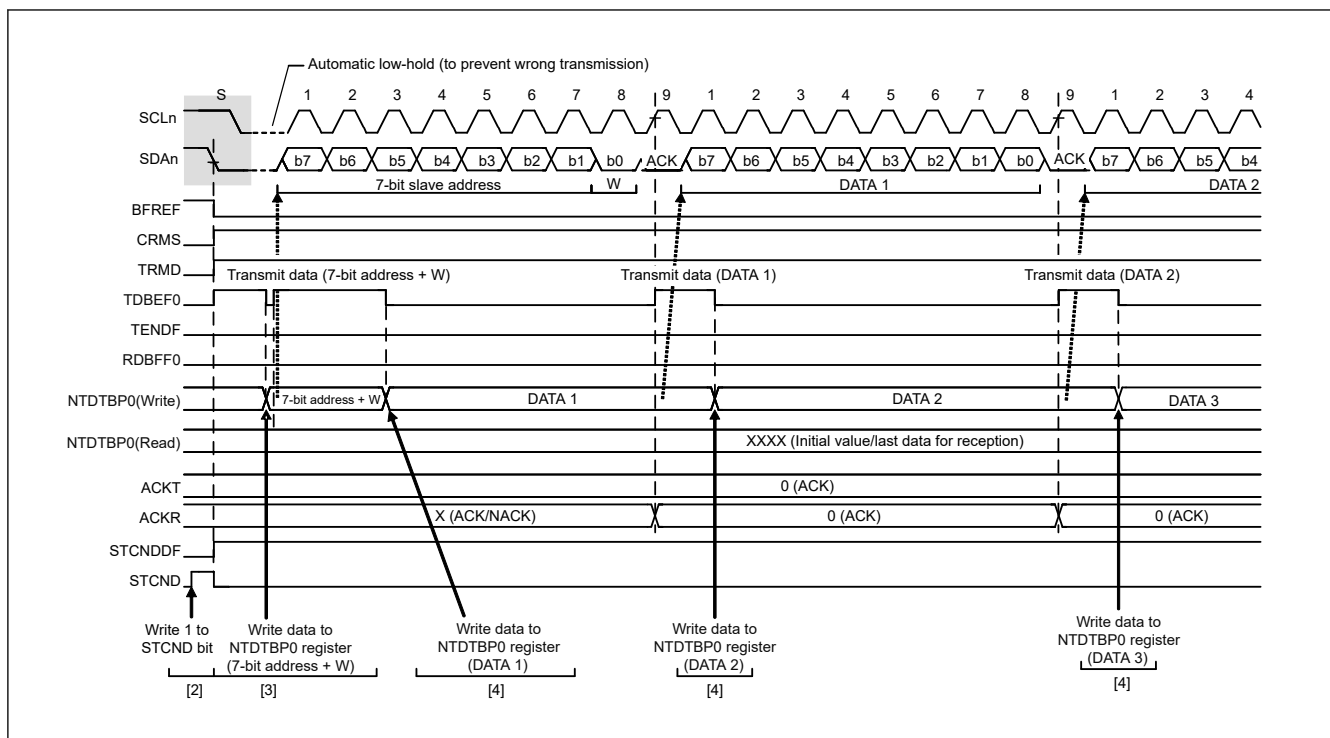


Figure 40.6 Master transmit operation timing (1) (7-bit address format)

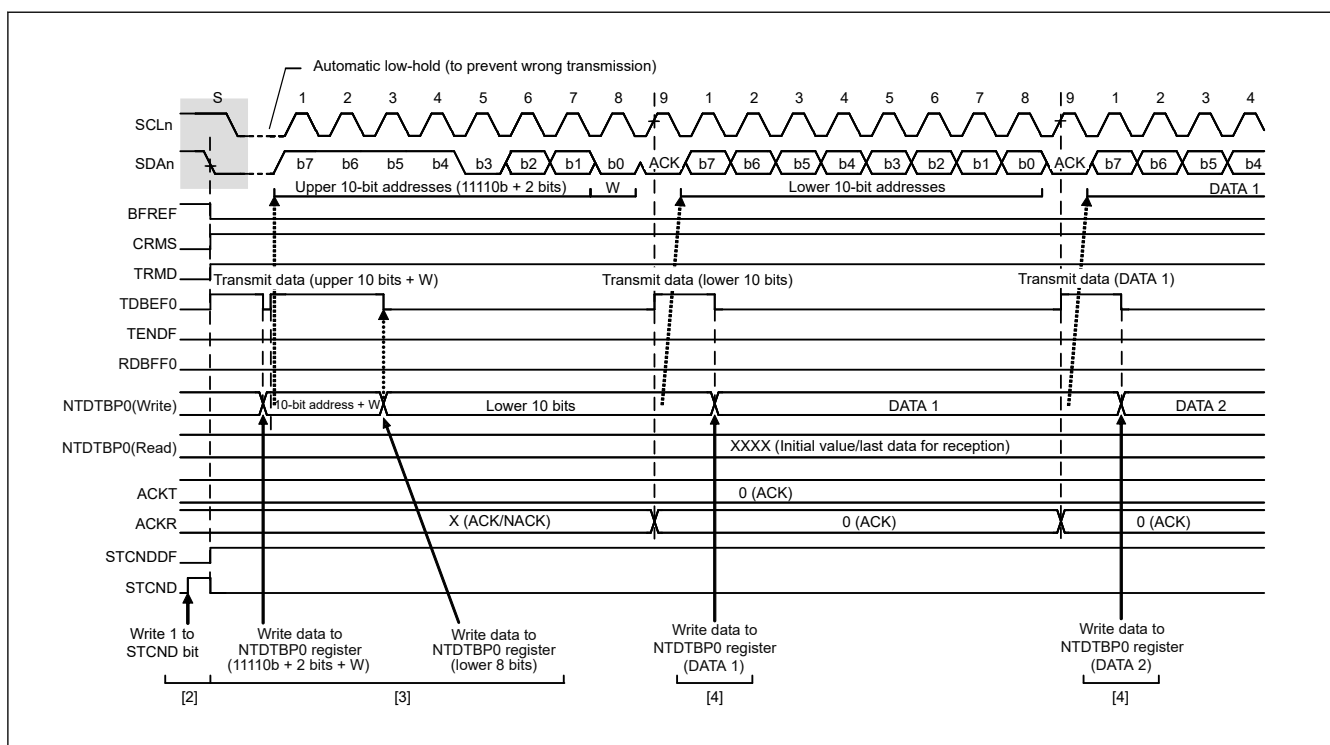


Figure 40.7 Master transmit operation timing (2) (10-bit address format)

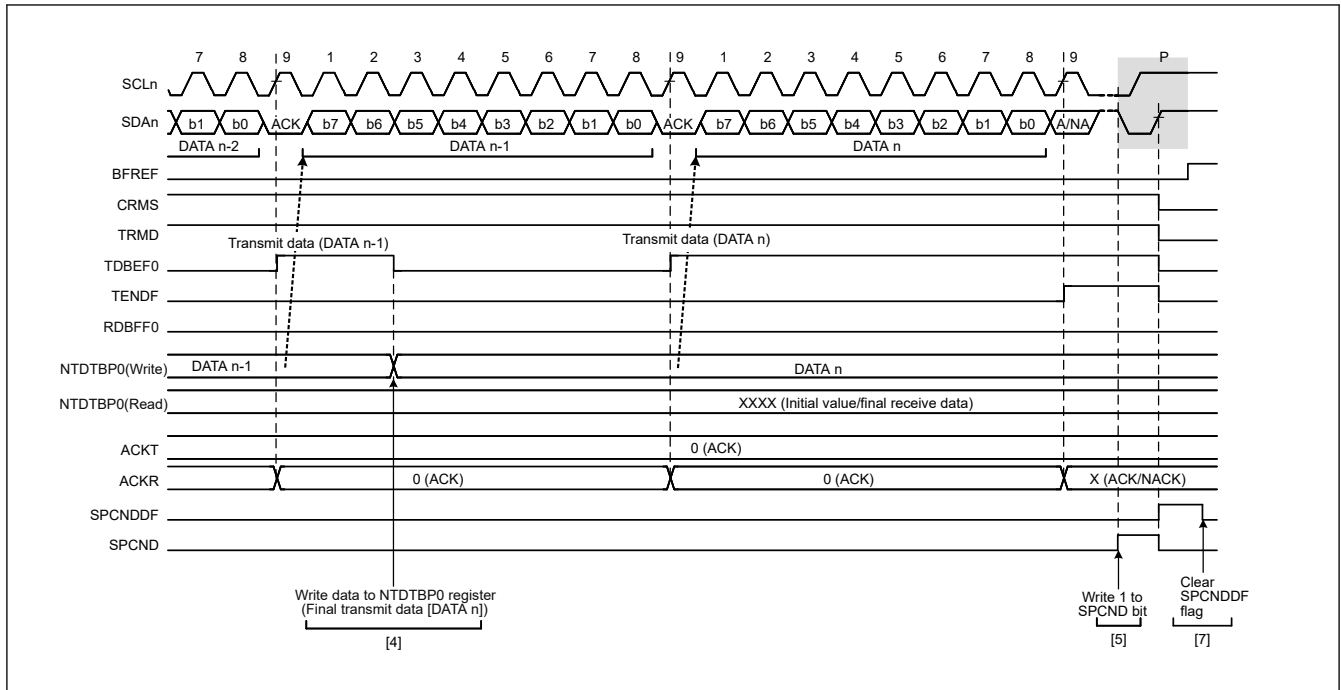


Figure 40.8 Master transmit operation timing (3)

(b) Data Read Transfer (Single Buffer transfer)

In master receive operation, I3C as a master device outputs the SCL clock, receives data from the slave device, and returns acknowledgments. Because I3C must start by sending a slave address to the corresponding slave device, this part of the procedure is performed in master transmit mode, but the subsequent steps are in master receive mode.

Figure 40.144 and Figure 40.145 show examples of usage of master reception (7-bit address format) and Figure 40.9 to Figure 40.11 show the timing of operations in master reception.

The following describes the procedure and operations for master reception.

1. Initial settings. For details, see [section 40.3.3.1. Initial Setting Flow](#).
2. Read the BCST.BFREF flag to check that the bus is open, and then set the CNDCTL.STCND bit to 1 (START condition issuance request). Upon receiving the request, I3C issues a START condition. When I3C detects the START condition, the BFREF flag is automatically set to 0 and the BST.STCNDDF flag is automatically set to 1 and the STCND bit is automatically set to 0. At this time, if the START condition is detected and the levels for the SDA output and the levels on the I3C_SDA line have matched while the STCND bit = 1, I3C recognizes that issuing of the START condition as requested by the STCND bit has been successfully completed, and bits CRMS and TRMD in the PRSST register are automatically set to 1, placing I3C in master transmit mode. The NTST.TDBEF0 flag is also automatically set to 1 in response to setting of the TRMD bit to 1.
3. Check that the NTST.TDBEF0 flag = 1, and then write the value for transmission (the first byte indicates the slave address and value of the R/W# bit) to the NTDTBP0 register. Once the data for transmission are written to the NTDTBP0 register, the TDBEF0 flag is automatically set to 0, the data are transferred from the Normal Tx Data Buffer to the Shift Register, and the TDBEF0 flag is again set to 1. Once the byte containing the slave address and R/W# bit has been transmitted, the value of the PRSST.TRMD bit is automatically updated to select transmit or receive mode in accord with the value of the transmitted R/W# bit. If the value of the R/W# bit was 1, the TRMD bit is set to 0 on the rising edge of the ninth cycle of SCL clock, placing I3C in master receive mode. At this time, the TDBEF0 flag is set to 0. The NTST.RDBFF0 flag is automatically set to 1 when ACK response is received from the slave device. If the slave device is not recognized or a communication failure occurs, the BST.NACKDF flag will be set to 1. At this time, set 1 to the CNDCTL.SPCND bit to issue a STOP condition. For master reception from a device with a 10-bit address, start by using master transmission to issue the 10-bit address, and then issue a Repeated START condition. After that, transmitting 1111 0, the two higher-order bits of the slave address, and the R bit places I3C in master receive mode.
4. Dummy read the NTDTBP0 register after confirming that the NTST.RDBFF0 flag = 1; this makes I3C start output of the SCL clock and start data reception.
5. After 1 byte of data has been received, the NTST.RDBFF0 flag is set to 1 on the rising edge of the eighth or ninth cycle of SCL clock (the clock signal) as selected by the SCSTRCTL.ACKTWE bit. Reading the NTDTBP0 register at

this time will produce the received data, and the RDBFF0 flag is automatically set to 0 at the same time. Furthermore, the value of the acknowledgment field received during the ninth cycle of SCL clock is returned as the value set in the ACKCTL.ACKT bit. Furthermore, if the next byte to be received is the next to last byte, set the SCSTRCTL.RWE bit to 1 (for wait insertion) before reading the NTDTBP0 register (containing the second byte from last). As well as enabling NACK output even in the case of delays in processing to set the ACKCTL.ACKT bit to 1 (NACK) in step 6, due to other interrupts, etc., this fixes the I3C_SCL line to the low level on the falling edge of the ninth clock cycle in reception of the last byte, so the state is such that issuing a STOP condition is possible.

6. When the SCSTRCTL.ACKTWE bit = 0 and the slave device must be notified that it is to end transfer for data reception after transfer of the next (final) byte, set the ACKCTL.ACKT bit to 1 (NACK).
7. After reading the byte before last from the NTDTBP0 register, if the value of the NTST.RDBFF0 flag is confirmed to be 1, write 1 to the CNDCTL.SPCND bit (STOP condition issuance request) and then read the last byte from the NTDTBP0 register. When 1 is written to the CNDCTL.SPCND bit, I3C is released from the wait state and issues the STOP condition after low-level output in the ninth clock cycle is completed or the I3C_SCL line is released from the low-hold state.
8. Upon detecting the STOP condition, I3C automatically sets bits CRMS and TRMD in the PRSST register to 00 and enters slave receive mode. Furthermore, detection of the STOP condition leads to setting of the BST.SPCNDDF flag to 1.
9. After checking that the BST.SPCNDDF flag = 1, set the BST.NACKDF and SPCNDDF flags to 0 for the next transfer operation.

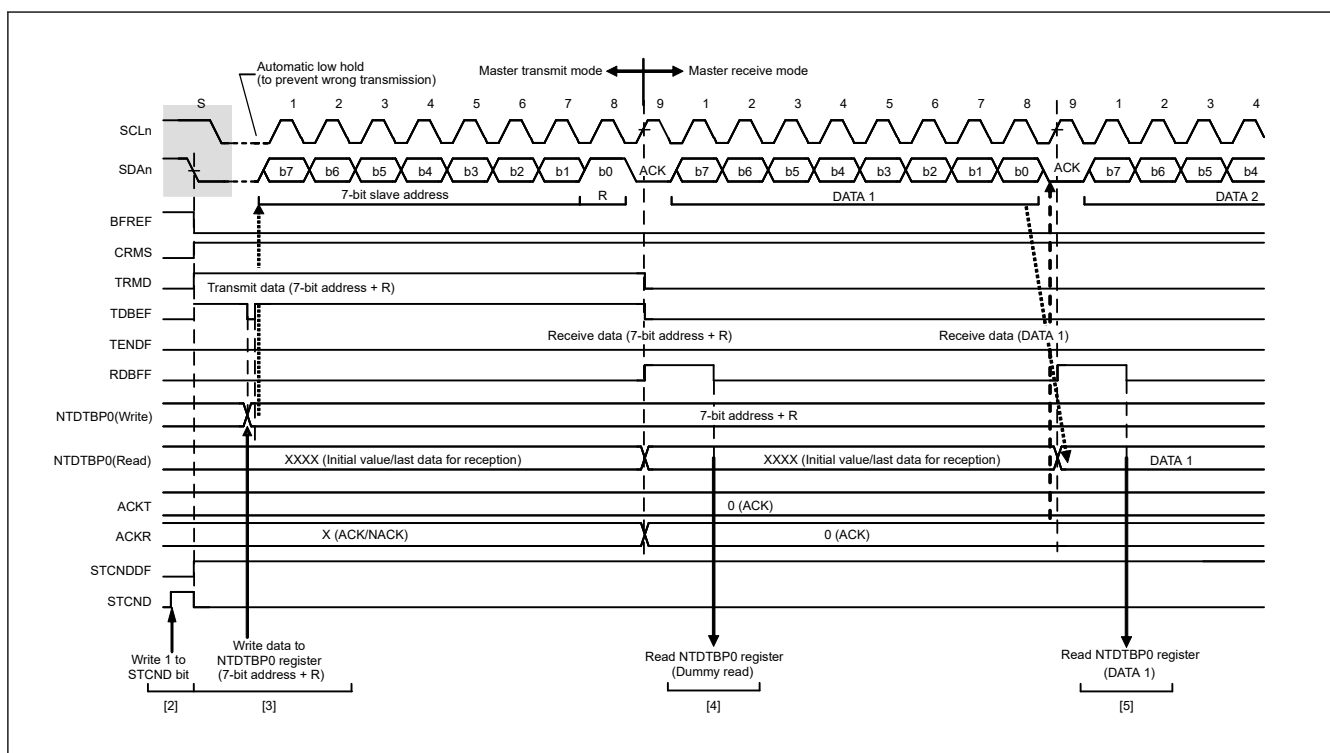


Figure 40.9 Master receive operation timing (1) (7-bit address format, when ACKTWE = 0)

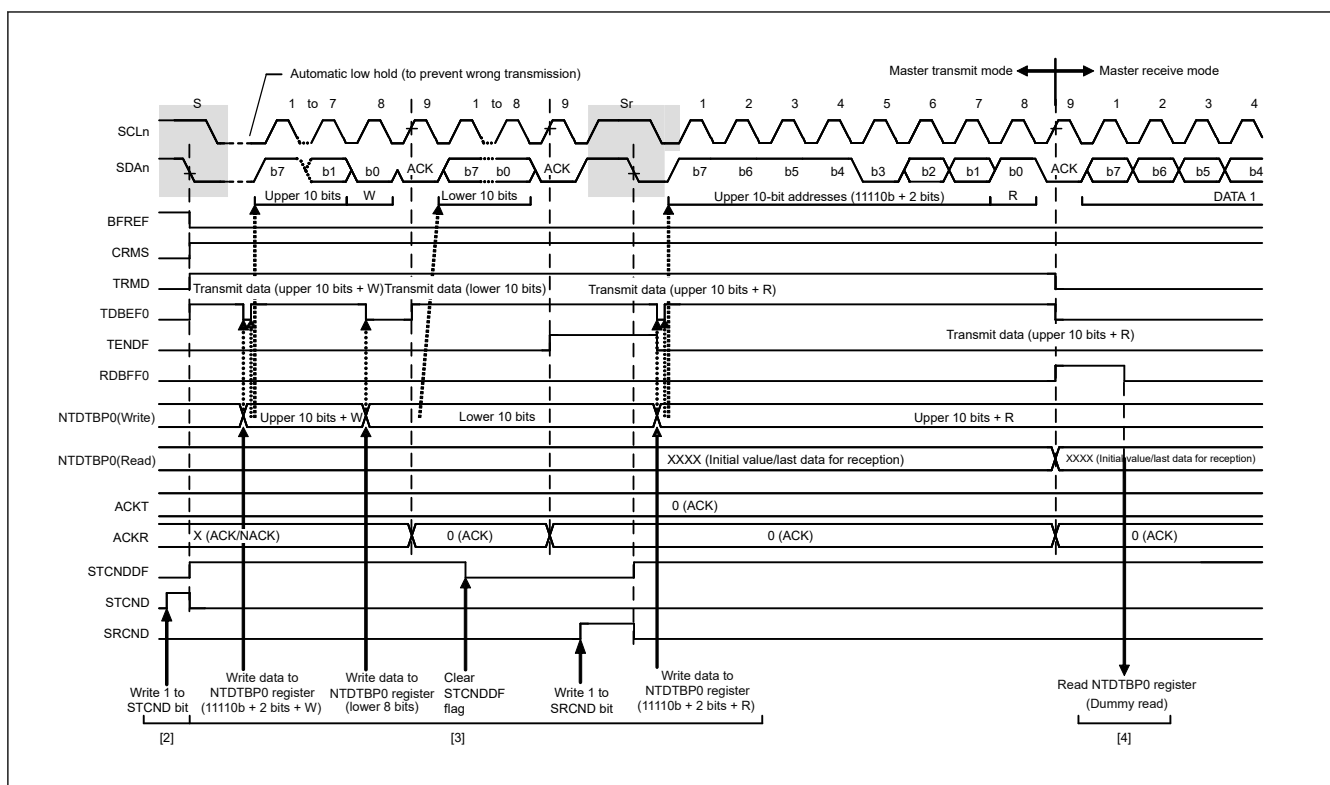


Figure 40.10 Master receive operation timing (2) (10-bit address format, when ACKTWE = 0)

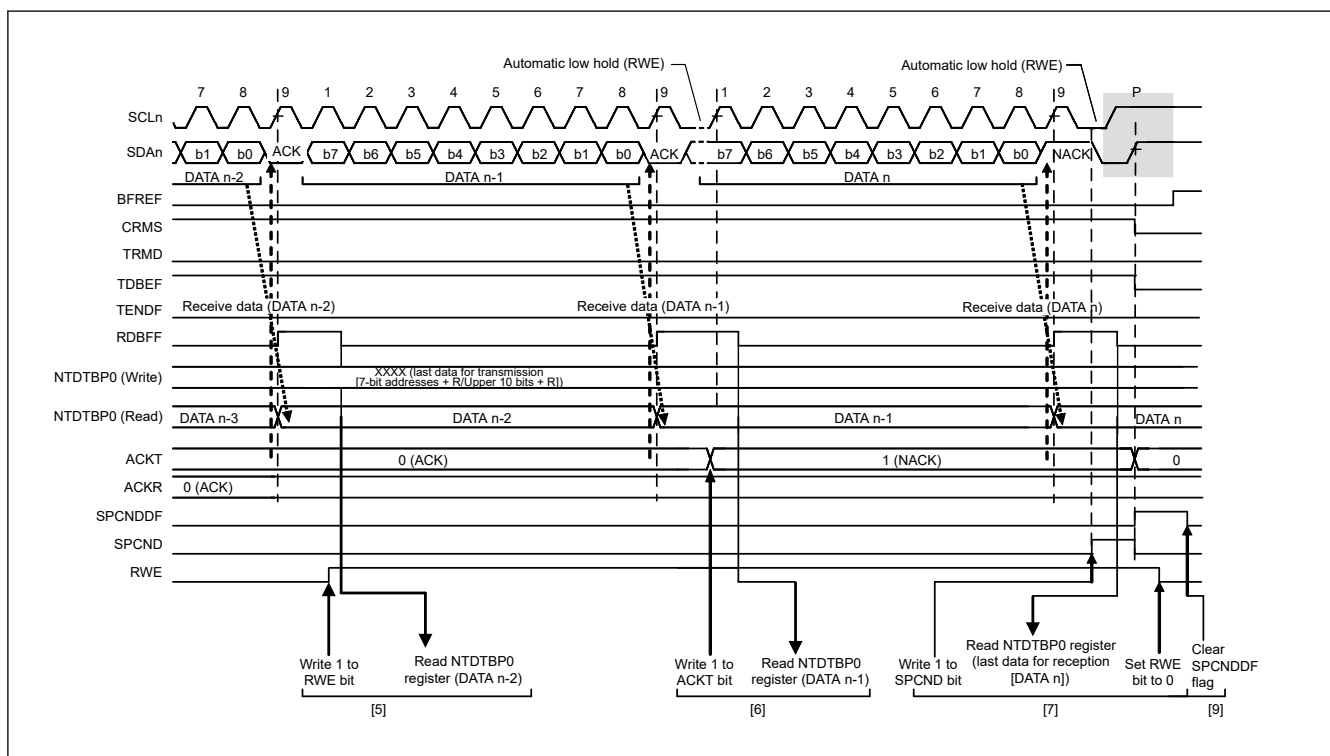


Figure 40.11 Master receive operation timing (3) (when ACKTWE = 0)

(2) I3C Master Operation

(a) Dynamic Address Assign Procedure

After initializing I3C, first execute Dynamic Address Assign Procedure for I3C Slave connected on the I3C Bus. The following describes the procedure.

1. Initial setting (see [section 40.3.3.1.2. I3C Initial Setting Flow](#) for details)
2. Execute Dynamic Address Assign with ENTDAAs or SETDASA Common Command Code (CCC) for I3C Slave set in DAT (DATBASm register).
Write Command Descriptor (Address Assign Command) to Command Buffer via the NCMDQP register.
3. When Command Descriptor is written to Command Buffer, Transaction is issued on I3C Bus.
4. When ENTDAAs is specified for CMD[7:0] of Address Assign Command:
Execute Dynamic Address Assign for I3C Slave for the number of DATs specified by DEV_COUNT[3:0] starting with DAT specified by DEV_INDEX[4:0] of Address Assign Command.
When SETDASA is specified for CMD[7:0] of Address Assign Command:
Execute Dynamic Address Assign for I3C Slave indicated by DAT specified by DEV_INDEX[4:0] of Address Assign Command.
5. In case of ENTDAAs, the Provisional ID, BCR, DCR transmitted from I3C Slave is stored in Receive Data Buffer (BCR is also automatically stored in the MSDCTm register).
Read the Provisional ID, BCR, and DCR from the Receive Data Buffer via the NTDTBPn register with an interrupt by RDBFF0 = 1.
6. When execution of Dynamic Address Assign is completed, issue STOP condition and store the Response Descriptor into the Response Buffer.
7. Read the Response Descriptor via the NRSPQP register and check the status.
8. Check whether the value of the DATA_LENGTH[15:0] bits of the Response Descriptor matches the value of DEV_COUNT[3:0] of the Address Assign Command.

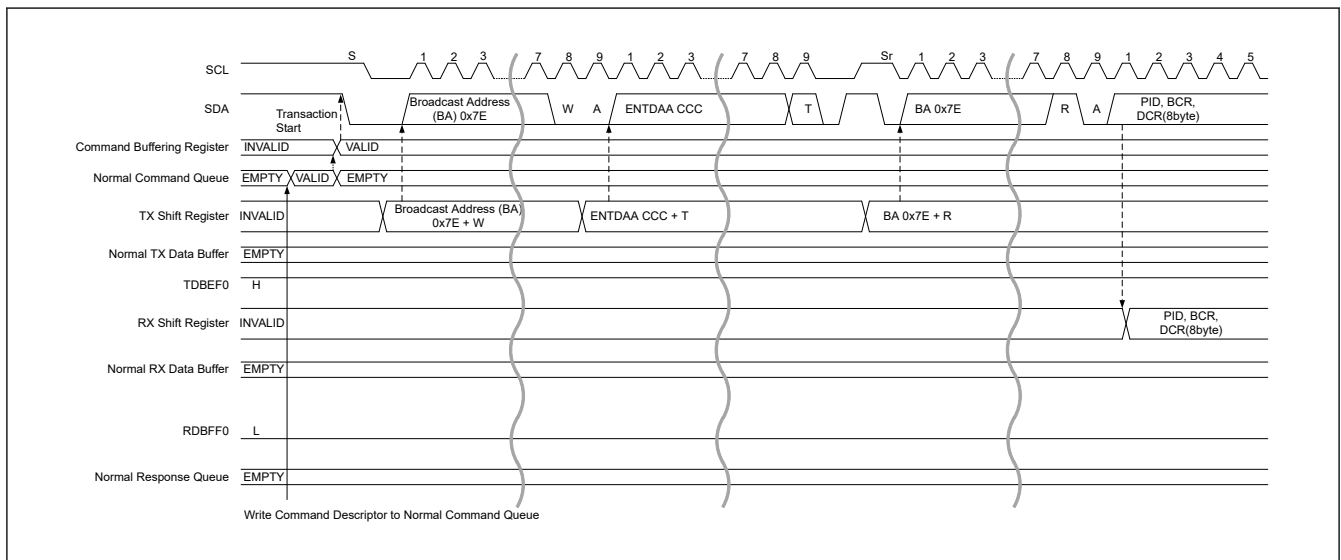


Figure 40.12 Dynamic address assign procedure (ENTDAA CCC) timing (1/3)

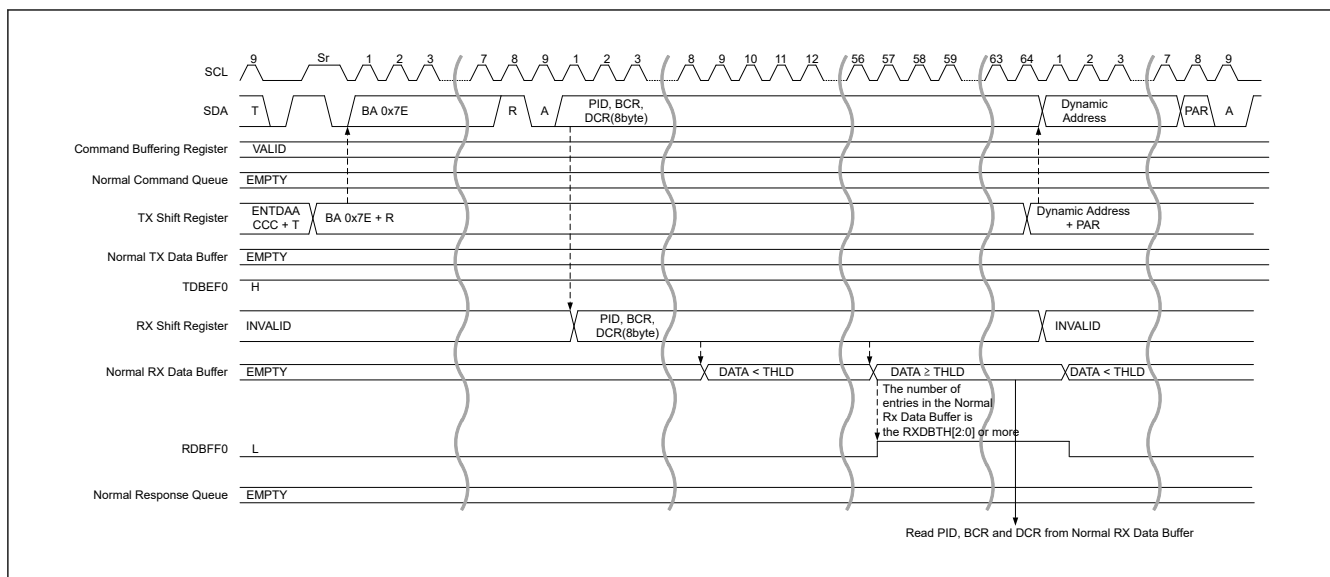


Figure 40.13 Dynamic address assign procedure (ENTDAA CCC) timing (2/3)

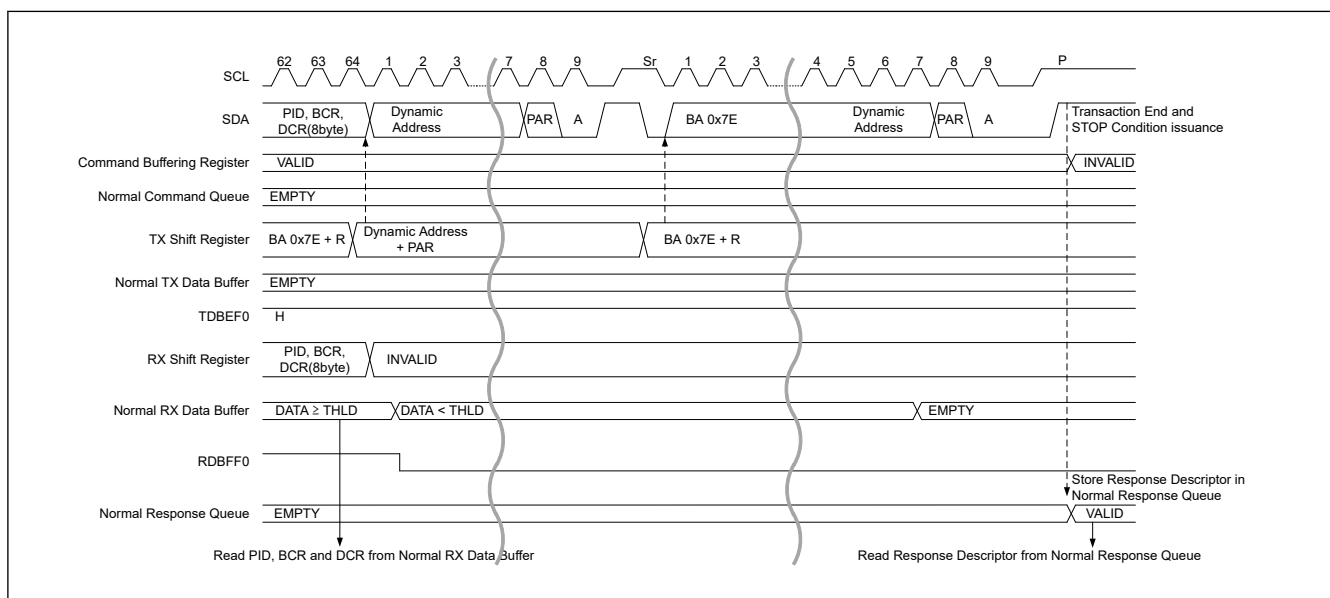


Figure 40.14 Dynamic address assign procedure (ENTDAA CCC) timing (3/3)

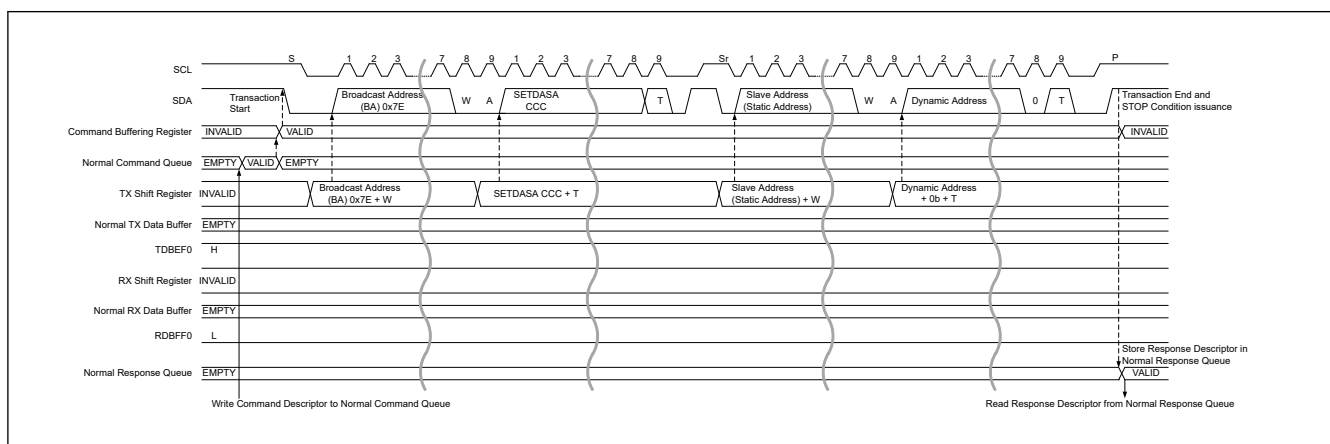


Figure 40.15 Dynamic address assign procedure (SETDASA CCC) timing

(b) SDR Data Write Transfer

1. Write data for transmission to the Tx Data Buffer via the NTDTBPn register.
2. Write a Command Descriptor (Immediate Transfer Command or Regular Transfer Command or Combo Transfer Command) for Data Transfer to the Command Buffer via the NCMDQP register.
3. When Command Descriptor is written to Command Buffer, transaction is issued on I3C Bus.
When NACK is received with the Address Header, transaction of the same command is automatically issued according to the NACK Retry Count value (DATBASm.DVNACK) of DAT.
4. If data for transmission still remain, write data for transmission by an interrupt with TDBEF0 = 1 to the Tx Data Buffer via the NTDTBPn register.
5. When data transmission for the number of Data Length specified by the DATA_LENGTH[15:0] bits of the Command Descriptor is completed, the Repeated START condition or STOP condition is issued and the Response Descriptor is stored in the Response Buffer.
6. Read the Response Descriptor via the NRSPQP register and check the status.
7. Check that the value of the DATA_LENGTH[15:0] bits of the Response Descriptor is 0.

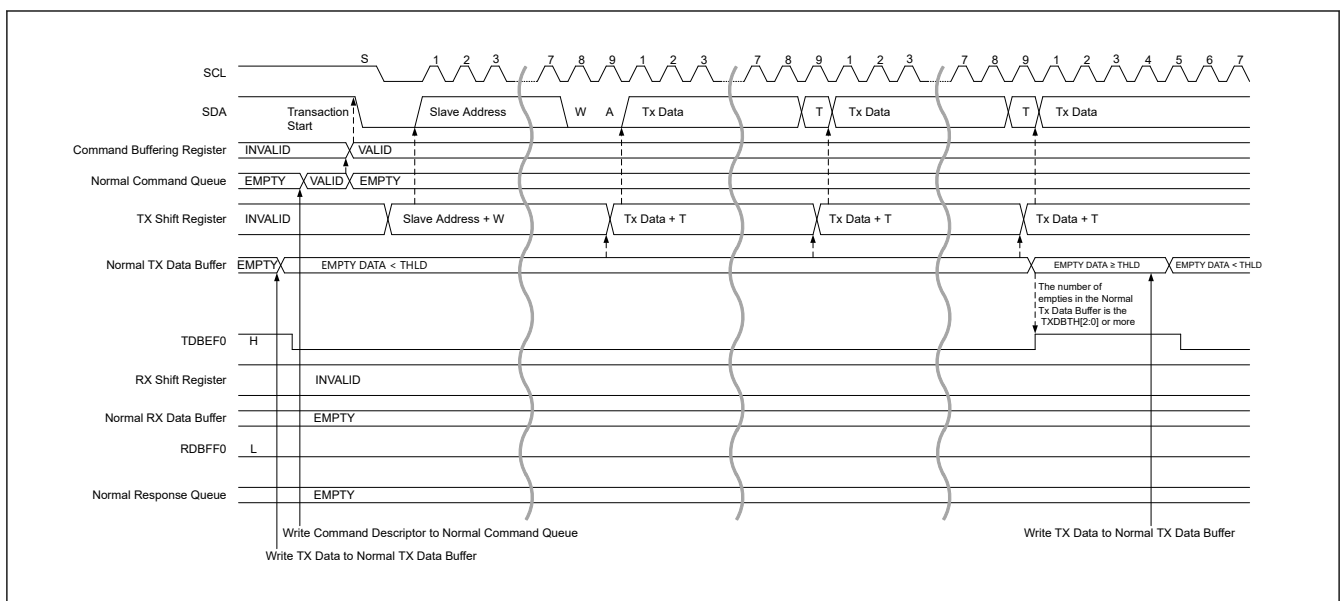


Figure 40.16 SDR data write transfer timing (1/2)

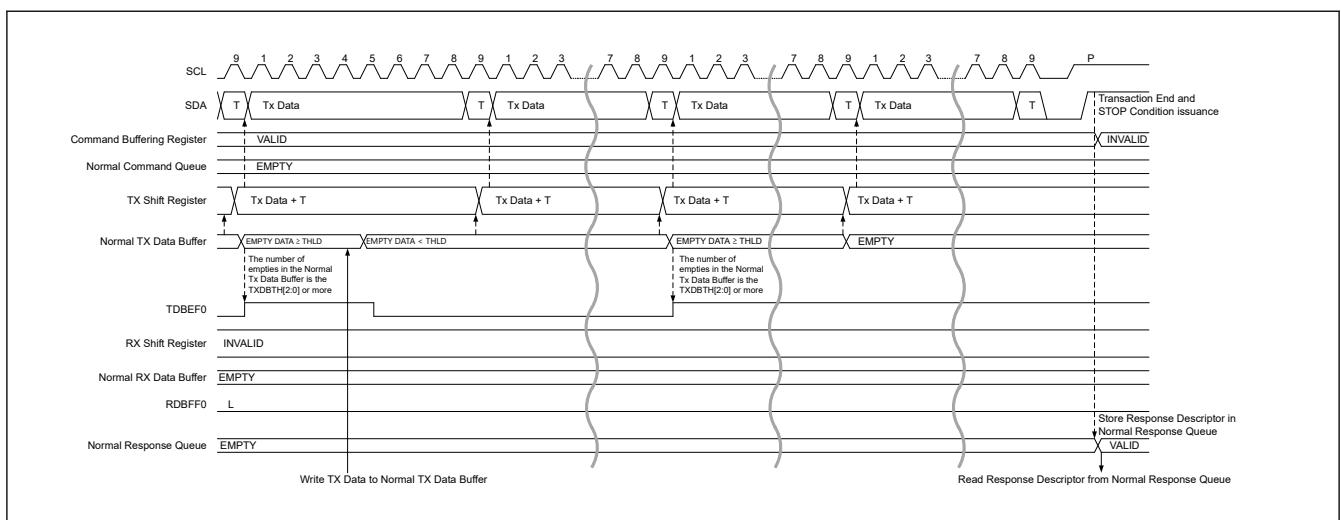


Figure 40.17 SDR data write transfer timing (2/2)

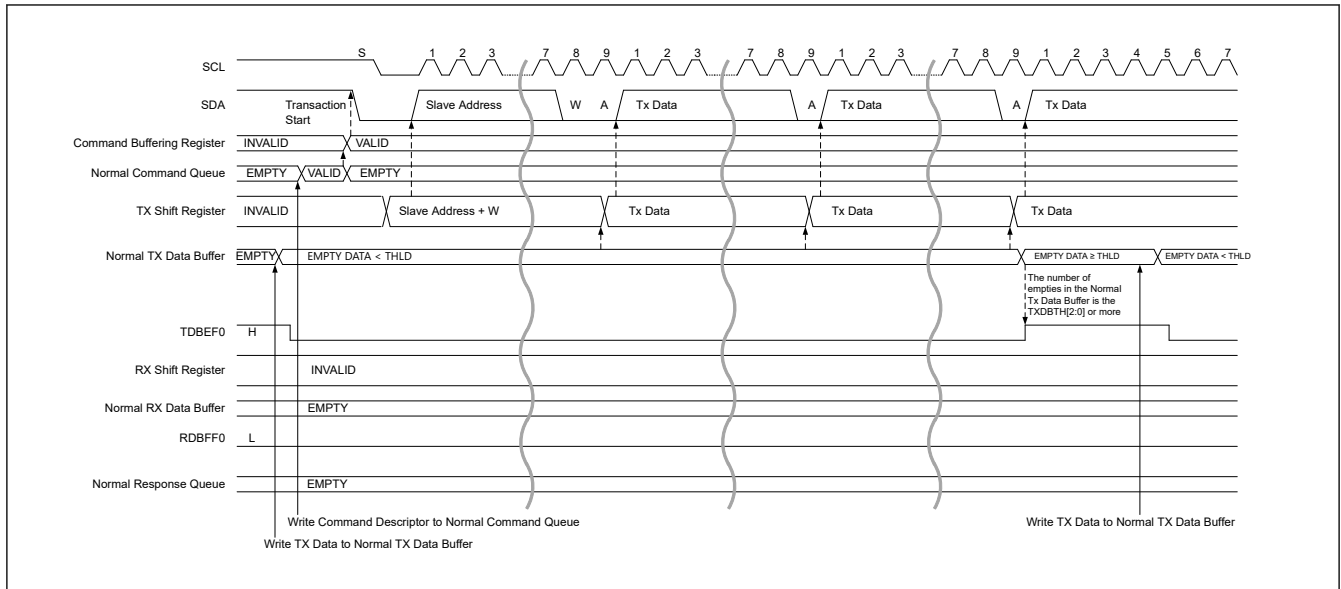


Figure 40.18 Legacy I2C message data write timing (1/2)

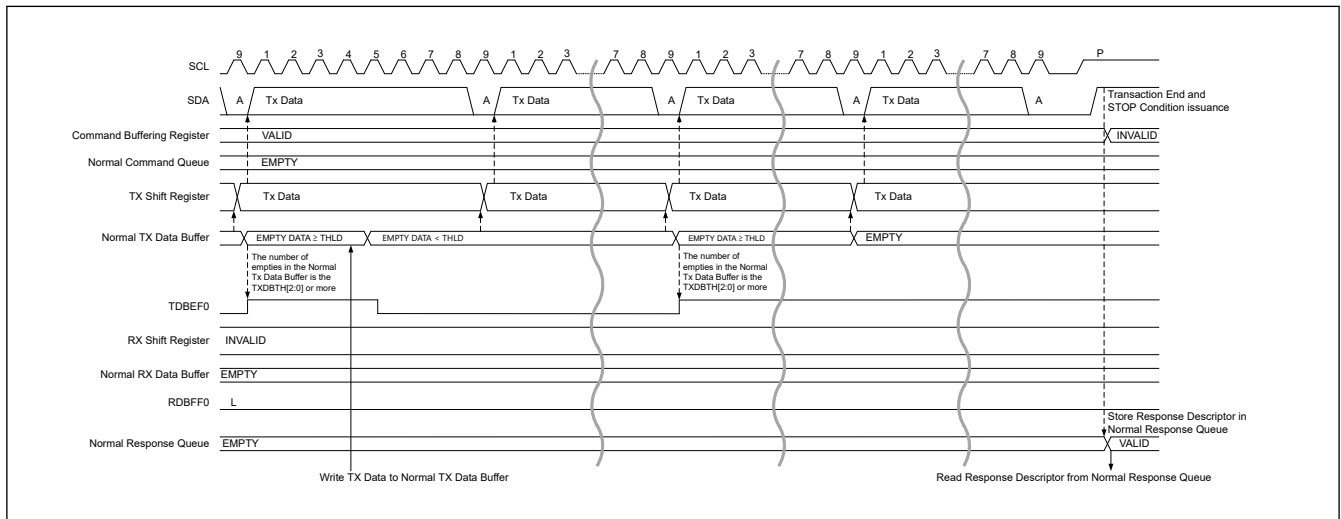


Figure 40.19 Legacy I2C message data write timing (2/2)

(c) SDR Data Read Transfer

1. Write a Command Descriptor (Immediate Transfer Command or Regular Transfer Command or Combo Transfer Command) for Data Transfer to the Command Buffer via the NCMDQP register.
2. When Command Descriptor is written to Command Buffer, transaction is issued on I3C Bus.
When NACK is received with the Address Header, transaction of the same command is automatically issued according to the NACK Retry Count value (DATBASm.DVNACK) of DAT.
3. Data received from the I3C Slave is stored in the Receive Data Buffer.
4. With the RDBFF0 = 1 interrupt, the received data is read from the Receive Data Buffer via the NTDTBPn register.
5. SDR:
Detecting Low in T-bit or receiving Data for the number of Data Length specified by the DATA_LENGTH[15:0] bits of Command Descriptor is completed, issue Repeated START condition or STOP condition and store the Response Descriptor into the Response Buffer.
Legacy I2C Message:
When data reception for the number of Data Length specified by the DATA_LENGTH[15:0] bits of Command Descriptor is completed, NACK is issued. After that, issue a Repeated START condition or STOP condition and store the Response Descriptor into the Response Buffer.
6. Read the Response Descriptor via the NRSPQP register and check the status.

7. Check whether the value of the DATA_LENGTH[15:0] bits of the Response Descriptor matches the data length setting value of the Command Descriptor.

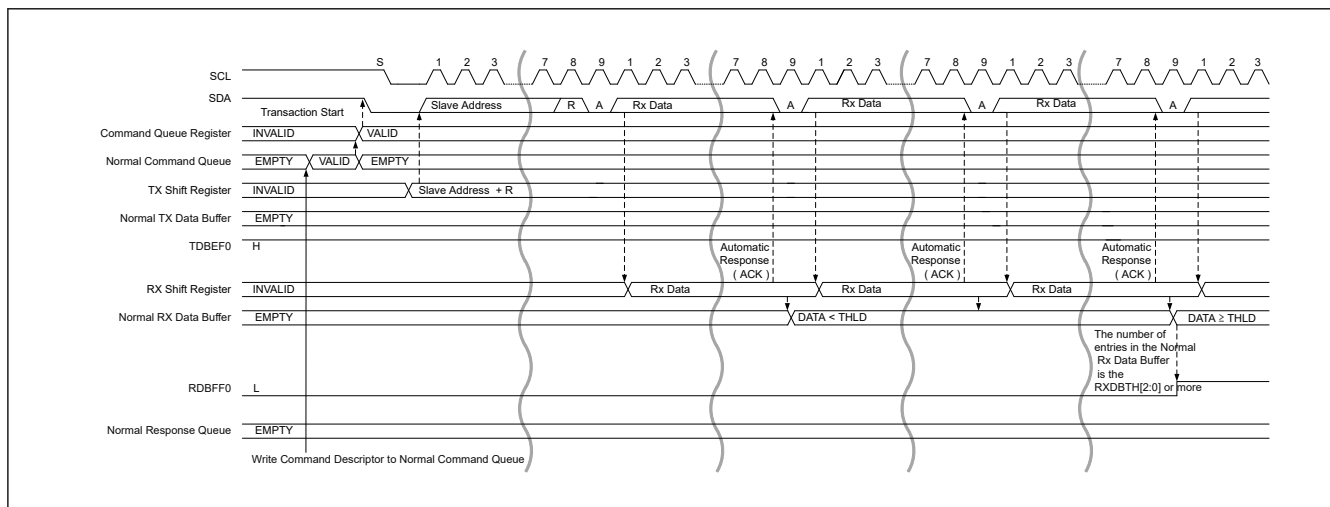


Figure 40.20 SDR data read transfer timing (1/2)

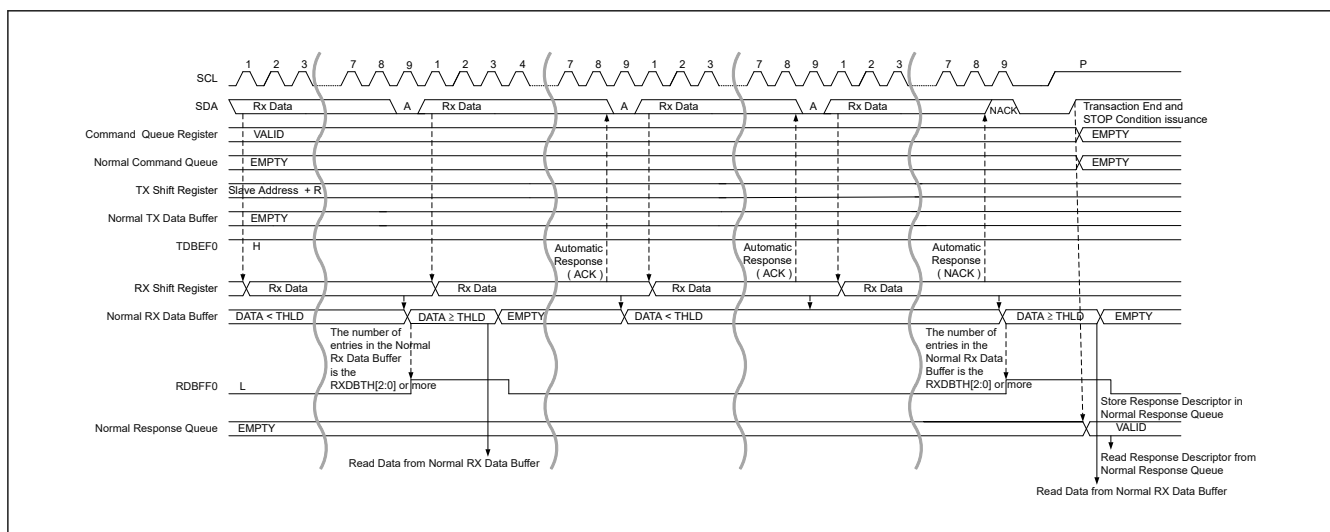


Figure 40.21 SDR data read transfer timing (2/2)

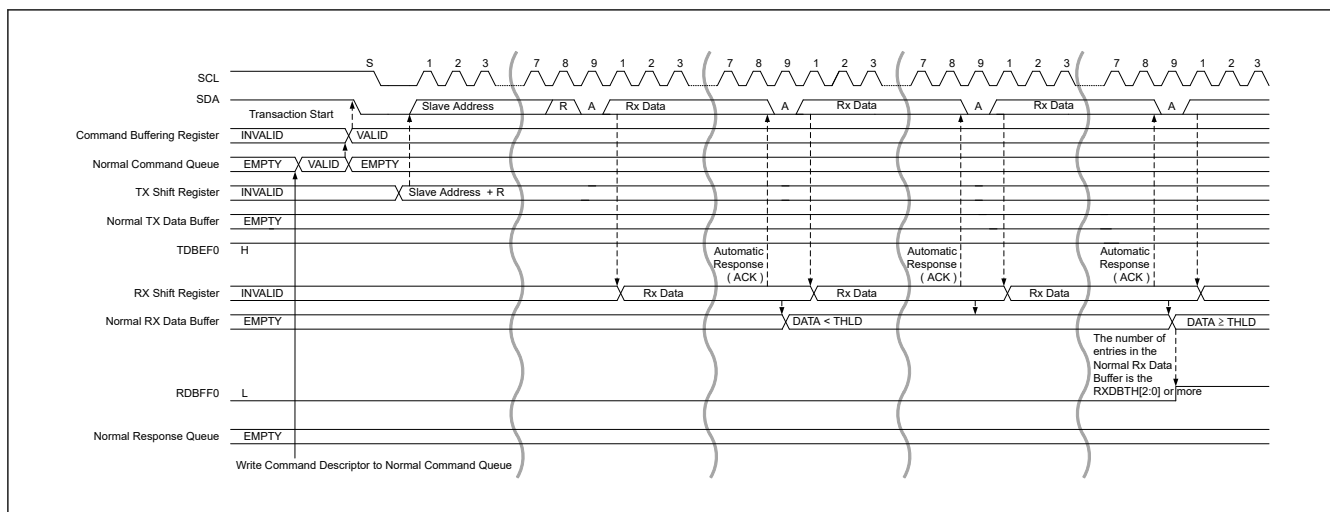


Figure 40.22 Legacy I2C message data read transfer timing (1/2)

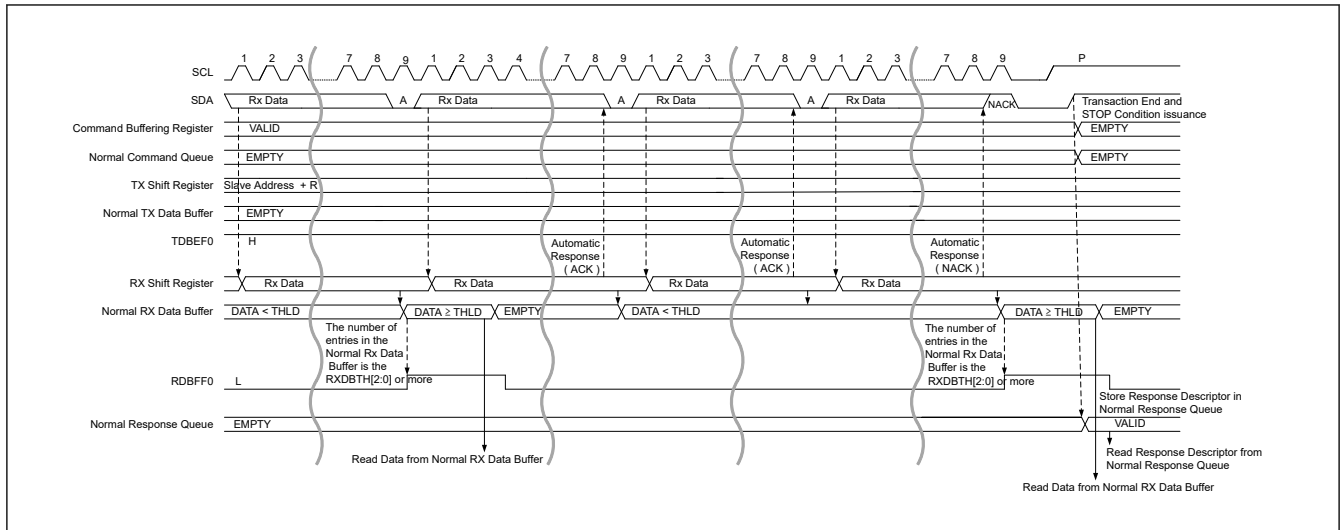


Figure 40.23 Legacy I²C message data read transfer timing (2/2)

(d) HDR Data Write Transfer

1. Write data for transmission to the Tx Data Buffer via the NTDTBPn register.
2. Write a Command Descriptor (Immediate Transfer Command or Regular Transfer Command or Combo Transfer Command) for Data Transfer to the Command Buffer via the NCMDQP register.
3. When Command Descriptor is written to Command Buffer, transaction is issued on I3C Bus.
When NACK is received with the Address Header, transaction of the same command is automatically issued according to the NACK Retry Count value (DATBASm.DVNACK) of DAT.
4. If data for transmission still remains, write data for transmission by an interrupt with TDBEF0 = 1 to the Tx Data Buffer via the NTDTBPn register.
5. HDR-DDR:
When the data transmission for the number of Data Length specified by the DATA_LENGTH[15:0] bits of the Command Descriptor is completed, CRC Word is transmitted. After that, issue HDR Restart Pattern or HDR Exit Pattern + STOP condition and store the Response Descriptor into Response Buffer.
HDR-Ternary:
When the data transmission for the number of Data Length specified by the DATA_LENGTH[15:0] bits of the Command Descriptor is completed, the HDR Restart Pattern or HDR Exit Pattern + STOP condition is issued and the Response Descriptor is stored in the Response Buffer.
6. Read the Response Descriptor via the NRSPQP register and check the status.
7. Check that the value of the DATA_LENGTH[15:0] bits of the Response Descriptor is 0.

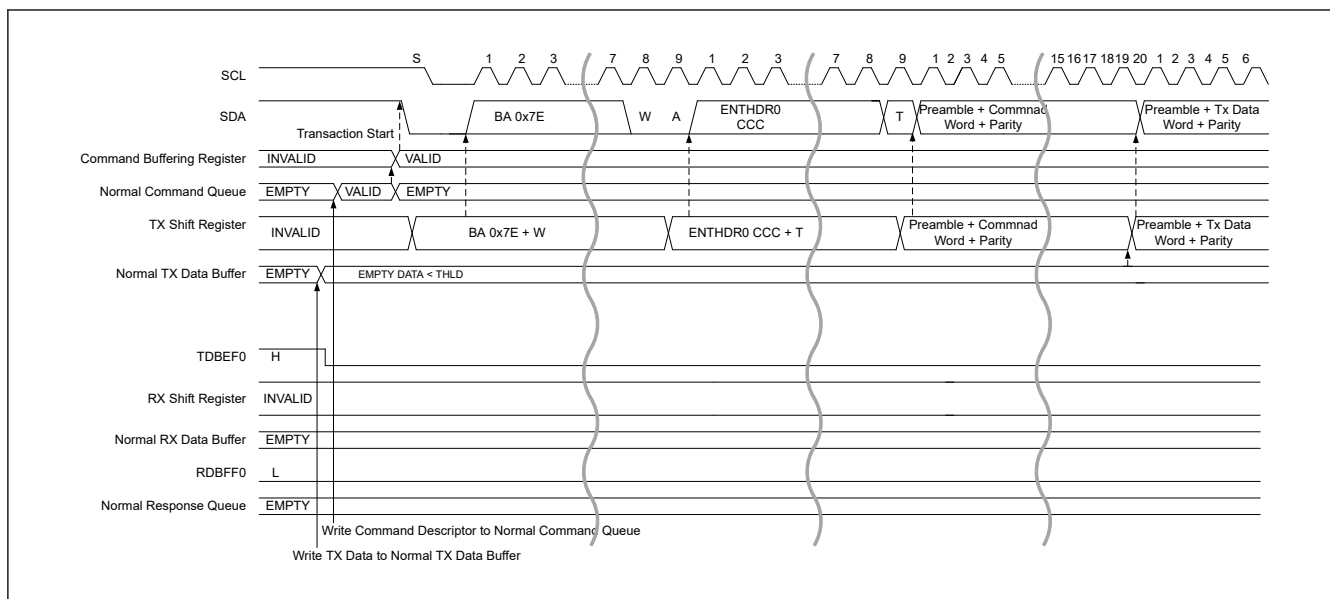


Figure 40.24 HDR data write transfer (HDR-DDR) timing (1/4)

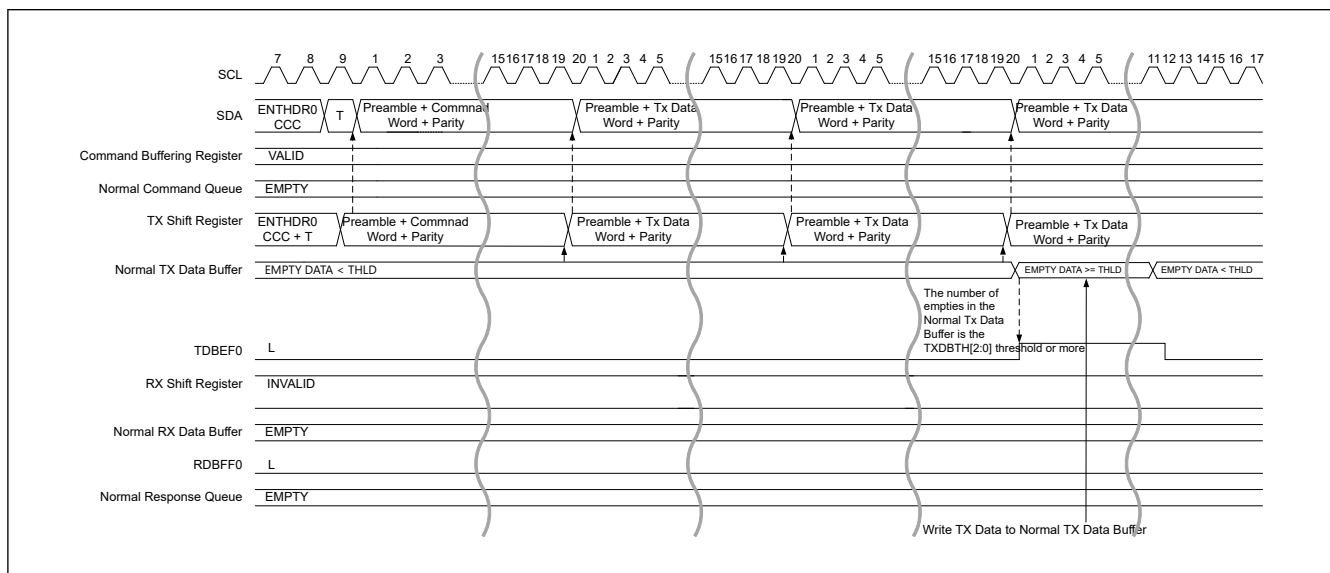


Figure 40.25 HDR data write transfer (HDR-DDR) timing (2/4)

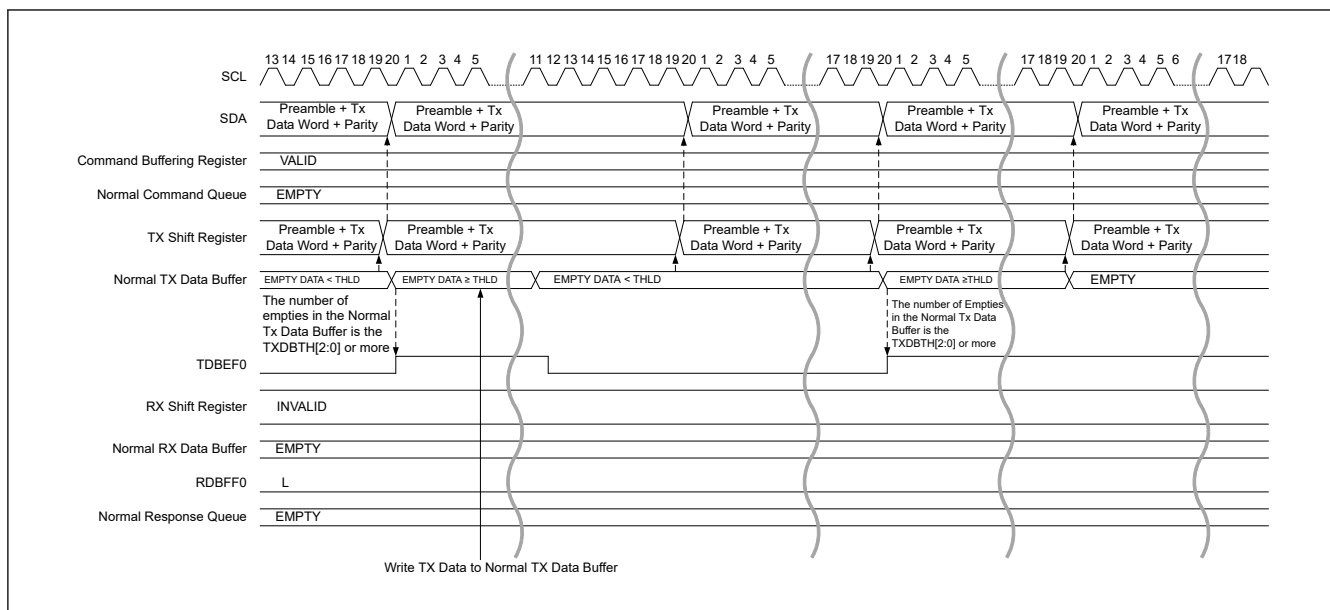


Figure 40.26 HDR data write transfer (HDR-DDR) timing (3/4)

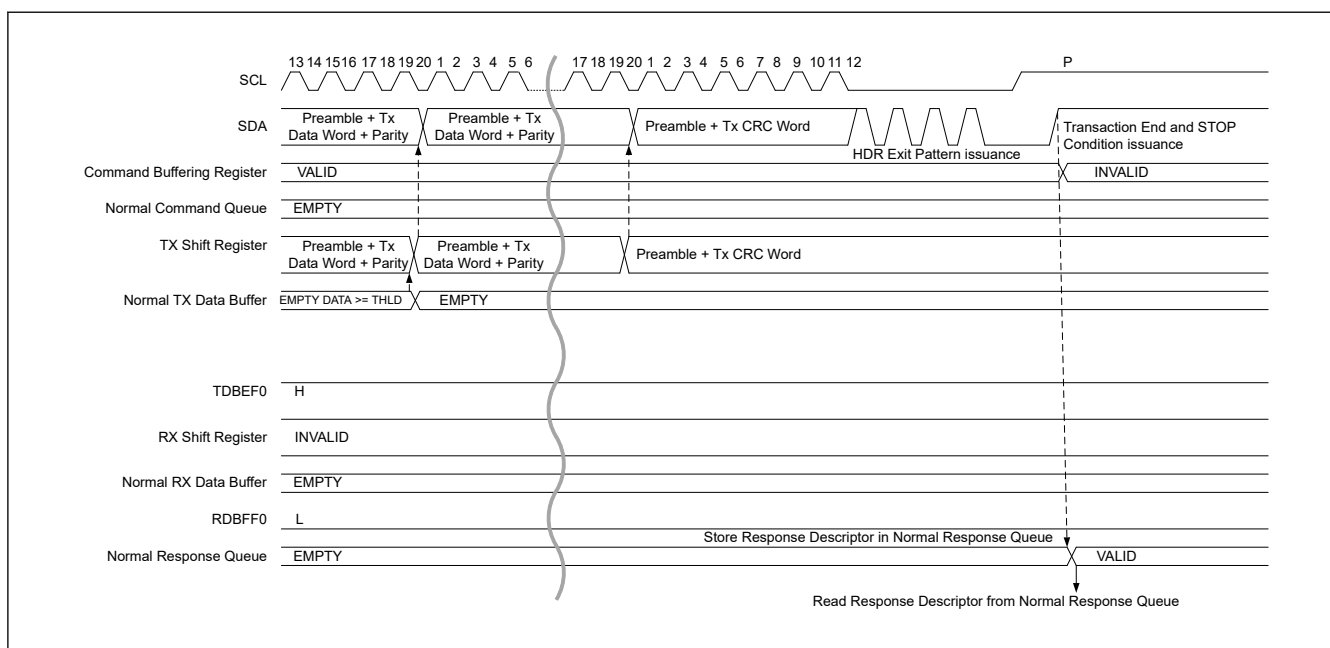


Figure 40.27 HDR data write transfer (HDR-DDR) timing (4/4)

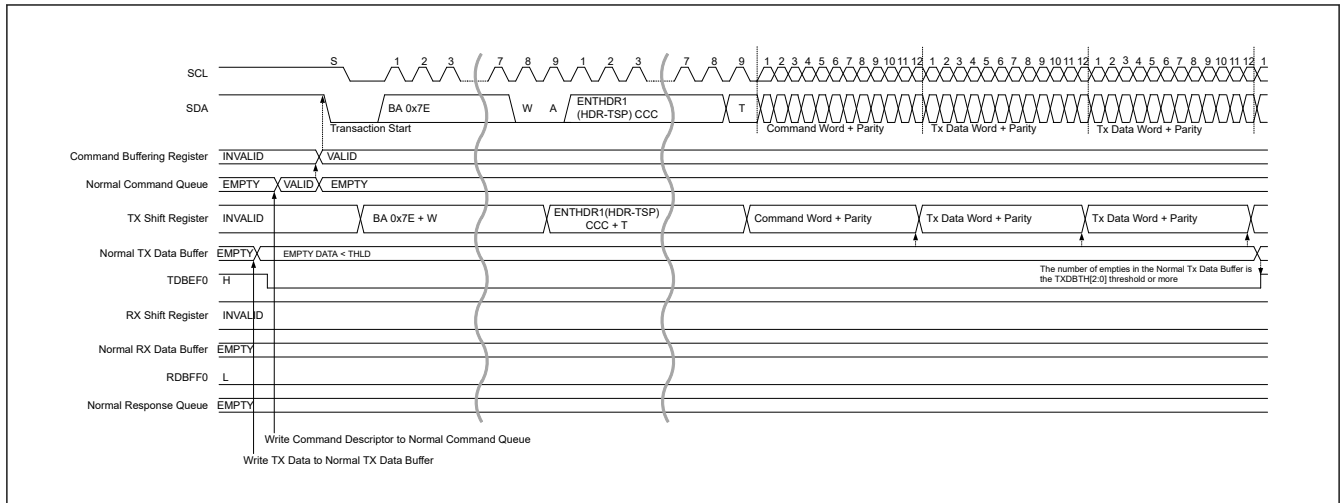


Figure 40.28 HDR data write transfer (HDR-TSP, TSL) timing (1/2)

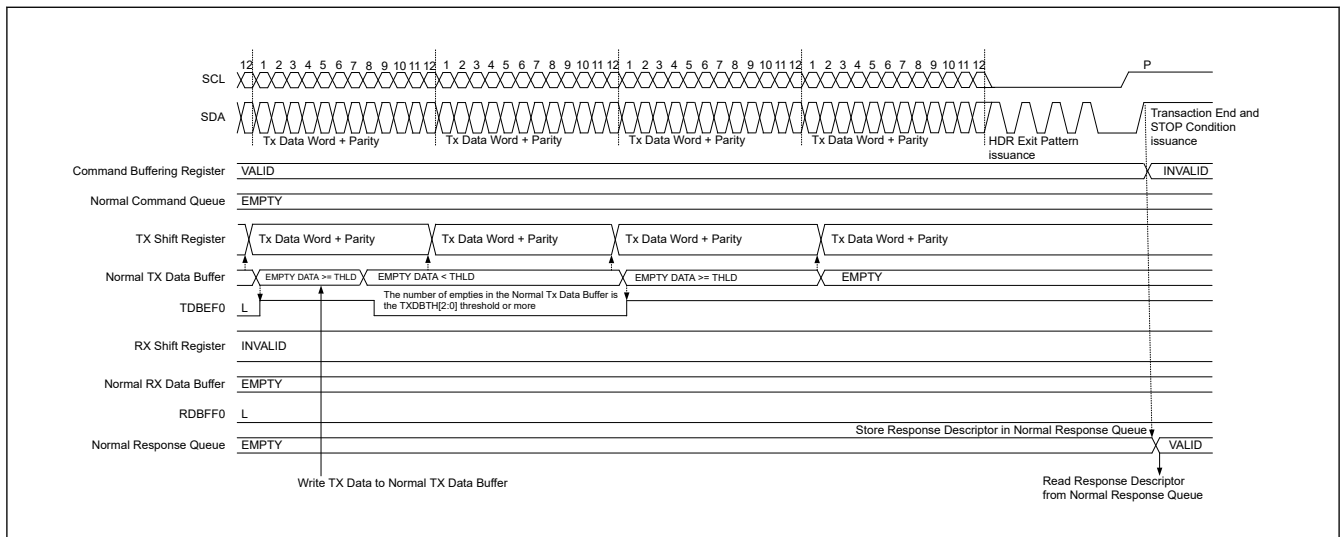


Figure 40.29 HDR data write transfer (HDR-TSP, TSL) timing (2/2)

(e) HDR Data Read Transfer

1. Write a Command Descriptor (Immediate Transfer Command or Regular Transfer Command or Combo Transfer Command) for Data Transfer to the Command Buffer via the NCMDQP register.
2. When Command Descriptor is written to Command Buffer, transaction is issued on I3C Bus.
When NACK is received with the Address Header, transaction of the same command is automatically issued according to the NACK Retry Count value (DATBASm.DVNACK) of DAT.
3. Data received from the I3C Slave is stored in the Receive Data Buffer.
4. With the RDBFF0 = 1 interrupt, the received data is read from the Receive Data Buffer via the NTDTBPn register.
5. HDR-DDR:
 - (a) When CRC Word is received, it issues HDR Restart Pattern or HDR Exit Pattern + STOP condition and stores the Response Descriptor into the Response Buffer.
 - (b) When the preamble following the received data for the number of Data Length specified by the DATA_LENGTH[15:0] bits of Command Descriptor indicates Data Word (11), drive (Abort) low to the second bit of the preamble. After that, issue HDR Restart Pattern or HDR Exit Pattern + STOP condition and store the Response Descriptor into the Response Buffer.

HDR-Ternary:

When detecting the start of the HDR Restart Pattern or HDR Exit Pattern from I3C Slave, issue HDR Restart Pattern or HDR Exit Pattern + STOP condition and store the Response Descriptor into the Response Buffer.

6. Read the Response Descriptor via the NRSPQP register and check the status.
7. Check whether the value of the DATA_LENGTH[15:0] bits of the Response Descriptor matches the data length setting value of the Command Descriptor.

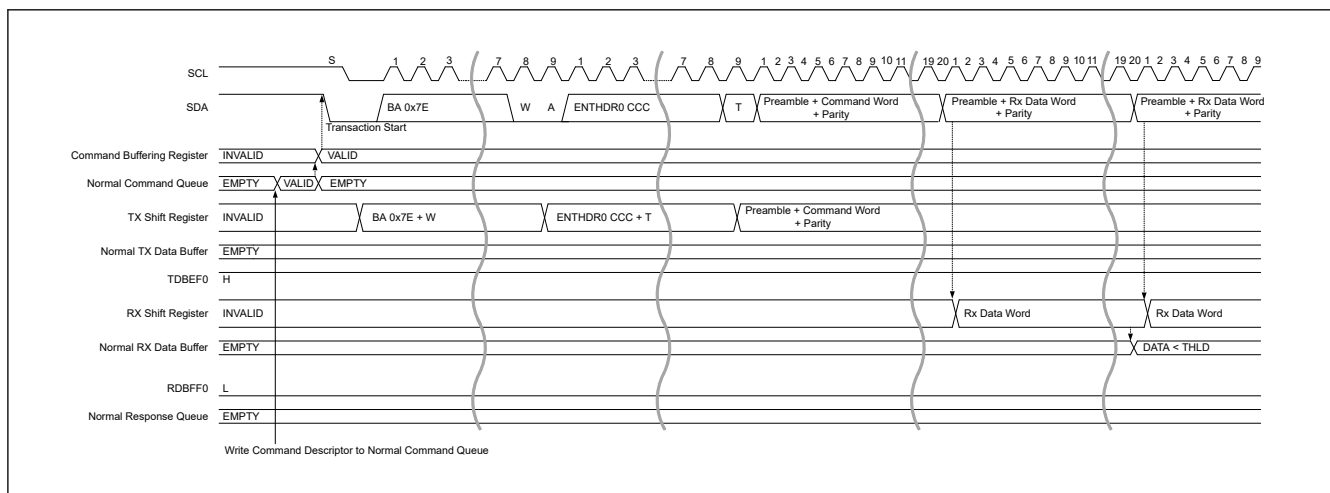


Figure 40.30 HDR data read transfer (HDR-DDR) timing (1/2)

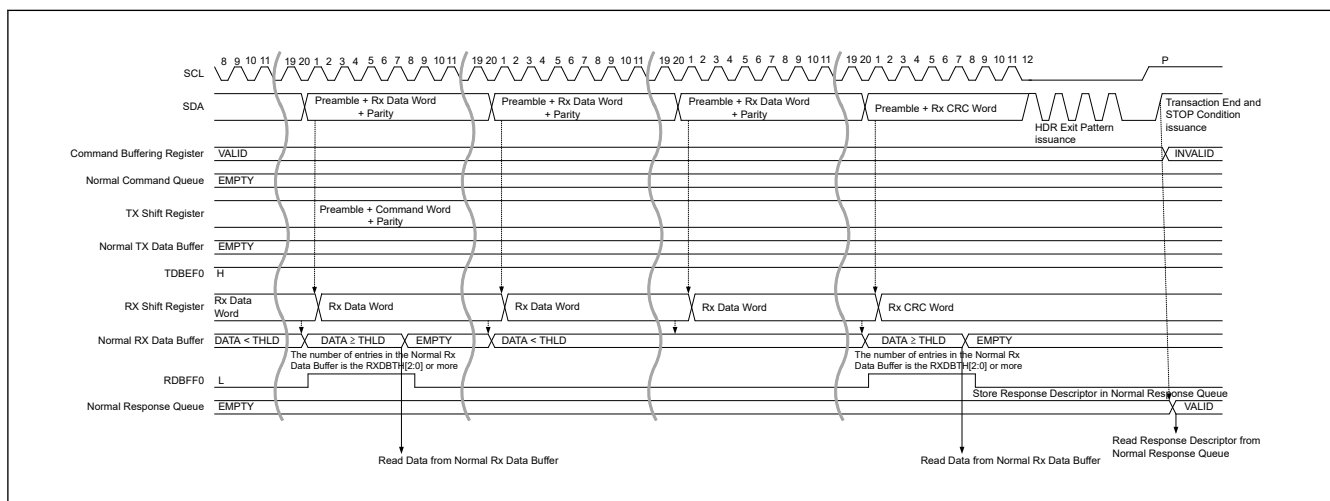


Figure 40.31 HDR data read transfer (HDR-DDR) timing (2/2)

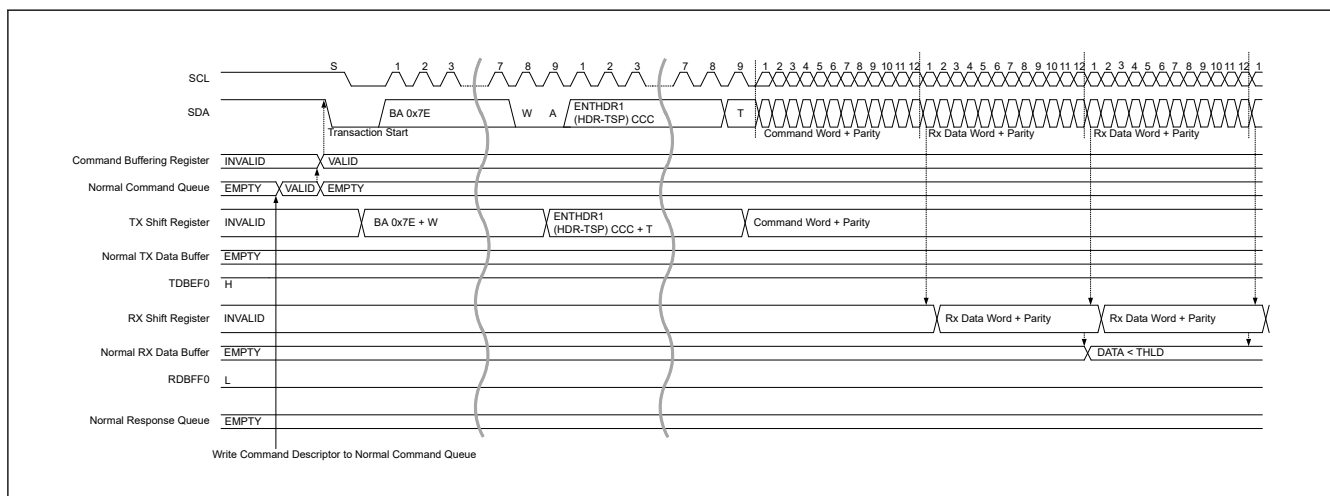


Figure 40.32 HDR data read transfer (HDR-TSP, TSL) timing (1/2)

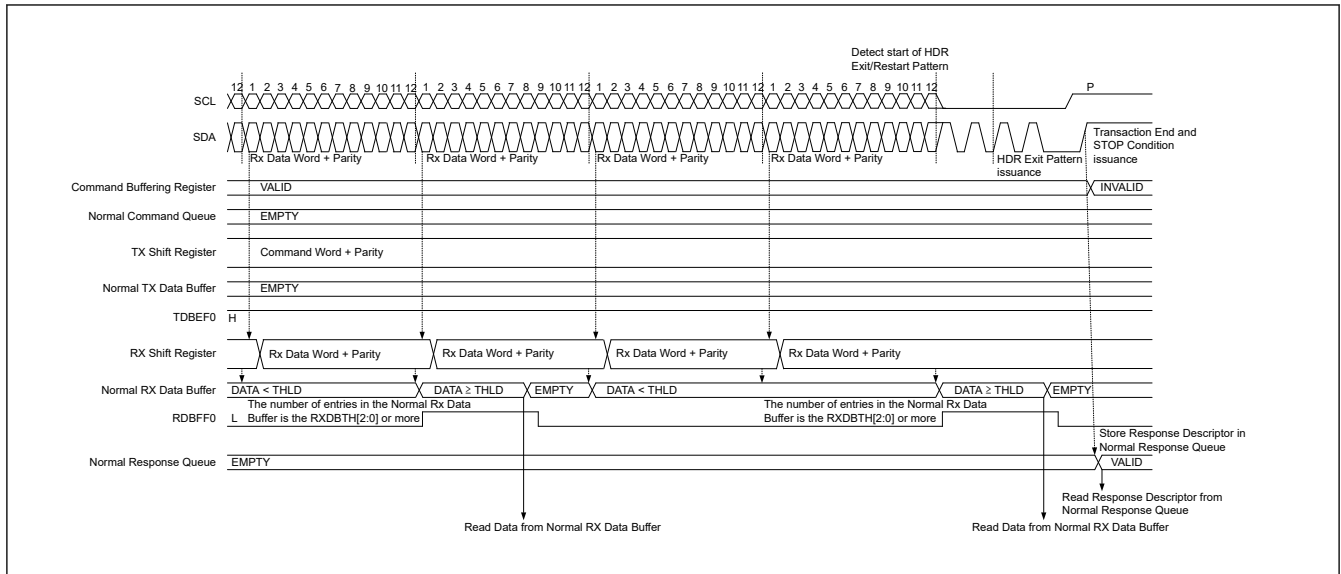


Figure 40.33 HDR data read transfer (HDR-TSP, TSL) timing (2/2)

(f) IBI Transfer

1. Write Command Descriptor to the Command Buffer and issue Transaction on I3C Bus.
If START Request (SDA Low Drive) is issued from the slave device, I3C drives SCL to Low and completes START condition.
Thereafter, the SCL is supplied and In-Band Interrupt Request is received.
2. In Slave Address with RnW of the Address Header, if losing Arbitration by issuing In-Band Interrupt from I3C Slave, stop issuing Transaction.
3. According to [section 40.3.2.3.8. In-Band Interrupt \[I3C Mode\]](#), detect In-Band Interrupt and process.
4. In the interrupt with IBIQEFF = 1, read the IBI Status Descriptor from the IBI Queue via the NIBIQP register and check the status.
When detected a Slave Interrupt Request and responded with ACK, Read the IBI Data for the Data Length indicated by the DATA_LENGTH[15:0] bits of the IBI Status Descriptor from the IBI Data Buffer via the NIBIQP register.
5. Restart issuing Transaction of Command of Step1.

An example of the processing procedure after detection of In-Band Interrupt is shown below.

Processing procedure for detecting Mastership Request and transferring master right to Secondary Master

1. If the I3C Secondary Master wins the Arbitration, issue a DEFSLVS CCC and notify Slave information to Secondary Master.
2. Issue a GETACCMST CCC and complete CCC by a STOP condition.

The Mastership processing flow is shown in [Figure 40.36](#)

- Note:
- After transferring master right to Secondary Master, to get master right again, issue a Mastership Request according to (f) IBI Transfer of [\(2\) I3C Slave Operation](#).
 - After Mastership Request is accepted by the Current Master, to get master right again at receiving the GETACCMST CCC and complete CCC by a STOP condition.

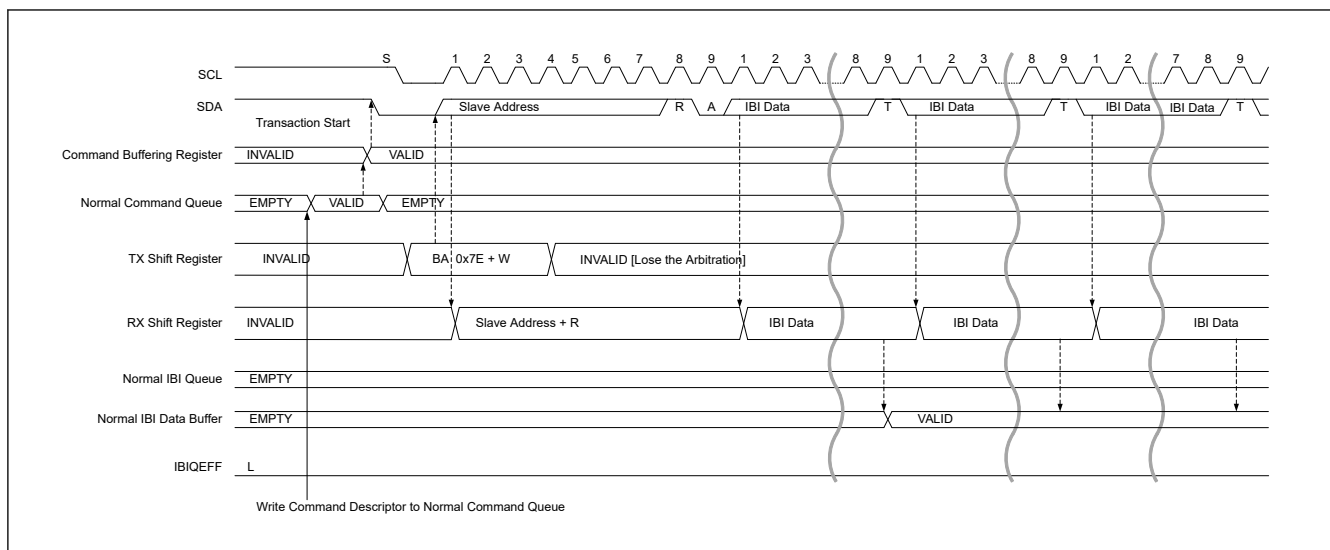


Figure 40.34 I3C master IBI transfer timing (1/2)

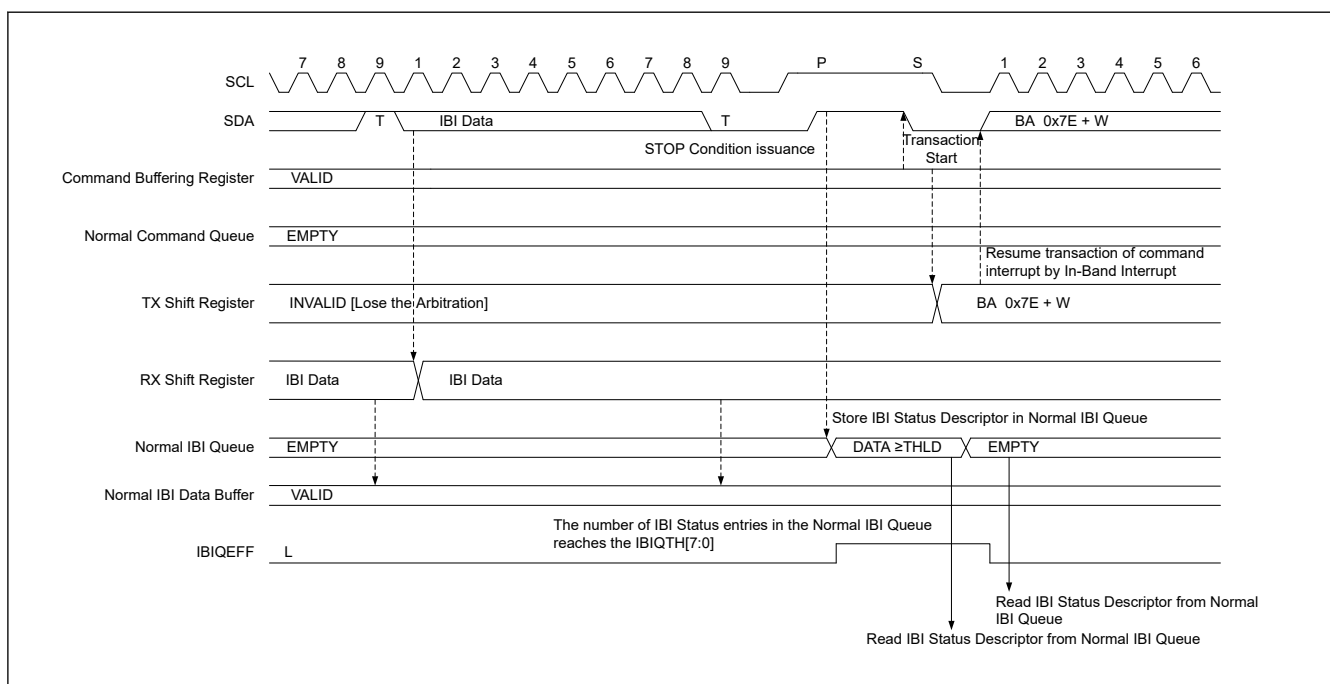


Figure 40.35 I3C master IBI transfer timing (2/2)

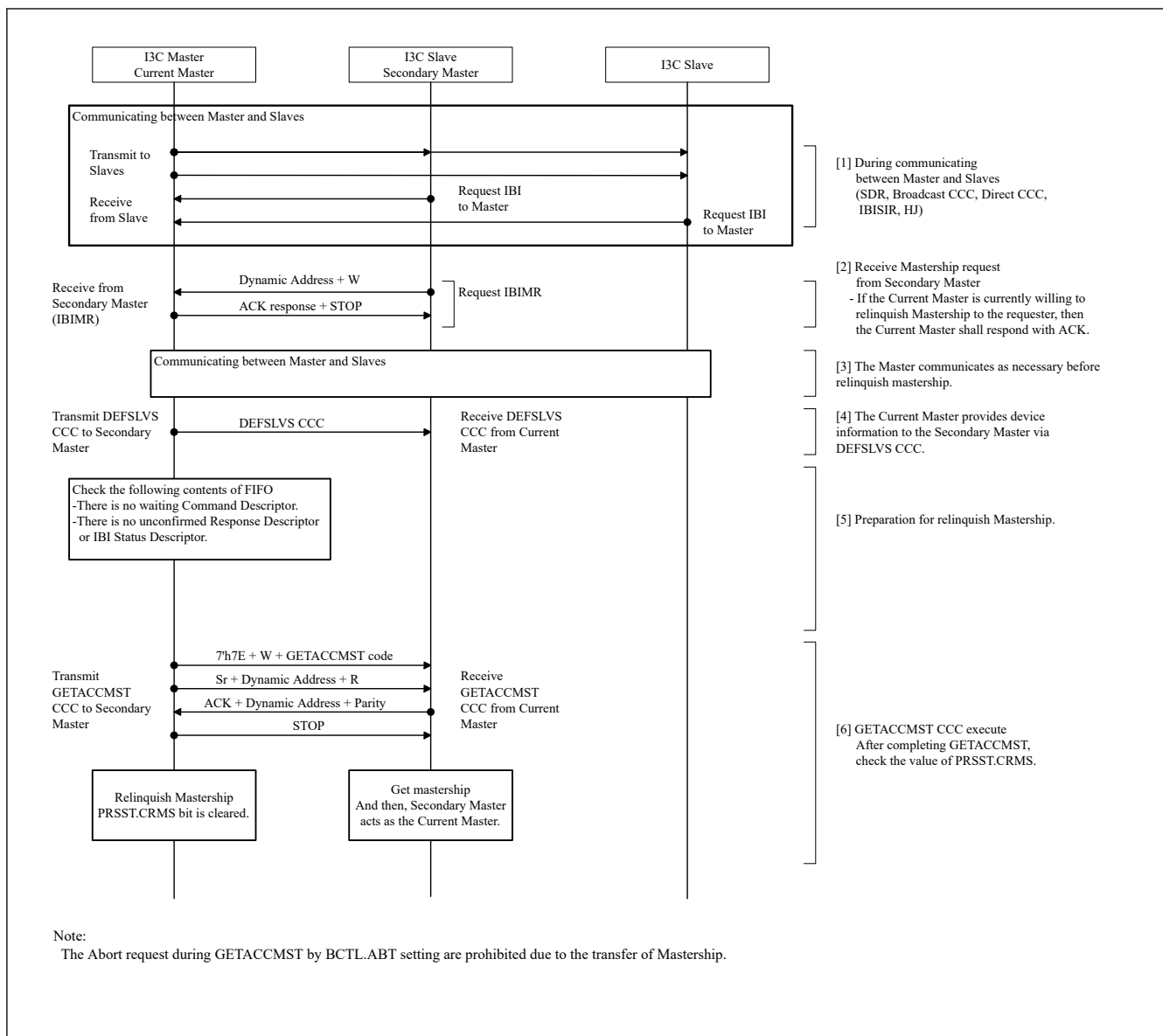


Figure 40.36 I3C Master Mastership processing flow

40.3.2.1.2 Slave Mode Operation

(1) I²C Slave Operation

(a) Data Write Transfer (Single Buffer transfer)

In slave receive operation, the master device outputs the SCL clock and transmit data, and I3C returns acknowledgments as a slave device.

Figure 40.153 shows an example of usage of slave reception and Figure 40.37 and Figure 40.38 show the timing of operations in slave reception.

The following describes the procedure and operations for slave reception.

1. Initial settings. For details, see [section 40.3.3.1. Initial Setting Flow](#). After initial settings, I3C will stay in the standby state until it receives a slave address that it matches.
2. After receiving a matching slave address, I3C sets one of the corresponding bits SVST.HOAF, GCAF, and SVAF[n] (n = 0 to 2) to 1 on the rising edge of the ninth cycle of SCL clock (the clock signal) and outputs the acknowledge bit (ACK) on the ninth cycle of SCL clock. If the value of the R/W# bit that was also received at this time is 0, I3C continues to place itself in slave receive mode and sets the NTST.RDBFF0 flag to 1.

3. After the BST.SPCNDDF flag is confirmed to be 0 and the NTST.RDBFF0 flag to be 1, dummy read the NTDTBP0 register (the dummy value consists of the slave address and R/W# bit when the 7-bit address format is selected, or the lower 8 bits when the 10-bit address format is selected).
4. When the NTDTBP0 register is read, I3C automatically sets the NTST.RDBFF0 flag to 0. If reading of the NTDTBP0 register is delayed and a next byte is received while the RDBFF0 flag is still set to 1, I3C holds the I3C_SCL line low from one SCL cycle before the timing with which RDBFF0 should be set. In this case, reading the NTDTBP0 register releases the I3C_SCL line from being held at the low level. When the BST.SPCNDDF flag = 1 and the NTST.RDBFF0 flag is also 1, read the NTDTBP0 register until all the data is completely received.
5. Upon detecting the STOP condition, I3C automatically clears bits SVST.HOAF, GCAF, and SVAF[n] (n = 0 to 2) to 0.
6. After checking that the BST.SPCNDDF flag = 1, set the BST.SPCNDDF flag to 0 for the next transfer operation.

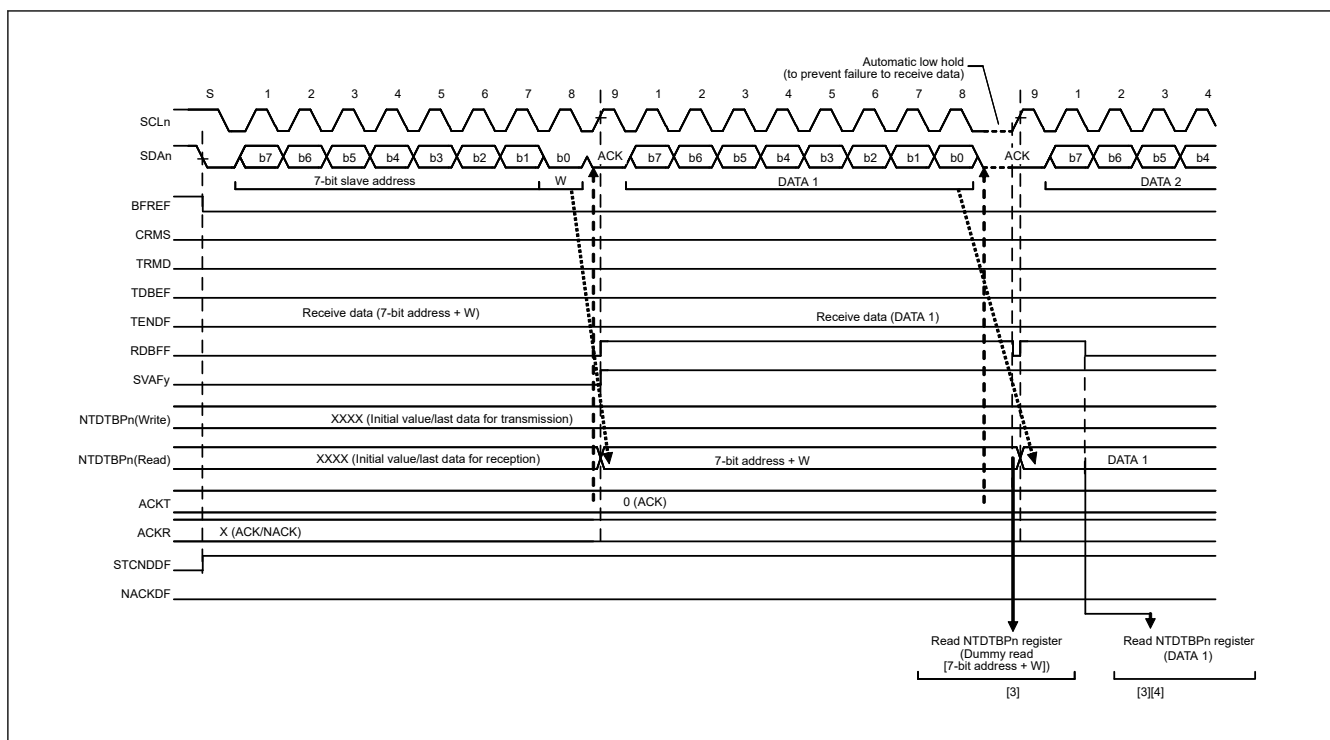


Figure 40.37 Slave receive operation timing (1) (7-bit address format, when ACKTWE = 0)

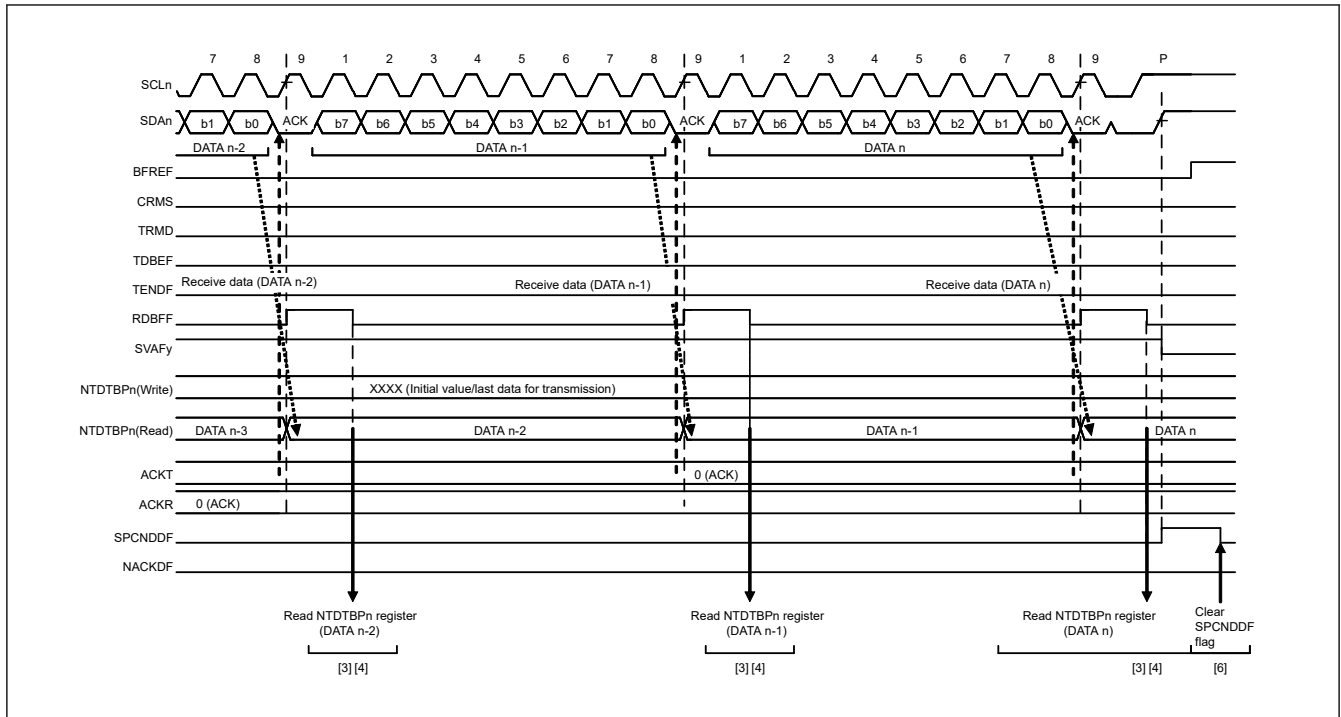


Figure 40.38 Slave receive operation timing (2) (when ACKTWE = 0)

(b) Data Read Transfer (Single Buffer transfer)

In slave transmit operation, the master device outputs the SCL clock, I3C transmits data as a slave device, and the master device returns acknowledgments.

Figure 40.152 shows an example of usage of slave transmission and Figure 40.39 and Figure 40.40 show the timing of operations in slave transmission.

The following describes the procedure and operations for slave transmission.

1. Initial settings. For details, see [section 40.3.3.1. Initial Setting Flow](#).
After initial settings, I3C will stay in the standby state until it receives a slave address that it matches.
2. After receiving a matching slave address, I3C sets one of the corresponding bits SVST.HOAF, GCAF, and SVAF[n] ($n = 0$ to 2) to 1 on the rising edge of the ninth cycle of SCL clock (the clock signal) and outputs the acknowledge bit (ACK) on the ninth cycle of SCL clock. If the value of the R/W# bit that was also received at this time is 1, I3C automatically places itself in slave transmit mode by setting both the PRSST.TRMD bit and the NTST.TDBEF0 flag to 1.
3. After the NTST.TDBEF0 flag is confirmed to be 1, write the data for transmission to the NTDTBP0 register.
At this time, if I3C does not receive acknowledge from the master device (receives a NACK signal) while the BSTE.NACKDE bit = 1, I3C aborts transfer of the next data.
4. Wait until the following (a) or (b) condition.
 - (a) The BST.NACKDF flag is set to 1.
 - (b) The BST.TENDF flag is set to 1 while the NTST.TDBEF0 flag = 1, after the last byte for transmission is written to the NTDTBP0 register.
5. When the BST.NACKDF flag or the BST.TENDF flag = 1, dummy read the NTDTBP0 register to complete the processing. This releases the I3C_SCL line.
6. Upon detecting the STOP condition, I3C automatically sets bits SVST.HOAF, GCAF, and SVAF[n] ($n = 0$ to 2), flags NTST.TDBEF0 and BST.TENDF, and the PRSST.TRMD bit to 0, and enters slave receive mode.
7. After checking that the BST.SPCNDDF flag = 1, set the BST.NACKDF and SPCNDDF flags to 0 for the next transfer operation.

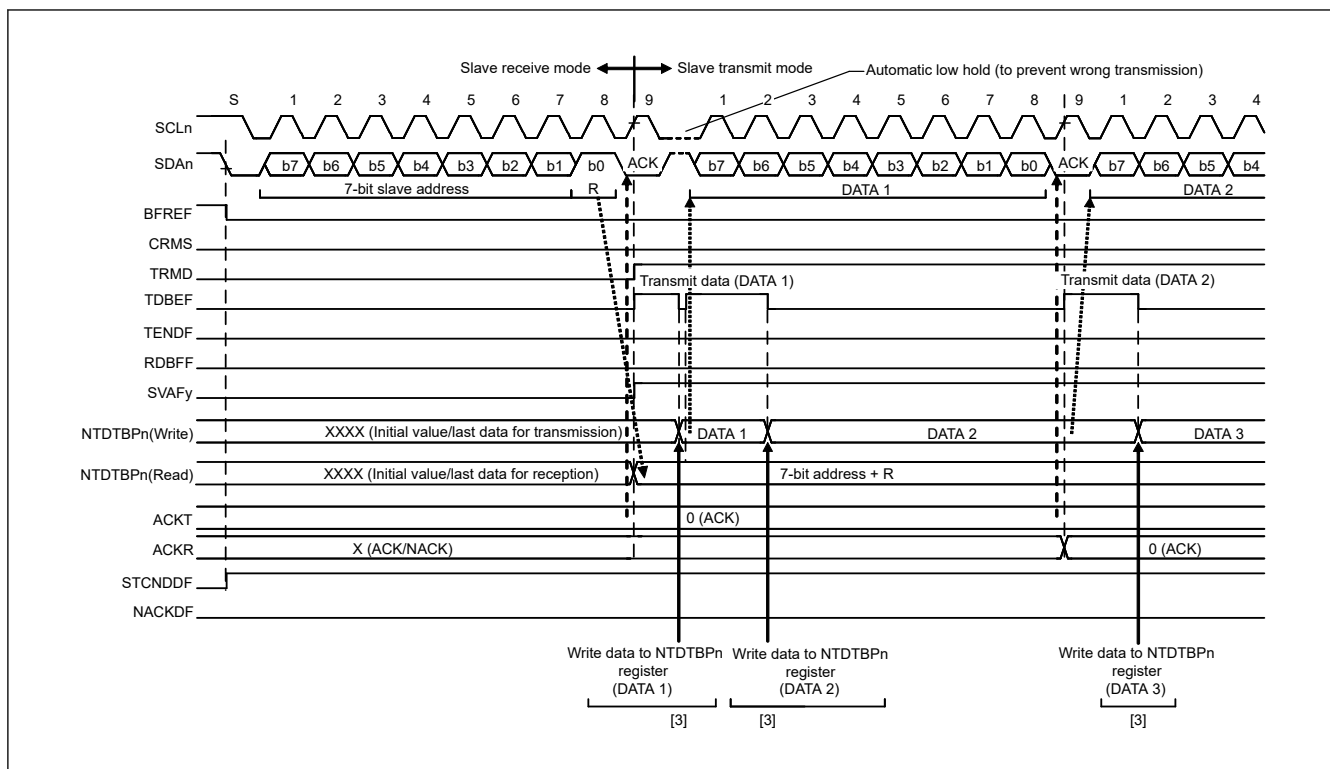


Figure 40.39 Slave transmit operation timing (1) (7-bit address format)

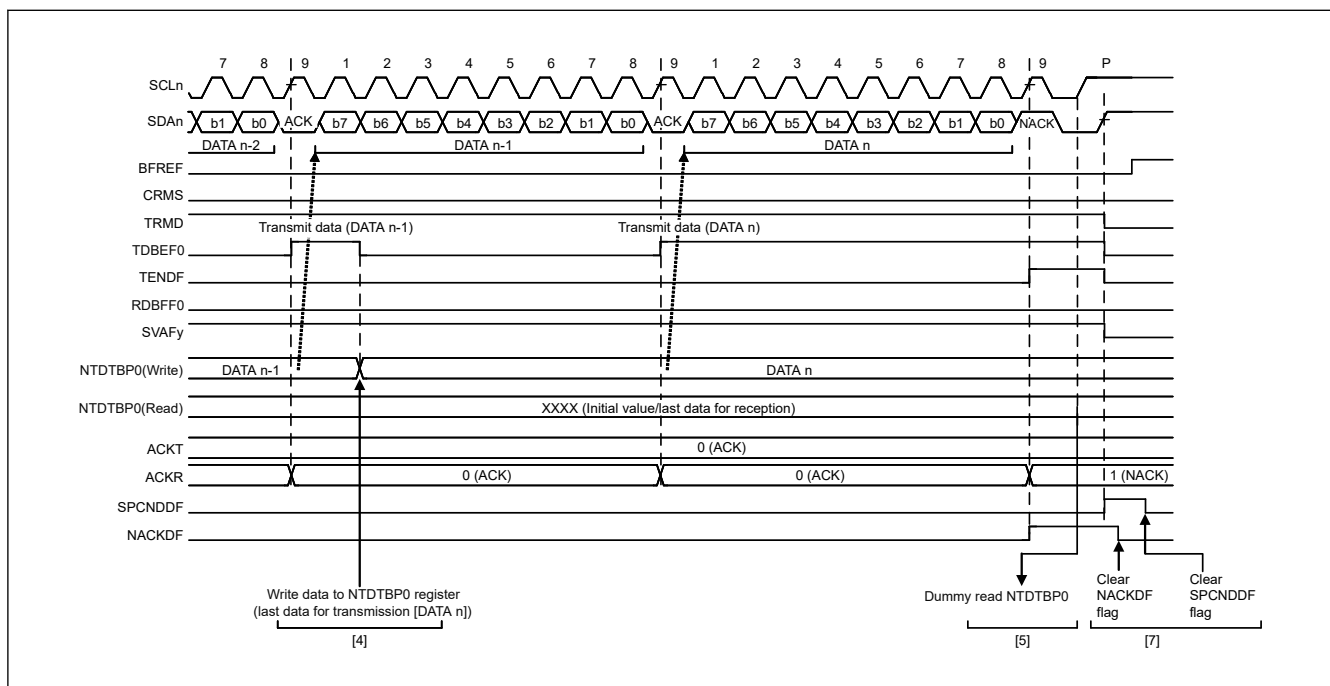


Figure 40.40 Slave transmit operation timing (2)

(2) I3C Slave Operation

(a) Dynamic Address Assign Procedure

After initializing I3C, the I3C master first performs Dynamic Address Assign Procedure.

The operation of R-I3 during the Dynamic Address Assign Procedure by ENTDAACCC is described below.

1. Initial setting (For details, see [section 40.3.3.1.2. I3C Initial Setting Flow](#))

- When ENTDAACCC is received, I3C transmits Provisional ID (SDCTPIDH[31:0], SDCTPIDL[15:0]), BCR (SVDCT.TBCRN), DCR (SVDCT.TDCR[7:0]) until a dynamic address is assigned. (For details, see "In case of Broadcast CCC (ENTDAA)" of (6) CCC detection function [I3C mode].)
- When ENTDAACCC is completed and a STOP condition is detected, Receive Status Descriptor is stored in Receive Status Queue.
- Read Receive Status Descriptor via NRSQP register and check the status.
- Read the data for the Data Length indicated by the DATA_LENGTH[15:0] bits of the Receive Status Descriptor from the Receive Data Buffer via the NTDTBP0 register.

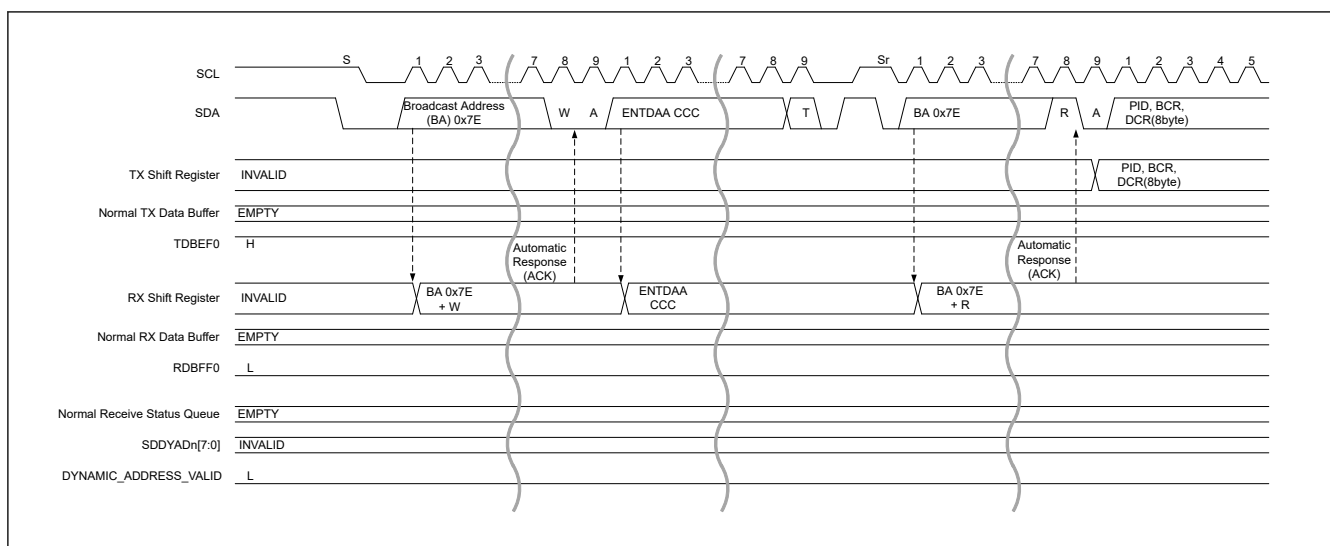


Figure 40.41 Dynamic address assign procedure (ENTDAA CCC) timing (1/3)

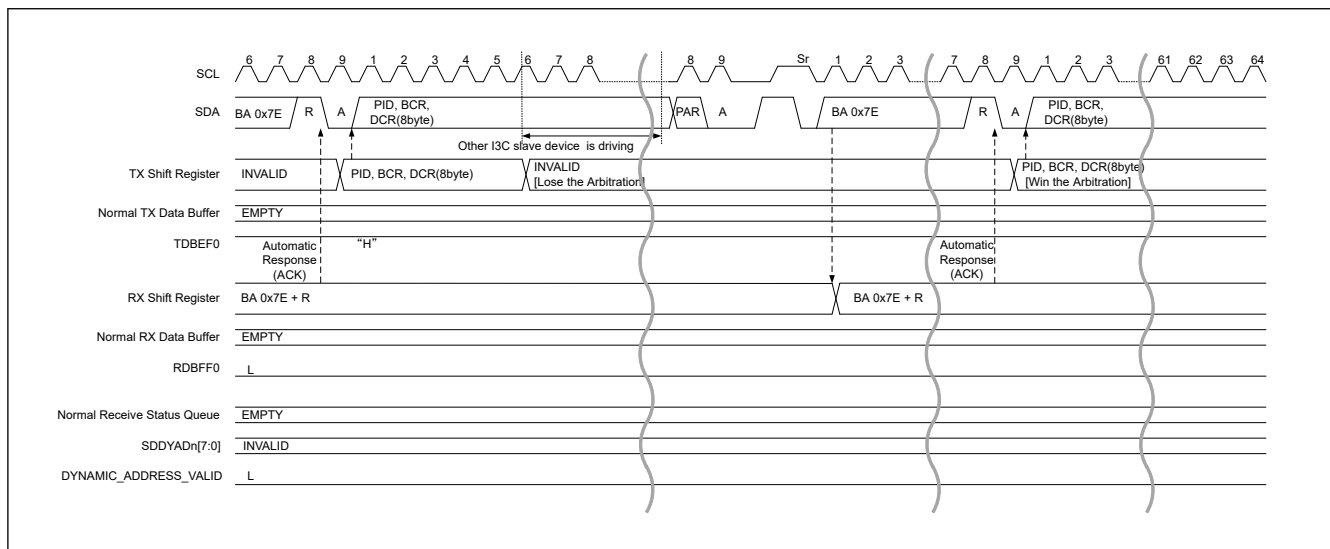


Figure 40.42 Dynamic address assign procedure (ENTDAA CCC) timing (2/3)

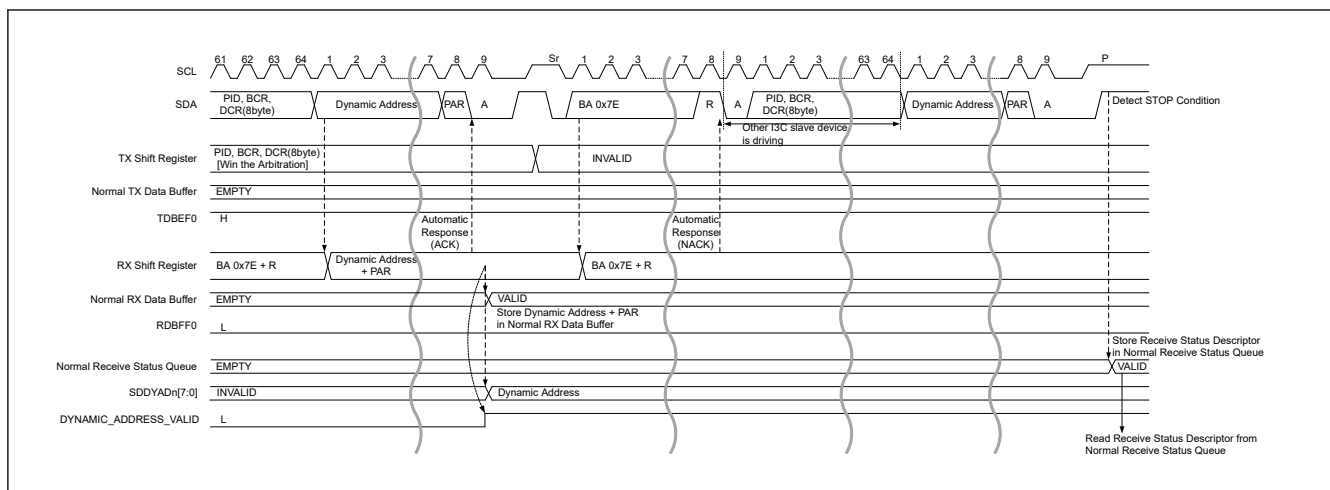


Figure 40.43 Dynamic address assign procedure (ENTDAA CCC) timing (3/3)

When communicating with a Static Address until the Dynamic Address is assigned from the I3C Master, by setting to the DVSTAD[6:0] bit of DAT (SDATBASn register), the SSTADV bit of the SVDVADn register is set to 1 and the Static Address Will be effective.

If the I3C Slave has a Static Address and the I3C Master executes the Dynamic Address Assign Procedure, it is possible to assign a Dynamic Address with SETDASA CCC.

The operation of I3C during SETDASA CCC Dynamic Address Assign Procedure is described below.

1. Initial setting (For details, see [section 40.3.3.1.2. I3C Initial Setting Flow](#))
2. When SETDASA CCC which agrees with its own Static Address is received, the SDDYAD [7:0] bit of DAT (SDATBAS0 register) is renewed and SDYADV bit of SVDVAD0 register is set in 1. (For details, see "In case of Direct Write CCC" of [\(6\) CCC detection function \[I3C mode\]](#).)
3. When SETDASA CCC is completed and a STOP condition is detected, Receive Status Descriptor is stored in Receive Status Queue.
4. Read Receive Status Descriptor via NRSQP register and check the status.

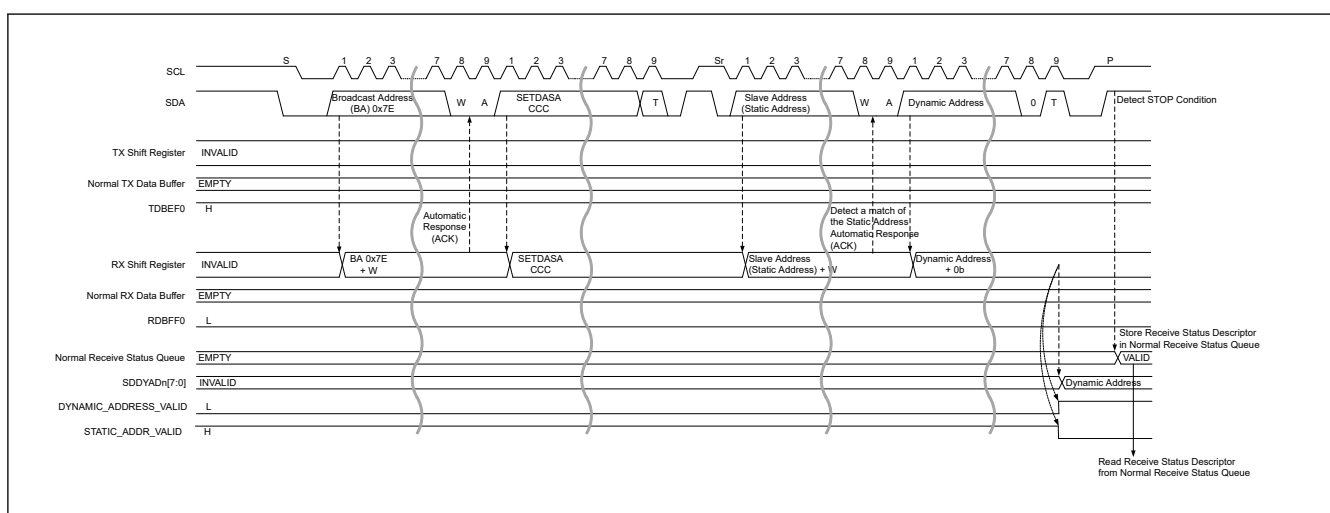


Figure 40.44 Dynamic address assign procedure (SETDASA CCC) timing

(b) SDR Data Write Transfer

1. When Transaction is issued from the I3C Master, it compares the Slave Address of Address Header with its own Slave Address, and if it matches, I3C responds with ACK.
When a Transaction is received, if the Receive Data Buffer is full, the I3C Slave will respond with NACK in the Address Header.
In preparation for retrying the I3C Master, read the data from the Receive Data Buffer via the NTDTBPn register, and empty the Receive Data Buffer.

2. Data received from I3C Master is stored in the Receive Data Buffer.
3. With the RDBFF0 = 1 interrupt, the received data is read from the Receive Data Buffer via the NTDTBPn register.
4. When Repeated START condition or STOP condition is detected, the Receive Status Descriptor is stored in the Receive Status Queue.
5. Read Receive Status Descriptor via NRSQP register and check the status.

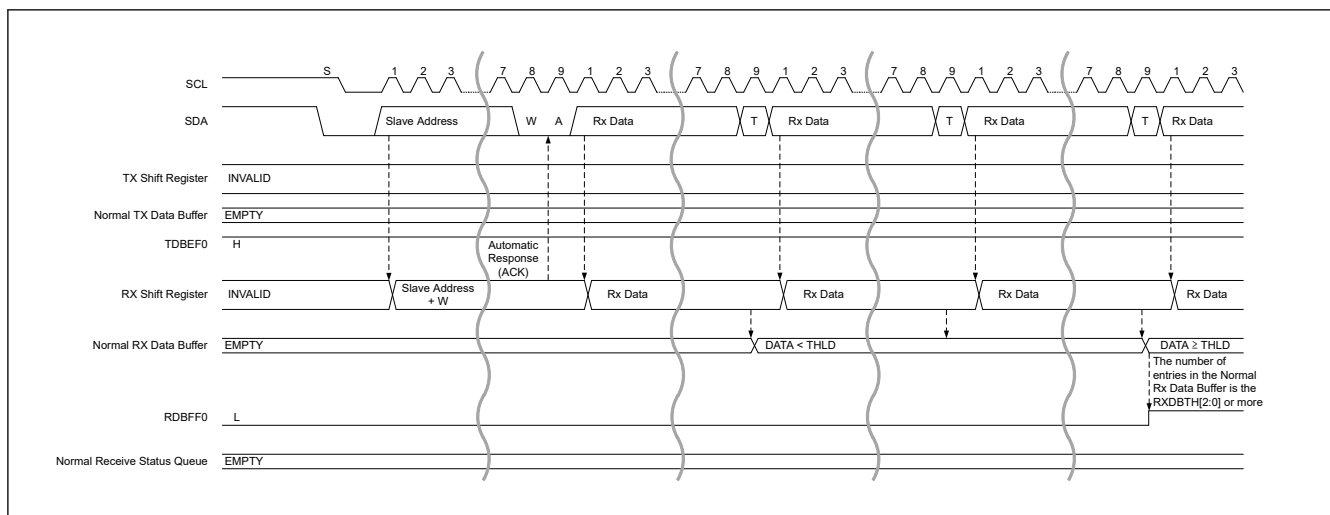


Figure 40.45 SDR data write transfer timing (1/2)

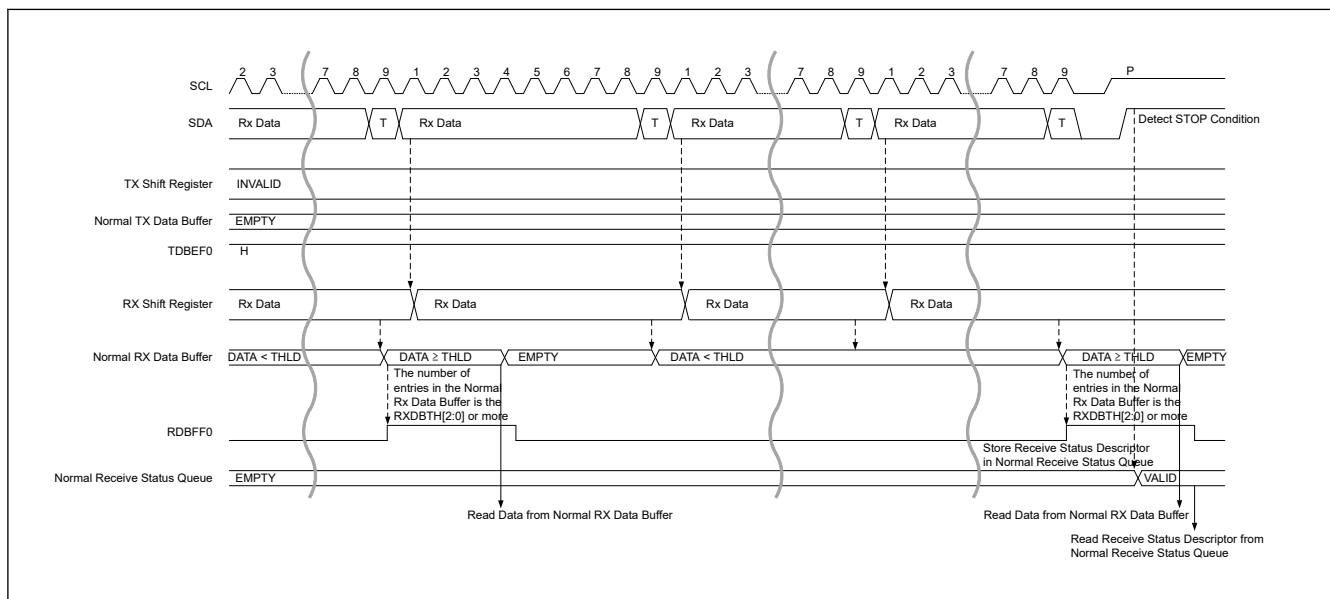


Figure 40.46 SDR data write transfer timing (2/2)

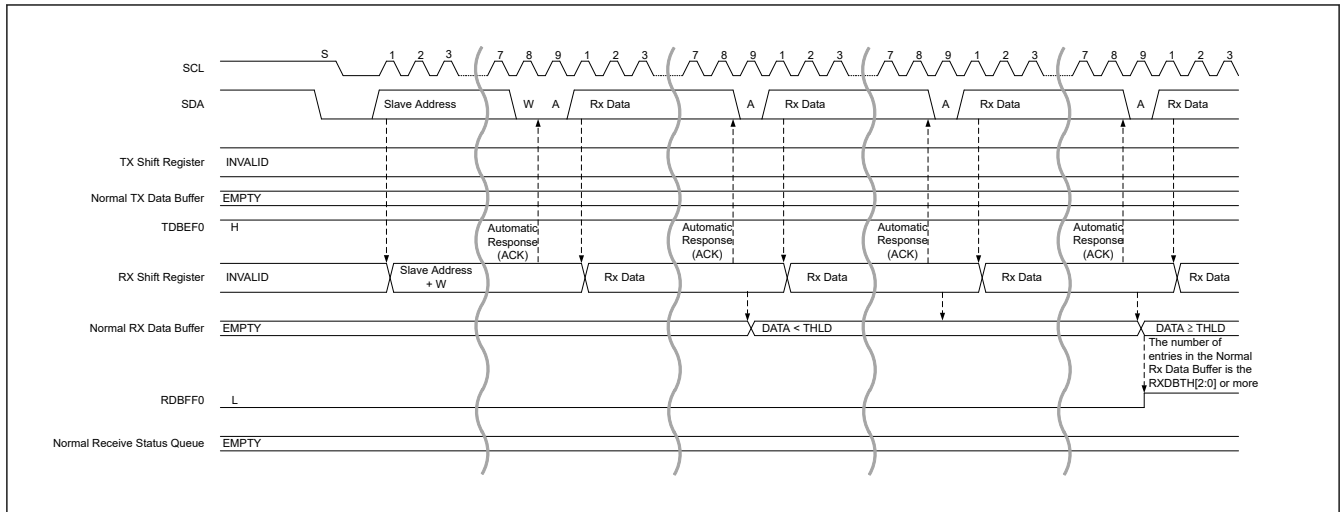


Figure 40.47 Legacy I²C message data write transfer timing (1/2)

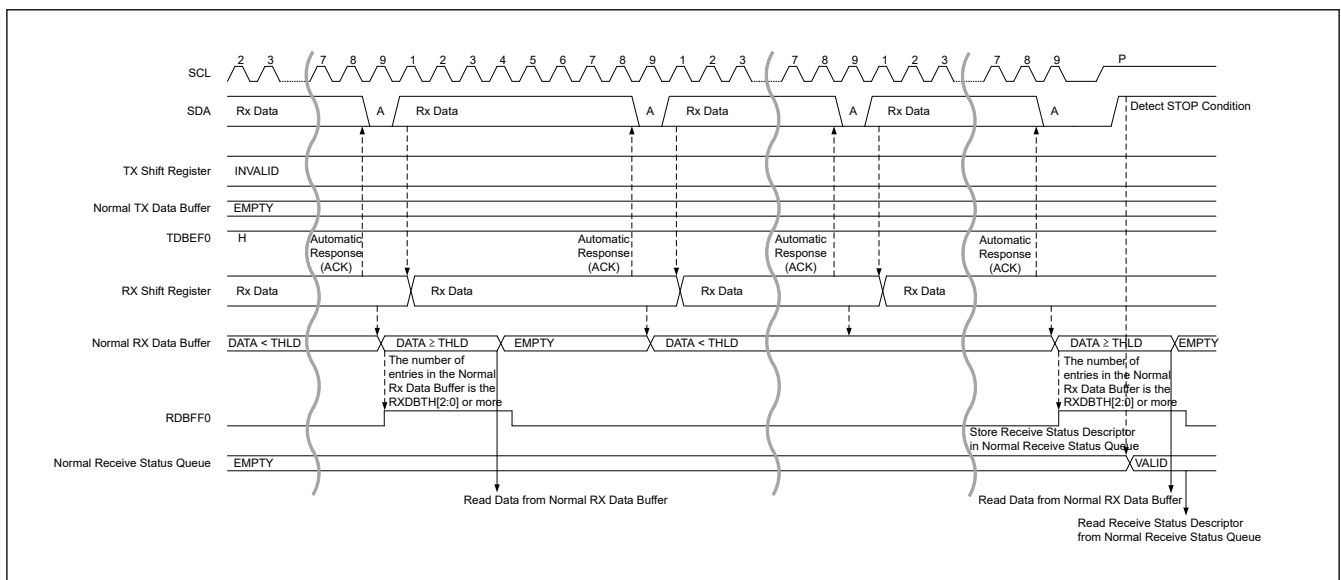


Figure 40.48 Legacy I²C message data write transfer timing (2/2)

(c) SDR Data Read Transfer

1. Write the data requested from the I3C Master to the Tx Data Buffer via the NTDTBPn register.
2. When Transaction is issued from the I3C Master, it compares the Slave Address of Address Header with its own Slave Address, and if it matches, I3C responds with ACK.
When a Transaction is received, if the Tx Data Buffer is EMPTY, I3C Slave responds with NACK with the Address Header.
In preparation for retrying the I3C Master, write data to the Tx Data Buffer via the NTDTBPn register.
3. Transmit the data stored in the Tx Data Buffer.
4. If data to be transmitted still remains, write the data to be transmitted with an interrupt by TDBEF0 = 1 to the Tx Data Buffer via the NTDTBPn register.
5. SDR:
When the transmission of the data stored in the Tx Data Buffer is completed, Low is output to the T-bit following Data, and it is notified to the I3C Master that it is the final data.
Legacy I²C Message:
When NACK is detected, data transmission is terminated.
6. When a Repeated START condition or STOP condition is detected, the Receive Status Descriptor is stored in the Receive Status Queue.

7. Read the Receive Status Descriptor via NRSQP and check the status.

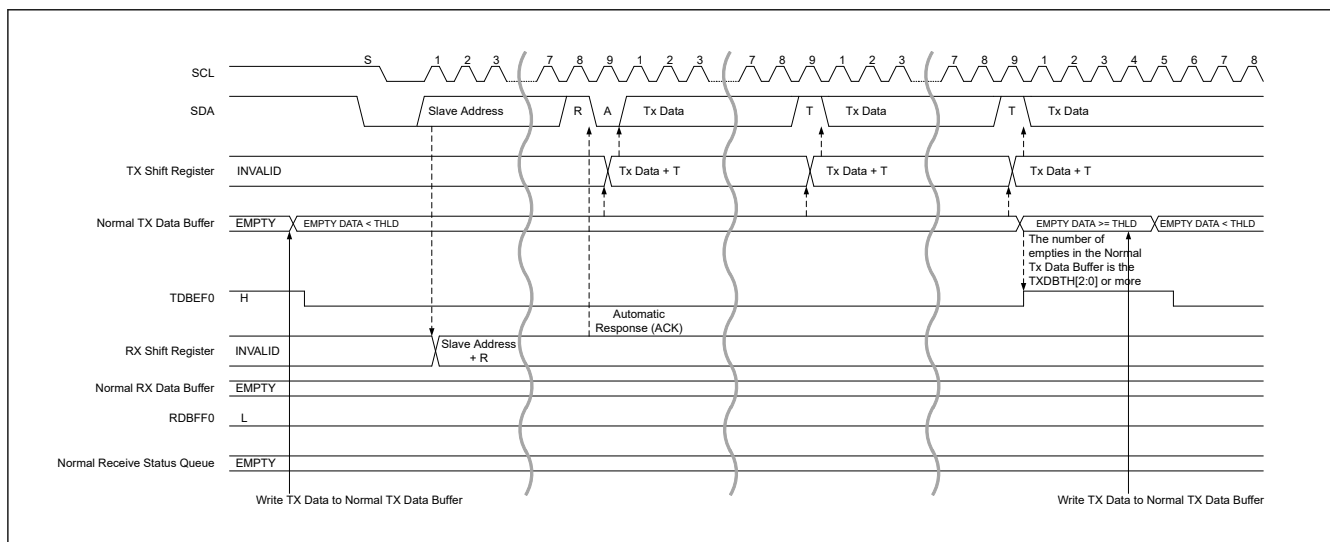


Figure 40.49 SDR data read transfer timing (1/2)

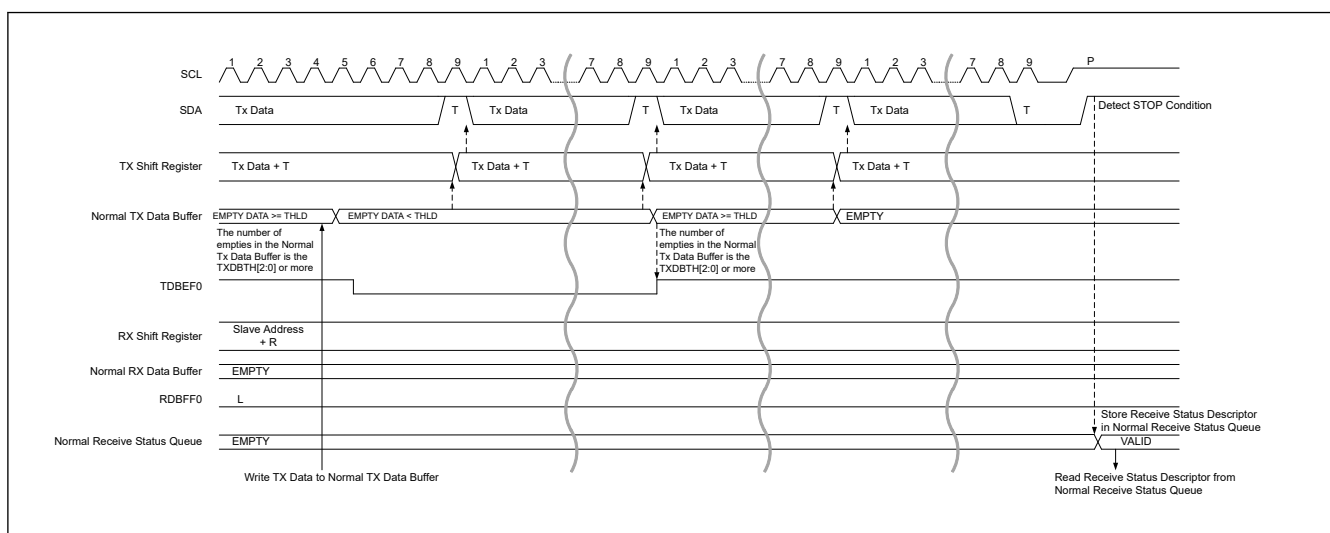


Figure 40.50 SDR data read transfer timing (2/2)

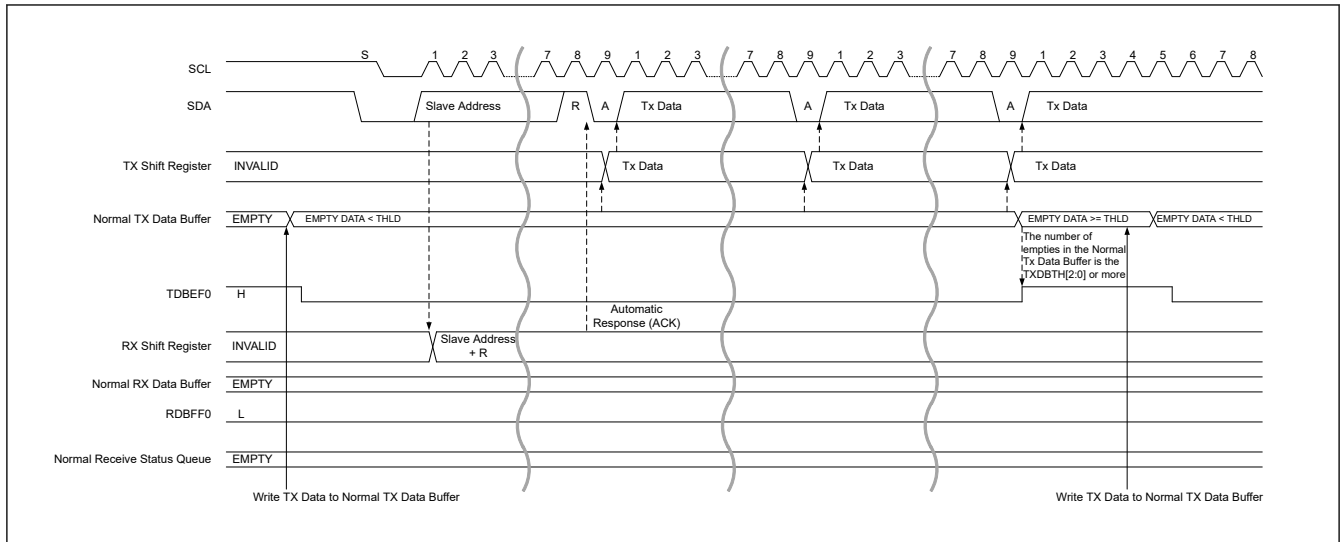


Figure 40.51 Legacy I²C message data read transfer timing (1/2)

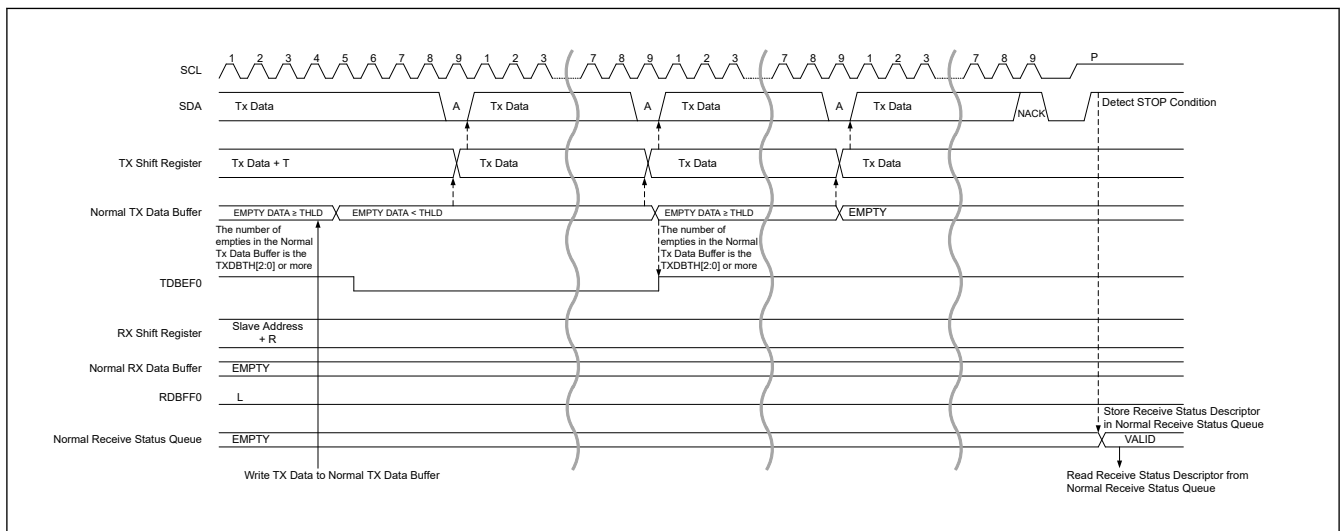


Figure 40.52 Legacy I²C message data read transfer timing (2/2)

(d) HDR Data Write Transfer

1. Upon receipt of ENTHDR * CCC from the I3C Master, it transits to HDR mode (For details, see "In case of Broadcast CCC (ENTHDR *)" of (6) CCC detection function [I3C mode]).
2. Compare the Slave Address of the HDR Command Word issued from the I3C Master with its own Slave Address, and if it matches, receive the following Data Word.
3. Data received from the I3C Master is stored in the Receive Data Buffer.
4. With the RDBFF0 = 1 interrupt, the received data is read from the Receive Data Buffer via the NTDTBPn register.
5. When detecting HDR Restart Pattern or HDR Exit Pattern + STOP condition, it stores the Receive Status Descriptor into the Receive Status Queue.
6. Read the Receive Status Descriptor via NRSQP and check the status.

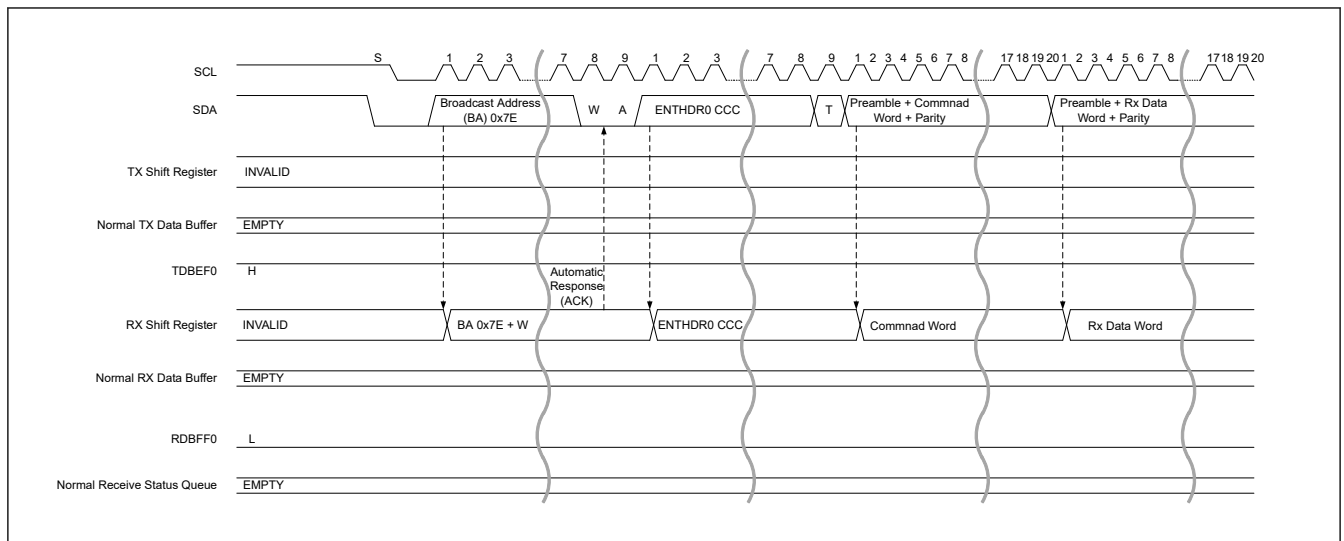


Figure 40.53 HDR data write transfer (HDR-DDR) timing (1/3)

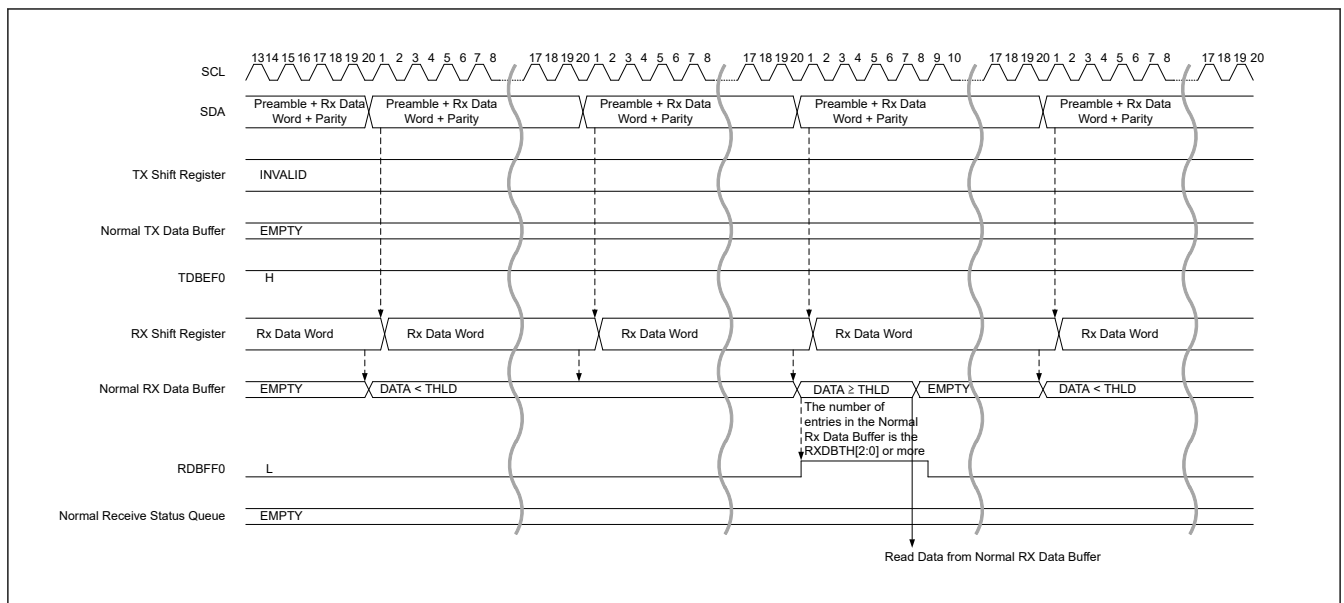


Figure 40.54 HDR data write transfer (HDR-DDR) timing (2/3)

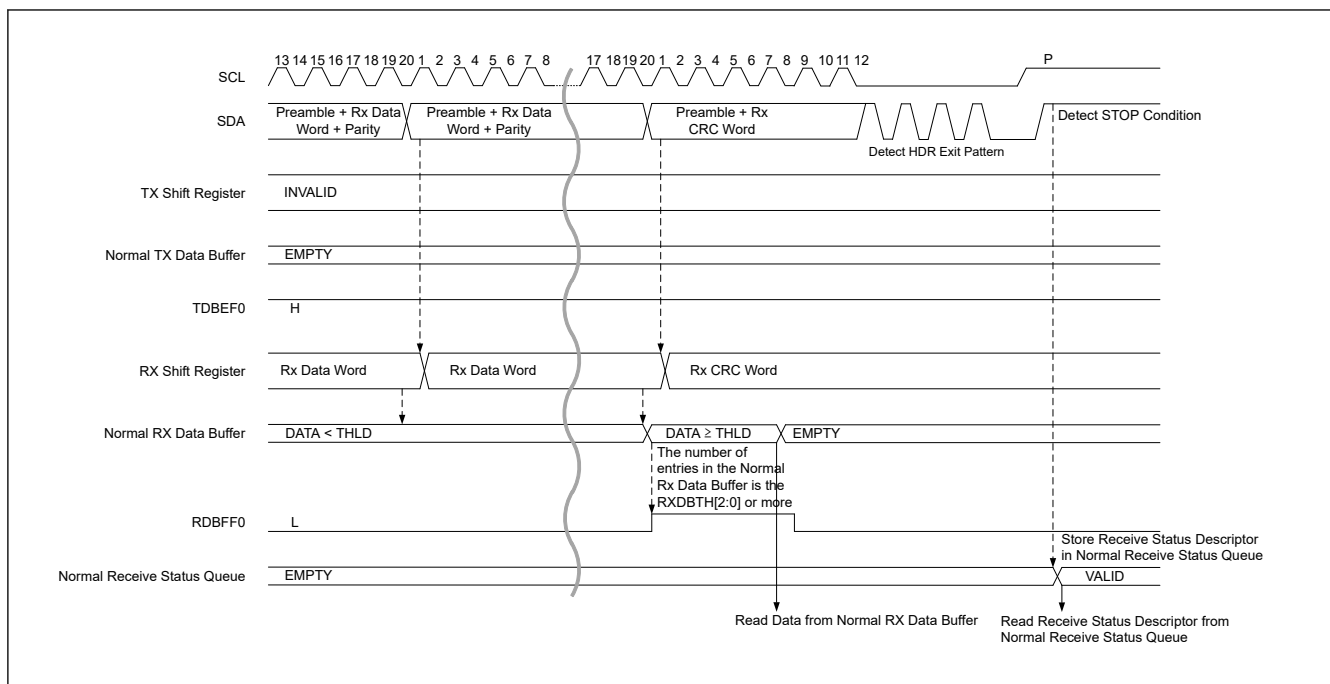


Figure 40.55 HDR data write transfer (HDR-DDR) timing (3/3)

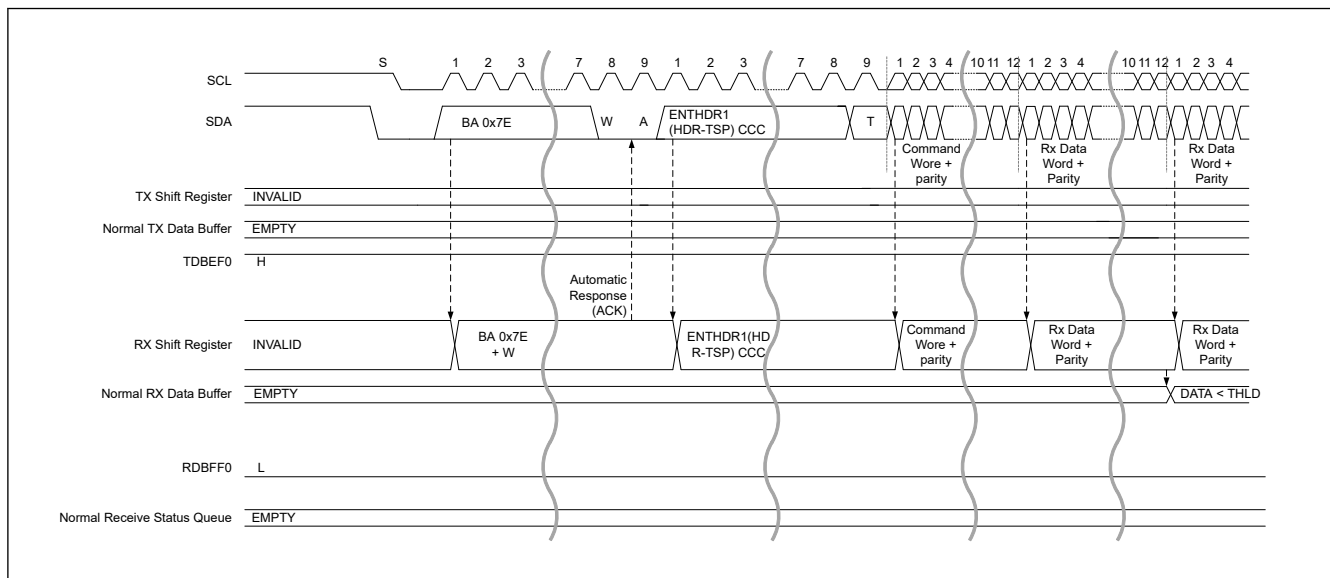


Figure 40.56 HDR data write transfer (HDR-TSP, TSL) timing (1/2)

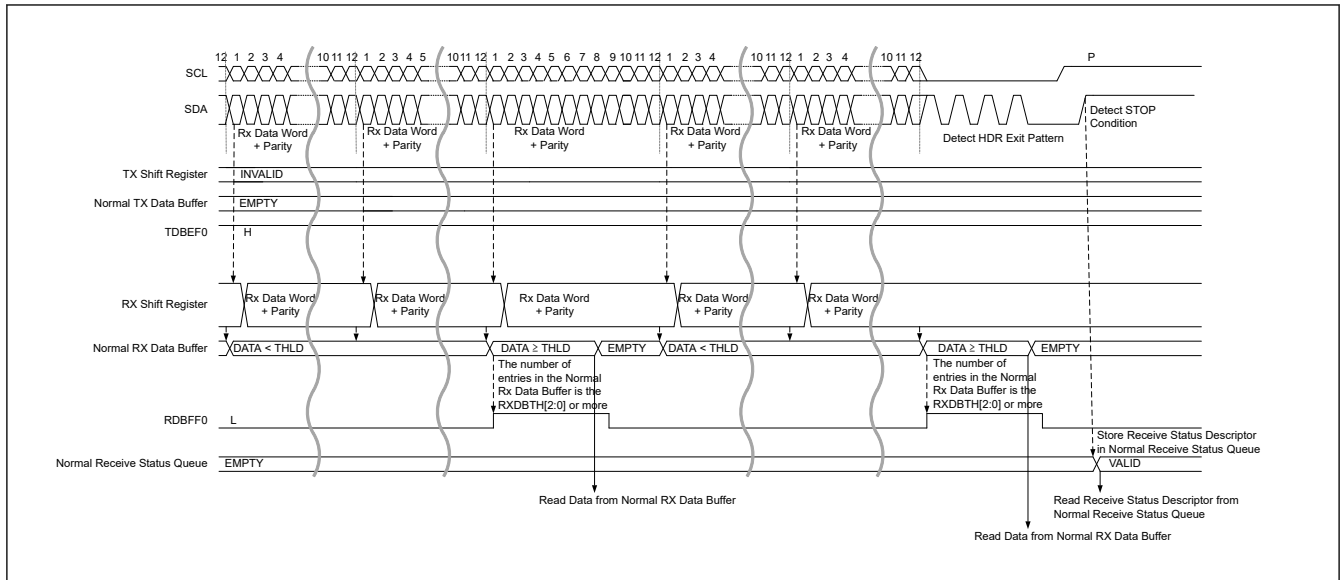


Figure 40.57 HDR data write transfer (HDR-TSP, TSL) timing (2/2)

(e) HDR Data Read Transfer

1. Write the data requested by the I3C Master to the Tx Data Buffer via the NTDTBPn register.
2. Upon receipt of ENTHDR * CCC from the I3C Master, it transits to HDR mode (For details, see "In case of Broadcast CCC (ENTHDR *)" of (6) CCC detection function [I3C mode]).
3. When comparing the Slave Address of the HDR Command Word issued from the I3C Master with its own Slave Address, if it matches, the Data stored in the Tx Data Buffer is transmitted.
If the Tx Data Buffer is EMPTY when the Slave Address of the HDR Command Word matches its own Slave Address, a NACK response will be sent.
NACK response :
 - HDR-DDR :
Send 11 followed by Command Word followed by Preamble.
 - HDR-Ternary :
Send the beginning of HDR Restart and HDR Exit Pattern (in English translation: "start of the HDR Restart Pattern and HDR Exit Pattern." In MIPI Spec)
4. If data to be transmitted still remains, Write the data to be transmitted with an interrupt by TDBEF0 = 1 to the Tx Data Buffer via the NTDTBPn register.
5. HDR-DDR:
When the transmission of the data stored in the Tx Data Buffer is completed, the CRC Word is sent continuously to notify the I3C Master that it is the final data.
HDR-Ternary:
When the transmission of the data stored in the Tx Data Buffer is completed, the beginning of the HDR Restart or HDR Exit Pattern is sent continuously to notify the I3C Master that it is the final data.
6. When detecting HDR Restart Pattern and HDR Exit Pattern + STOP condition, it stores the Receive Status Descriptor into the Receive Status Queue.
7. Read the Receive Status Descriptor via NRSQP and check the status.

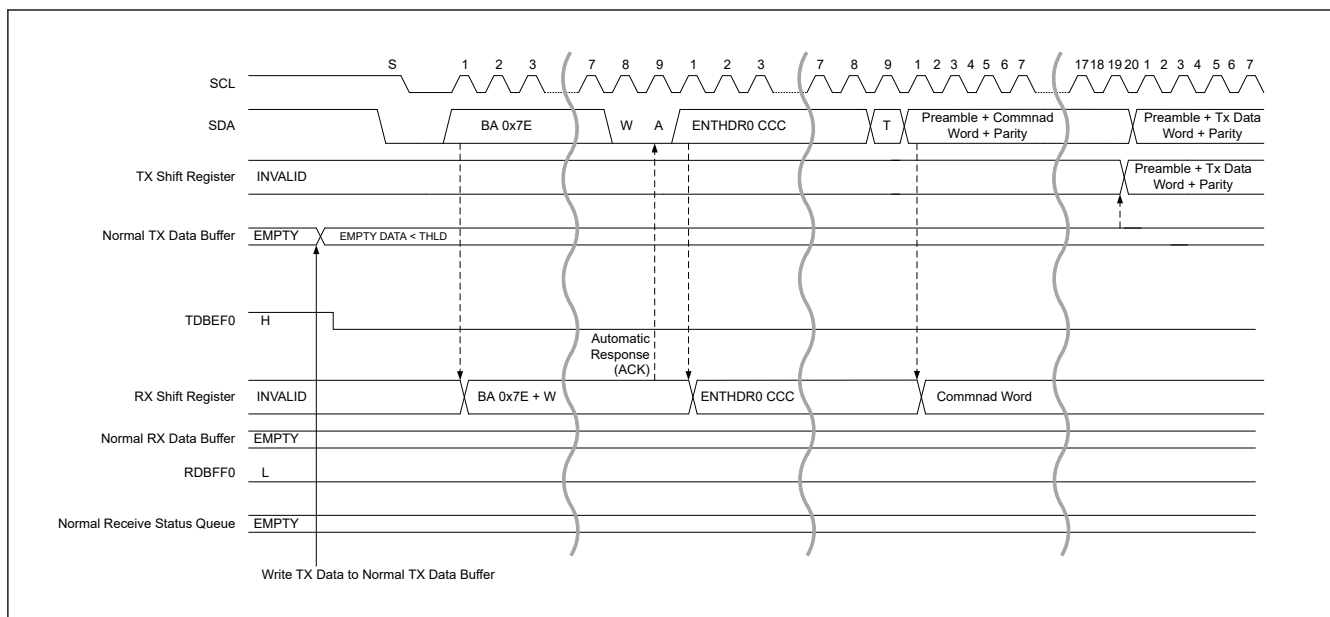


Figure 40.58 HDR data read transfer (HDR-DDR) timing (1/3)

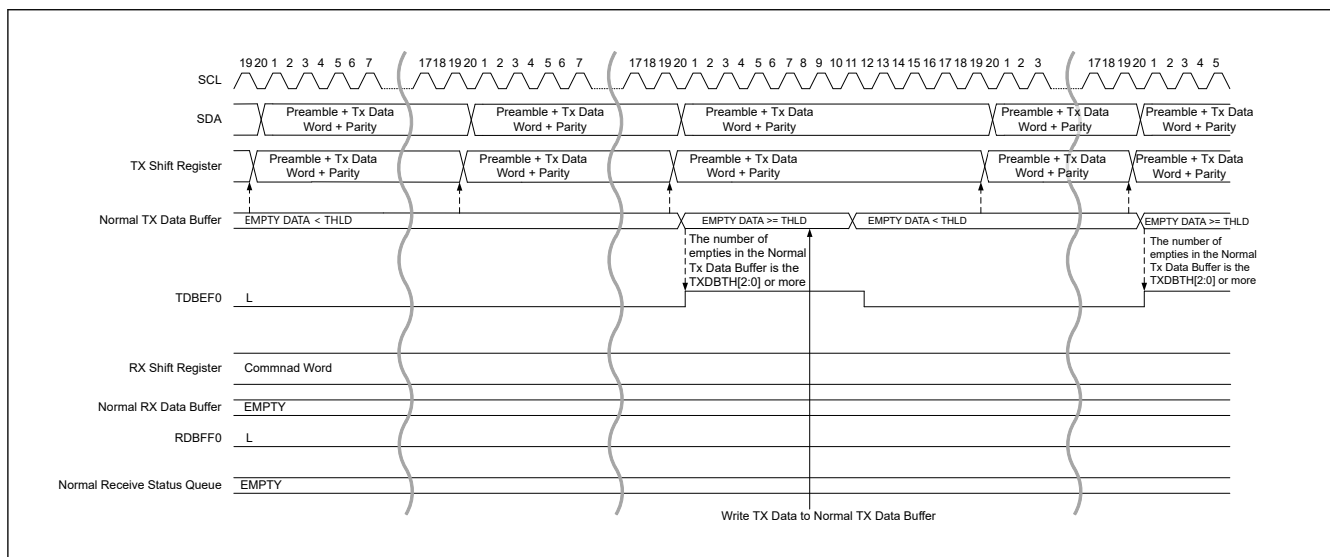


Figure 40.59 HDR data read transfer (HDR-DDR) timing (2/3)

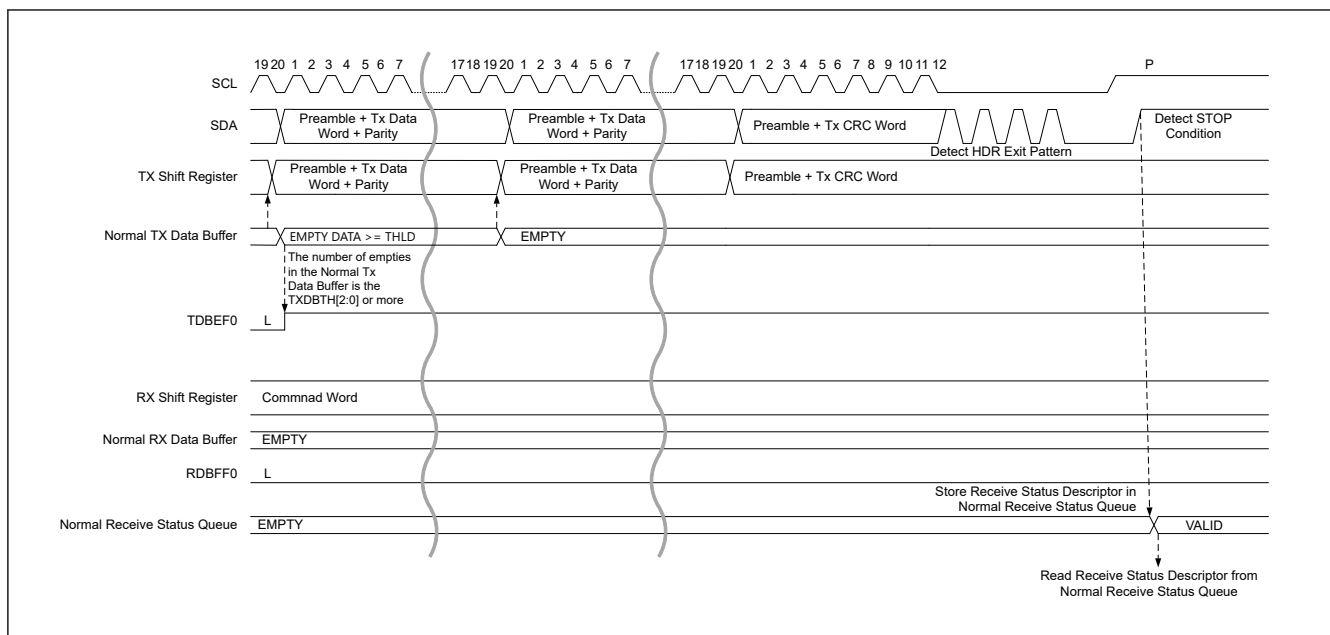


Figure 40.60 HDR data read transfer (HDR-DDR) timing (3/3)

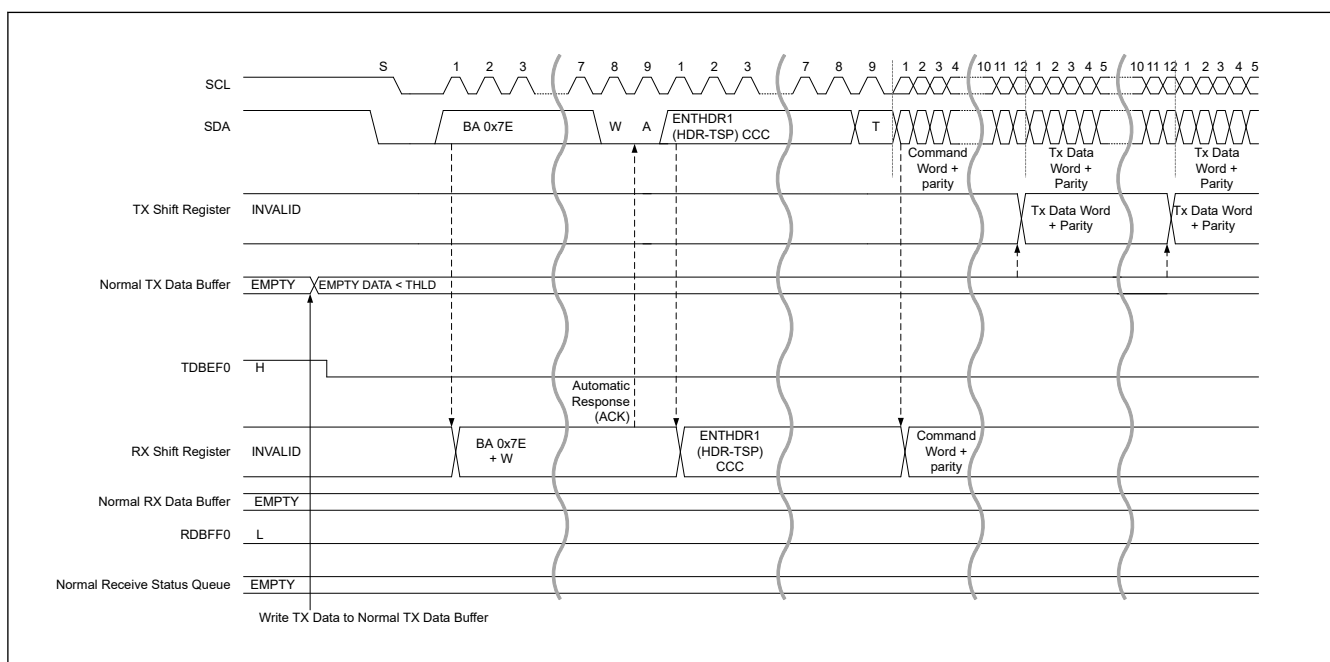


Figure 40.61 HDR data read transfer (HDR-TSP, TSL) timing (1/2)

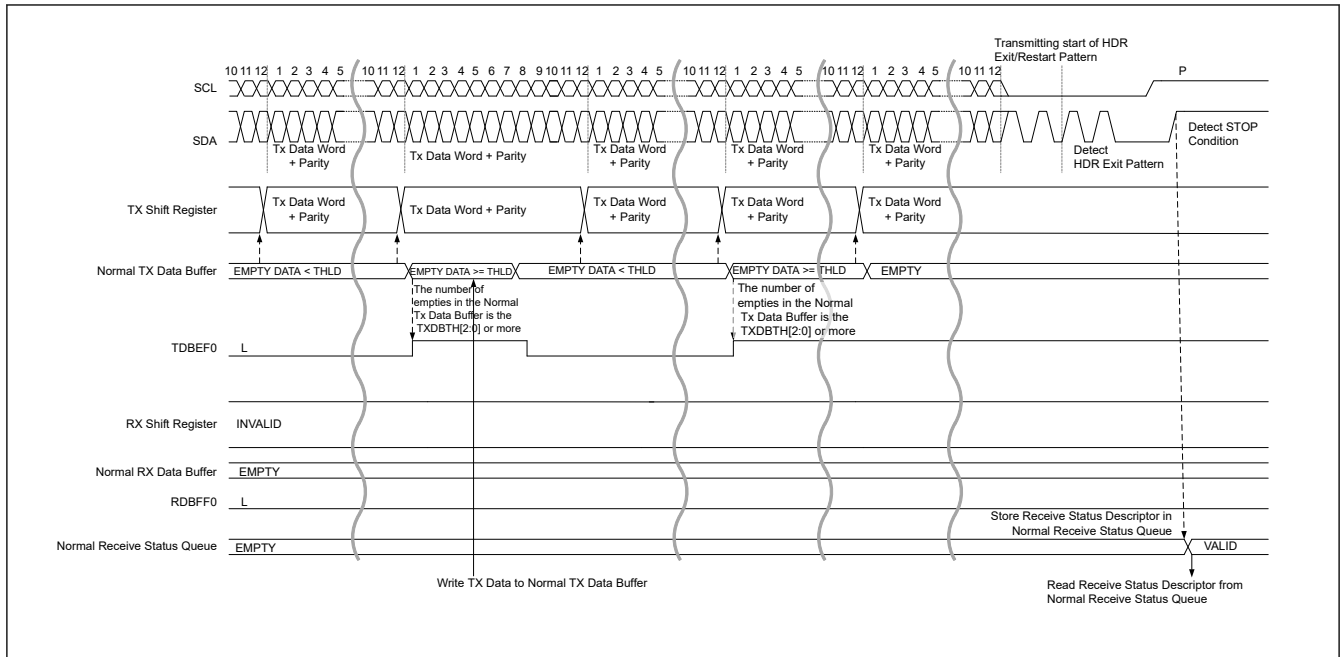


Figure 40.62 HDR data read transfer (HDR-TSP, TSL) timing (2/2)

(f) IBI Transfer

- When sending Slave Interrupt Request.
When transmitting IBI Data, write IBI Data to the IBI Data Buffer via the NIBIQP register.
- Write Command Descriptor (Immediate Transfer Command or Regular Transfer Command) to the Command Buffer for IBI Transfer via the NCMDQP register.
- When Command Descriptor is written to Command Buffer, IBI Transaction is issued under the following conditions.
 - When START condition is detected in Slave Interrupt Request or Mastership Request. (Does not apply a Repeated START condition)
 - If no START is forthcoming within the following Bus Condition, then this module issue a START Request by pulling the I3C_SDA line Low.
 - (a) Slave Interrupt Request, Mastership Request : Bus Available
- In Slave Address with RnW of the Address Header, if losing Arbitration by issuing a Transaction from I3C Master, stop issuing Transaction.
When detecting Repeated START condition or STOP condition, store the Response Descriptor into the Response Buffer.
- When sending Slave Interrupt Request:
 - When IBI data for transmission still remain, write IBI data with an interrupt by IBIQEFF = 1 to the IBI Data Buffer via the NIBIQP register.
 - When the transmission of IBI Data for the number of Data Length specified by the DATA_LENGTH[15:0] bits of the Command Descriptor is completed, output Low to the T-bit following IBI Data and notify the I3C Master that it is the final IBI Data.
- When detecting Repeated START condition or STOP condition, store the Response Descriptor into the Response Buffer.
- Read the Response Descriptor from the Response Buffer with the NRSPQP register and check the status. If NACK is responded, repeat steps 1 to 7.
- When sending Slave Interrupt Request:
Check that the value of the DATA_LENGTH[15:0] bit of the Response Descriptor is 0.

The Mastership processing flow is shown in [Figure 40.65](#).

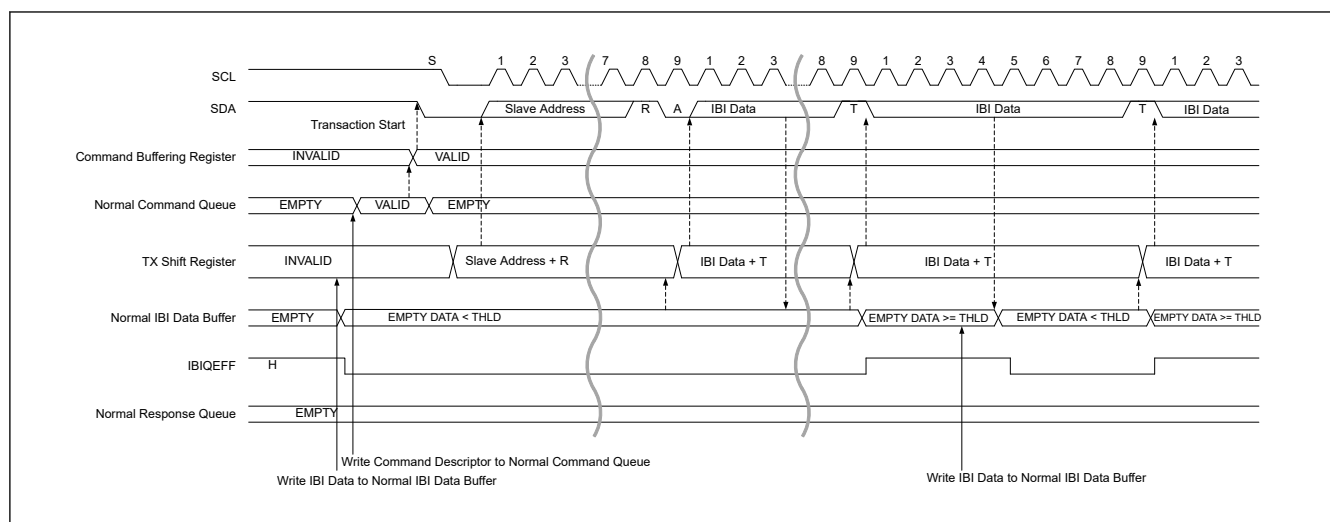


Figure 40.63 I3C slave IBI transfer timing (1/2)

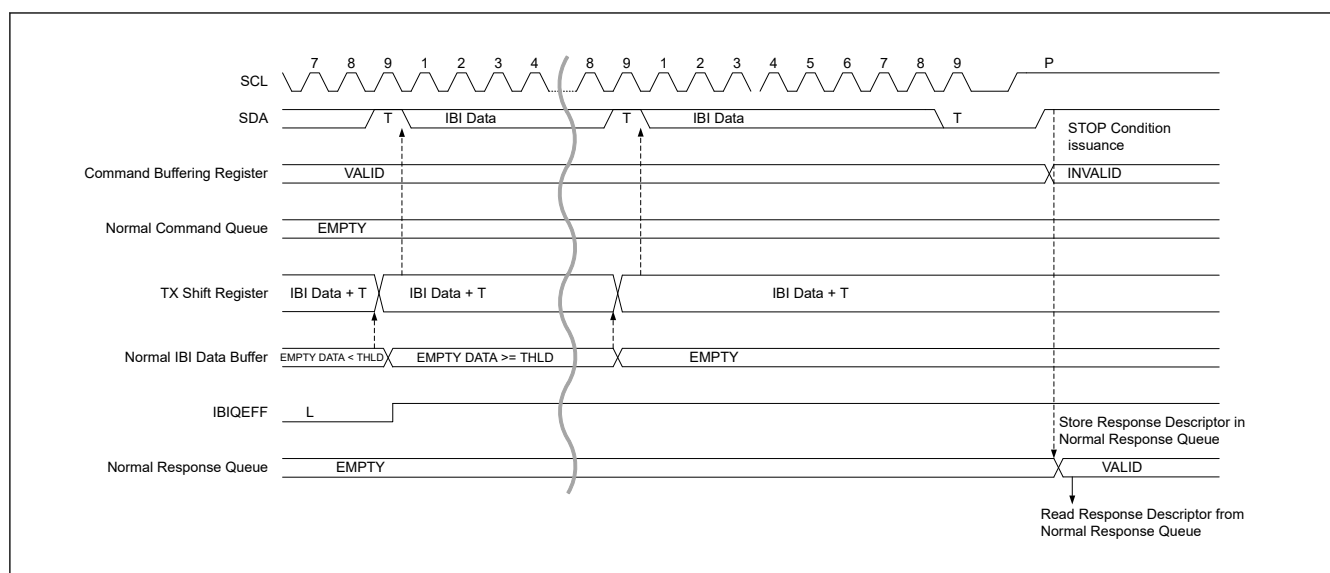


Figure 40.64 I3C slave IBI transfer timing (2/2)

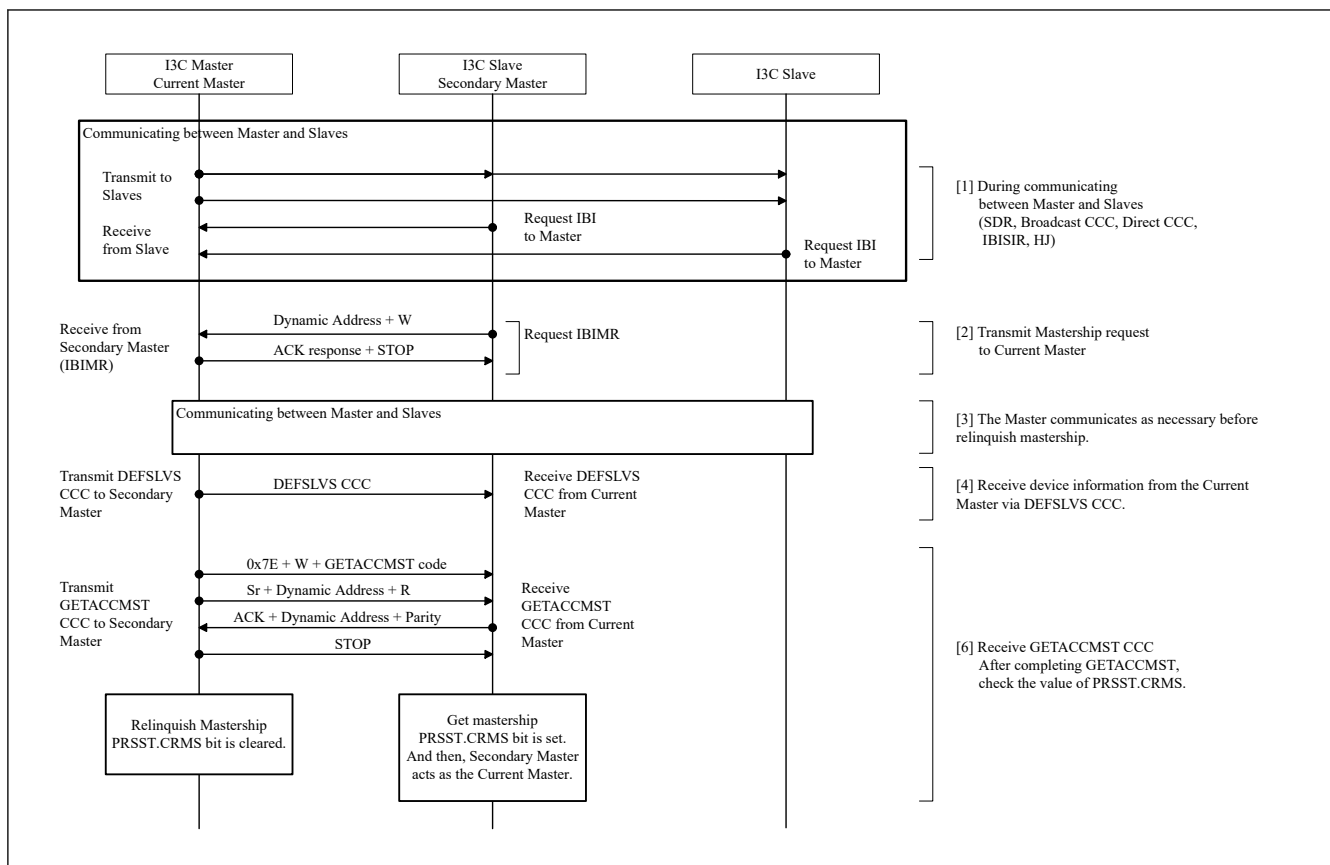


Figure 40.65 I3C Slave Mastership processing flow

40.3.2.2 Data Handler

The relationship between the transfer method and the queue is shown in [Table 40.10](#).

Table 40.10 Transfer method and queue

Protocol	Transfer method	Queue/Buffer	size	Master	Slave	Secondary Master
I ² C Mode	Single buffer transfer	Normal Tx Data	1 byte	✓	✓	—
		Normal Rx Data	1 byte	✓	✓	—
I3C Mode	Normal FIFO buffer transfer	Normal Command	4 QUEUES	✓	✓	✓
		Normal Response	4 QUEUES	✓	✓	✓
		Normal Tx Data	16 DWORDs	✓	✓	✓
		Normal Rx Data	16 DWORDs	✓	✓	✓
		Normal Receive Status	2 QUEUES	—	✓	✓
		Normal IBI Status	2 QUEUES	✓	—	✓
		Normal IBI Data	8 DWORDs	✓	✓	✓
	High Priority FIFO buffer transfer (in Master Mode only)	High Priority Command	2 QUEUES	✓	—	✓
		High Priority Response	2 QUEUES	✓	—	✓
		High Priority Tx Data	2 DWORDs	✓	—	✓
		High Priority Rx Data	2 DWORDs	✓	—	✓

40.3.2.2.1 Transfer Method in I²C Mode

(1) Single Buffer transfer

Each process (condition issue, data transfer, ACK / NACK response) is controlled by software.

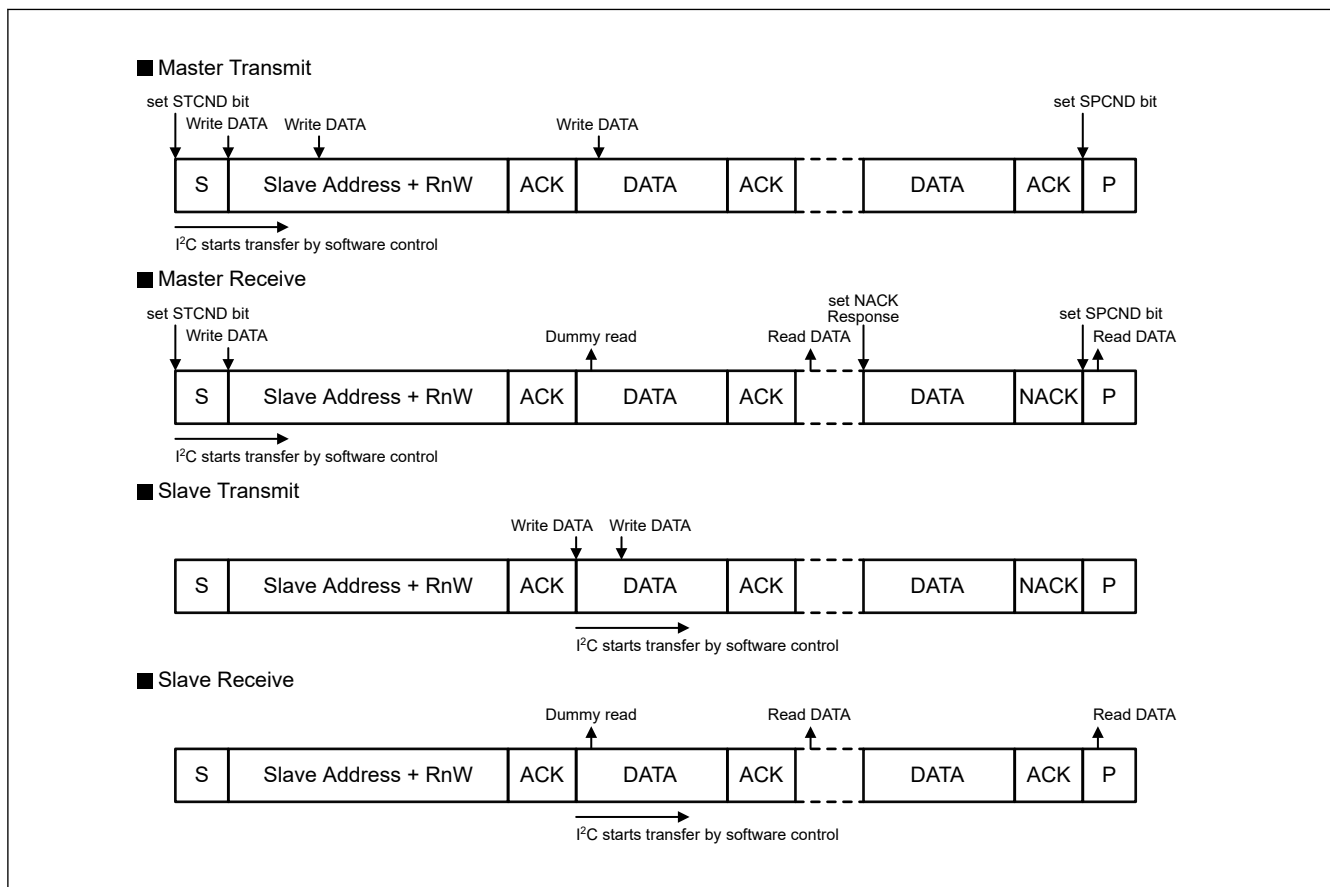


Figure 40.66 Data handler with single buffer transfer

40.3.2.2.2 Transfer Method in I3C Mode

(1) Normal FIFO Buffer Transfer

I3C autonomously starts transfer when data and command are written.

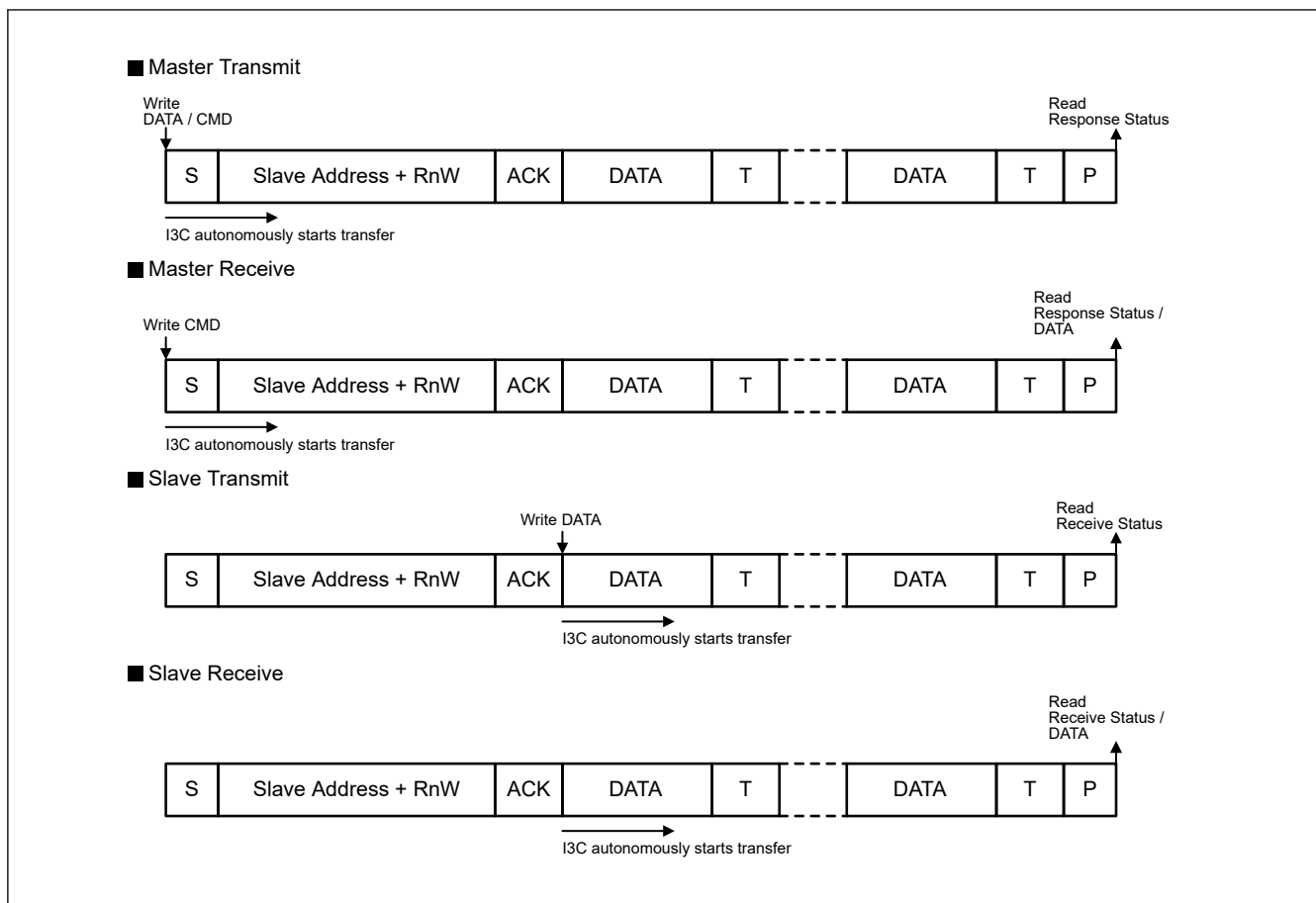


Figure 40.67 Data handler with normal FIFO buffer transfer

(2) High Priority FIFO Buffer Transfer

I3C handles the command of the High Priority FIFO buffer transfer higher priority than the command of the normal FIFO buffer transfer.

If data and commands are written to the High Priority FIFO buffer during normal FIFO buffer transfer, I3C waits for the STOP condition and then processes the command in the High Priority FIFO buffer.

After the command processing in the High Priority FIFO buffer is completed, if the command remains in the normal FIFO buffer, the I3C resumes processing the command in the normal FIFO buffer.

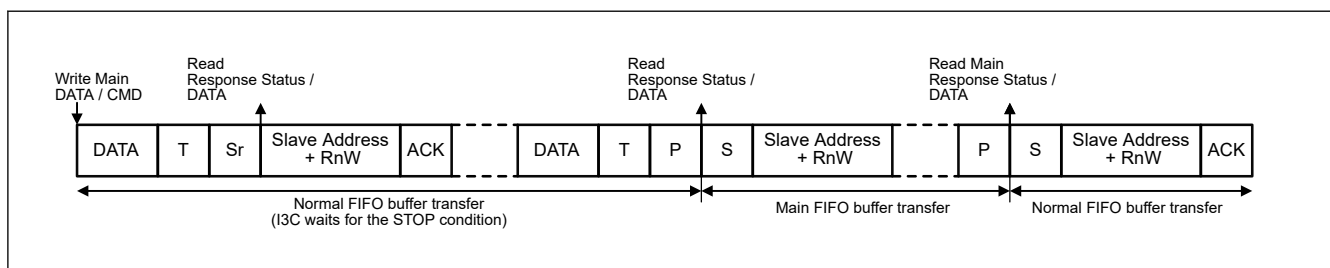


Figure 40.68 Data handler with high priority FIFO buffer transfer

40.3.2.3 I²C/I3C Protocol

40.3.2.3.1 Communication Protocol

(1) I²C Communication Data Format

The I²C bus format consists of 8-bit data and 1-bit acknowledge. The frame following a START condition or Repeated START condition is an address frame used to specify a slave device with which the master device communicates. The specified slave is valid until a new slave is specified or a STOP condition is issued.

Figure 40.69 shows the I²C bus format, and Figure 40.70 shows the I²C bus timing.

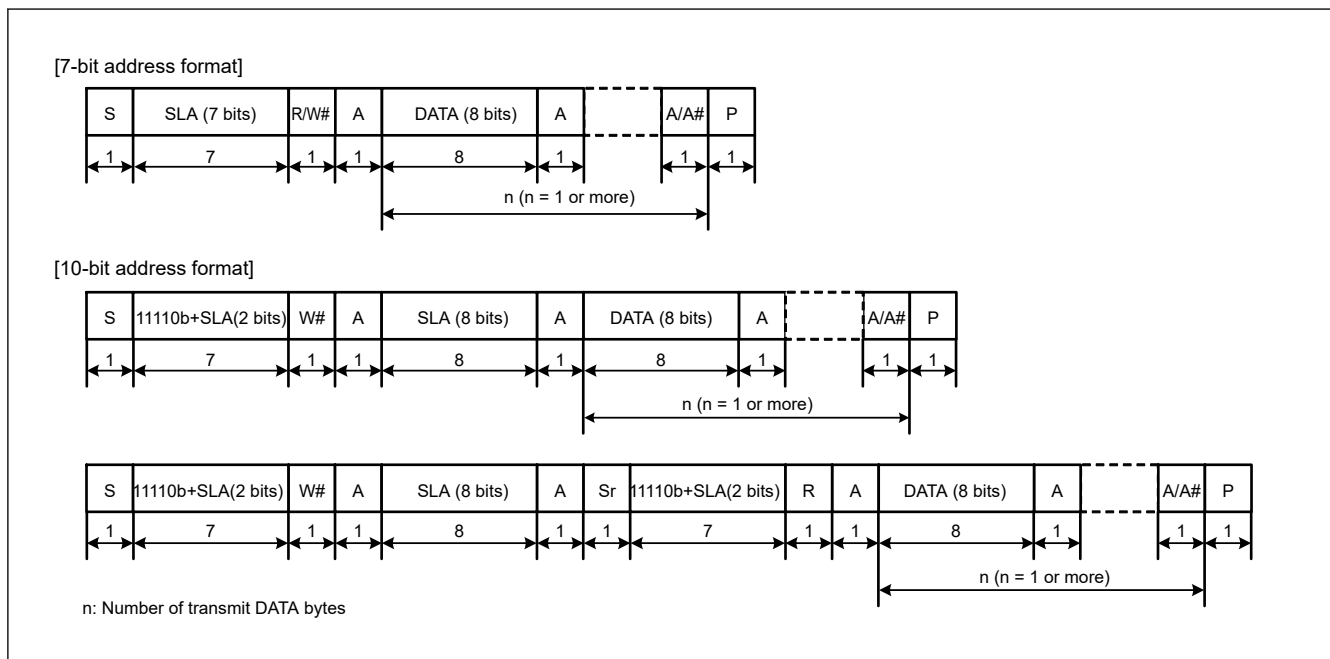


Figure 40.69 I²C bus format

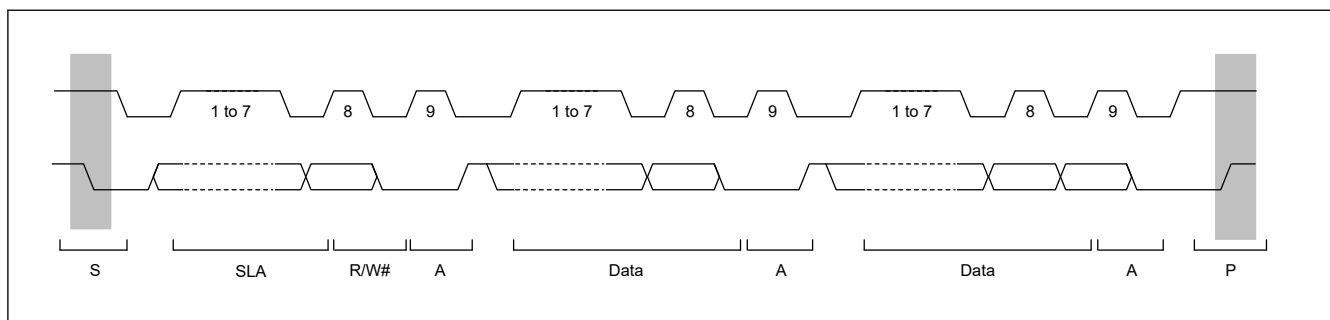


Figure 40.70 I²C bus timing (SLA = 7 bits)

- S: START condition. The master device drives the I3C_SDA line low from high level while the I3C_SCL line is at a high level.
- SLA: Slave address, by which the master device selects a slave device.
- R/W#: Indicates the direction of data transfer: from the slave device to the master device when R/W = 1, or from the master device to the slave device when R/W = 0.
- A: Acknowledge. The receive device drives the I3C_SDA line low. (In master transmit mode, the slave device returns acknowledge. In master receive mode, the master device returns acknowledge.)
- A#: Not Acknowledge. The receive device drives the I3C_SDA line high.
- Sr: Repeated START condition. The master device drives the I3C_SDA line low from the high level after the setup time has elapsed with the I3C_SCL line at the high level.

DATA: Transmitted or received data

P: STOP condition. The master device drives the I3C_SDA line high from low level while the I3C_SCL line is at a high level.

(2) I3C Communication Data Format

Figure 40.71 through Figure 40.77 illustrate a typical communication for each of the six I3C Protocols. While these diagrams do not exhaustively illustrate all possible I3C communications, they do serve as useful introductions to the signaling and transmission formatting used in each I3C Protocol.

Figure 40.71 illustrates example communication using I3C Single Data Rate (SDR) coding with Broadcast (0x7E). It shows the Master reading a byte of data from the Slave at Address 0x2B in SDR Mode. From the Bus Free Condition, the Master issues a START by driving the I3C_SDA line Low while keeping the I3C_SCL line High. It then issues the Broadcast Address (0x7E) followed by RnW (0 for Write). Then the Master turns on a pull-up resistor and goes to Open Drain.

All Slaves ACK by pulling the I3C_SDA line Low (in the Figure, pink fill means the Slave is in control of the I3C_SDA line at this time). The Master then issues a Repeated START, then the Address of the Slave (0x2B) it wants to read followed by RnW (1 for Read). The Master then turns on a pull-up resistor and goes to Open Drain, allowing the Slave to acknowledge by pulling the I3C_SDA line Low. At this point, the Master continues to toggle the I3C_SCL line and release the I3C_SDA line, allowing the Slave to drive SDA to send one byte of data (0x4A) followed by T. T = 1 informs the Master that there is additional data, whereas T = 0 signals the end. Here there is additional data, so the Slave drives SDA High until SCL goes High, at which time it releases SDA. The Master has the option of holding SDA High with a weak pullup, which signals to the Slave that the Master allows another byte to be transmitted, or to pull SDA Low (while SCL is High – hence a Repeated START), which would signal to the Slave that the Master has terminated the Read and is taking over.

SDR Mode is backwards compatible with Legacy I2C Devices, because the High time of an SCL pulse is always less than 50ns and therefore SCL will always appear to be Low because of the I2C 50ns Spike Filter.

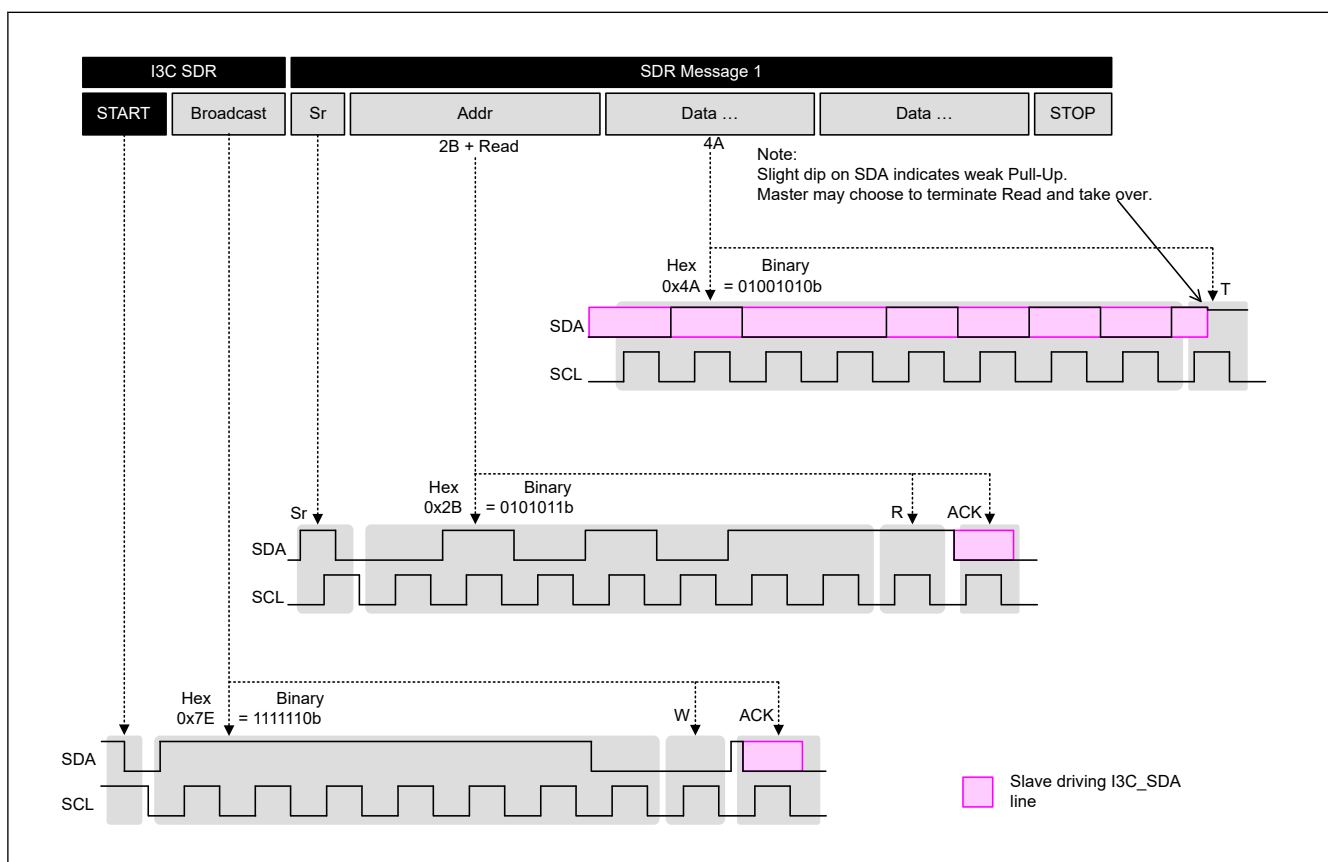


Figure 40.71 Example communication using I3C coding SDR with broadcast (0x7E)

Figure 40.72 illustrates example communication using I3C Single Data Rate (SDR) coding without Broadcast (0x7E). It shows the Master reading a byte of data from the Slave at Address 0x2B in SDR Mode. From the Bus Free Condition, The Master then issues a START, then the Address of the Slave (0x2B) it wants to read followed by RnW (1 for Read).

The Master then turns on a pull-up resistor and goes to Open Drain, allowing the Slave to acknowledge by pulling the I3C_SDA line Low. At this point, the Master continues to toggle the I3C_SCL line and release the I3C_SDA line, allowing the Slave to drive SDA to send one byte of data (0x4A) followed by T. T = 1 informs the Master that there is additional data, whereas T = 0 signals the end. Here there is additional data, so the Slave drives SDA High until SCL goes High, at which time it releases SDA. The Master has the option of holding SDA High with a weak pull-up, which signals to the Slave that the Master allows another byte to be transmitted, or to pull SDA Low (while SCL is High – hence a Repeated START), which would signal to the Slave that the Master has terminated the Read and is taking over.

SDR Mode is backwards compatible with Legacy I2C Devices, because the High time of an SCL pulse is always less than 50ns and therefore SCL will always appear to be Low because of the I2C 50ns Spike Filter.

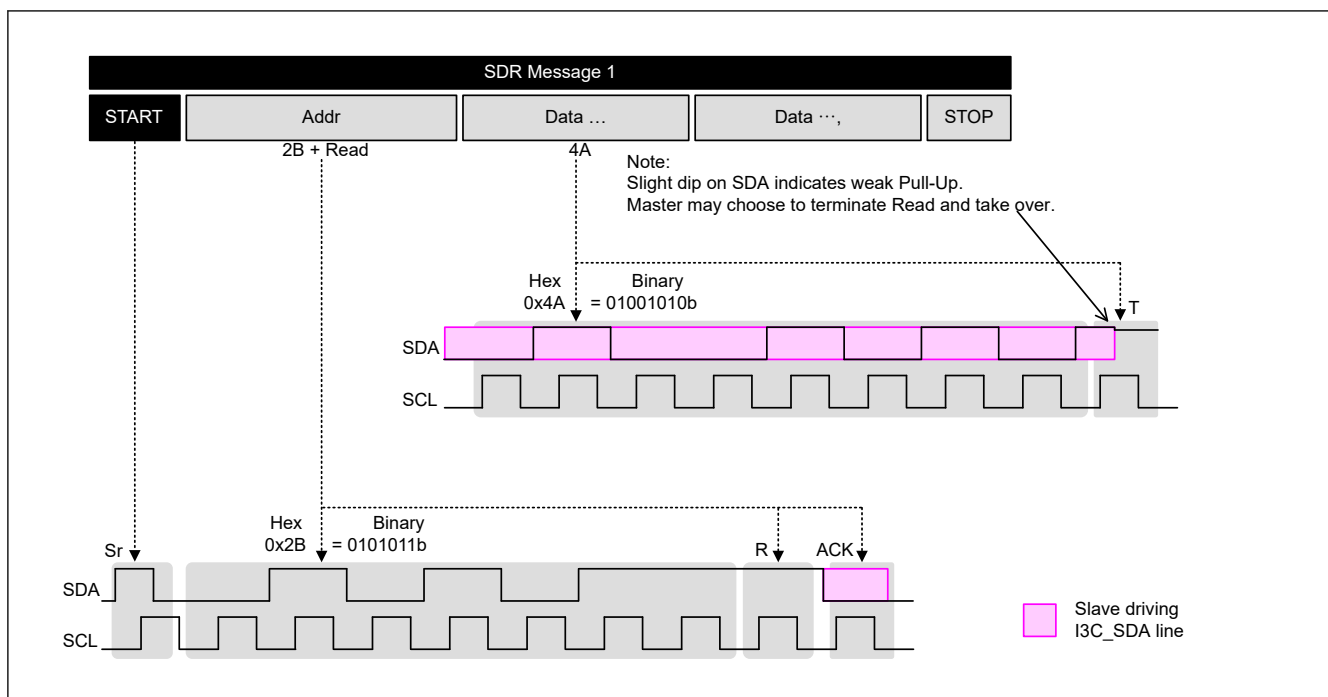


Figure 40.72 Example communication using I3C coding SDR without broadcast (0x7E)

Figure 40.73 shows the Master issuing a CCC Direct Command to a single Slave. This particular command (GETPID) reads the Provisional ID of a Slave.

From the Bus Free Condition, the Master issues a START by driving the I3C_SDA line Low while keeping the I3C_SCL line High. It then issues the Broadcast Address (0x7E) followed by RnW (0 for Write). Then the Master turns on a pull-up resistor and goes to Open Drain. All Slaves ACK by pulling SDA Low (in the Figure, pink fill means the Slaves are in control of SDA at this time). The Master then issues the Direct Common Command Code for GETPID (0x8C) followed by parity bit T (odd parity = 0 for 0x8C) then the 7-bit Dynamic Address of the Slave (chosen arbitrarily here to be 0x2B) followed by a RnW bit (1 for Read). Then the Master turns on a pull-up resistor and goes to Open Drain, allowing the Slave at Address 0x2B to ACK by pulling SDA Low, which tells the Master that the Slave Acknowledges the command and will comply. (Alternatively, the Slave may NACK by not pulling SDA Low, which would inform the Master that the Slave will not comply – in this case, that an error occurred.) Following the ACK the Slave outputs its 48-bit PID one byte at a time, and then the Master issues a Repeated START (this part of the waveform sequence is not shown in the Figure).

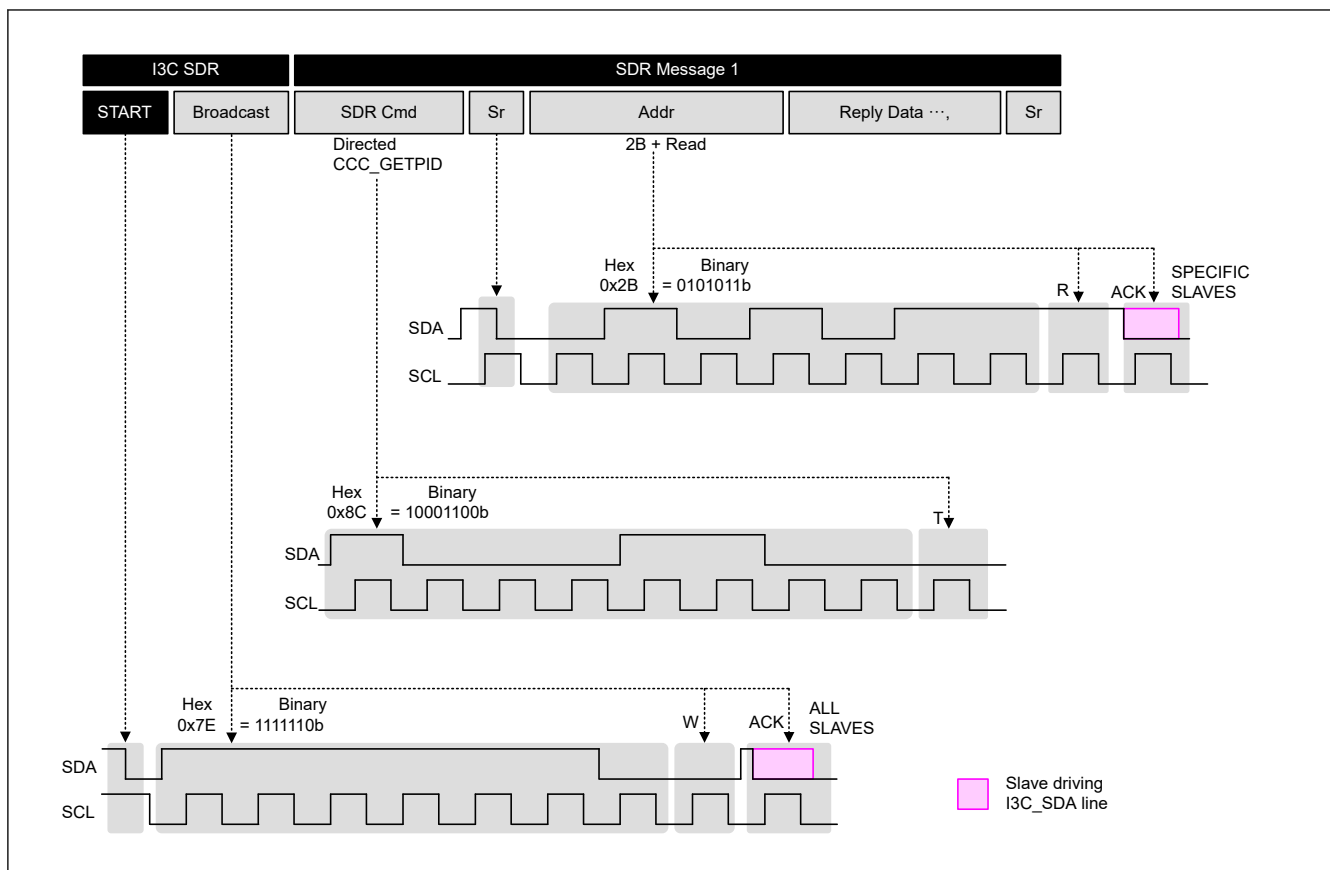


Figure 40.73 Example communication using I3C coding SDR with CCC direct addressing

Figure 40.74 illustrates example SDR communication with a CCC Broadcast command. The command used in this example sets the Maximum Read Length of all Slaves to 43 bytes (0x002B).

From the Bus Free Condition, the Master issues a START by driving the I3C_SDA line Low while keeping the I3C_SCL line High. It then issues the Broadcast Address (0x7E) followed by RnW (0 for Write). Then the Master turns on a pull-up resistor and goes to Open Drain. All Slaves ACK by pulling SDA Low (in the Figure, pink fill means the Slaves are in control of SDA at this time). The Master then issues the Broadcast Common Command Code for SETMRL (0x09) followed by parity bit T (odd parity = 1 for 0x09), and then 2 data bytes (MSB first) to define the maximum number of bytes which can be read from a Slave in a single read operation. Each data byte is followed by a T bit (parity bit – odd parity). After this the Master issues a Repeated START.

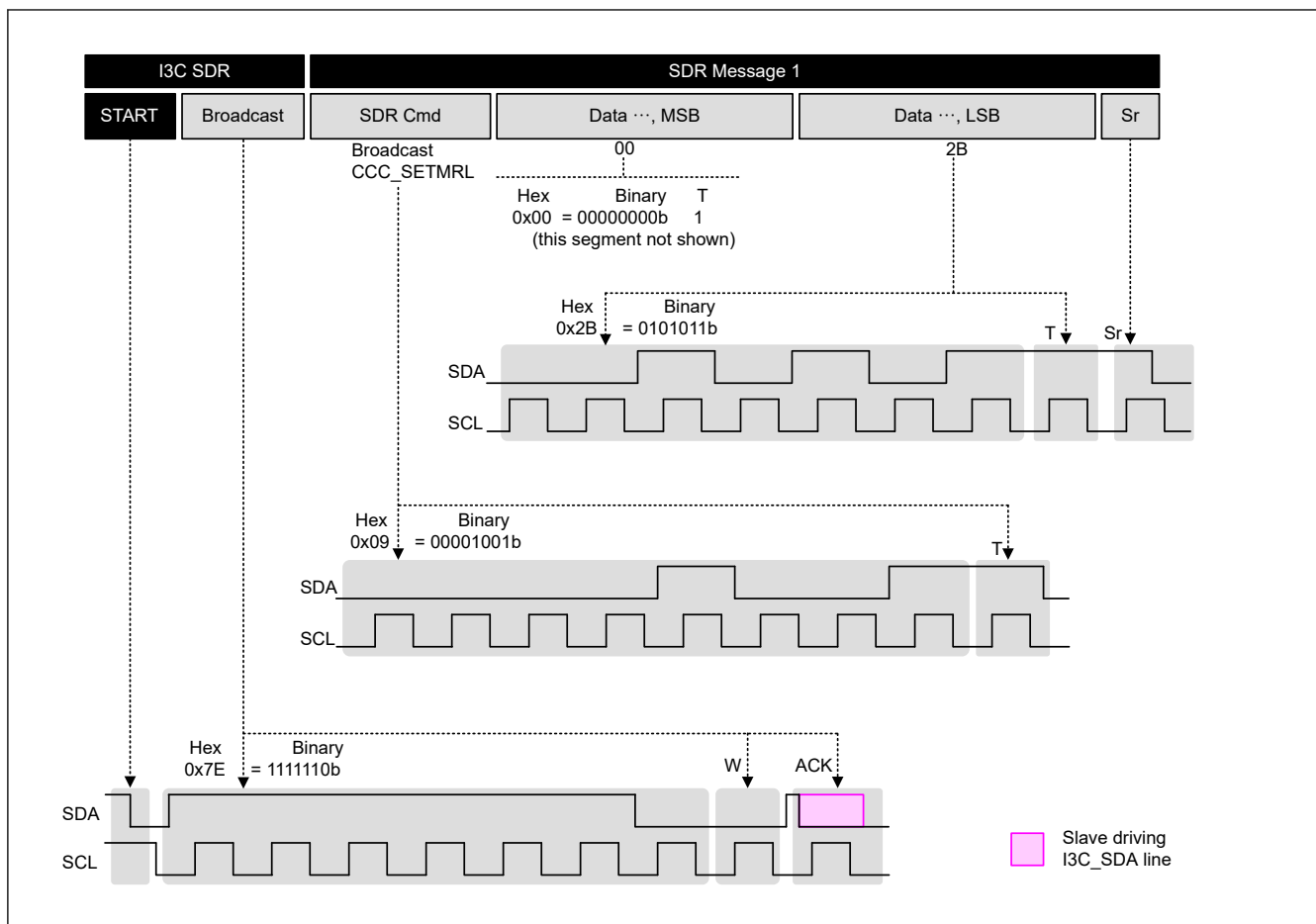


Figure 40.74 Example communication using I3C coding SDR with CCC broadcast

Figure 40.75 illustrates use of the HDR-DDR (Double Data Rate) Mode. It shows how the Master can change the Mode from SDR (Single Data Rate) Mode to HDR-DDR Mode, and a sample of the HDR-DDR data format.

From the Bus Free Condition, the Master issues a START by driving the I3C_SDA line Low while keeping the I3C_SCL line High. The Master then issues the Broadcast Address (0x7E) followed by RnW (0 for Write). Then the Master turns on a pull-up resistor and goes to Open Drain. All Slaves 'ACK' by pulling SDA Low (in the Figure, pink fill means the Slaves are in control of SDA at this time). The Master then issues the Broadcast Common Command Code for ENTHDR0 (0x20), followed by parity bit T (odd parity = 0 for 0x20). At this point, the Bus is in HDR-DDR Mode. In the HDR-DDR protocol, the I3C_SDA line is sampled on every SCL edge (both Low-to-High and High-to-Low transitions of SCL). The HDR-DDR Word consists of a 2-bit Preamble, followed by two bytes of data, followed by two parity bits. The waveform for the 5-bit CRC and following traffic is not shown in the Figure.

HDR-DDR Mode is backwards compatible with Legacy I²C Devices, because the High time of an SCL pulse is always less than 50ns and as a result SCL will always appear to be Low because of the I²C 50ns Spike Filter.

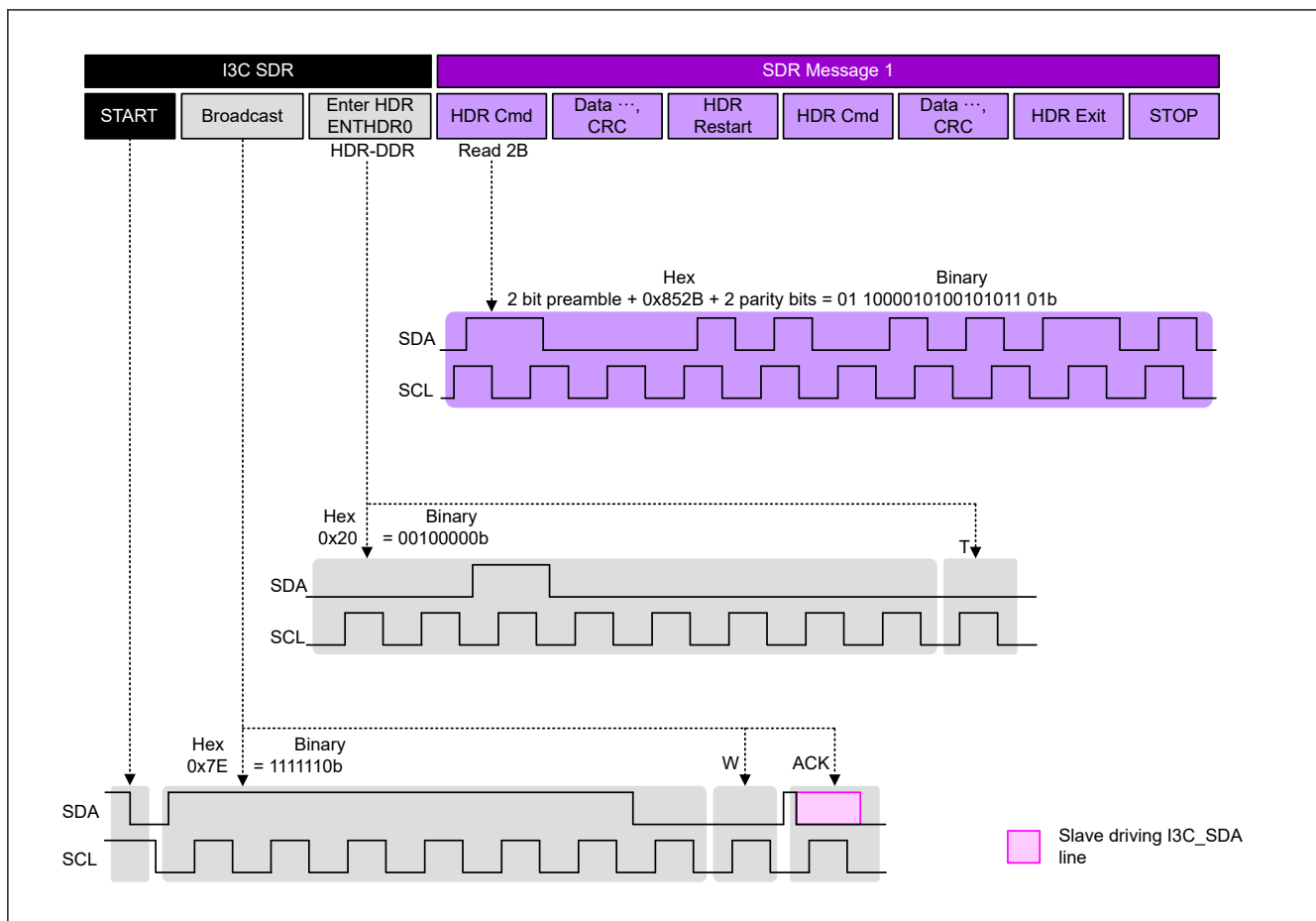


Figure 40.75 Example communication using HDR-DDR protocol

Figure 40.76 illustrates use of the HDR-TSL (Ternary Symbol Legacy) Mode, which is only used if a Legacy I²C Device is present on the Bus. It shows how the Master can change the Mode from SDR (Single Data Rate) Mode to HDR-TSL Mode, and a sample of the HDR-TSL data format.

From the Bus Free Condition, the Master issues a START by driving the I3C_SDA line Low while keeping the I3C_SCL line High. It then issues the Broadcast Address (0x7E) followed by RnW (0 for Write). Then the Master turns on a pull-up resistor and goes to Open Drain. All Slaves ACK by pulling SDA Low (in the Figure, pink fill means the Slaves are in control of SDA at this time). The Master then issues the Broadcast Common Command Code for ENTHDR2 (0x22) followed by parity bit 'T' (odd parity = 1 for 0x22). At this point, the Bus is in HDR-TSL Mode. In this protocol, the I3C_SCL line is more than just the clock. Both SDA transitions and SCL transitions carry data. The ternary conversion can be understood by considering that each octal digit is equivalent to two ternary Symbols. For example, for octal sequence 412255 we have: 4 = 3 + 1, 1 = 0 + 1, 2 = 0 + 2, 2 = 0 + 2, 5 = 3 + 2, 5 = 3 + 2. The clock pulse occurs when SCL transitions, when SDA transitions, or when both lines transition simultaneously.

HDR-TSL Mode is backwards compatible with Legacy I²C Devices, because the High time of an SCL pulse is always less than 50ns, and as a result SCL will always appear to be Low due to the I²C 50ns Spike Filter. Note that in the Figure, the thick blue High-to-Low transitions of SCL are dummy edges added in order to make sure that SCL pulses are always less than 50ns. High-to-Low SCL transitions that occur without a simultaneous SDA transition are always dummy transitions.

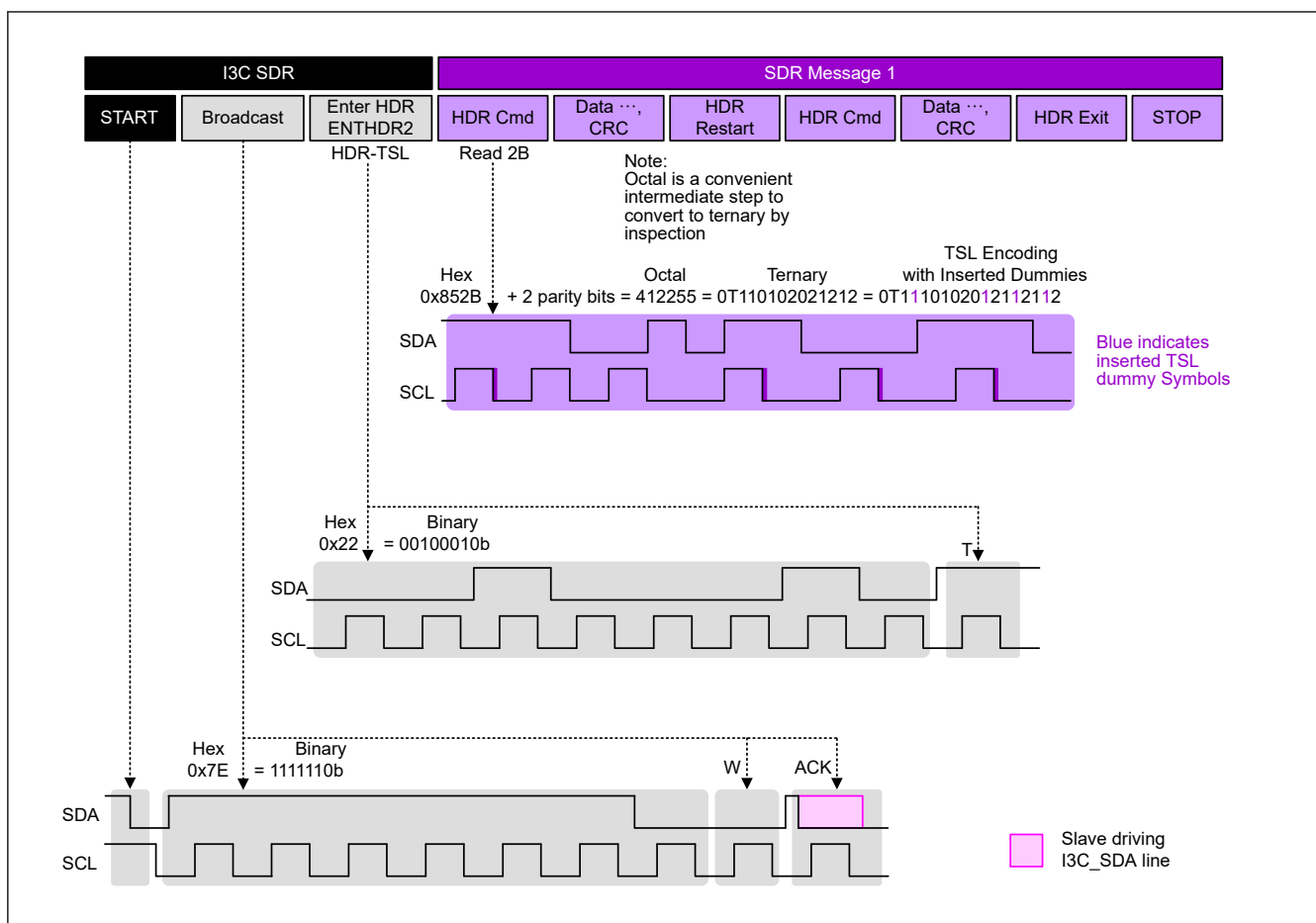


Figure 40.76 Example communication using HDR-TSL protocol

Figure 40.77 illustrates use of HDR-TSP (High Data Rate – Ternary Symbol for Pure Bus) Mode. This protocol is only used when there are no I²C Devices on the Bus. The Figure shows how the Master can change the Mode from SDR (Single Data Rate) to HDR-TSP Mode, and a sample of the TSL data format.

From the Bus Free Condition, the Master issues a START by driving the I3C_SDA line Low while keeping the I3C_SCL line High. The Master then issues the Broadcast Address (0x7E) followed by RnW (0 for Write). Then the Master turns on a pull-up resistor and goes to Open Drain. All Slaves ACK by pulling SDA Low (in the Figure, pink fill means the Slaves are in control of SDA at this time). The Master then issues the Broadcast Common Command Code for ENTHDR1 (0x21) followed by parity bit 'T' (odd parity = 1 for 0x21). At this point, the Bus is in HDR-TSP Mode. In the HDR-TSP protocol, the SCL is more than just the clock. Both SDA transitions and SCL transitions carry data. The ternary conversion can be understood by considering that each octal digit is equivalent to two ternary Symbols. For example, for octal sequence 412255 (4 = 3 + 1, 1 = 0 + 1, 2 = 0 + 2, 2 = 0 + 2, 5 = 3 + 2, 5 = 3 + 2). The clock pulse occurs when SCL transitions, when SDA transitions, or when both lines transition simultaneously.

HDR-TSP Mode supports the highest data-rate and uses the least energy per byte, however it is not backwards compatible with Legacy I²C Devices since the High duration of an SCL pulse can exceed 50ns.

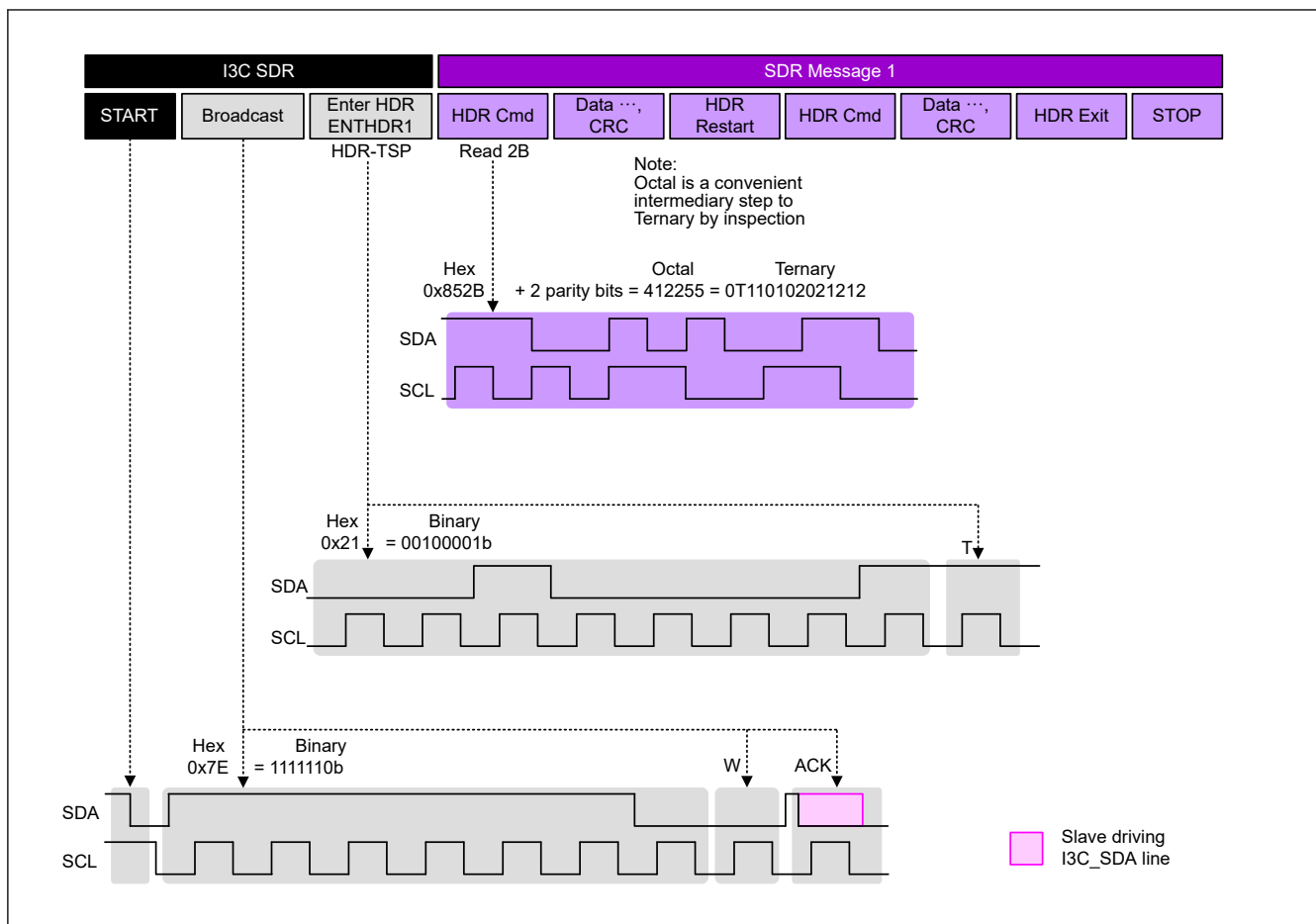


Figure 40.77 Example communication using HDR-TSP protocol

40.3.2.3.2 Bus Conditions

I3C defines three distinct conditions in which the I3C Bus shall be considered inactive: Bus Free, Bus Available, and Bus Idle (see [Figure 40.78](#)).

(1) Bus Free Condition

State on the I3C Bus where both the I3C_SCL line and the I3C_SDA line are High for at least the period set by BFRECDT.FRECYC[8:0] bit.

(2) Bus Available Condition [I3C mode]

State on the I3C Bus where both the I3C_SCL line and the I3C_SDA line are High for at least the period set by BAVLCDT.AVLCYC[8:0] bit.

A Slave may only issue a START Request (For example, for an In-Band Interrupt, or for a Master Handoff Request) after a Bus Available Condition.

(3) Bus Idle Condition [I3C mode]

State on the I3C Bus where both the I3C_SCL line and the I3C_SDA line are High for at least the period set by BIDLCDT.IDLCYC[17:0] bit.

A Slave may only issue a START Request after a Bus Idle Condition.

Specifications are as follows. IDLE needs to be the largest.

$\text{BFRECDT.FRECYC}[8:0] < \text{BAVLCDT.AVLCYC}[8:0] < \text{BIDLCDT.IDLCYC}[17:0]$

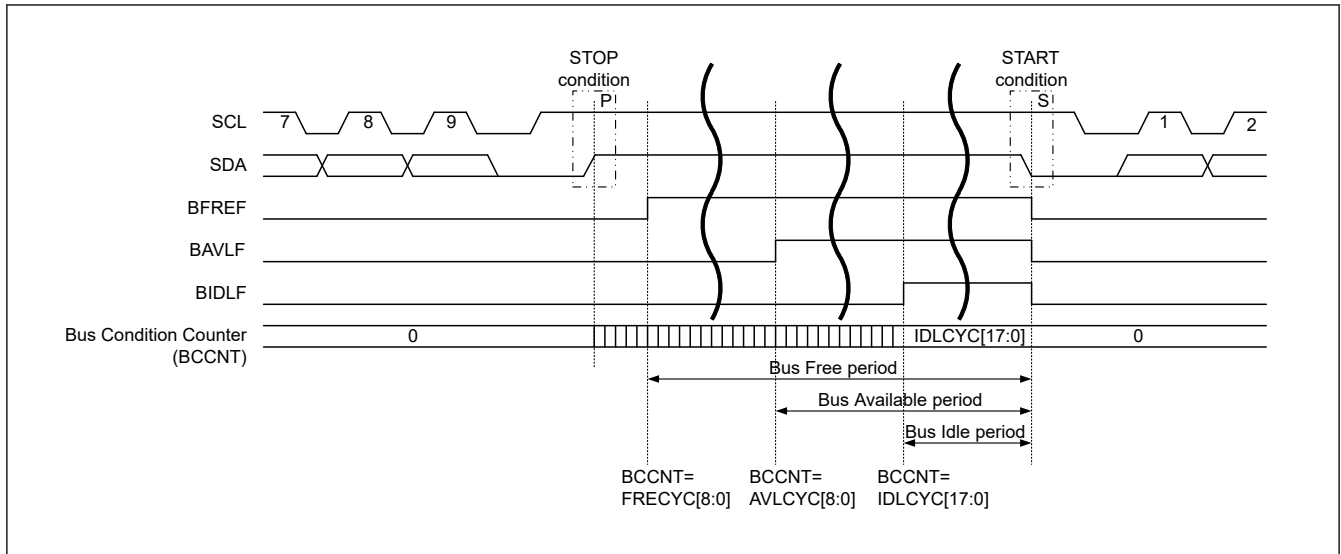


Figure 40.78 Bus conditions

40.3.2.3.3 START Condition / Repeated START Condition / STOP Condition Issuing Function

(1) Issuing a START Condition

I3C issues a START condition when the CNDCTL.STCND bit is set to 1.

Set the STCND bit to 1 (START condition issuance request) when the BCST.BFREF flag is set to 1 (bus free state).

I3C issues a START condition.

When a START condition is issued normally, I3C automatically shifts to the master transmit mode. A START condition is issued in the following sequence.

[START condition issuance]

- Drive the I3C_SDA line low (high level to low level).
- Ensure the time set in STDBR.SBRHO[7:0] and the START condition hold time.
- Drive the I3C_SCL line low (high level to low level).
- Detect low level of the I3C_SCL line and ensure the low-level period of I3C_SCL line set in STDBR.SBRLO[7:0].

(2) Issuing a Repeated START Condition

I3C issues a Repeated START condition when the CNDCTL.SRCND bit is set to 1.

When the SRCND bit is set to 1, a Repeated START condition issuance request is made and I3C issues a Repeated START condition when the BCST.BFREF flag = 0 (bus busy state) and the PRSST.CRMS bit = 1 (master mode).

A Repeated START condition is issued in the following sequence.

[Repeated START condition issuance]

- Release the I3C_SDA line.
- Ensure the low-level period of I3C_SCL line set in STDBR.SBRLO[7:0] or EXTBR.EBRLO[7:0].
- Release the I3C_SCL line (low level to high level).
- Detect a high level of the I3C_SCL line and ensure the time set in STDBR.SBRLO[7:0] or EXTBR.EBRLO[7:0] and the Repeated START condition setup time.
- Drive the I3C_SDA line low (high level to low level).
- Ensure the time set in STDBR.SBRHO[7:0] or EXTBR.EBRHO[7:0] or EXTBR.EBRHO[7:0] and the Repeated START condition hold time.
- Drive the I3C_SCL line low (high level to low level).

- Detect a low level of the I3C_SCL line and ensure the low-level period of I3C_SCL line set in STDBR.SBRLO[7:0] or EXTBR.EBRLO[7:0].

Note: When issuing Repeated START conditions request, write the slave address to NTDTBP0 after confirming CNDCTL.SRCND = 0. Data written in the period of CNDCTL.SRCND = 1 is not forwarded because retransmission condition before the occurrence.

To issue a Repeated START condition in Hs-mode, follow the steps below.

1. Wait for PRSTDBG.SCOLV=0.
2. Set EXTBR.EBRHO[7:0] to satisfy the hold time of the Repeated START condition.
3. Set the CNDCTL.SRCND bit to 1.
4. After confirming CNDCTL.SRCND=0, wait for PRSTDBG.SCOLV=0.
5. Set EXTBR.EBRHO [7: 0] according to the High period of the SCL clock in Hs-mode.
6. Write the slave address to NTDTBP0.

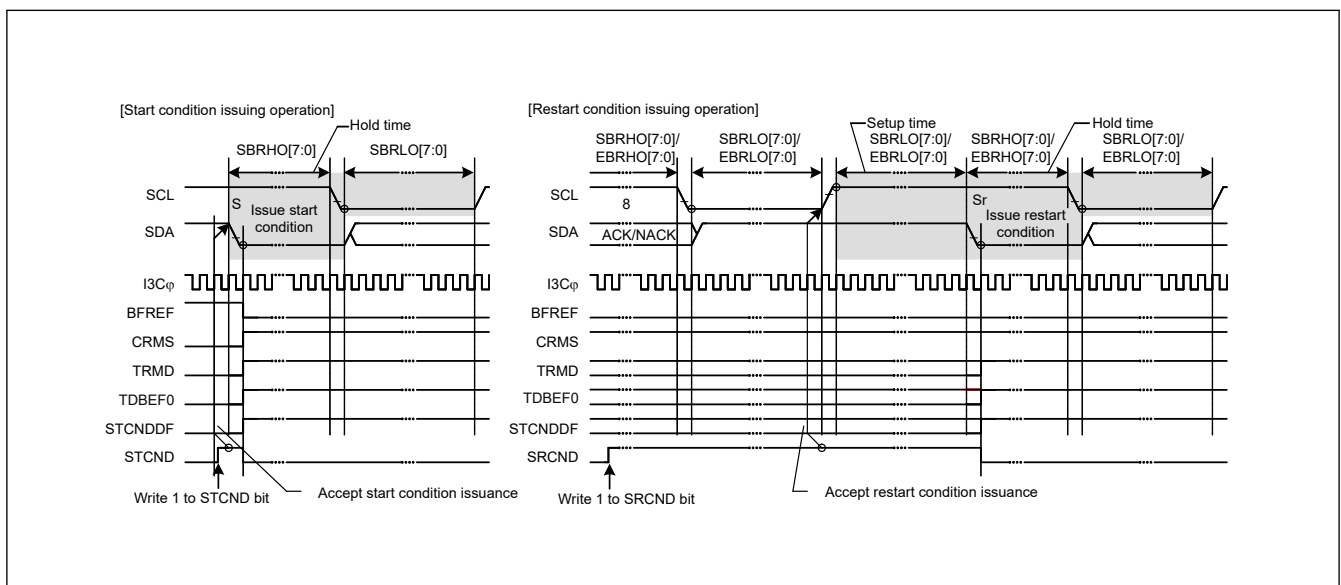


Figure 40.79 START condition / repeated START condition issue timing (STCND and SRCND bits)

Figure 40.80 shows the operation to issue a Repeated START condition after the master transmission.

[Repeated START condition issuance after the master transmission]

- Initial setting. For details, see [section 40.3.3.1. Initial Setting Flow](#).
- Read the BFREF flag in BCST to check that the bus is open, and then set the STCND bit in CNDCTL to 1 (START condition issuance request). Upon receiving the request, I3C issues a START condition. At the same time, the BFREF flag is automatically set to 0 and the STCNDFF flag in BST is automatically set to 1 and the STCND bit is automatically set to 0. At this time, if the START condition is detected and the internal levels for the SDA output state and the levels on the I3C_SDA line have matched while the STCND bit = 1, I3C recognizes that issuing of the START condition as requested by the STCND bit has been successfully completed, and CRMS and TRMD bits in PRSST is automatically set to 1, placing I3C in master transmit mode. The NTST.TDBEF0 flag is also automatically set to 1 in response to setting of the TRMD bit to 1.
- Check that the NTST.TDBEF0 flag = 1, and then write the value for transmission (the slave address and the R/W# bit) to NTDTBP0. Once the data for transmission are written to NTDTBP0, the TDBEF0 flag is automatically set to 0, the data are transferred from NTDTBP0, and the TDBEF0 flag is again set to 1. After the byte containing the slave address and R/W# bit has been transmitted, the value of the TRMD bit is automatically updated to select master transmit or master receive mode in accord with the value of the transmitted R/W# bit. If the value of the R/W# bit was 0, I3C continues in master transmit mode. Since the BST.NACKDF flag being 1 at this time indicates that no slave device recognized the address or there was an error in communications, write 1 to CNDCTL.SPCND bit to issue a STOP condition. For data transmission with an address in the 10-bit format, start by writing 1111 0, the 2 higher-order bits of the slave address, and W to NTDTBP0 as the first address transmission. Then, as the second address transmission, write the 8 lower-order bits of the slave address to NTDTBP0.

- After confirming that the NTST.TDBEF0 flag = 1, write the data for transmission to the NTDTBP0 register. I3C automatically holds the I3C_SCL line low until the data for transmission are ready, a Repeated START condition is issued or a STOP condition is issued.
- After all bytes of data for transmission have been written to the NTDTBP0 register, wait until the value of the BST.TENDF flag returns to 1, and then, after check that the BST.STCNDDF flag = 1, set the BST.STCNDDF flag to 0.
- Set the SRCND bit in CNDCTL to 1 (Repeated START condition issuance request). Upon receiving the request, I3C issues a Repeated START condition.
- After check that the BST.STCNDDF flag = 1, write the value for transmission (the slave address and the R/W# bit) to NTDTBP0.

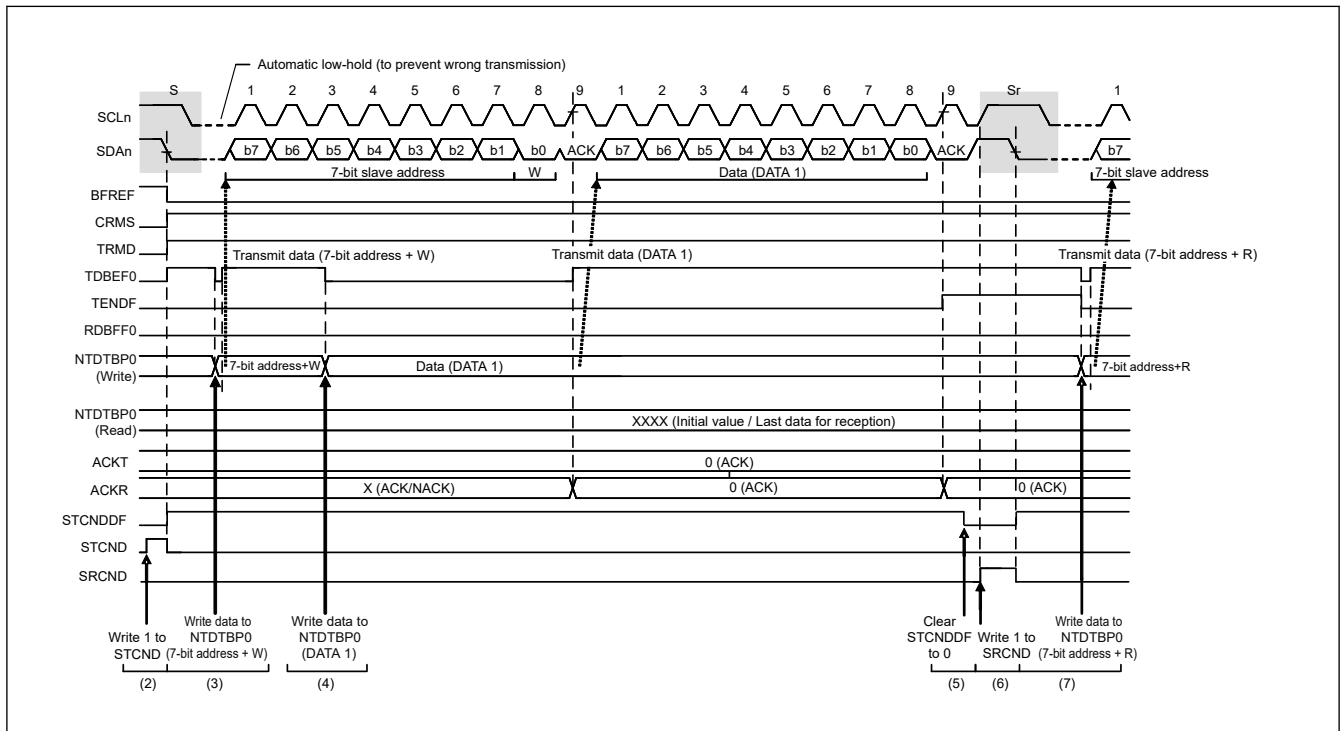


Figure 40.80 Repeated START condition issuance after the master transmission timing

(3) Issuing a STOP Condition

I3C issues a STOP condition when the SPCND bit in CNDCTL is set to 1.

When the SPCND bit is set to 1, a STOP condition issuance request is made and I3C issues a STOP condition when the BCST.BFREF flag = 0 (bus busy state) and the PRSST.MST bit = 1 (master mode).

A STOP condition is issued in the following sequence.

[STOP condition issuance]

- Drive the I3C_SDA line low (high level to low level).
- Ensure the low-level period of I3C_SCL line set in STDBR.SBRLO[7:0] or EXTBR.EBRLO[7:0].
- Release the I3C_SCL line (low level to high level).
- Detect a high level of the I3C_SCL line and ensure the time set in STDBR.SBRHO[7:0] or EXTBR.EBRHO[7:0] and the STOP condition setup time.
- Release the I3C_SDA line (low level to high level).
- Ensure the time set in BFRECDT.FRECYC[8:0] and the bus free time.
- Set the BFREF flag to 1 (to release the bus mastership).

Note: To issue a STOP condition in Hs-mode, follow the steps below.

1. Wait for PRSTDBG.SCOLV=0.

2. Set EXTBR.EBRHO[7:0] to satisfy the setup time for the STOP condition.
3. Set the CNDCTL.SPCND bit to 1.
4. Wait for CNDCTL.SPCND=0.
5. Set EXTBR.EBRHO [7:0] according to the High period of the SCL clock in Hs-mode.

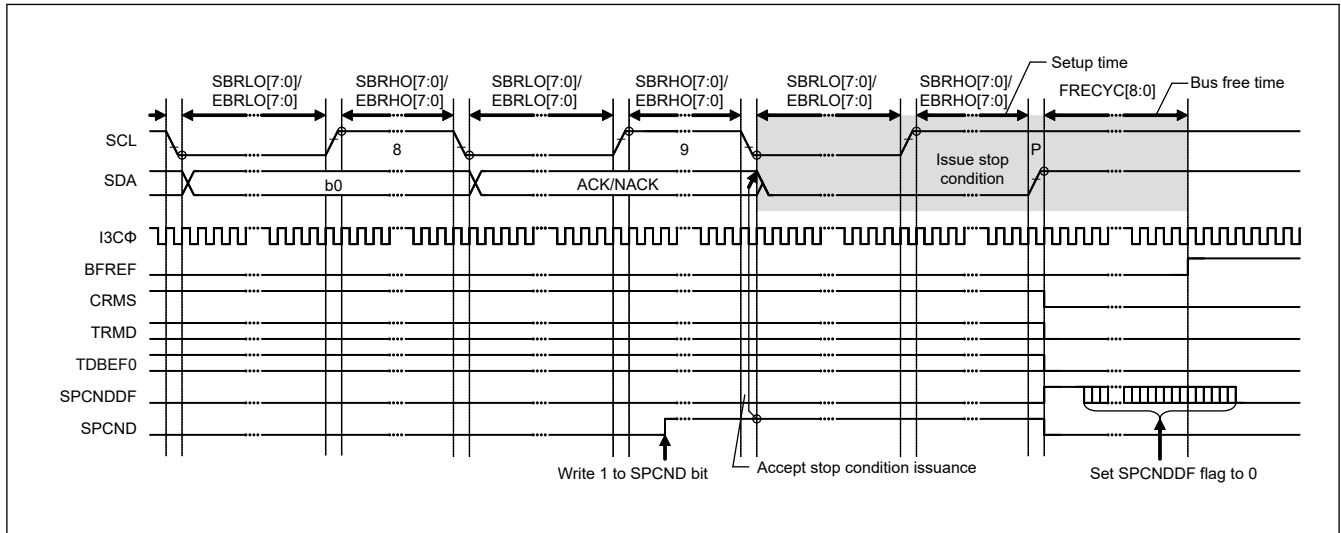


Figure 40.81 STOP condition issue timing (SPCND bit)

40.3.2.3.4 Address Match Detection

I3C can set three unique slave addresses in addition to the general call address and host address, and also can set 7-bit or 10-bit slave addresses.

(1) Slave-Address Match Detection [I²C mode]

I3C can set three unique slave addresses, and has a slave address detection function for each unique slave address.

When the SVCTL.SVAEy bit ($y = 0$ to 2) is set to 1, the slave addresses set in the SVDVADy register ($y = 0$ to 2) can be detected.

When I3C detects a match of the set slave address, the corresponding SVST.SVAF[y] flag ($y = 0$ to 2) is set to 1 at the rising edge of the ninth SCL clock cycle, and the NTST.RDBFF0 flag or the NTST.TDBEF0 flag is set to 1 by the following R/W# bit. This causes a Normal Rx Data buffer full interrupt (I3C_RX) or Normal Tx Data buffer empty interrupt (I3C_TX) to be generated. The SVAF[y] flag is used to identify which slave address has been specified.

Figure 40.82 to Figure 40.84 show the SVAF[y] flag set timing in three cases.

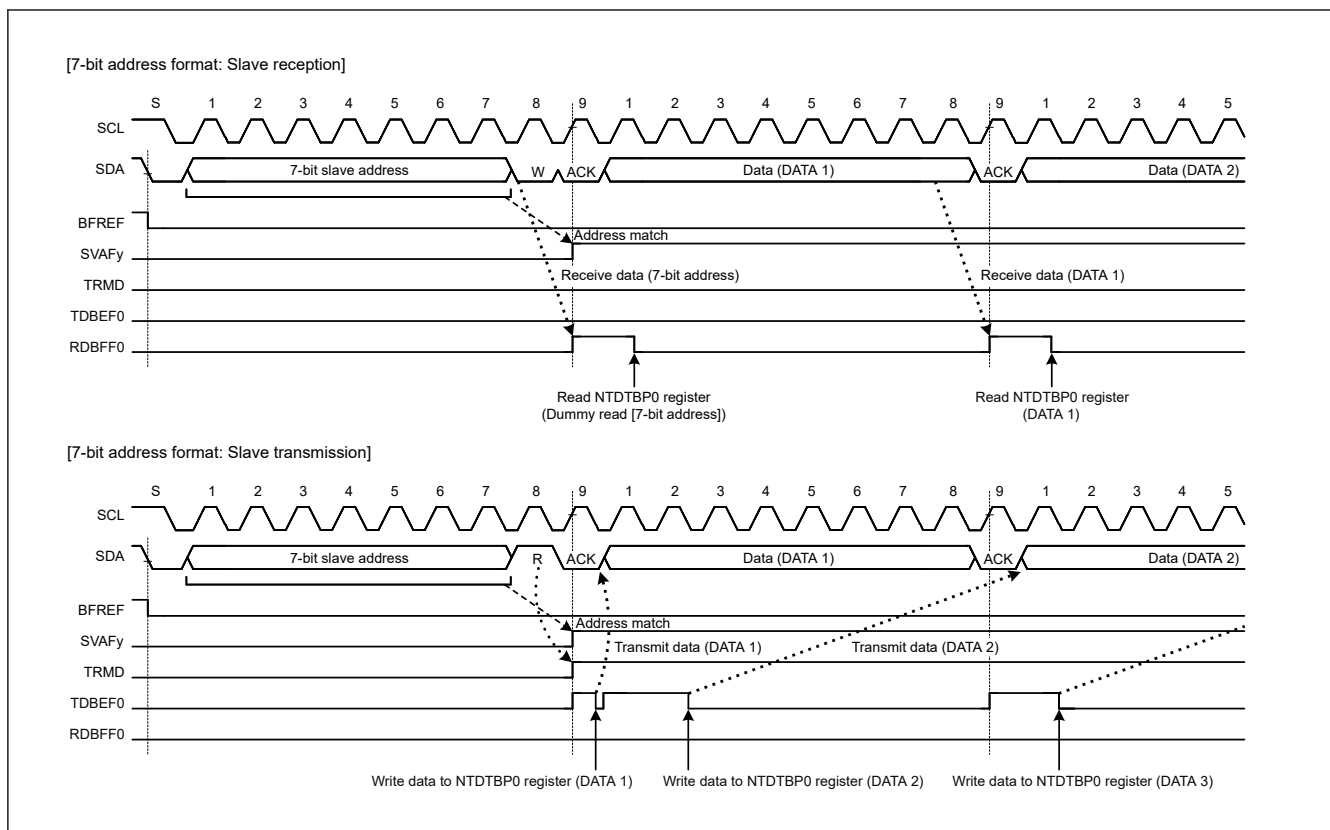


Figure 40.82 SVAfy flag set timing with 7-bit address format selected

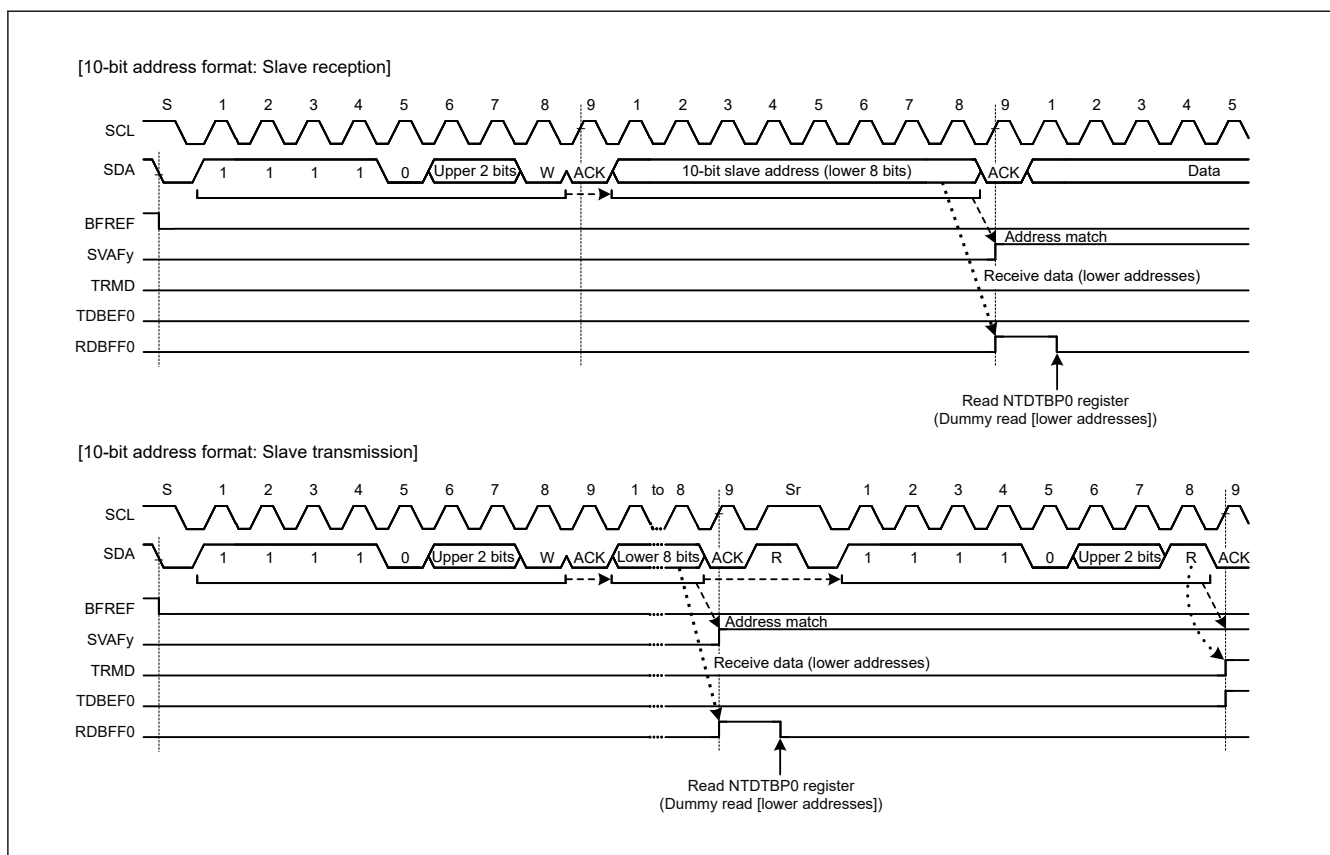


Figure 40.83 SVAfy flag set timing with 10-bit address format selected

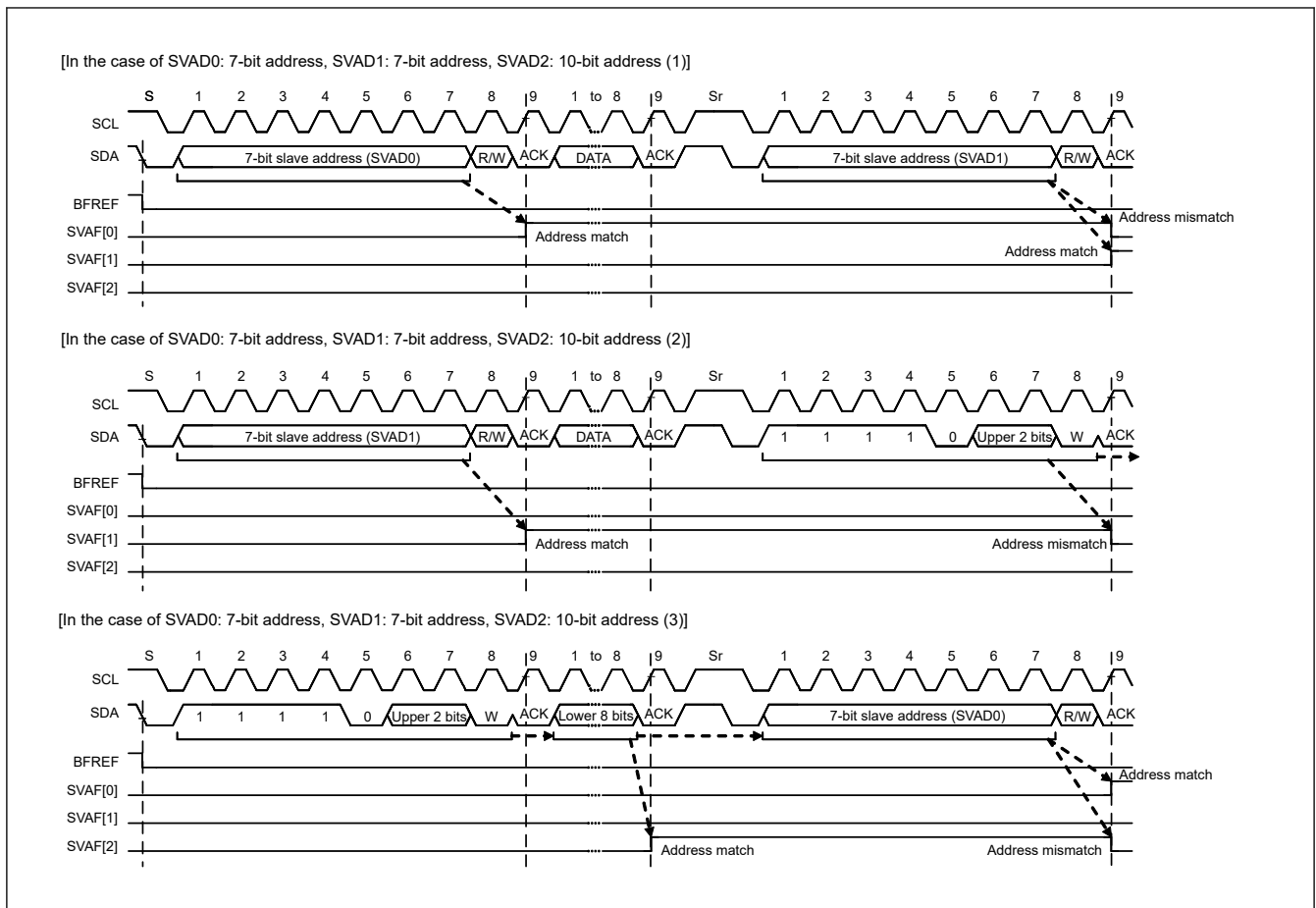


Figure 40.84 SVAFy flag set/clear timing with 7-bit/10-bit address formats mixed

(2) Detection of the General Call Address [I²C mode]

I3C has a facility for detecting the general call address (0000 000 + 0 (write)). This is enabled by setting the SVCTL.GCAE bit to 1.

If the address received after a START or Repeated START condition is issued is 0000 000 + 1 (read) (start byte), I3C recognizes this as the address of a slave device with an all-zero address but not as the general call address.

When I3C detects the general call address, both the SVST.GCAF flag and the NTST.RDBFF0 flag are set to 1 on the rising edge of the ninth cycle of SCL clock. This leads to the generation of a Normal Rx Data buffer full interrupt (I3C_RX). The value of the GCAF flag can be confirmed to recognize that the general call address has been transmitted.

Operation after detection of the general call address is the same as normal slave receive operation.

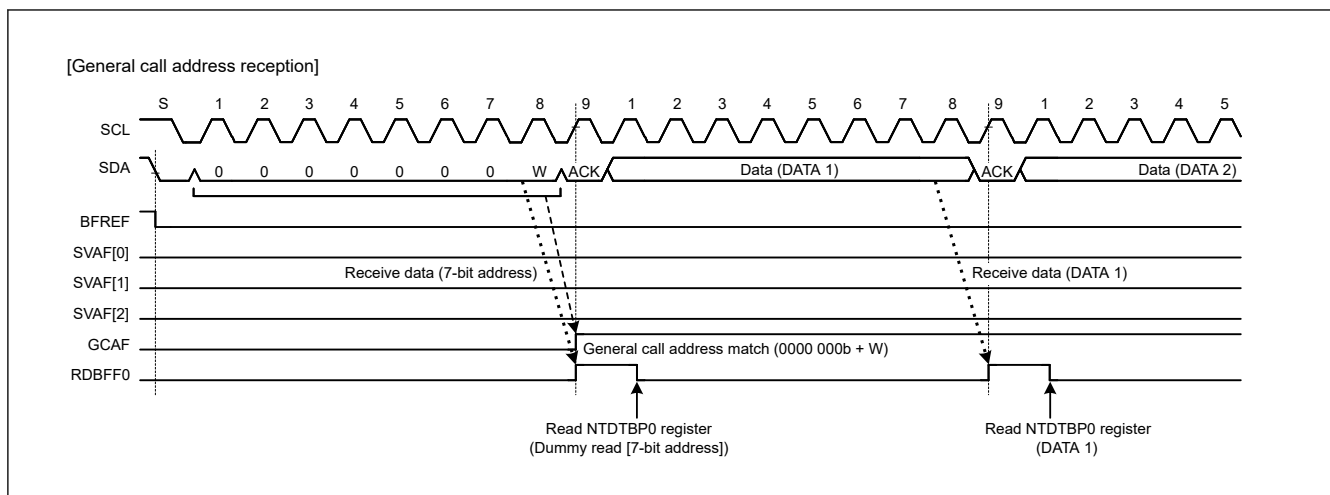


Figure 40.85 Timing of GCAF flag setting during reception of general call address

(3) Device-ID Address Detection [I²C mode]

I3C module has a facility for detecting device-ID addresses conformant with the I²C-bus specification (Rev.03). When I3C receives 1111 100 as the first byte after a START condition or Repeated START condition was issued with the SVCTL.DVIDE bit set to 1, I3C recognizes the address as a device ID, sets the SVST.DVIDF flag to 1 on the rising edge of the ninth SCL clock cycle when the following R/W# bit = 0, and then compares the second and subsequent bytes with its own slave address. If the address matches the value in the slave address register, I3C sets the corresponding SVST.SVA[y] flag (y = 0 to 2) to 1.

After that, when the first byte received after a START or Repeated START condition is issued matches the device ID address (1111 100) again and the following R/W# bit = 1, I3C does not compare the second and subsequent bytes and sets the NTST.TDBEF0 flag to 1.

In the device-ID address detection function, I3C sets the DVIDF flag to 0 if a match with I3C's own slave address is not obtained or a match with the device ID address is not obtained after a match with I3C's own slave address and the detection of a Repeated START condition. If the first byte after detection of a START or Repeated START condition matches the device ID address (1111 100) and the R/W# bit = 0, I3C sets the DVIDF flag to 1 and compares the second and subsequent bytes with I3C's slave address. If the R/W# bit = 1, the DVIDF flag holds the previous value and I3C does not compare the second and subsequent bytes. Therefore, the reception of a device-ID address can be checked by reading the DVIDF flag after confirming that TDBEF0 flag = 1.

Furthermore, prepare the device-ID fields (3 bytes: 12 bits indicating the manufacturer + 9 bits identifying the part + 3 bits indicating the revision) that must be sent to the host after reception of a continuous device-ID field as normal data for transmission. For details of the information that must be included in device-ID fields, contact NXP Semiconductors.

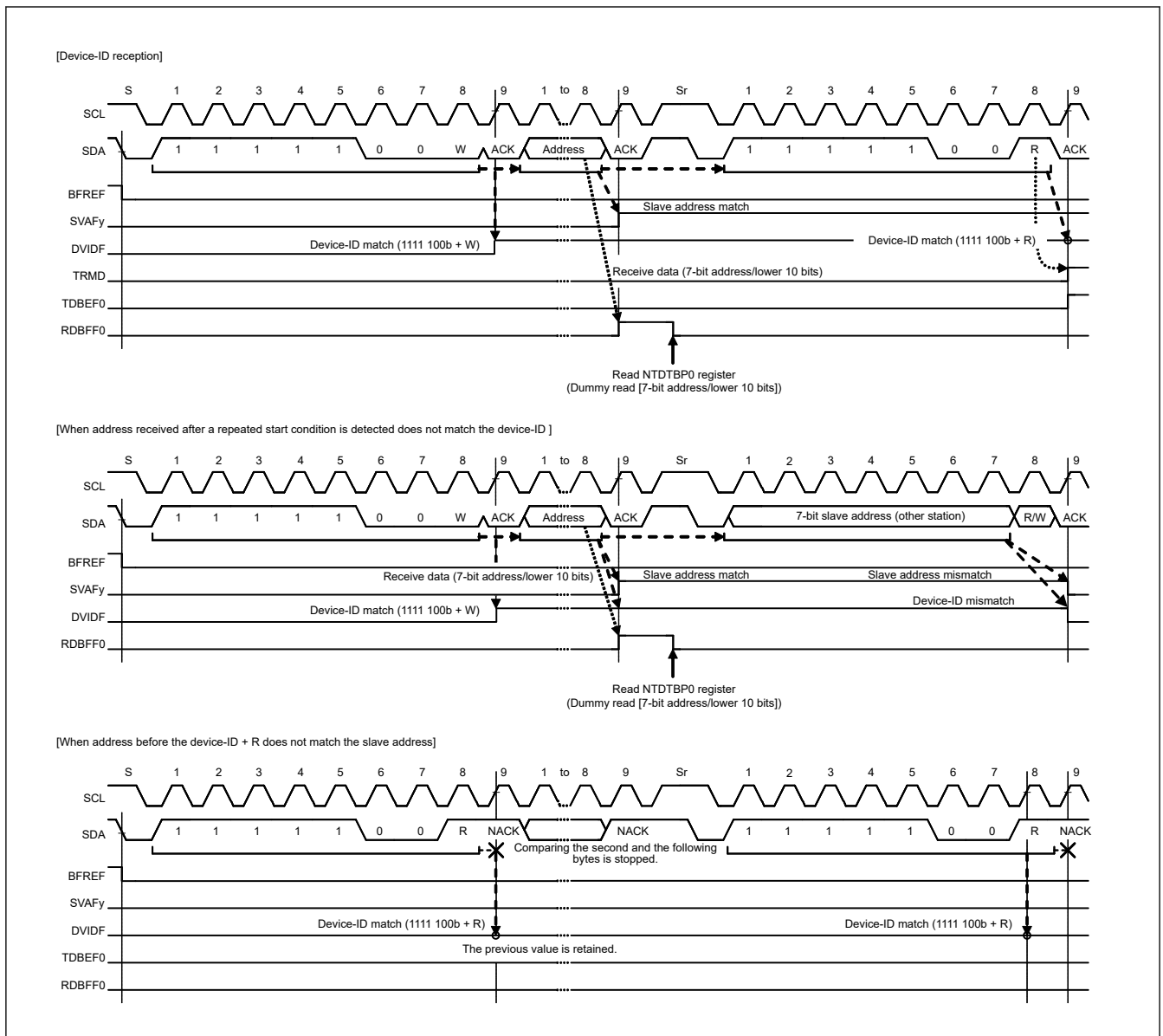


Figure 40.86 SVAfY/DVIDF flag set/clear timing during reception of device-ID

(4) Host Address Detection [I²C mode]

I3C has a function to detect the host address while the SMBus is operating. When the SVCTL.HOAE bit is set to 1 while the BFCTL.SMBS bit = 1, I3C can detect the host address (0001 000) in slave receive mode (bits CRMS and TRMD in the PRSST register = 00).

When I3C detects the host address, the SVST.HOAF flag is set to 1 at the rising edge of the ninth SCL clock cycle, and at the same time, the NTST.RDBFF0 flag is set to 1 when the R/W# bit = 0 (Wr bit). This causes a Normal Rx Data buffer full interrupt (I3C_RX) to be generated. The HOAF flag is used to recognize that the host address was sent from the smart battery or other devices.

If the bit following the host address (0001 000) is an Rd bit (R/W# bit = 1), I3C can also detect the host address. After the host address is detected, I3C operates in the same manner as normal slave operation.

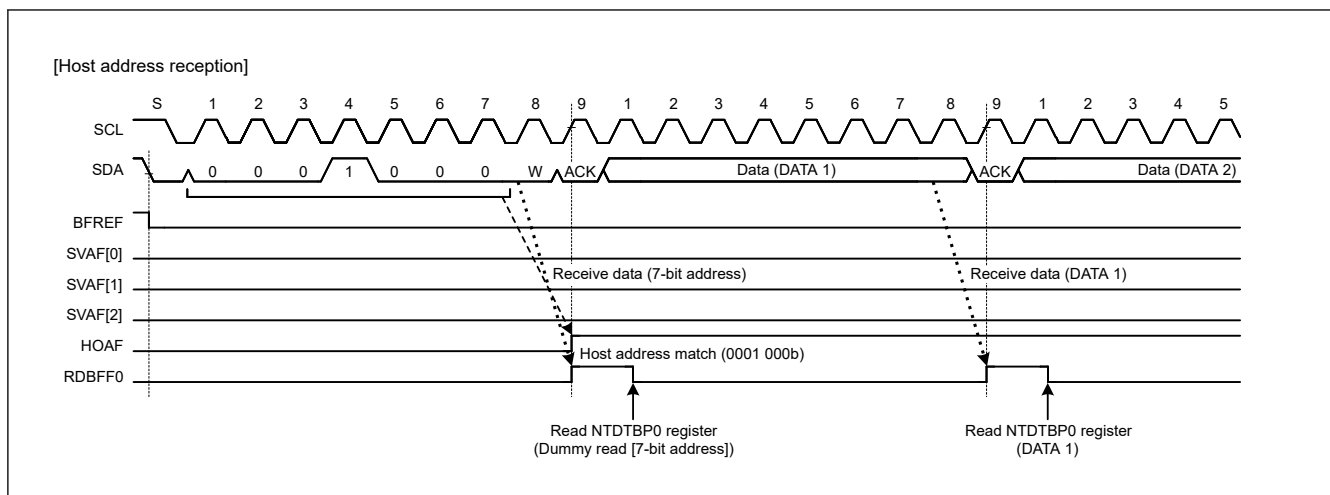


Figure 40.87 HOAF flag set timing during reception of host address

(5) Hs-mode master code Detection [I²C mode]

IIC has a facility for detecting the Hs-mode master code (0000 1XXXb). When IIC receives the Hs-mode master code (0000 1XXXb) as the first byte after a START condition was issued with the SVCTL.HSMCE bit set to 1, this module recognizes the address as the Hs-mode master code, sets the SVST.HSMCF flag to 1 on the rising edge of the ninth SCL clock cycle. The first byte after Repeated START after NACK response to Hs-mode master code is recognized as a slave address and compared with the slave address set by SVDVADy.SVAD[9:0] (y = 0 to 2). When IIC detects a match of the set slave address, the corresponding SVST.SVA[y] flag (y = 0 to 2) is set to 1 at the rising edge of the ninth SCL clock cycle, and the NTST.RDBFF0 flag or the NTST.TDBEF0 flag is set to 1 by the following R/W# bit. This causes a Normal Rx Data buffer full interrupt (I3C_RX) or Normal Tx Data buffer empty interrupt (I3C_TX) to be generated. The SVA[y] flag is used to identify which slave address has been specified. The SVST.HSMCF flag is cleared to 0 when the STOP condition is detected.

Note: If the Hs-mode master code (0000 1XXXb) is received with the SVCTL.HSMCE bit set to 0, other patterns are ignored until the STOP condition is detected.

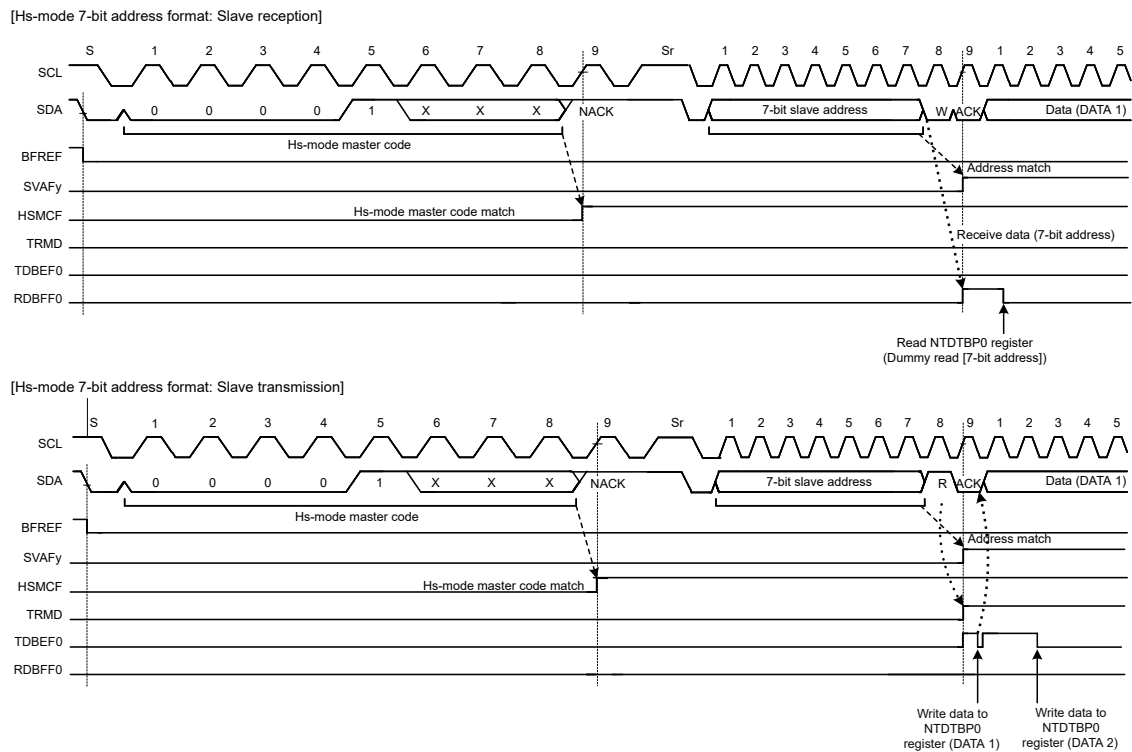


Figure 40.88 SVAfY/HSMCF Flag Set Timing during Reception of Hs-mode master code

(6) CCC detection function [I3C mode]

- In case of Broadcast CCC
 1. It receives Broadcast Address (0x7E) and R/W# = 0 after START condition or Repeated START.
 2. Respond to ACK.
 3. Receive Common Command Code (CCC).
 4. In accordance with the CCC, the following data is stored. (Storage destination: see [Table 40.11](#))
 5. Store the Receive Status Descriptor into the Receive Status Queue.
- In case of Broadcast CCC (ENTDAA)
 1. It receives Broadcast Address (0x7E) and R/W# = 0 after START condition.
 2. Respond to ACK.
 3. Receive ENTDAA.
 4. If receives Broadcast Address (0x7E) and R/W# = 1 after Repeated START.
 5. When the Dynamic Address is not assigned, ACK response is done.
 6. This Provisional ID (SDCTPIDH[31:0], SDCTPIDL[15:0]), BCR (SVDCT.TBCRn) and DCR (SVDCT.TDCR[7:0]) are transmitted.
 7. When winning the arbitration in a transmission of the above Step 6, the dynamic address following that is received. When losing arbitration in a transmission of the above Step 6, processing of Step 6 is repeated from Step 4.
 8. When parity of the Dynamic Address is valid, ACK response is done.
 9. When parity of the Dynamic Address is invalid, NACK replies, and repeat the process from Steps 4 to 7.
 10. SDATBAS0.SDDYAD[7:0] is renewed and the SVDVAD0.SDYADV bit is set to 1.

11. Upon detecting the STOP condition, Store the Receive Status Descriptor into the Receive Status Queue.

- In case of Broadcast CCC (ENTHDR*)

1. It receives Broadcast Address (0x7E) and R/W# = 0 after START condition or Repeated START.
2. Respond to ACK.
3. Receive ENTHDR*.
4. Transit to HDR mode according to ENTHDR*.
5. Receives the HDR command word.
6. Compare the Dynamic Address in the HDR command word with the assigned Dynamic Address, and if they match, transmit / receive according to the Read / Write bit.
If they do not match, wait for HDR Restart Pattern or HDR Exit Pattern + STOP.
7. When HDR Restart Pattern or HDR Exit Pattern + STOP is received, Receive Status Descriptor is stored in Receive Status Queue.
For HDR Restart Pattern, repeat Steps 5 to 6.

- In case of Direct Write CCC

1. It receives Broadcast Address (0x7E) and R/W# = 0 after START condition or Repeated START.
2. Respond to ACK.
3. Receive Common Command Code (CCC).
4. Receive Dynamic Address and R/W# = 0 after Repeated START.
5. Compare the received Dynamic Address with the assigned Dynamic Address, and if it matches, I3C responds with ACK.
If they do not match, it responds with NACK and waits for Repeated START or STOP.
6. In accordance with the CCC, the following data is stored. (Storage destination: see [Table 40.11](#))
7. Store the Receive Status Descriptor into the Receive Status Queue.

- In case of Direct Read CCC

1. It receives Broadcast Address (0x7E) and R/W# = 1 after START condition or Repeated START.
2. Respond to ACK.
3. Receive Common Command Code (CCC).
4. Receive Dynamic Address and R/W# = 1 after Repeated START.
5. Compare the received Dynamic Address with the assigned Dynamic Address, and if it matches, I3C responds with ACK.
If they do not match, it responds with NACK and waits for Repeated START or STOP.
6. Respond from SFR according to CCC. (Responding CCC: see [Table 40.11](#))
7. Store the Receive Status Descriptor into the Receive Status Queue.

Table 40.11 Common command code operation (1 of 2)

Command Code	CCC Type	Command Name	With Data	Auto Response	Storage
0x00	Broadcast	ENEC	Yes	—	SFR
0x01	Broadcast	DISEC	Yes	—	SFR
0x02	Broadcast	ENTAS0	No	—	SFR
0x03	Broadcast	ENTAS1	No	—	SFR
0x04	Broadcast	ENTAS2	No	—	SFR
0x05	Broadcast	ENTAS3	No	—	SFR
0x06	Broadcast	RSTDAA	No	—	SFR
0x07	Broadcast	ENTDAA	Yes	Yes	SFR
0x08	Broadcast	DEFSLVS	Yes	—	FIFO

Table 40.11 Common command code operation (2 of 2)

Command Code	CCC Type	Command Name	With Data	Auto Response	Storage
0x09	Broadcast	SETMWL	Yes	—	SFR
0x0A	Broadcast	SETMRL	Yes	—	SFR
0x0B	Broadcast	ENTTM	Yes	—	SFR
0x20	Broadcast	ENTHDR0	No	—	n/a
0x21	Broadcast	ENTHDR1	No	—	n/a
0x22	Broadcast	ENTHDR2	No	—	n/a
0x28	Broadcast	SETXTIME	Yes	—	FIFO
0x29	Broadcast	SETAASA	No	—	SFR
0x80	Direct Write	ENEC	Yes	—	SFR
0x81	Direct Write	DISEC	Yes	—	SFR
0x82	Direct Write	ENTAS0	No	—	SFR
0x83	Direct Write	ENTAS1	No	—	SFR
0x84	Direct Write	ENTAS2	No	—	SFR
0x85	Direct Write	ENTAS3	No	—	SFR
0x86	Direct Write	RSTDAA	No	—	SFR
0x87	Direct Write	SETDASA	Yes	—	SFR
0x88	Direct Write	SETNEWDA	Yes	—	SFR
0x89	Direct Write	SETMWL	Yes	—	SFR
0x8A	Direct Write	SETMRL	Yes	—	SFR
0x8B	Direct Read	GETMWL	—	Yes	SFR
0x8C	Direct Read	GETMRL	—	Yes	SFR
0x8D	Direct Read	GETPID	—	Yes	SFR
0x8E	Direct Read	GETBCR	—	Yes	SFR
0x8F	Direct Read	GETDCR	—	Yes	SFR
0x90	Direct Read	GETSTATUS	—	Yes	SFR
0x91	Direct Read	GETACCMST	—	Yes	SFR
0x94	Direct Read	GETMXDS	—	Yes	SFR
0x95	Direct Read	GETHDCAP	—	Yes	SFR
0x98	Direct Write	SETXTIME	Yes	—	FIFO
0x99	Direct Read	GETXTIME	—	Yes	SFR

40.3.2.3.5 Arbitration-Lost Detection [I²C Mode]

In addition to the normal arbitration-lost detection function defined by the I²C-bus specification, the I3C has functions to prevent double-issue of a start condition, to detect arbitration-lost during transmission of NACK, and to detect arbitration-lost in slave transmit mode.

(1) Master Arbitration-Lost Detection (MALE Bit)

The I3C drives the I3C_SDA line low to issue a start condition. However, if the I3C_SDA line has already been driven low by another master device issuing a start condition, this module causes arbitration to be lost, so priority is given to transfer by the other master device. Similarly, if the CNDCTL.STCND bit is set to 1 while the BCST.BFREF flag is 0 (bus busy state), arbitration is lost, so priority is given to transfer by the other master device. No start condition is issued in this case.

When a start condition is issued successfully, if the data for transmission including the address bits (the internal SDA output level) and the level on the I3C_SDA line do not match (the high output as the internal SDA output, that is, the I3C_SDA pin is in the high-impedance state) and the low level is detected on the I3C_SDA line, the I3C loses in arbitration.

I3C detects master arbitration-lost when the following conditions are met while the BSTE.ALE bit = 1 and the BFCTL.MALE bit = 1 (master arbitration-lost detection enabled).

If arbitration of mastership is lost, I3C immediately enters slave receive mode.

If a slave address (including the general call address) matches its own address at this time, I3C continues in slave operation.

[Conditions for master arbitration-lost]

- Non-matching of the internal level for output on SDA and the level on the I3C_SDA line after a START condition was issued by setting the CNDCTL.STCND bit to 1 while the BCST.BFREF flag was set to 1 (erroneous issuing of a START condition)
- Setting of the CNDCTL.STCND bit to 1 (START condition double-issue error) while the BFREF flag is set to 0

Note: I3C does not issue a START condition.

- When the transmit data excluding acknowledge (internal SDA output level) does not match the level on the I3C_SDA line in master transmit mode (bits CRMS and TRMD in the PRSST register = 11b)

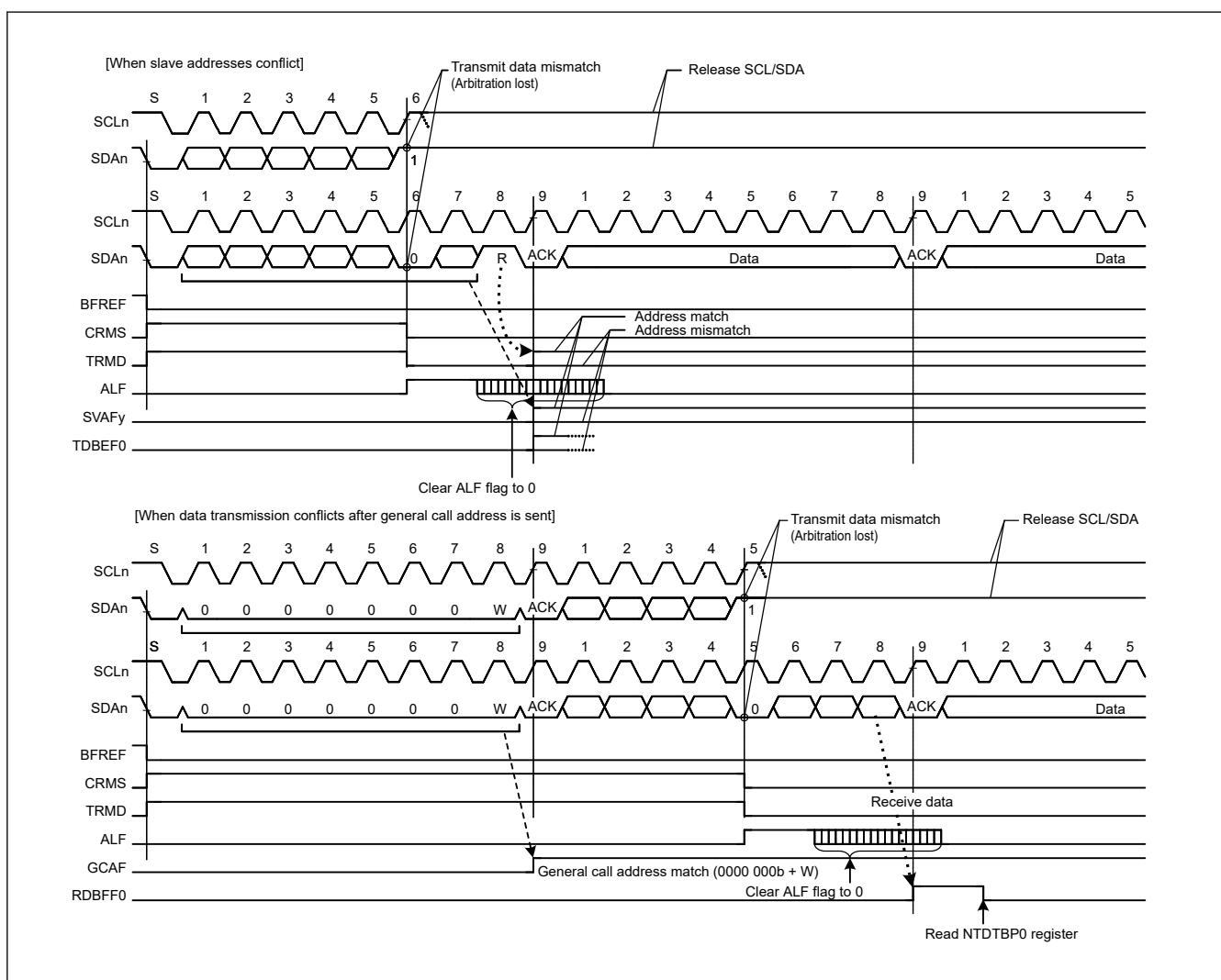


Figure 40.89 Examples of master arbitration-lost detection (MALE = 1)

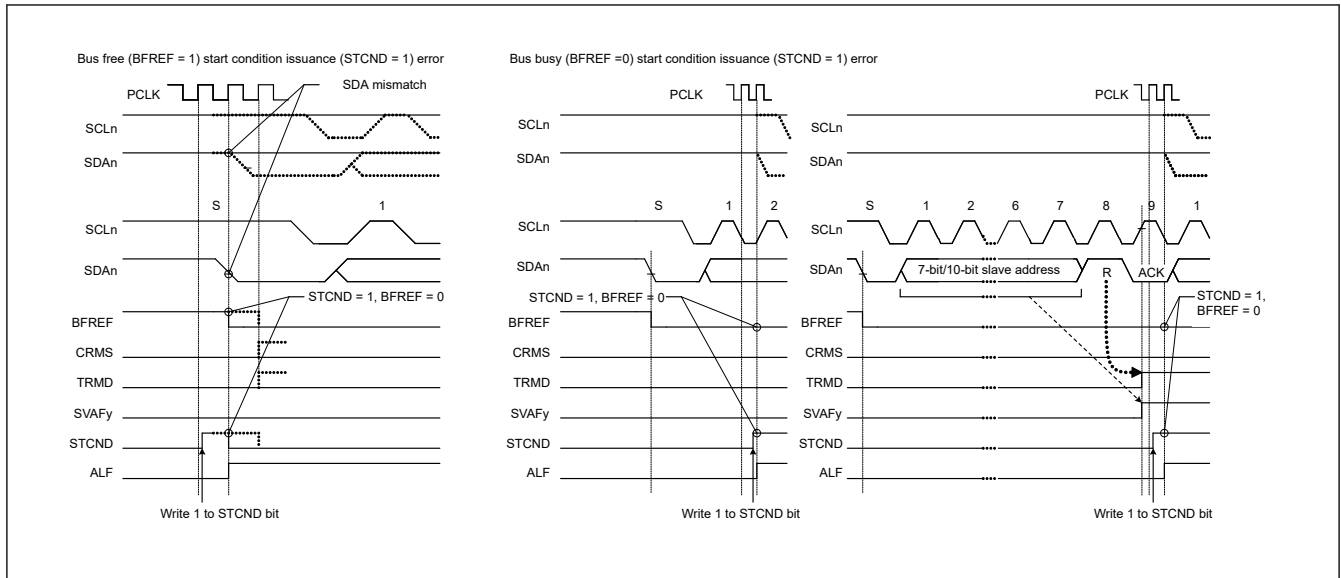


Figure 40.90 Arbitration-lost detection when a START condition is issued (MALE = 1)

(2) Arbitration-Lost Detection during NACK Transmission (NALE Bit)

The I3C has a function to cause arbitration to be lost if the internal SDA output level does not match the level on the I3C_SDA line (the high output as the internal SDA output; i.e. the I3C_SDA pin is in the high-impedance state) and the low level is detected on the I3C_SDA line during transmission of NACK in receive mode. Arbitration is lost due to a conflict of NACK transmission and ACK transmission when two or more master devices receive data from the same slave device simultaneously in a multi-master system. Such conflict occurs when multiple master devices send/receive the same information through a single slave device.

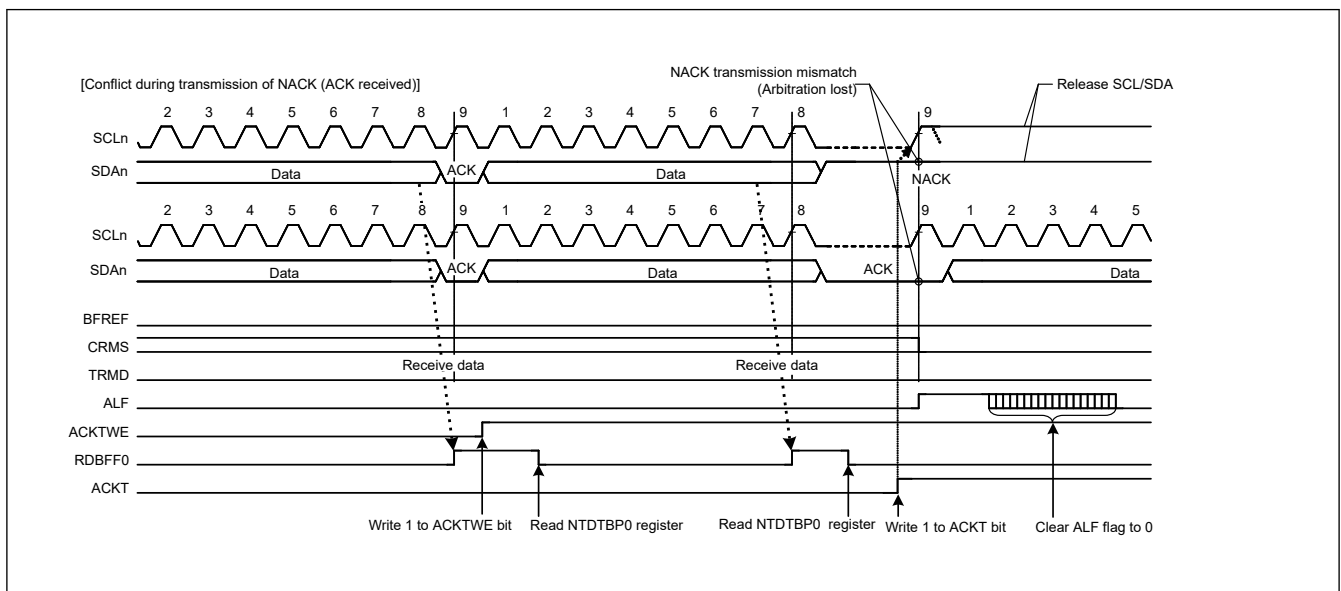


Figure 40.91 Example of arbitration-lost detection during transmission of NACK (NALE = 1)

The following section explains arbitration-lost detection using an example where two master devices (master A and master B) and a single slave device are connected through the bus. In this example, master A receives 2 bytes of data from the slave device, and master B receives 4 bytes of data from the slave device.

If master A and master B access the slave device simultaneously, because the slave address is identical, arbitration is not lost in both master A and master B during access to the slave device. Therefore, both master A and master B recognize that they have obtained the bus mastership and operate as such. In this example,, master A sends NACK when it has received 2 final bytes of data from the slave device. Meanwhile, master B sends ACK because it has not received the necessary 4 bytes of data. At this time, the NACK transmission from master A and the ACK transmission from master B conflict. In general, if

a conflict like this occurs, master A cannot detect ACK transmitted by master B and issues a stop condition. Therefore, the issuance of the stop condition conflicts with the SCL clock output of master B, which disturbs communication.

When this module receives ACK during transmission of NACK, it detects a defeat in conflict with other master devices and causes arbitration to be lost.

If arbitration is lost during transmission of NACK, this module is immediately released from the slave-matched state and enters slave receive mode. This prevents a stop condition from being issued, preventing a communication failure on the bus.

Similarly, in the ARP command processing of SMBus, the function to detect loss of arbitration during transmission of NACK is also available for eliminating the extra clock cycle processing (such as 0xFF transmission processing) necessary if the UDID (Unique Device Identifier) of assign address does not match in the Get UDID (general) processing after the Assign Address command.

The I3C detects arbitration-lost during transmission of NACK when the following condition is met while the BSTE.ALE bit = 1 and the BFCTL.NALE bit = 1 (arbitration-lost detection during NACK transmission enabled).

[Condition for arbitration-lost during NACK transmission]

- When the internal SDA output level does not match the I3C_SDA line (ACK is received) during transmission of NACK (ACKCTL.ACKT bit = 1)

(3) Slave Arbitration-Lost Detection (SALE Bit)

The I3C has a function to cause arbitration to be lost if the data for transmission (the internal SDA output level) and the level on the I3C_SDA line do not match (the high output as the internal SDA output, that is, the I3C_SDA pin is in the high impedance state) and the low level is detected on the I3C_SDA line in slave transmit mode. This arbitration-lost detection function is mainly used when transmitting a UDID (Unique Device Identifier) over an SMBus.

If arbitration is lost during transmission of DATA, this module is immediately released from the slave-matched state and enters slave receive mode. This function can detect conflicts of data during transmission of UDIDs over an SMBus and eliminate subsequent redundant processing (processing for the transmission of 0xFF).

The I3C detects slave arbitration-lost when the following condition is met while the BSTE.ALE bit = 1 and the BFCTL.SALE bit = 1 (slave arbitration-lost detection enabled).

[Condition for slave arbitration-lost]

- When the transmit data excluding acknowledge (internal SDA output level) does not match the level on the I3C_SDA line in slave transmit mode (bits CRMS and TRMD in the PRSST register = 01b).

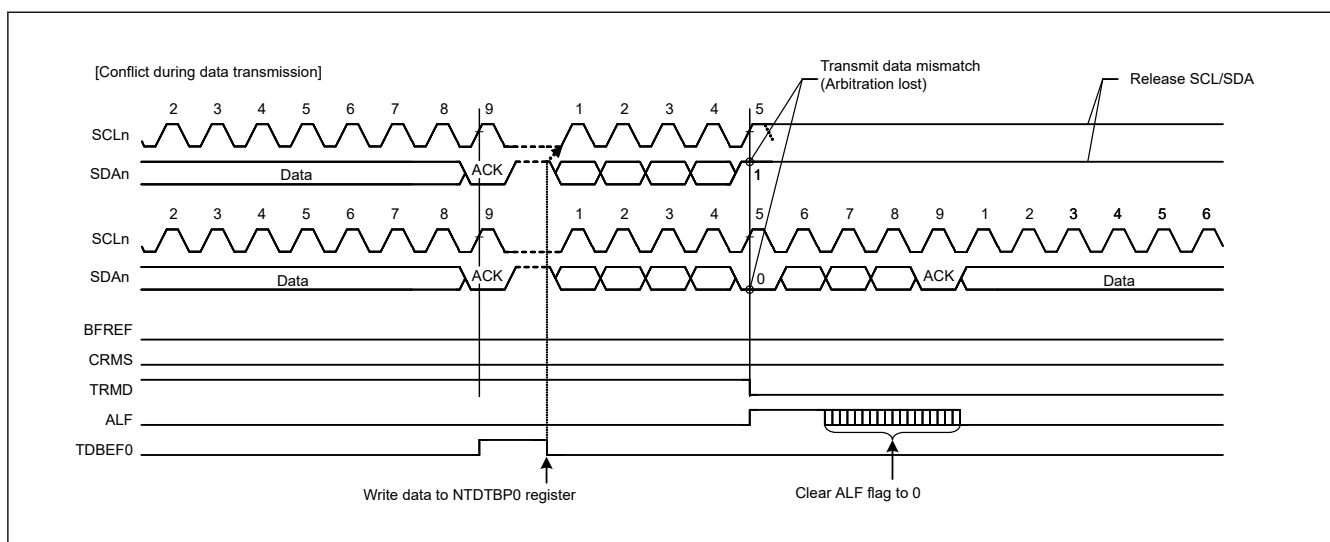


Figure 40.92 Example of slave arbitration-lost detection (SALE = 1)

40.3.2.3.6 Clock Stretching [I²C Mode]

(1) Function to Prevent Wrong Transmission of Transmit Data

When data have not been written to the Normal Transfer Data Buffer Port Register 0 (NTDTBP0) with I3C in transmission mode (PRSS.TRMD = 1), the I3C_SCL line is automatically held at the low level over the intervals shown below. This low-hold period is extended until data for transmission have been written, which prevents the unintended transmission of erroneous data.

Master transmit mode

- Low-level interval after a START condition or Repeated START condition is issued
- Low-level interval between the ninth clock cycle of one transfer and the first clock cycle of the next

Slave transmit mode

- Low-level interval between the ninth clock cycle of one transfer and the first clock cycle of the next

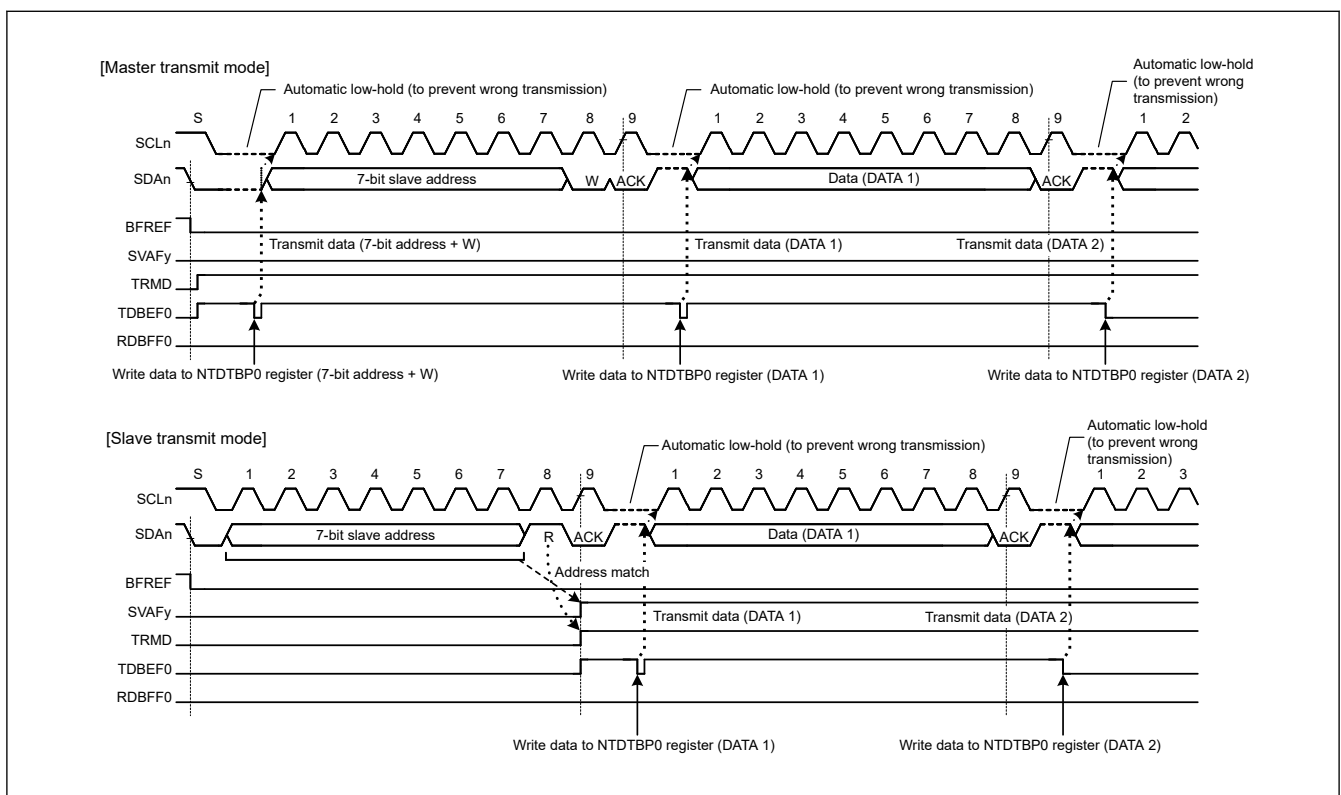


Figure 40.93 Automatic low-hold operation in transmit mode

(2) NACK Reception Transfer Abort Function

I3C has a function to abort transfer operation when NACK is received in transmit mode (PRSS.TRMD = 1). This function is enabled when the BSTE.NACKDE bit is set to 1 (transfer abort enabled). If the next transmit data has already been written (NTST.TDBEF0 = 0) when NACK is received, next data transmission at the falling edge of the ninth SCL clock cycle is automatically aborted. This prevents the I3C_SDA line output level from being held low when the MSB of the next transmit data is 0.

If the transfer operation is aborted by this function (BST.NACKDF = 1), transmit operation and receive operation are discontinued. To restore transmit/receive operation, be sure to set the NACKDF flag to 0. In master transmit mode, restore operation using either of the methods below:

- After issuing a Repeated START condition, set the NACKDF flag to 0
- After issuing a STOP condition, set the NACKDF flag to 0 and then issue a START condition

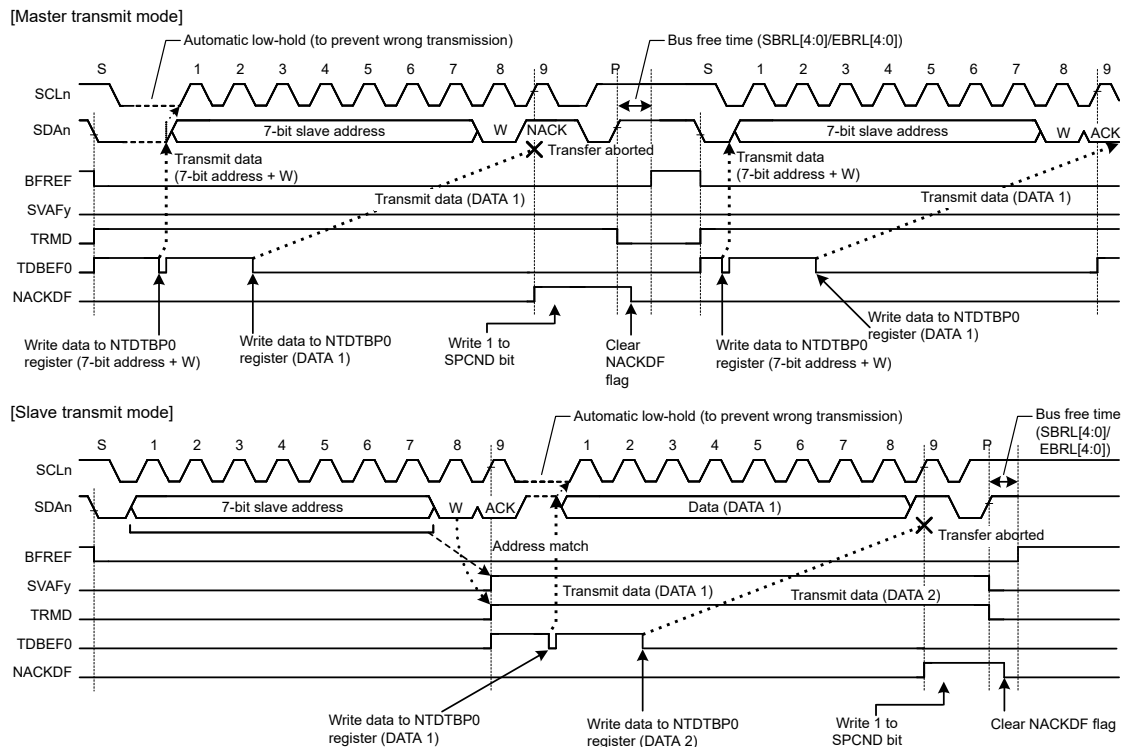


Figure 40.94 Abort of data transfer when NACK is received (NACK = 1)

(3) Function to Prevent Failure to Receive Data

If response processing is delayed when receive data (NTDTBP0) read is delayed for a period of one transfer frame or more with receive data full (NTST.RDBFF0 = 1) in receive mode (PRST.TRMD = 0), I3C holds the I3C_SCL line low automatically immediately before the next data is received to prevent failure to receive data.

This function to prevent failure to receive data using the automatic low-hold function is also enabled even if the read processing of the final receive data is delayed and, in the meantime, I3C's own slave address or another slave address is received after a STOP condition is issued.

Sections in which the I3C_SCL line is held low can be selected with a combination of the RWE and ACKTWE bits in SCSTRCTL.

(a) 1-Byte Receive Operation and Automatic Low-Hold Function Using the RWE Bit

When the SCSTRCTL.RWE bit is set to 1, I3C performs 1-byte receive operation using the RWE bit function.

Furthermore, when the SCSTRCTL.ACKTWE bit = 0, I3C automatically sends the ACKCTL.ACKT bit value for the acknowledge bit in the period from the falling edge of the eighth SCL clock cycle to the falling edge of the ninth SCL clock cycle, and automatically holds the I3C_SCL line low at the falling edge of the ninth SCL clock cycle using the RWE bit function. This low-hold is released by reading data from NTDTBP0, which enables byte-wise receive operation.

The RWE bit function is enabled for receive frames after a match with I3C's own slave address (including the general call address and host address) is obtained in master receive mode or slave receive mode.

(b) 1-Byte Receive Operation (ACK/NACK Transmission Control) and Automatic Low-Hold Function Using the ACKTWE Bit

When the SCSTRCTL.ACKTWE bit is set to 1, I3C performs 1-byte receive operation using the ACKTWE bit function.

When the ACKTWE bit is set to 1, the NTST.RDBFF0 flag (Normal Rx Data buffer full) is set to 1 at the rising edge of the eighth SCL clock cycle, and the I3C_SCL line is automatically held low at the falling edge of the eighth SCL clock

cycle. This low-hold is released by writing a value to the ACKCTL.ACKT bit, but cannot be released by reading data from NTDTBP0, which enables receive operation by the ACK/NACK transmission control according to the data received in byte units.

The ACKTWE bit function is enabled for receive frames after a match with I3C's own slave address (including the general call address and host address) is obtained in master receive mode or slave receive mode.

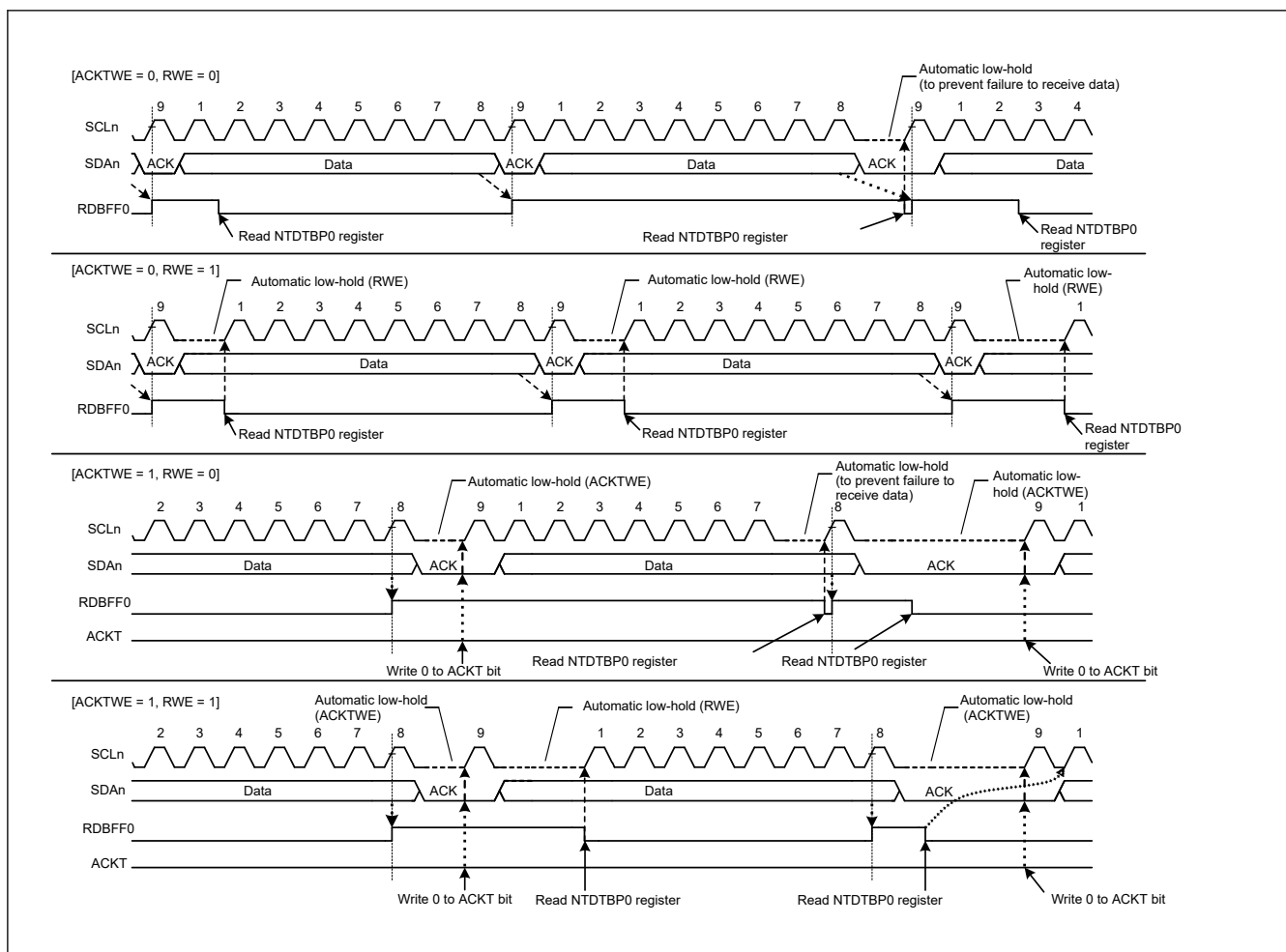


Figure 40.95 Automatic low-hold operation in receive mode (using ACKTWE and RWE bits)

40.3.2.3.7 Clock Stalling [I3C Mode]

I3C has the function of stalling the SCL during the SCL Low period.

The SCL stall control is described in the table below.

Table 40.12 I3C clock stalling (1 of 2)

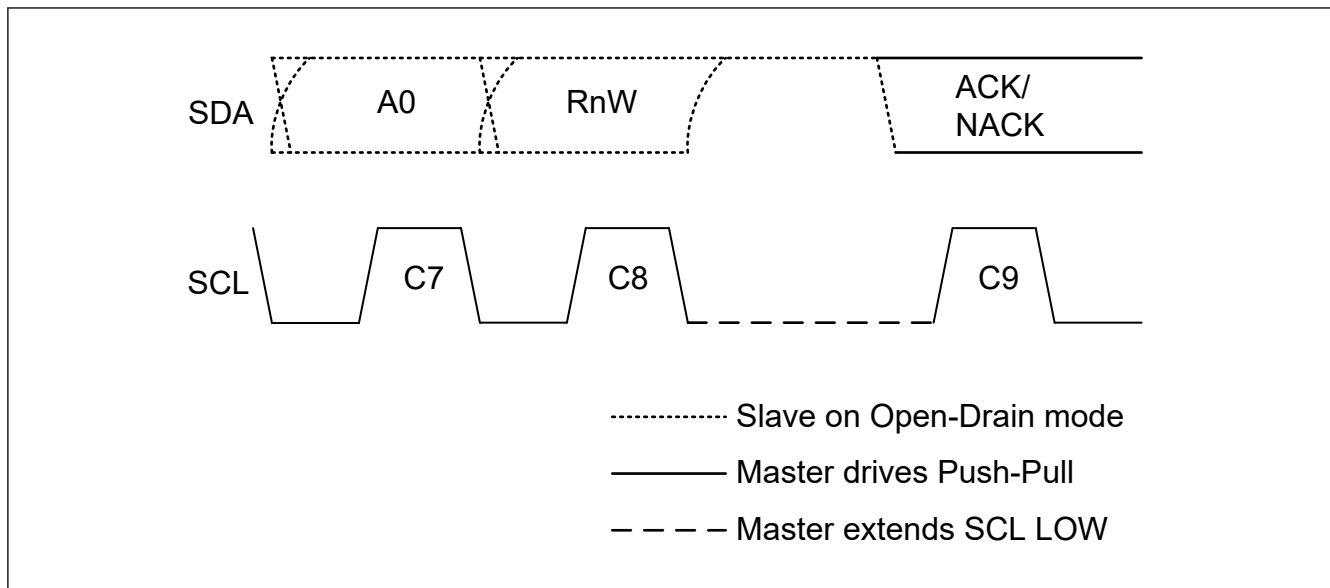
Clock stalling condition	Clock stalling control	Clock stalling period
I3C Transfer, ACK/NACK Phase	SCSTLCTL.ACKPE bit setting	During the count period of SCSTLCTL.STLCYC [15:0] value
	Tx Data Buffer Empty	Until data is written to the Tx Data Buffer
	Rx Data Buffer Full	Until data is read from the Rx Data Buffer
I3C Write Data Transfer, Parity Bit	SCSTLCTL.PARPE bit setting	During the count period of SCSTLCTL.STLCYC [15:0] value
	Tx Data Buffer Empty	Until data is written to the Tx Data Buffer
I3C Read Transfer, Transition Bit	Rx Data Buffer Full	Until data is read from the Rx Data Buffer

Table 40.12 I3C clock stalling (2 of 2)

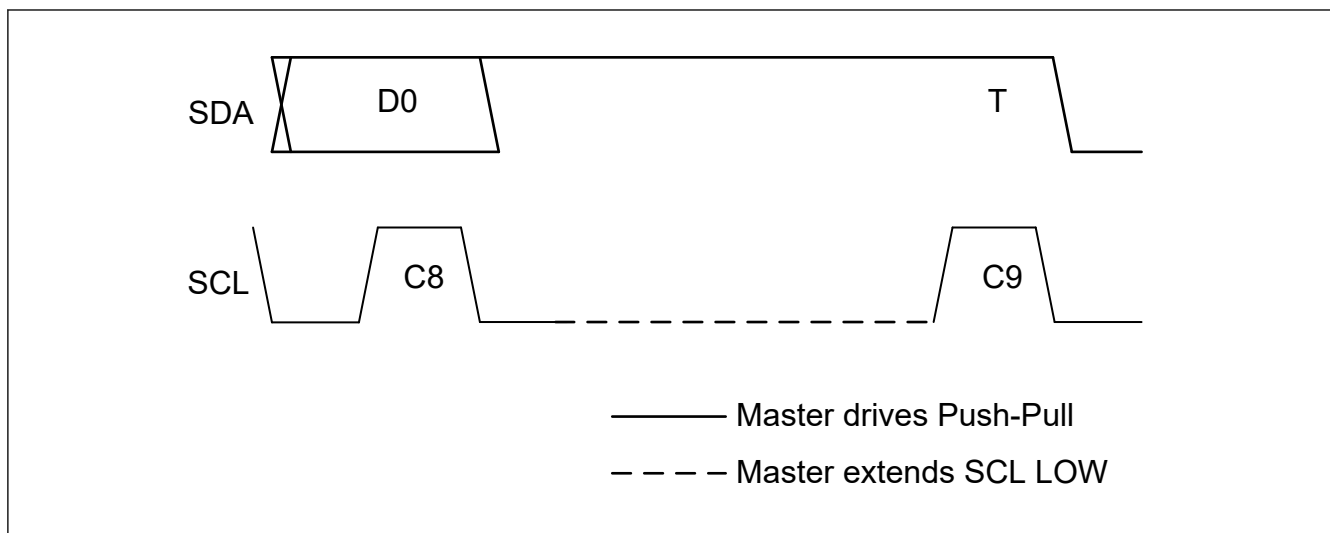
Clock stalling condition	Clock stalling control	Clock stalling period
Assigned Address Phase	SCSTLCTL.AAPE bit setting	During the count period of SCSTLCTL.STLCYC [15:0] value

The following figure shows the stalling timing of each Condition.

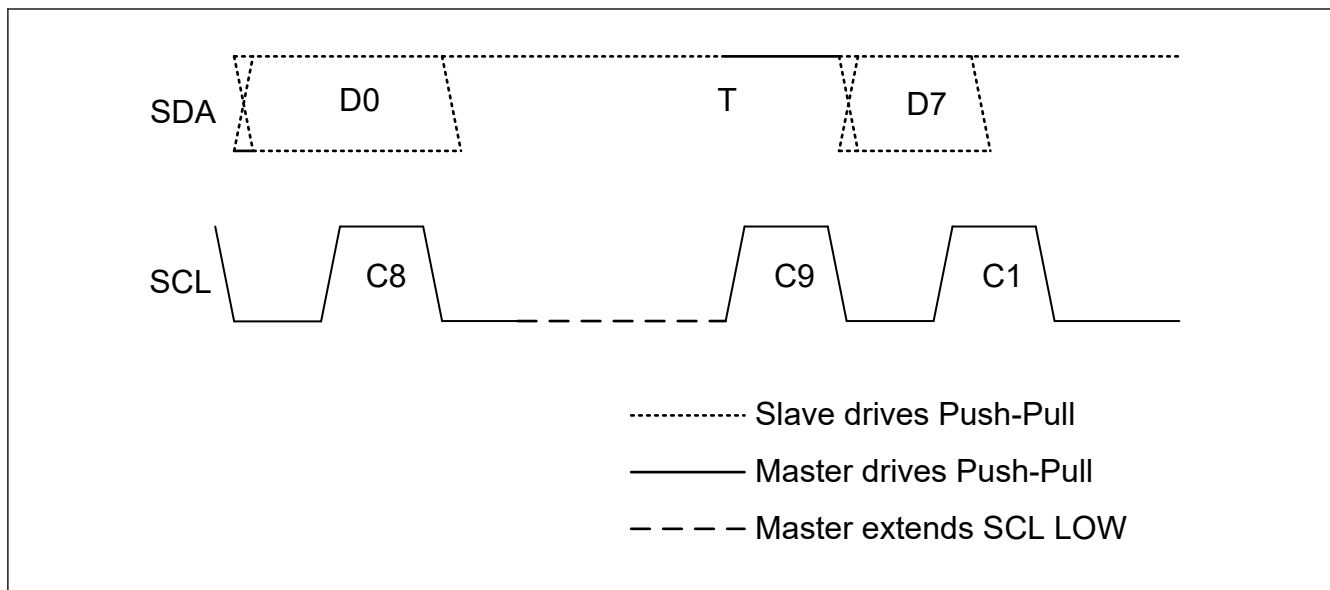
(1) I3C Transfer, ACK/NACK Phase

**Figure 40.96 Master clock stalling in ACK phase**

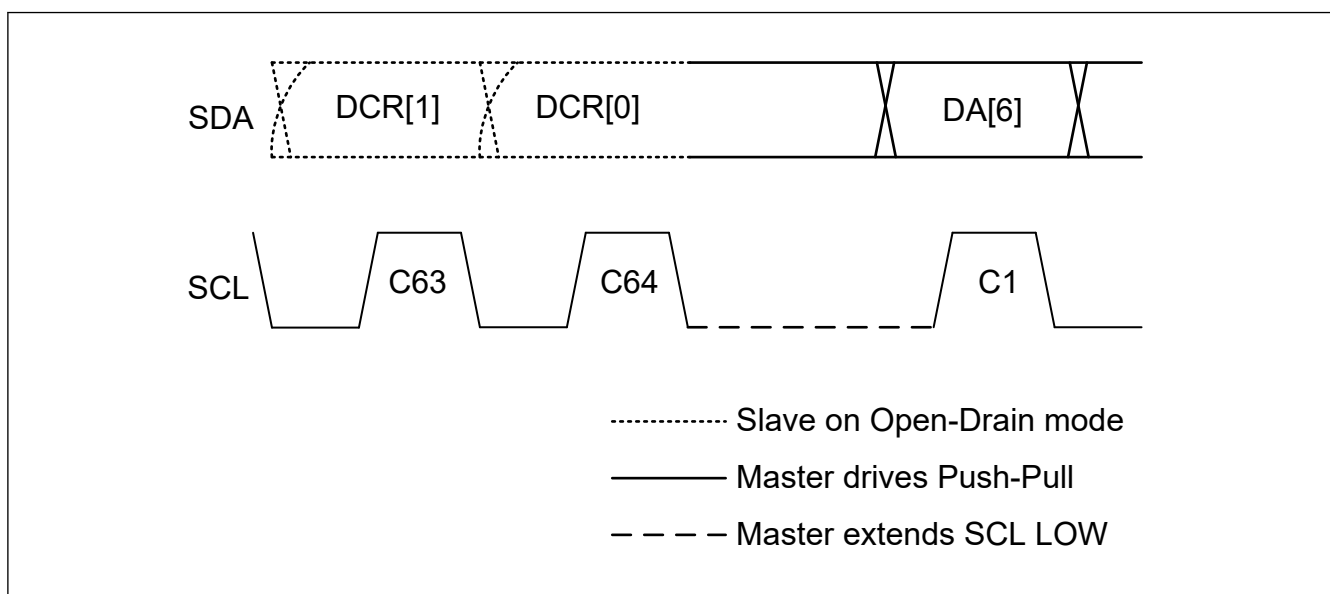
(2) I3C Write Data Transfer, Parity Bit

**Figure 40.97 Master clock stalling in write parity bit**

(3) I3C Read Transfer, Transition Bit

**Figure 40.98** Master clock stalling in T-bit before next read data

(4) Dynamic Address Assignment, First Bit of Assigned Address

**Figure 40.99** Master clock stalling in dynamic address first bit

40.3.2.3.8 In-Band Interrupt [I3C Mode]

I3C detects In-Band Interrupt in the arbitrated Address Header following a START condition (but not following a Repeated START). If START Request (SDA Low Drive) is issued from Slave Device, I3C drives SCL low and completes START condition. After that, it supplies SCL and receives In-Band Interrupt Request.

The In-Band Interrupt to be detected is classified into the following three types.

- Slave Interrupt Request
- Mastership Request

The operation when detecting each In-Band Interrupt is described below.

(1) Slave Interrupt Request

1. Detect Slave Address with RnW bit High in Address Header.
2. Compare the detected Slave Address with the DVDYAD[7:0] in each DAT (DATBASm register).
3. When it does not match DAT.DVDYAD[7:0]:
Responds NACK, then issues the STOP condition.
When it matches the DAT.DVDYAD[7:0] bits and the DAT.DVSIRRJ bit = 1:
It operates in the following order:
 - (a) Responds NACK.
 - (b) Issues Repeated START condition, then automatically issues Direct DISEC CCC to the detected Slave.
 - (c) Issues the STOP condition.

When it matches the DAT.DVDYAD[7:0] bits and the DAT.DVSIRRJ bit = 0:
Responds ACK.

4. When DAT.DVIBIPL = 0:
Issues the STOP condition.
When DAT.DVIBIPL = 1:
Drives the SCL to receive the IBI Data from the Slave following the ACK response and receives IBI Data.
It stores the received IBI Data into the IBI Data Queue.
Each time IBI Data of the size set by the NQTHCTL.IBIDSSZ[7:0] bits is received, the IBI Status Descriptor is stored in the IBI Queue.
5. After detection of Low of T-bit following IBI Data, issues STOP condition.
6. After issues of STOP condition
NACK response:
 - If IBINCTL.NRSIRCTL = 0, the IBI Status Descriptor is not stored into the IBI Queue.
 - If IBINCTL.NRSIRCTL = 1, the IBI Status Descriptor is stored into the IBI Queue.

ACK response:

Stores the IBI Status Descriptor into the IBI Queue.

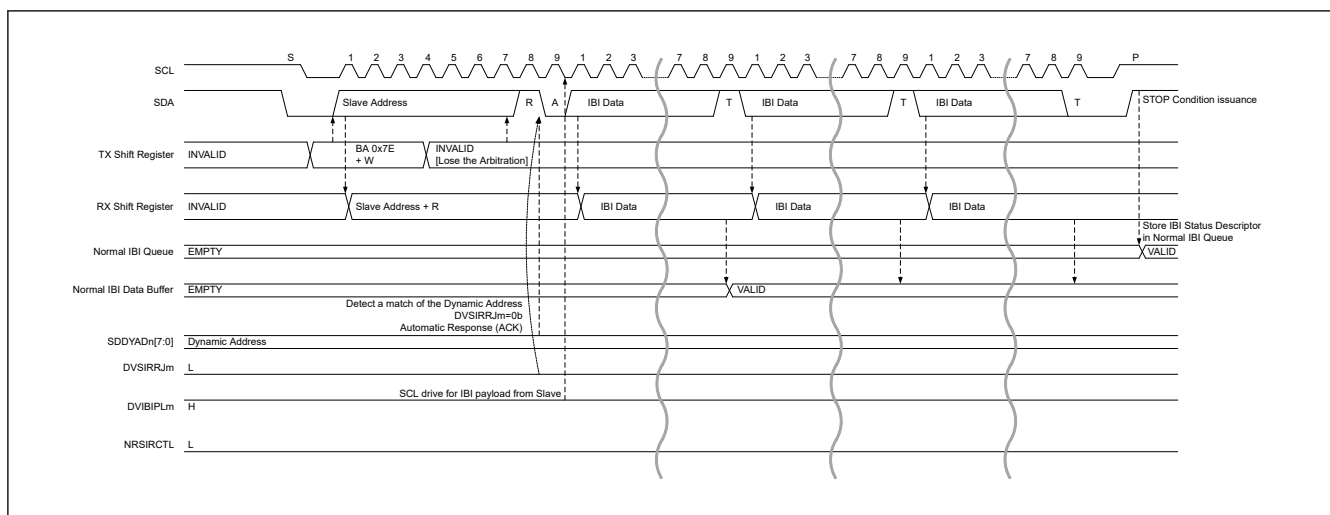


Figure 40.100 Slave interrupt request : ACK and DVIBIPL = 1

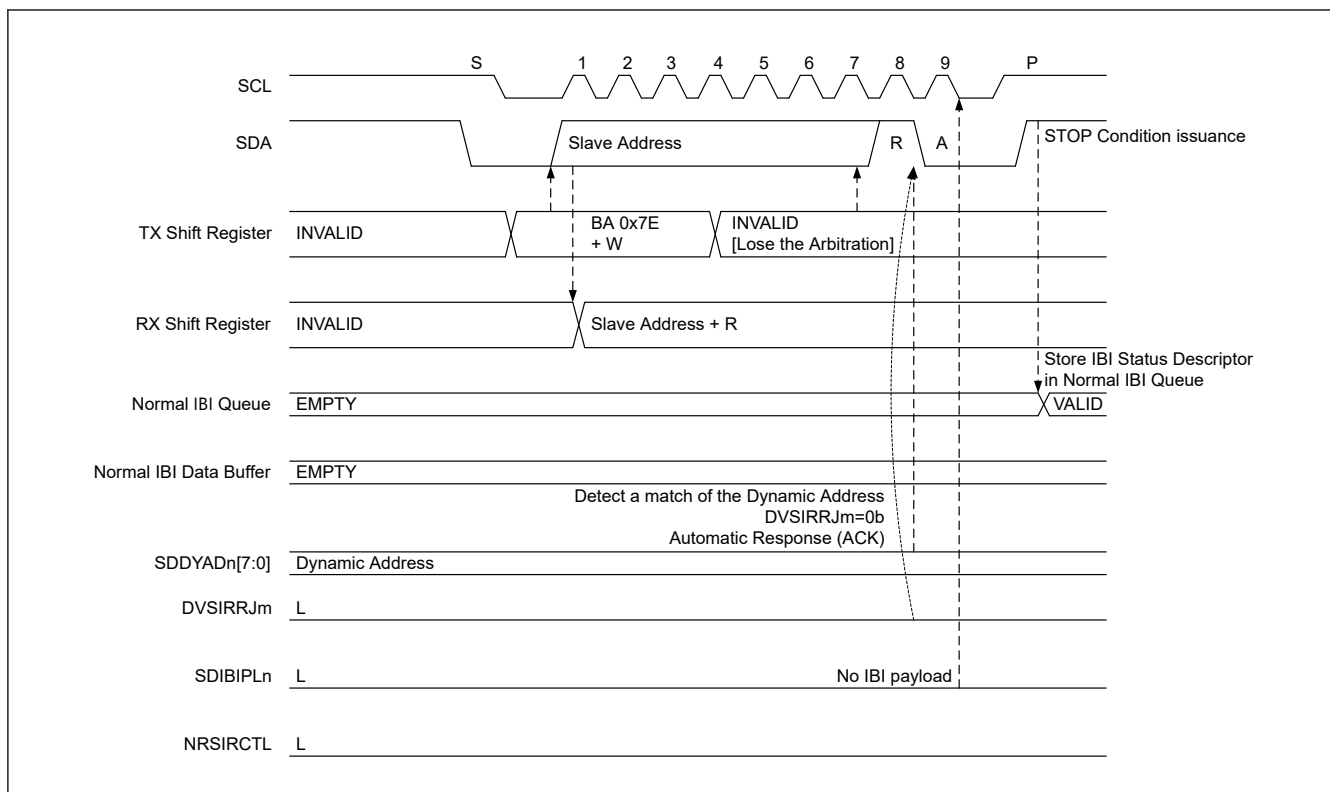


Figure 40.101 Slave interrupt request : ACK and DVIBIPL = 0

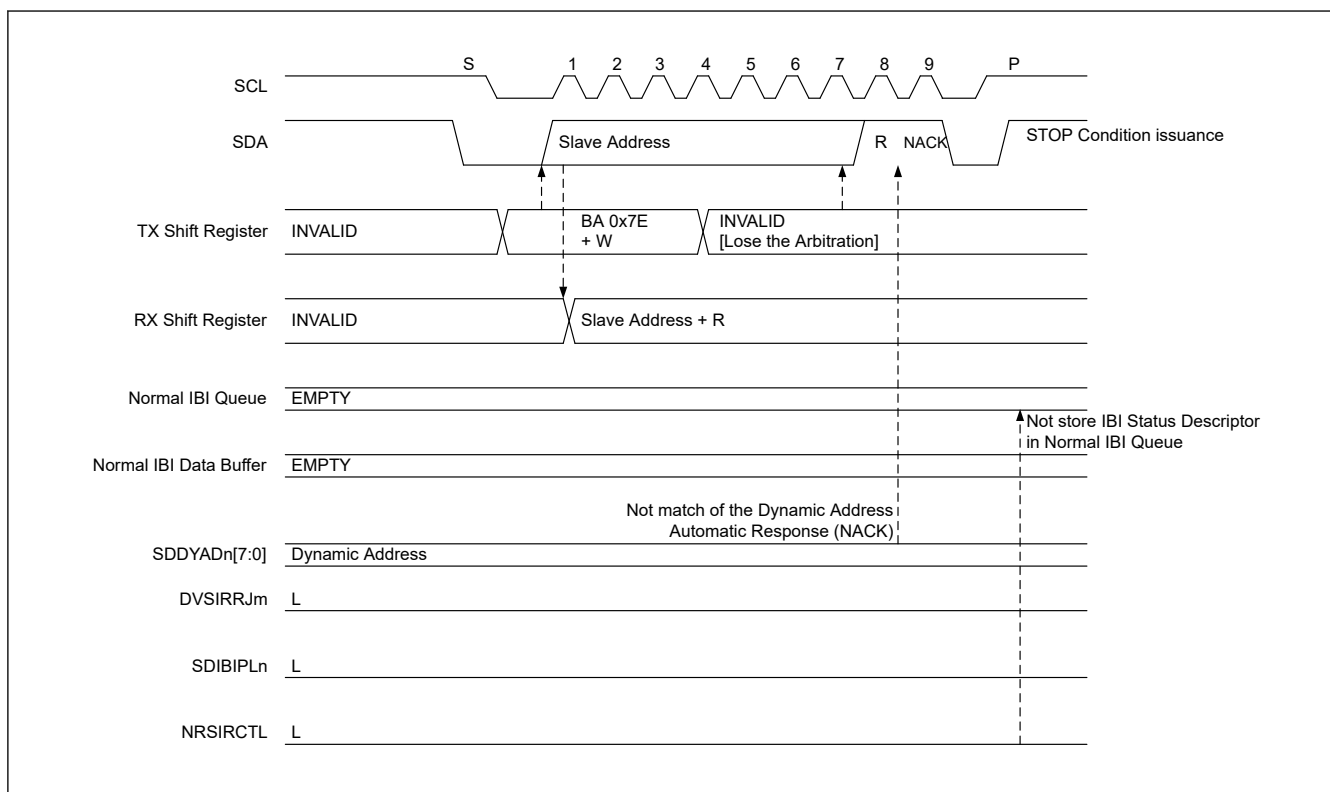


Figure 40.102 Slave interrupt request : NACK (not match the SDDYAD[7:0] of DAT) and NRSIRCTL = 0

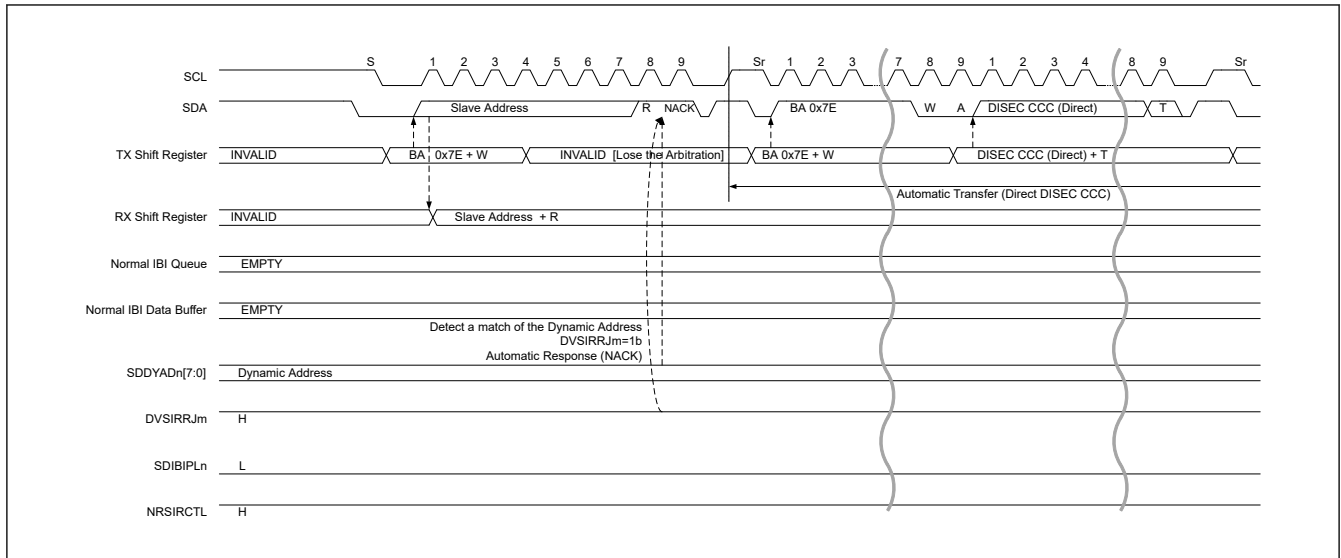


Figure 40.103 Slave interrupt request : NACK (DVSIRRJ = 1) and NRSIRCTL = 1 (1/2)

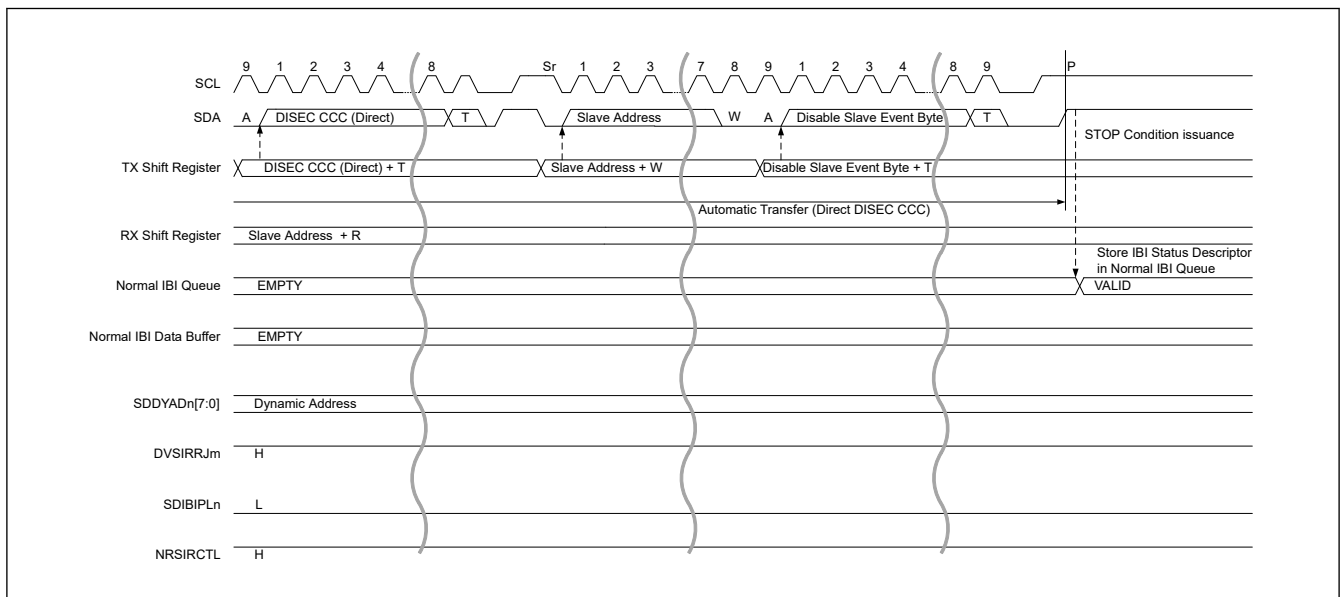


Figure 40.104 Slave interrupt request : NACK (DVSIRRJ = 1) and NRSIRCTL = 1 (2/2)

(2) Mastership Request

1. Detect Slave Address with RnW bit Low in Address Header.
2. Compare the detected Slave Address with the DVDYAD[7:0] in each DAT (DATBAS register).
3. When it does not match DAT.DVDYAD[7:0]:
Responds NACK, then issues the STOP condition.
When it matches the DAT.DVDYAD[7:0] bits and Device Role[1:0] in RBCR (MSDCTm) is other than I3C Master (01b):
Responds NACK, then issues the STOP condition.
When it matches the DAT.DVDYAD[7:0] bits and Device Role[1:0] in RBCR (MSDCTm) is I3C Master (01b):
 - When DAT.DVMRRJ = 1
It operates in the following order.
 - (a) Responds NACK.
 - (b) Issued Repeated START condition and automatically issues Direct DISEC CCC to the detected Slave.
 - (c) Issues the STOP condition.

- When $\text{DAT.DVMRRJ} = 0$
Responds ACK, then issues STOP condition.

4. After issues of STOP condition,
NACK response:

- If $\text{IBINCTL.NRMRCTL} = 0$, the IBI Status Descriptor is not stored into the IBI Queue.
- If $\text{IBINCTL.NRMRCTL} = 1$, the IBI Status Descriptor is stored into the IBI Queue.

ACK response:

Stores the IBI Status Descriptor into the IBI Queue.

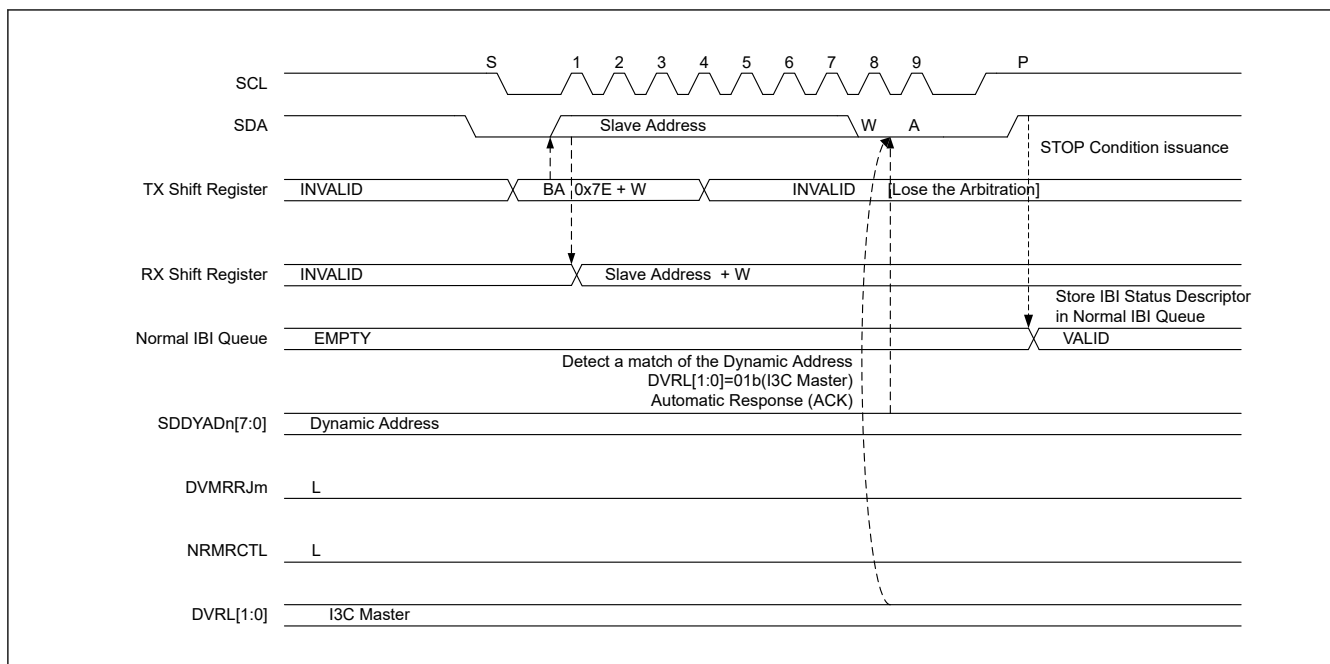


Figure 40.105 Mastership request : ACK

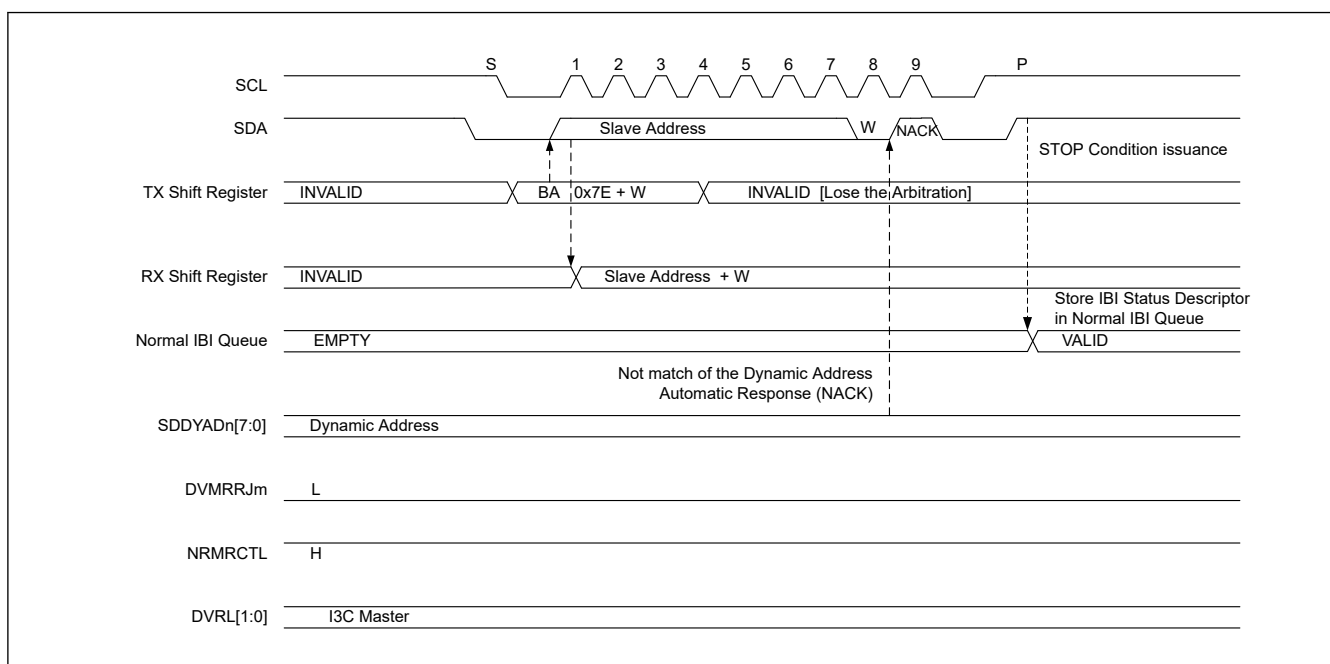


Figure 40.106 Mastership request : NACK (not match the DVDYAD[7:0] of DAT) and NRMCTL = 1

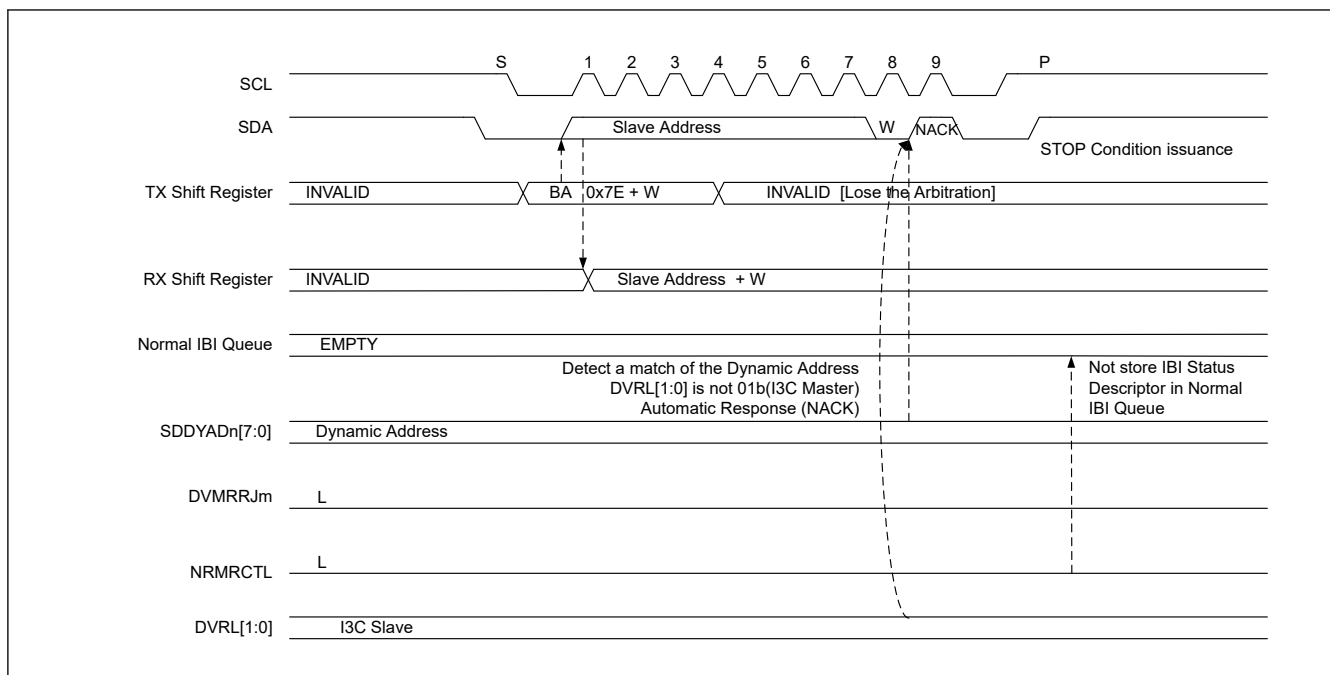


Figure 40.107 Mastership request : NACK (Device Role[1:0] is not 01 (I3C master) and NRMRCTL = 0

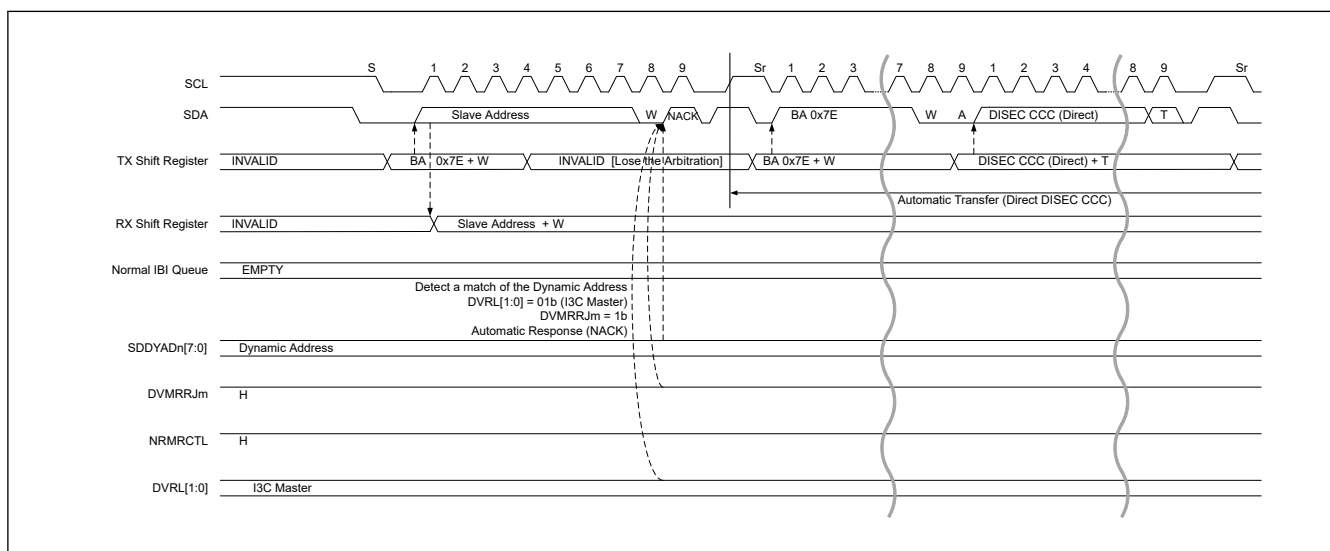


Figure 40.108 Mastership request : NACK (DVMRRJ = 1) and NRMRCTL = 1 (1/2)

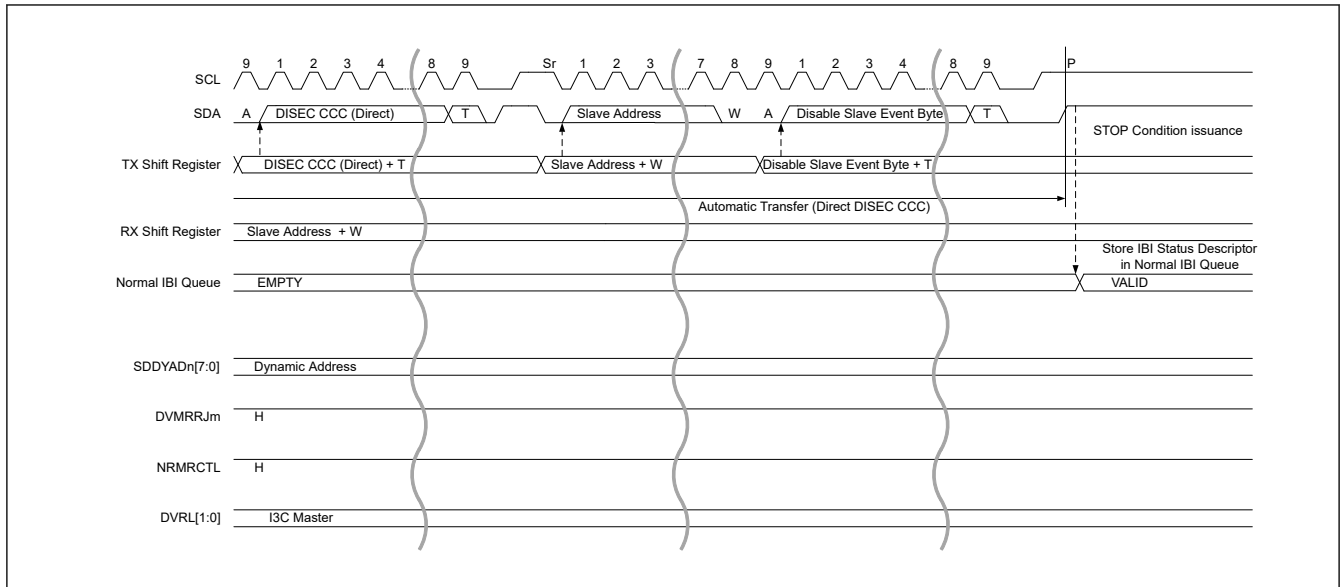


Figure 40.109 Mastership request : NACK (DVMRRJ = 1) and NRMRACTL = 1 (2/2)

40.3.2.3.9 Timing Control

Timing Control is a function that enables Master to efficiently read data from Slave by controlling and grasping the timing at which the Slave Device samples the sensor value.

For details, refer to 5.1.8 Timing Control of MIPI I3C Specification v1.0.

In I3C, timing Control supports the following three modes.

- Sync mode
- Async mode 0 (Asynchronous Basic mode)
- Async mode 1 (Asynchronous Advanced mode)

The resources for realizing Timing Control in each mode are described in the following sections.

(1) Sync Mode

1. I3C Master

- When STCTL.STOE is set to 1, when the master sends an ST message (SETXTIME CCC with ST subcommand), there is a function to issue the synchronous timing event under the START condition of the ST message. While measuring the T_{ph} period with an external timer, the start of T_{ph} and the Delay Time [DT] of the ST message can be measured by capturing the count value with the synchronous timing event. The measured value of Delay Time is sent as a DT message (SETXTIME CCC with DT subcommand) following the ST message.

2. I3C Slave

- When STCTL.STOE is set to 1, there is a function to issue the synchronous timing event for each START condition. STCTL.STOE is cleared when an ST message is received (SETXTIME CCC using the ST subcommand). Check the reception of the ST message with the Receive Status Descriptor, and use an external timer to correct the T_{ph} period based on the count value captured in the synchronous timing event and the Delay Time obtained from the DT message. While measuring the T_{ph} period with an external timer, the start of T_{ph} and the Delay Time [DT] of the ST message can be measured by capturing the count value with the synchronous timing event. The sampling timing is recalculated by the corrected T_{ph} .

(2) Async Mode 0 (Asynchronous Basic Mode)

For timing control in Async Mode 0, set the ATCTL register if necessary.

1. I3C Master

I3C has counters of MREFMREF(32bit) and MC2(16bit) for Async mode 0.

- **MREF Counter**
When ATCCNTE.ATCE is enabled, it starts counting.
It captures as MREF on the SCL rise edge next to ACK for the IBI transmitted from the I3C Slave.
- **MC2 Counter**
After enabling ATCCNTE.ATCE, it counts up from the SCL rise edge next to ACK for the IBI transmitted from the I3C Slave to the SCL rise edge next to the Tbit after Mandatory Byte, and capture it as MC2.

The MREF and MC2 capture values are stored next to the IBI Status Descriptor when IBI is received from the I3C Slave with the DATBASm.DVIBITS bit set to 1.

The MREF counter implemented in I3C is 32-bit counter.

However, if the 32-bit counter is insufficient due to system requirements, I3C has MREF counter overflow and MREF capture event for expansion. These events are enabled by setting ATCTL.MREFOE to 1.

MREF counter overflow is output when the internal MREF counter overflows.

The MREF counter can be extended by using it as a count event for an external timer. MREF capture event is output at the same timing as the capture timing of the internal MREF counter. By using it as the capture timing of the external timer, it can be used as an MREF counter concatenated with the value stored in the IBI Data buffer.

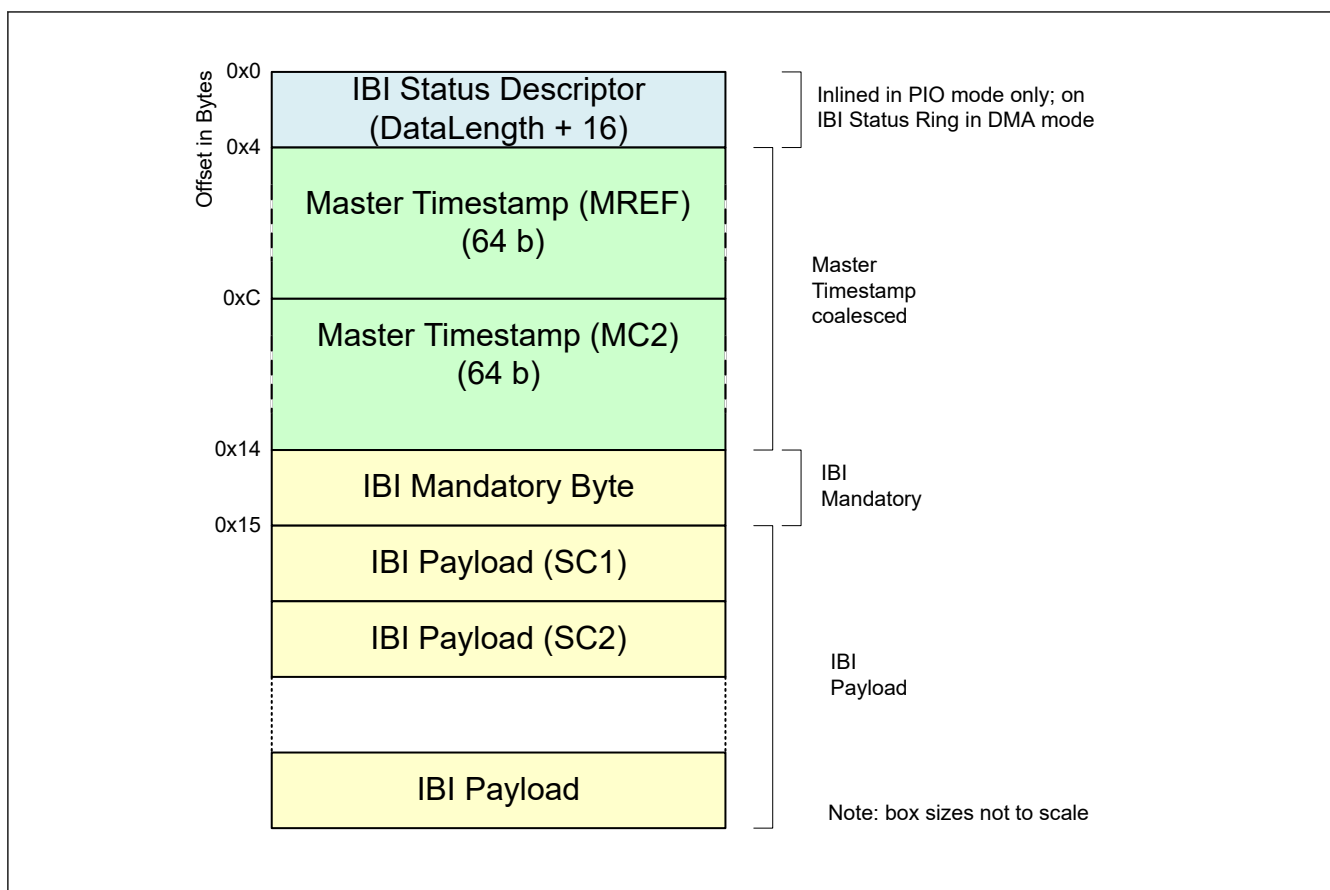


Figure 40.110 Master timestamp counters for IBI event

Note: Please evaluate the Sensor Event time of I3C Slave according to the calculation formula of the MIPI I3C specification v1.0 document.

2. I3C Slave

I3C has counters of SC1(16bit) and SC2(8bit) for Async mode 0.

- **SC1 Counter**

After enabling ATCCNTE.ATCE, it counts up from SC1 count trigger^{*1} to SCL rise edge next to ACK for the IBI, and capture it as SC1.

Note 1. SW or external trigger can be selected by selection bits.

- SC2 Counter

After enabling ATCCNTE.ATCE, it counts up from SCL rise edge next to ACK for the IBI transmitted from I3C Slave to SCL rise edge next to Tbit after Mandatory Byte, and capture it as SC2.

When the CETSS.ASYNE [0] bit = 1 and the ITS bit in Command Descriptor for issuing IBI is 1, the SC1 and SC2 capture values are transmitted following the IBI Mandatory Byte as shown in the following figure.

If the SC1 and SC2 counters overflow, 0xFFFF and 0xFF are captured and CETSS.ICOVF is set 1.

The DATA_LENGTH[15:0] bits value of the Command Descriptor sets a value obtained by adding the number of data of SC1 and SC2 to the number of transmission data.

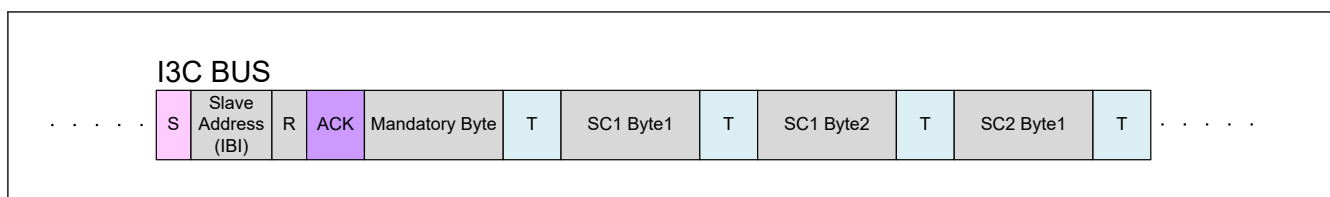


Figure 40.111 Example of asynchronous mode 0 timestamp data transfer

(3) Async Mode 1 (Asynchronous Advanced Mode)

For timing control in Async Mode 1, set the ATCTL register if necessary.

1. I3C Master

I3C has counters of MREF(32bit), MSyncCNT(32bit) and MC2(16bit) for Async mode 1.

- MREF Counter

When ATCCNTE.ATCE is enabled, it starts counting.

It captures as MREF at the SCL rise edge next to ACK for the IBI transmitted from the I3C Slave.

- MSyncCNT Counter

When ATCCNTE.ATCE is enabled, it starts counting.

It captures as MSyncCNT for each aME (SDA falling edge of START condition), and store it in the capture register.

- MC2 Counter

After enabling ATCCNTE.ATCE, it counts up from SCL rise edge next to ACK for the IBI transmitted from I3C Slave to SCL rise edge next to Tbit after Mandatory Byte, and capture it as MC2.

The MREF and MC2 capture values are stored next to the IBI Status Descriptor when the IBI is received from the I3C Slave with the DATBASm.DVIBITS bit set to 1 (same as Async mode 0).

When ATCTL.AMEOE is enabled, an aME Event is issued for each aME. Use that event as a trigger to read the MSyncCNT capture value from the MRCCPT register and hold it in an external memory.

2. I3C Slave

I3C has counters of SC1(16bit), SC2(8bit) and aME_TICK(8bit) for Async mode 1.

- SC1 Counter

After enabling ATCCNTE.ATCE, it counts up from SC1 count trigger^{*1} to the first aME, and capture it as SC1.

Note 1. SW or external trigger can be selected by selection bits.

- SC2 Counter

After enabling ATCCNTE.ATCE, it counts up from SCL rise edge next to ACK for the IBI transmitted from I3C Slave to SCL rise edge next to Tbit after Mandatory Byte, and capture it as SC2.

- aME_TICK Counter

After enabling ATCCNTE.ATCE, it counts every aME, and capture it as aME_TICK at the SCL rise edge next to ACK for the IBI.

The aME_TICK counter is cleared on the first aME after the SC1 count trigger.

When the CETSS.ASYNE[1] bit = 1 and the ITS bit in Command Descriptor for issuing IBI is 1, the SC1, SC2 and aME_TICK capture values are transmitted following the IBI Mandatory Byte as shown in the following figure.

If the SC1 and SC2 counters overflow, 0xFFFF and 0xFF are captured and CETSS.ICOVF is set 1.

The DATA_LENGTH[15:0] bits value of the Command Descriptor sets a value obtained by adding the number of data of SC1, SC2 and aME_TICK to the number of transmission data.

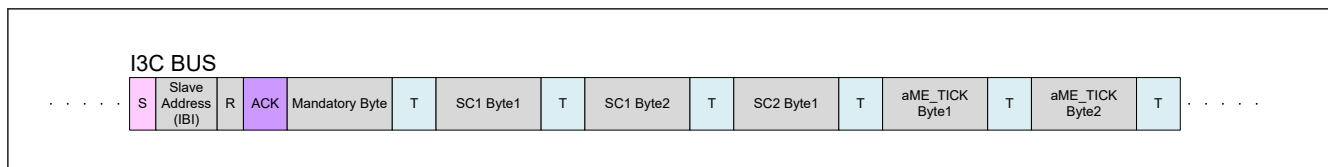


Figure 40.112 Example of asynchronous mode 1 timestamp data transfer

40.3.2.3.10 HDR-DDR CRC5 Algorithm

The CRC5 value is computed on a complete message, including the Command Word and all Data Words.

For a Command Word, the CRC5 is computed based on the value of the 16-bit payload, including:

- The Read vs. Write bit
- The Command Value
- The Slave Address, and
- The lowest-order bit (Write Reserved and Read Parity adjustment).

For Data Words, the CRC5 is computed based on all Data 16-bit values transmitted for that Command.

The CRC5 value is initialized to 0x1F. The CRC5 polynomial is below.

$$\text{CRC5} = X^5 + X^2 + X^0$$

40.3.2.3.11 Port Control

(1) Extra SCL Clock Cycle Output Function

In master mode, I3C module has a facility for the output of extra SCL clock cycles to release the I3C_SDA line of the slave device from being held at the low level due to the master being out of synchronization with the slave device.

This function is mainly used in master mode to release the I3C_SDA line of the slave device from the state of being fixed to the low level by including extra cycles of SCL output from I3C with single cycles of the SCL clock as the unit in the case of a bus error where I3C cannot issue a Repeated START condition or a STOP condition because the slave device is holding the I3C_SDA line at the low level. Do not use this facility in normal situations. Using it when communications are proceeding correctly will lead to malfunctions.

When the OUTCTL.EXCYC bit is set to 1, an additional clock pulse at the frequency set by the REFCKCTL.IREFCKS[2:0] bits and the STDBR.SBRHO[7:0] and STDBR.SBRLO[7:0] registers is output from the I3C_SCL pin. After output of this clock pulse, the EXCYC bit automatically becomes 0. After confirming that the EXCYC bit is 0, wait for the setup time of the Repeated START condition or STOP condition, and then confirm the detection of the Repeated START condition or STOP condition. If the Repeated START condition or STOP condition is not detected, consecutive additional clock pulses can be output by writing 1 to the EXCYC bit again.

When I3C module is in master mode and the slave device is holding the I3C_SDA line at the low level because synchronization with the slave device has been lost due to the effects of noise, etc., the output of a Repeated START condition or a STOP condition is not possible. The facility for output of an extra cycle of the SCL clock can be used to output extra cycles of SCL one by one to make the slave device release the I3C_SDA line from being held at the low level, thus recovering the bus from an unusable state. Release of the I3C_SDA line by the slave device can be monitored by reading the SDILV bit in PRSTDBG. After the I3C_SDA line has been released by the slave device, the preset of a Repeated START condition or a STOP condition is issued.

Use this function with the BFCTL.MALE bit set to 0 (master arbitration-lost detection is disabled).

[Output conditions for using the EXCYC bit in OUTCTL]

- When the bus is free (BFREF flag in BCST = 1) or in master mode (CRMS bit = 1 in PRSST and BFREF flag = 0 in BCST)
- When the communication device does not hold the I3C_SCL line low

Figure 40.113 shows the operation timing of the extra SCL clock cycle output function (EXCYC bit).

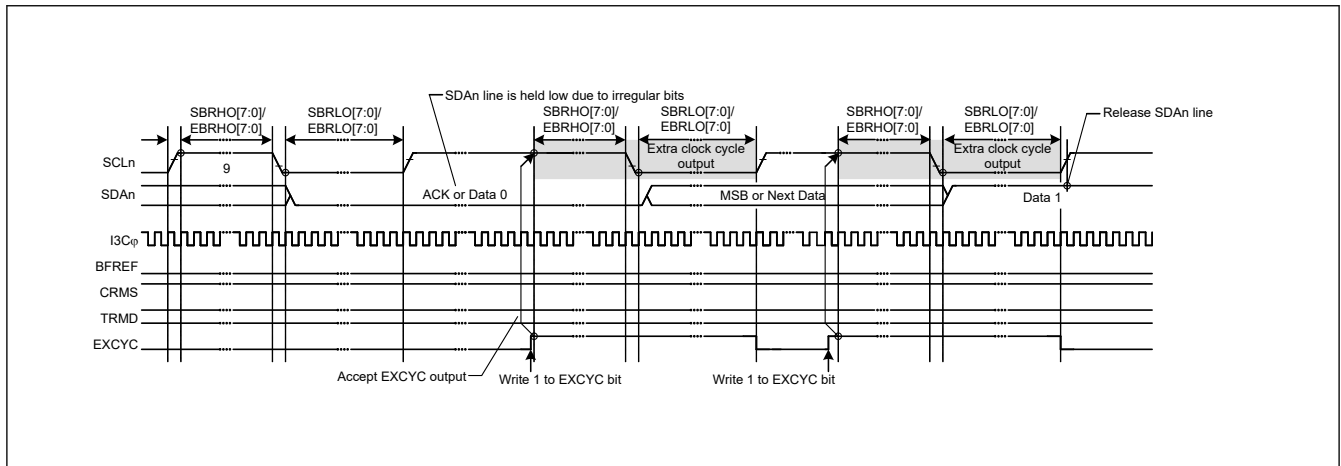


Figure 40.113 Extra SCL clock cycle output function (EXCYC bit)

40.3.2.3.12 SMBus Operation [I²C Mode]

I3C is available for data communication conforming to the SMBus (Version 2.0). To perform SMBus communication, set the BFCTL.SMBS bit to 1. To use the transfer rate within a range of 10 kbps to 100 kbps of the SMBus specification, set the REFCKCTL.IREFCKS[2:0] bits, the STDBR.SBRHO[7:0] bits, and the STDBR.SBRLO[7:0] bits. In addition, determine the values of the OUTCTL.SDODCS bit and the OUTCTL.SDOD[2:0] bits to meet the data hold time specification of 300 ns or more. If I3C is used only as an I²C slave device, the transfer rate setting is not necessary, whereas the STDBR.SBRLO[7:0] bits needs to be set to a value longer than the data setup time (250 ns).

For the SMBus device default address (1100 001), use one of the slave device address table basic registers 0 to 2 (SDATBASn.SDSTAD[6:0] bits) (y = 0 to 2), and set the corresponding SDATBASn.SDADLS bit (7-bit/10-bit address format select) (y = 0 to 2) to 0 (7-bit address format).

When transmitting the UDID (Unique Device Identifier), set the BFCTL.SALE bit to 1 to enable the slave arbitration lost detection function.

(1) SMBus Timeout Measurement

(a) Measuring timeout of slave device

The following period (timeout interval: $T_{\text{LOW:SEXT}}$) must be measured for slave devices in SMBus communication.

- From START condition to STOP condition

To measure timeout for slave devices, measure the period from START condition detection to STOP condition detection with the GPT timer using a START condition detection interrupt (I3C_EEI) and STOP condition detection interrupt (I3C_EEI) of I3C. The measured timeout period must be within the total clock low-level period [slave device] $T_{\text{LOW:SEXT}}$: 25 ms (max.) of the SMBus specification.

If the time measured with the GPT exceeds the clock low-level detection timeout T_{TIMEOUT} : 25 ms (min.) of the SMBus specification, the slave device must release the bus by writing 1 to the RSTCTL.INTLRST bit to issue an internal reset of I3C. When an internal reset is issued, I3C stops driving the bus for the I3C_SCL pin and I3C_SDA pin and make the I3C_SCL/I3C_SDA pin outputs high-impedance, which releases the bus.

(b) Measuring timeout of master device

The following periods (timeout interval: $T_{\text{LOW:MEXT}}$) must be measured for master devices in SMBus communication.

- From START condition to acknowledge bit
- Between acknowledge bits

- From acknowledge bit to STOP condition

To measure timeout for master devices, measure these periods with the GPT timer using a START condition detection interrupt (I3C_EEI), STOP condition detection interrupt (I3C_EEI), and transmit end interrupt (I3C_TEND) or Normal Rx Data buffer full interrupt (I3C_RX) of I3C. The measured timeout period must be within the total clock low-level extended period (master device) $T_{LOW:MEXT}$: 10 ms (max.) of the SMBus specification, and the total of all $T_{LOW:MEXT}$ from START condition to STOP condition must be within $T_{LOW:SEXT}$: 25 ms (max.).

For the ACK receive timing (rising edge of the ninth SCL clock cycle), monitor the BST.TENDF flag in master transmit mode (master transmitter) and the NTST.RDBFF0 flag in master receive mode (master receiver). For this reason, perform bitwise transmit operation in master transmit mode, and hold the SCSTRCTL.ACKTWE bit 0 until the byte just before reception of the final byte in master receive mode. While the ACKTWE bit = 0, the RDBFF0 flag is set to 1 at the rising edge of the ninth SCL clock cycle.

If the period measured with the GPT exceeds the total clock low-level extended period (master device) $T_{LOW:MEXT}$: 10 ms (max.) of the SMBus specification or the total of measured periods exceeds the clock low-level detection timeout $T_{TIMEOUT}$: 25 ms (min.) of the SMBus specification, the master device must stop the transaction by issuing a STOP condition. In master transmit mode, immediately stop the transmit operation (writing data to NTDTBP0).

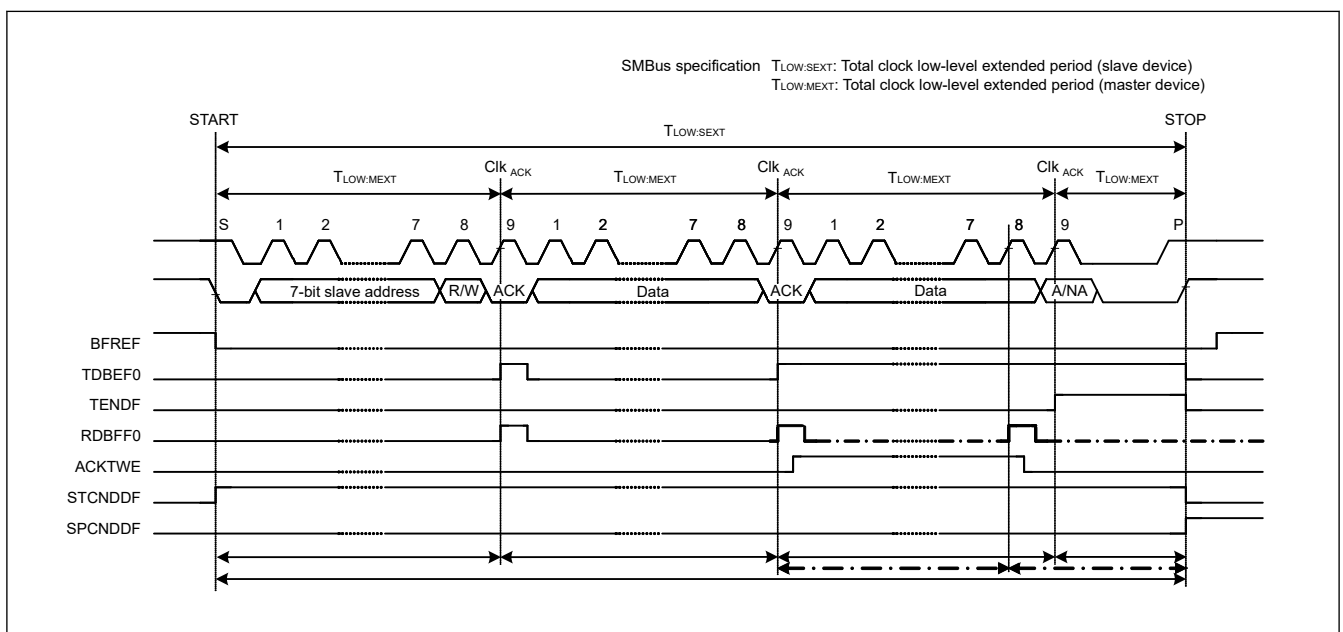


Figure 40.114 SMBus timeout measurement

(2) Packet Error Code (PEC)

This MCU incorporates a CRC calculator. The CRC calculator enables transmission of a packet error code (PEC) or checking the received data of the SMBus in data communication of I3C. For the CRC generating polynomials of the CRC calculator, see [section 47, Cyclic Redundancy Check \(CRC\)](#).

The PEC data in master transmit mode can be generated by writing all transmit data to the CRC data input register (CRCDIR) in the CRC calculator.

The PEC data in master receive mode can be checked by writing all receive data to CRCDIR in the CRC calculator and comparing the obtained value in the CRC data output register (CRCDOR) with the received PEC data.

To send ACK or NACK according to the match or mismatch result when the final byte is received as a result of the PEC code check, set the SCSTRCTL.ACKTWE bit to 1 before the rising edge of the eighth SCL clock cycle during reception of the final byte, and hold the I3C_SCL line low at the falling edge of the eighth clock cycle.

(3) SMBus Host Notification Protocol (Notify ARP Master Command)

In communications over an SMBus, a slave device can temporarily act as a master device to notify the SMBus host (or ARP master) of its own slave address or to request its own slave address from the SMBus host.

For a product of this MCU to operate as an SMBus host (or ARP master), the host address (0001 000) sent from the slave device must be detected as a slave address, so I3C has a function for detecting the host address. To detect the host address as a slave address, set the BFCTL.SMBS bit and the SVCTL.HOAE bit to 1. Operation after the host address has been detected is the same as normal slave operation.

40.3.2.3.13 Common Command Codes (CCC) [I3C Mode]

For the common command code (CCC), refer to 5.1.9 Common Command Codes (CCC) in MIPI I3C Specification v1.0. I3C is based on Table 15 I3C Common Command Codes in 5.1.9.3 Common Command Definitions of MIPI I3C Specification v1.0.

The MIPI Reserved area and Vendor Extension area of Command Code are described below.

I3C Master mode :

When sending CCCs in the MIPI Reserved area and Vendor Extension area from the I3C Master, only Broadcast/Direct SET CCCs using the Immediate Transfer Command can be sent.

Sending Direct GET CCC is not supported.

I3C Slave mode :

Only Broadcast/Direct SET CCC can be received for CCC in MIPI Reserved area and Vendor Extension area.

Receiving Direct GET CCC is not supported.

40.3.2.4 Error Detection

40.3.2.4.1 SDR Error Detection and Recovery Methods for I3C Slave Devices [I3C Mode]

The seven error types summarized in [Table 40.13](#) are supported for all I3C slave devices. Each error type is further explained below the table.

Table 40.13 SDR slave error types

Error type	Description	Error detection method	Error recovery method
S0	Broadcast address/W (= 0x7E/W) or Dynamic address/RW	Detect any of the following: 0x3E / W 0x5E / W 0x6E / W 0x76 / W 0x7A / W 0x7C / W 0x7F / W 0x7E / R	Enable HDR EXIT Detector and ignore all other patterns
S1	CCC code	Parity check, using T-Bit	Enable HDR EXIT detector and neglect other patterns
S2	Write data	Parity check, using T-Bit	Enable STOP detector and neglect other patterns
S3	Assigned address during Dynamic address arbitration	Parity check, using PAR Bit	Generate NACK (after PAR), then wait for another Repeated START and 7E/R to re-transmit the Provisional ID
S4	0x7E/R after Sr during Dynamic address arbitration	Detect any value other than 0x7E/R after Sr during Dynamic Address Arbitration	Generate NACK (after 0x7E/R), then enable STOP Detector and ignore all other patterns
S5	Transaction after detecting CCC	Detect illegally formatted CCC	Generate NACK (after Slave Address), then enable STOP Detector and ignore all other patterns
S6 (optional)	Monitoring error	Slave detects (through monitoring) that transmitted Data differs from what it intended to transmit (Does not apply during Dynamic address arbitration)	Stop the transmission, then enable STOP Detector and ignore all other patterns

40.3.2.4.2 SDR Error Detection and Recovery Methods for I3C Master Devices [I3C Mode]

The two error types summarized in [Table 40.14](#) are supported for all I3C master devices. Each error type is further explained below the table.

Table 40.14 SDR master error types

Error type	Description	Error detection method	Error recovery method
M0	Transaction after sending CCC	Detect illegally formatted CCC	Stop the transmission, then send STOP and retry the transmission.
M1 (optional)	Monitoring error	Master detects (through monitoring) transmitted data different from what it intended to transmit (Does not apply during Dynamic address arbitration)	Stop the transmission, then send STOP and retry the transmission.
M2	No response to Broadcast address (0x7E)	Master detects NACK after Broadcast address (0x7E) transmission	Upon detection of NACK, master transmits HDR exit pattern followed by STOP

40.3.2.4.3 HDR-DDR Error Detection

Four error types are defined for HDR-DDR mode.

Table 40.15 HDR-DDR error types

Error type	Description	Error detection method	Error recovery method
Framing	The Preamble 2-bits before command and data	The Slave detects that the Preamble 2-bits before command and data is not valid values.* ¹	Master: Issue SCL clocks until 19 SCL clocks (38 bits) have been seen with SDA High. The Master shall then Park SCL Low, and emit an HDR Exit Pattern using SDA. Slave: Wait for HDR EXIT Pattern
Parity checking	Parity error	The Master and the Slave detects mismatched parity.* ²	
CRC5 checking	CRC5 error	The Master and the Slave detects mismatched CRC.* ³	
NACK receiving	NACK by the Slave on a Read command	The Master detects a NACK by the Slave on a Read command.* ⁴	Issue SCL clocks until 19 SCL clocks (38 bits) have been seen with SDA high. The Master shall then Park SCL low, and emit either an HDR exit pattern or an HDR restart pattern using SDA.
Monitoring	Monitoring error	Both the Master and the Slave should always monitor the data that they each transmit, and they detect the monitored data differs from the data that the Master or Slave intended to send.* ⁵	Both the Master and the Slave can stop the transmission and then Master and Slave should each do the following. Master: Issue SCL clocks until 19 SCL clocks (38 bits) have been seen with SDA High. The Master shall then Park SCL Low, and emit an HDR Exit Pattern using SDA. After the Master sends HDR EXIT, it can retry the transmission. Slave: Wait for HDR Exit Pattern

Note 1. This supports a positional error detection mechanism:

- A Command Word shall always follow the Enter HDR CCC and HDR Restart Pattern, and shall never appear in any other position. It is an error condition for a Command Word to appear in any other position, or to be missing where expected.
- A Data Word shall always follow either a Command Word or another Data Word, and shall never appear in any other position. It is an error condition for a Data Word to appear in any other position, or to be missing where expected.
- A single CRC Word shall always follow the last Data Word for a Command, such that it ends the Message. As a result, a CRC Word shall always be followed by either the HDR Restart Pattern or the HDR Exit Pattern. It is error condition for a CRC Word to appear in any other position, or to be missing where expected.
- The first nibble of a valid CRC shall contain the allowed value 0xC. Any other value in the first nibble should be considered a framing error.

- Note 2. Parity encoding (for transmitters) parity checking (for receivers) shall be performed on all Command Words and Data Words. Mismatched parity is an error condition.
- Note 3. CRC5 encoding (for transmitters) and CRC5 checking (for receivers) shall be performed on all payload bits for Command Words and Data Words. Mismatched CRC is an Error.
- Note 4. NACK by the Slave on a Read command is not a normal behavior (ACK is normal). The Master may choose to treat a NACK on a Read command as a possible framing error. The Master can choose to treat the NACK of a Read command as a possible line error, and therefore use the same approach in order to ensure that the Slave is not driving the Bus.
- Note 5. If the Master or Slave performs such monitoring, and if the monitored Data differs from the Data that the Master or Slave intended to send (except for Data transferred during a Dynamic Address Arbitration procedure), then this shall be considered an error.

40.3.2.4.4 HDR-TSP/TSL Error Detection

Two error types are defined for HDR-TSP/TSL Mode.

Table 40.16 HDR-TSP/TSL error types

Error type	Description	Error detection method	Error recovery method
Symbol 2 checking	Continuous error of symbol 2	More than one symbol 2 in a row. (In symbol 2 the I3C_SCL line does not change, and the I3C_SDA line does change.) *1	Master: Wait until the Slave stops transitioning the Bus for 2x the maximum edge-to-edge duration used by that Slave. Then the Master shall force out the HDR Exit Pattern. Slave: Wait for HDR Exit Pattern.
Parity checking	Parity errors	The Slave detects mismatched parity. *2	
Monitoring	Monitoring error	Both the Master and the Slave should always monitor the data that they each transmit, and they detect the monitored data differs from the data that the Master or Slave intended to send.*3	Both the Master and the Slave can stop the transmission and then Master and Slave should each do the following. Master: Wait until the Slave stops transitioning the Bus for 2x the maximum edge-to-edge duration used by that Slave. Then the Master shall force out the HDR Exit Pattern. After the Master sends HDR EXIT, it can retry the transmission. Slave: Wait for HDR EXIT Pattern.

Note 1. Exception: More than one Symbol 2 in a row is allowed for HDR Restart or HDR Exit, but only from a known starting state (SCL Low and SDA High), and only at a Data Word boundary (allowing for one Symbol to set up that state). Thus the Symbol pattern 2,2 is only allowed under the following situations:

- As part of the HDR Exit Pattern, which uses 3 or 4 such Symbol pairs and involves SCL Low
- As part of the HDR Restart pattern, which uses 2 such Symbol pairs and involves SCL Low
- As the result of one normal Symbol ending in 2, and the next Symbol starting with 2

Note 2. An error may be seen by an HDR Slave, or by an HDR Master when a Slave is driving the I3C Bus:

- A Slave that sees an error shall stop tracking Symbols and use HDR Exit and HDR Restart Pattern detection. This means to wait until safe. A Slave shall enable the HDR Exit and HDR Restart Pattern Detector upon entry to any HDR Mode, ensuring that it is always safe.
- A Master that sees an error shall wait until the Slave stops transitioning the Bus for 2x the maximum edge-to-edge duration used by that Slave. Then the Master shall force out the HDR Exit Pattern.

Note 3. If the Master or Slave performs such monitoring, and if the monitored Data differs from the Data the Master or Slave intended to send (except for data transferred during the Dynamic Address Arbitration procedure), then this shall be considered an error.

Note: It could be appropriate to wait for the Symbol 2,2, but the Master would still need to see quiescence. If the Master is seeing errors, then the results of even a Symbol 2,2 are questionable.

40.3.2.4.5 Timeout Error Detection

I3C includes a timeout function for detecting when the I3C_SCL line has been stuck longer than the predetermined time. I3C can detect an abnormal bus state by monitoring that the I3C_SCL line is stuck low or high for a predetermined time.

The timeout function monitors the I3C_SCL line state and counts the low-level period or high-level period using the internal counter. The timeout function resets the internal counter each time the I3C_SCL line changes (rising or falling), but continues to count unless the I3C_SCL line changes. If the internal counter overflows due to no I3C_SCL line change, I3C can detect the timeout and report the bus hung state.

This timeout function is enabled when BSTE.TODE = 1. It detects a hung state that the I3C_SCL line is stuck low or high during the following conditions: (When TMOCTL.TOMDS[1:0] = 00b)

- The bus is busy (BCST.BFREF = 0) in master mode (PRSSST.CRMS = 1).
- I3C's own slave address is detected (SVST register is not 0x0000) and the bus is busy (BCST.BFREF = 0) in slave mode (PRSSST.CRMS = 0).
- The bus is free (BCST.BFREF = 1) while generation of a START condition is requested (CNDCTL.STCND = 1).

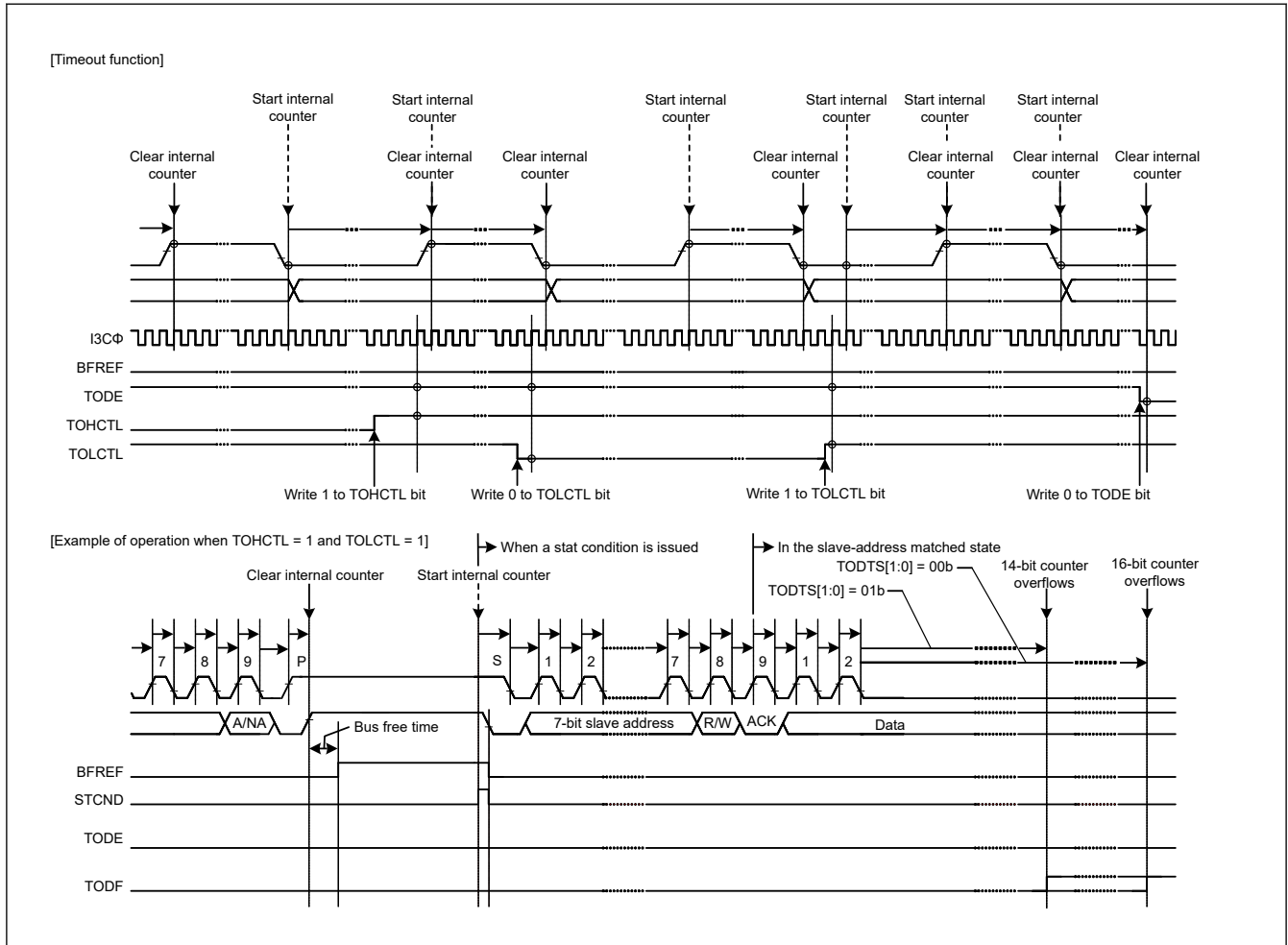


Figure 40.115 Timeout error detection (TODE, TODTS[1:0], TOHCTL, and TOLCTL bits)

40.3.2.4.6 Resume Operation [I3C Mode]

I3C enters the Halt state as a result of any type of error occurring in a transfer.

The error type is indicated by the field ERR_STATUS in Response Descriptor or Receive Status Descriptor. After I3C has entered the Halt state, the user must write the value 1 to the RSM bit to resume operation. I3C shall auto-clear the RSM bit once it has initiated the next Command transfer or detected the START condition.

40.3.2.4.7 Abort Operation [I3C Mode]

When the BCTL.ABT bit is set to 1, I3C relinquish control of the bus before completing the currently issued transfer. In response to an abort request, I3C issues the STOP condition on the bus after the complete data byte is transferred or received. After I3C has aborted, the user shall clear the BCTL.ABT bit to allow operation on the bus.

Note: For Read transaction, when BCTL.ABT is set to 1, that receive data is stored in Rx data buffer. However, for HDR-TSP / TSL, receive data received after that is not stored.

Abbreviations

Pa: Parity

Pr: Preamble

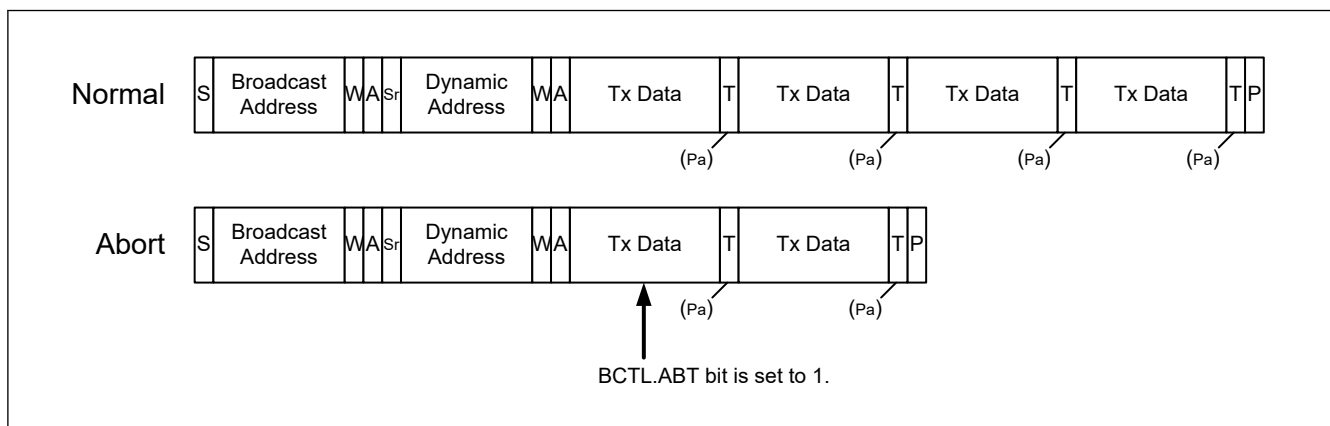


Figure 40.116 Abort operation of SDR write transfer

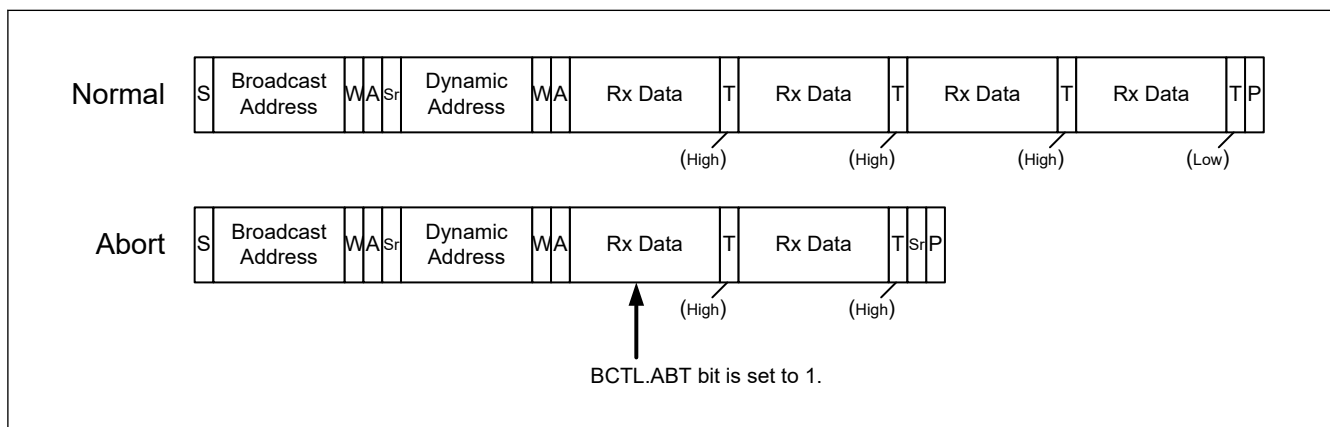


Figure 40.117 Abort operation of SDR read transfer

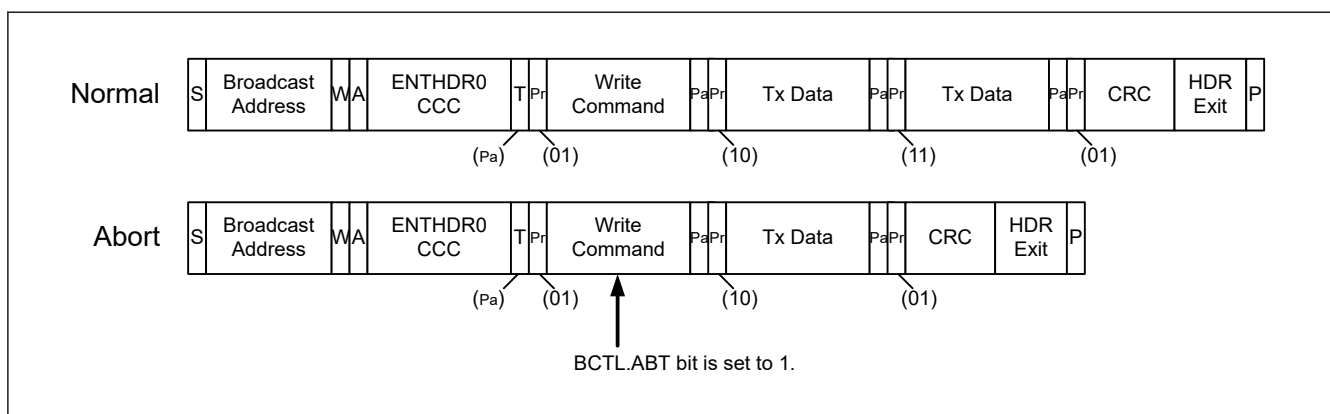


Figure 40.118 Abort operation of HDR-DDR write transfer

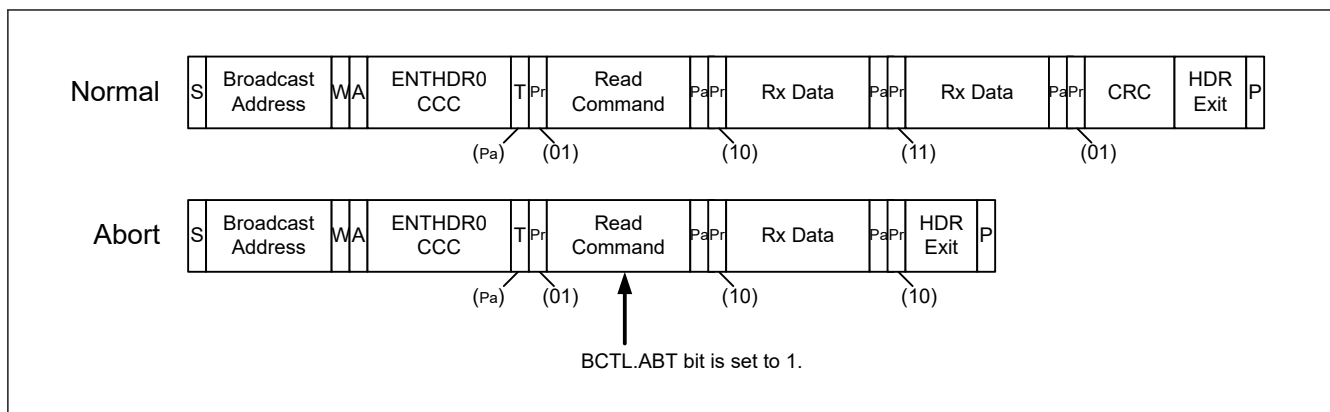


Figure 40.119 Abort operation of HDR-DDR read transfer

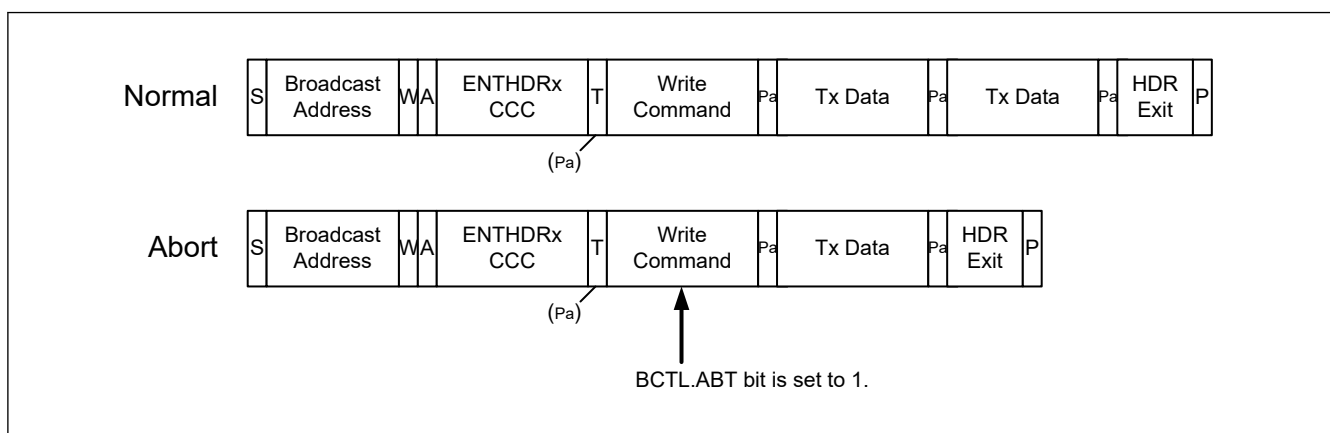


Figure 40.120 Abort operation of HDR-TSP/TSL write transfer

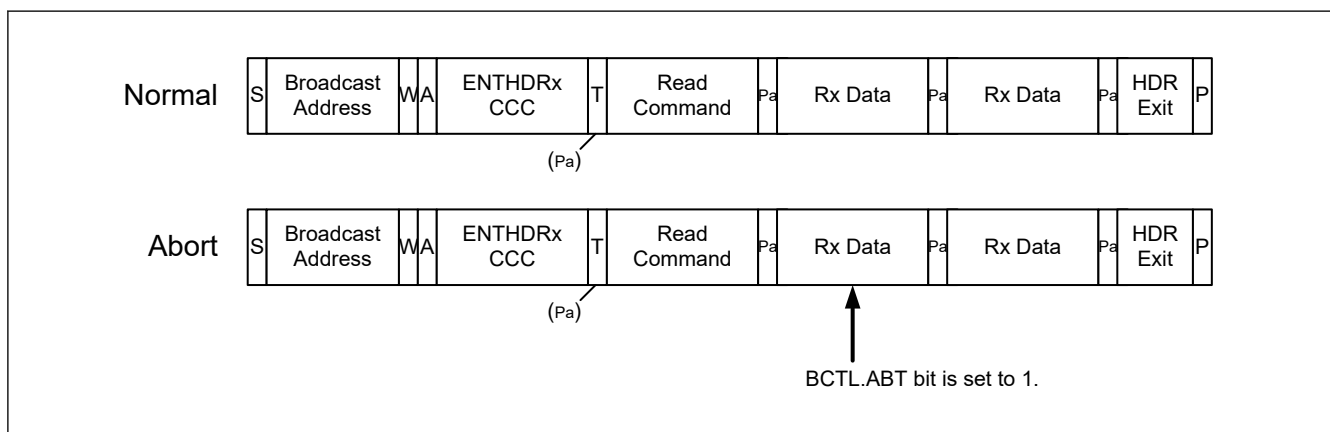


Figure 40.121 Abort operation of HDR-TSP/TSL read transfer

40.3.2.4.8 Error Recovery Operation

(1) Error Recovery Operation

When an error occurs, the INST.INEF, NTST.TEF, NTST.TABTF, HTST.TEF and HTST.TABTF flags are set to 1 according to the cause of the error, or the interrupts associated with each flag are asserted (when detection and interrupts are enabled.)

There is a possibility of communication error or internal module error.

Note: If an error occurs, I3C will be suspended (BCTL.RSM becomes 1). After I3C is suspended, the application must write the value 1 to the BCTL.RSM bit to resume I3C operation and recover from the suspended state.

Figure 40.122 and Figure 40.123 show the error recovery flow.

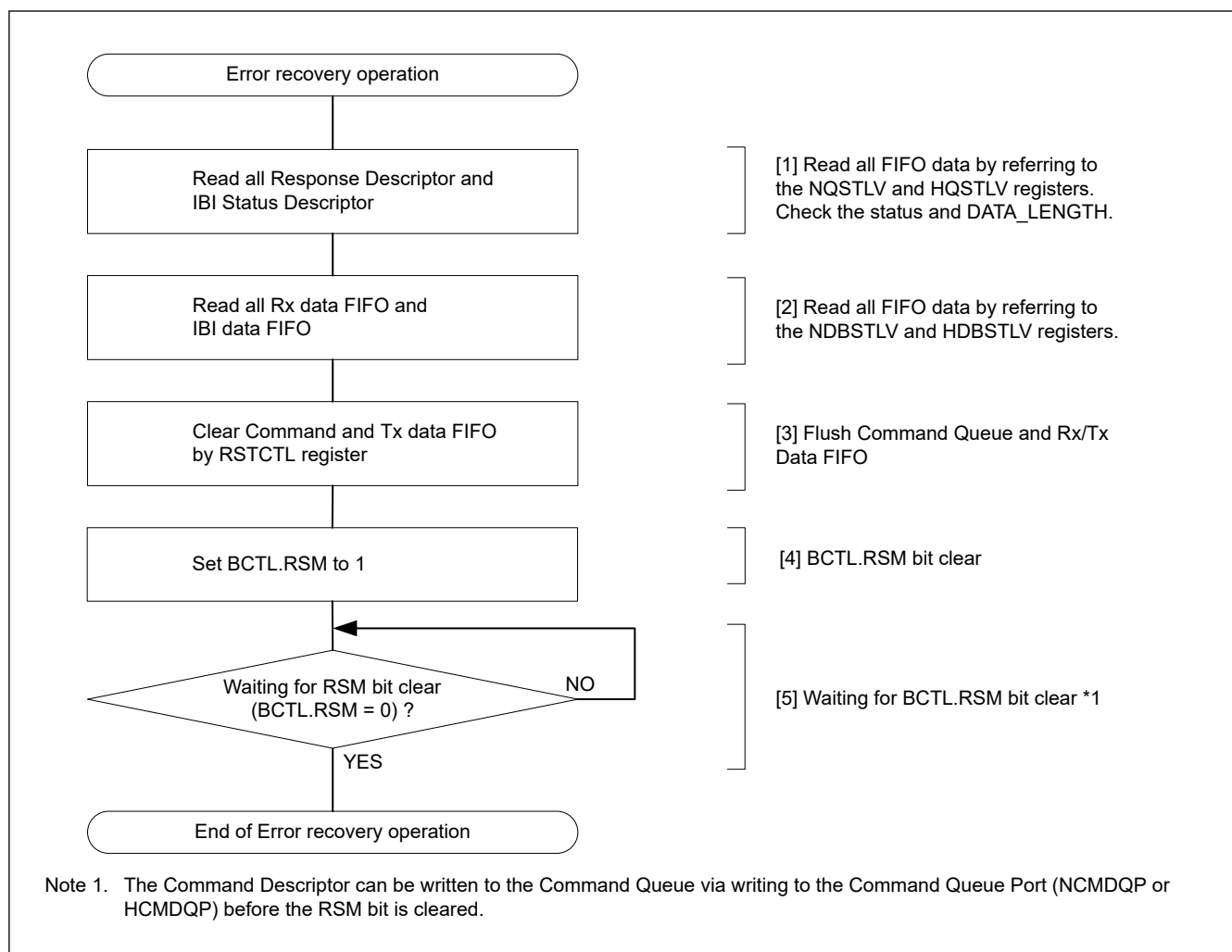


Figure 40.122 Example of error recovery operation flowchart for I3C master

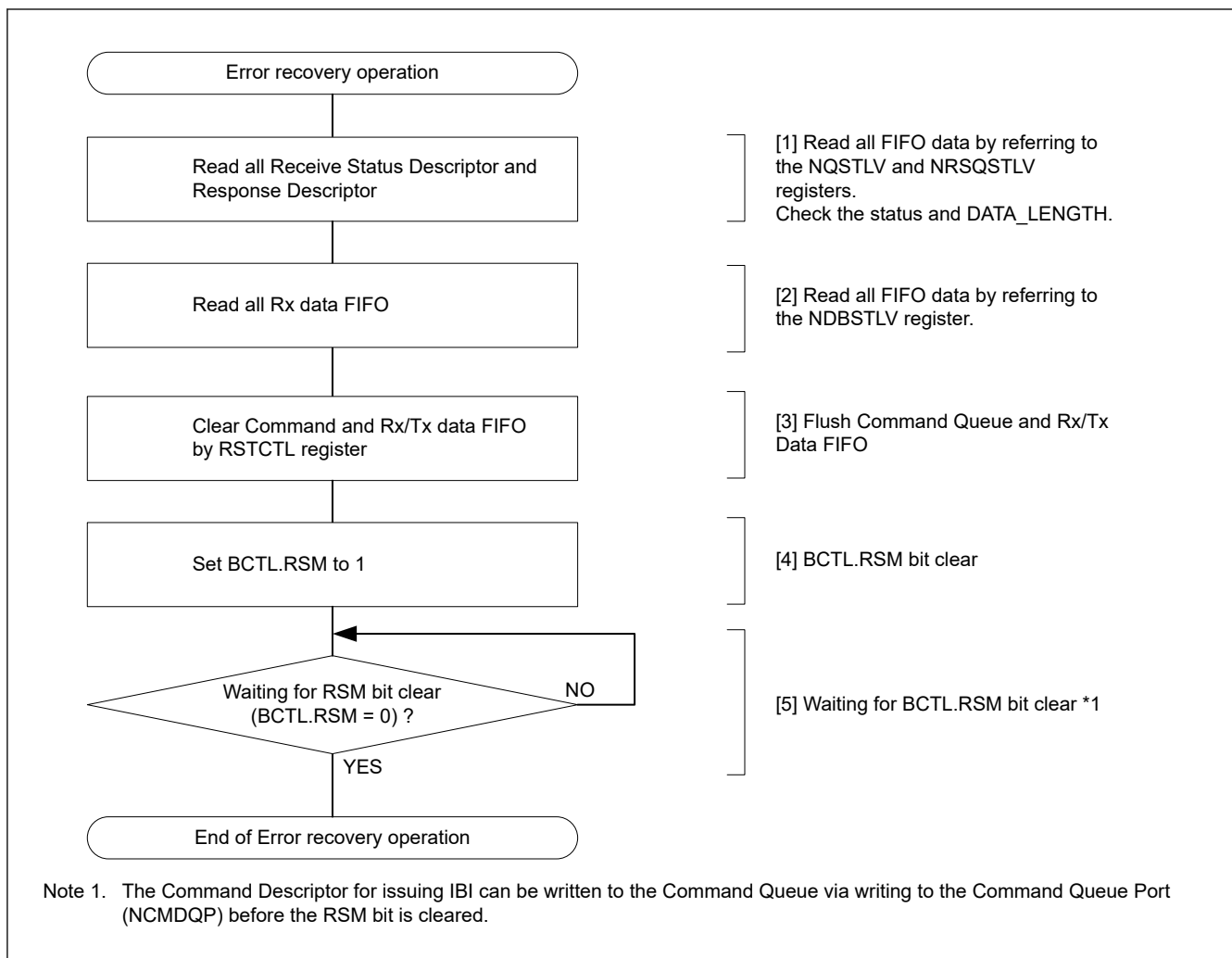


Figure 40.123 Example of error recovery operation flowchart for I3C slave

When I3C Slave recovers from an error according to the error recovery flow, after setting BCTL.RSM to 1, BCTL.RSM becomes 0 after detecting a state in which Bus Available period communication is not performed on I3C Bus.

If communication occurs on the I3C bus within the Bus Available period, BCTL.RSM will not be set to 0 and error recovery will not be completed, NACK response will be made to the communication.

(2) Master Error Detection and Escalation Handling

If the Master does not receive an ACK of a transmitted private Message to a Slave and Steps 1 and 2 described in Chapter 5.1.10.2.4 of MIPI I3C Spec v1.0 fail, the processing flow of Step 3 is shown in [Figure 40.124](#) and [Figure 40.125](#).

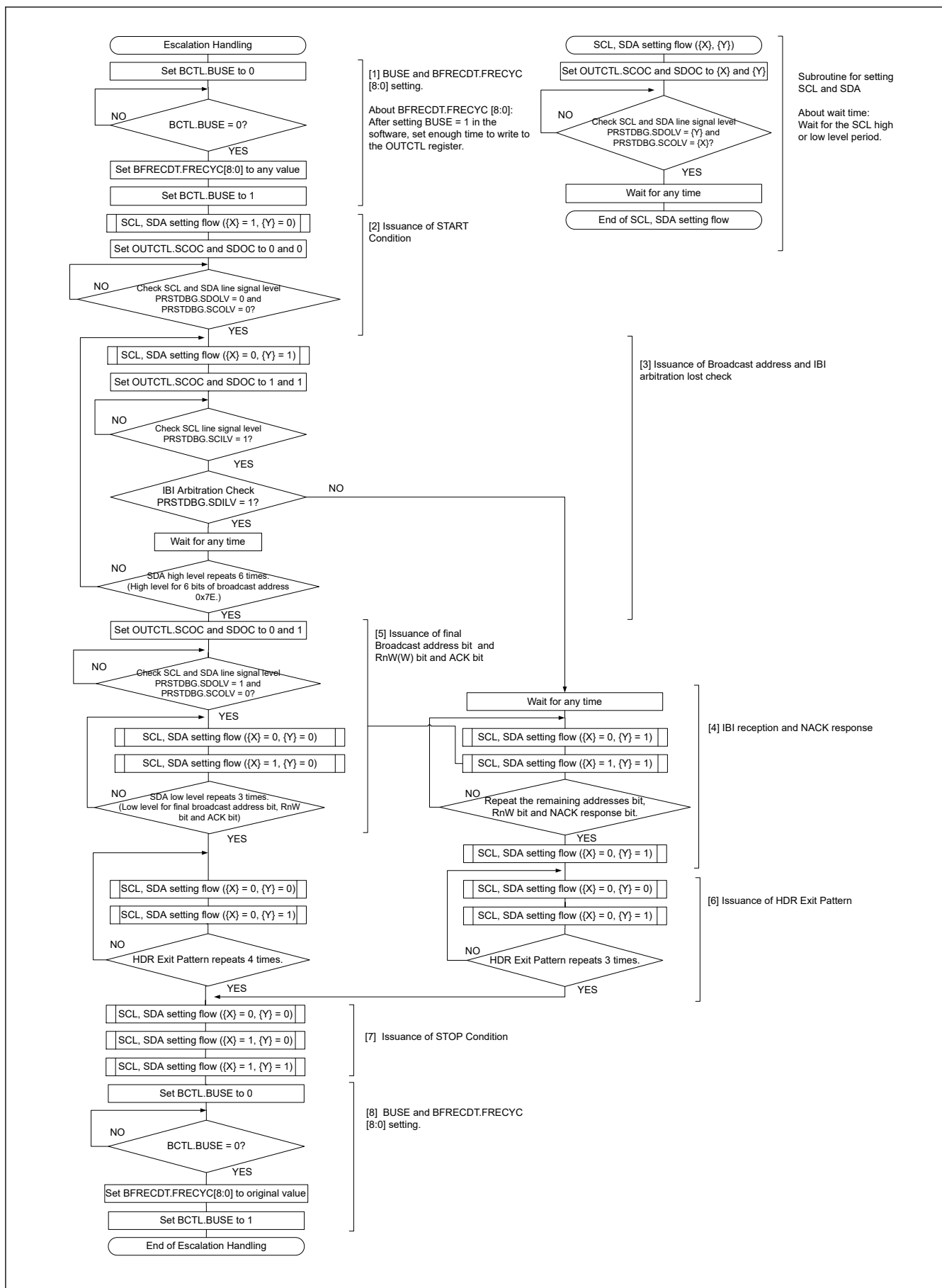


Figure 40.124 Escalation handling flowchart for I3C master

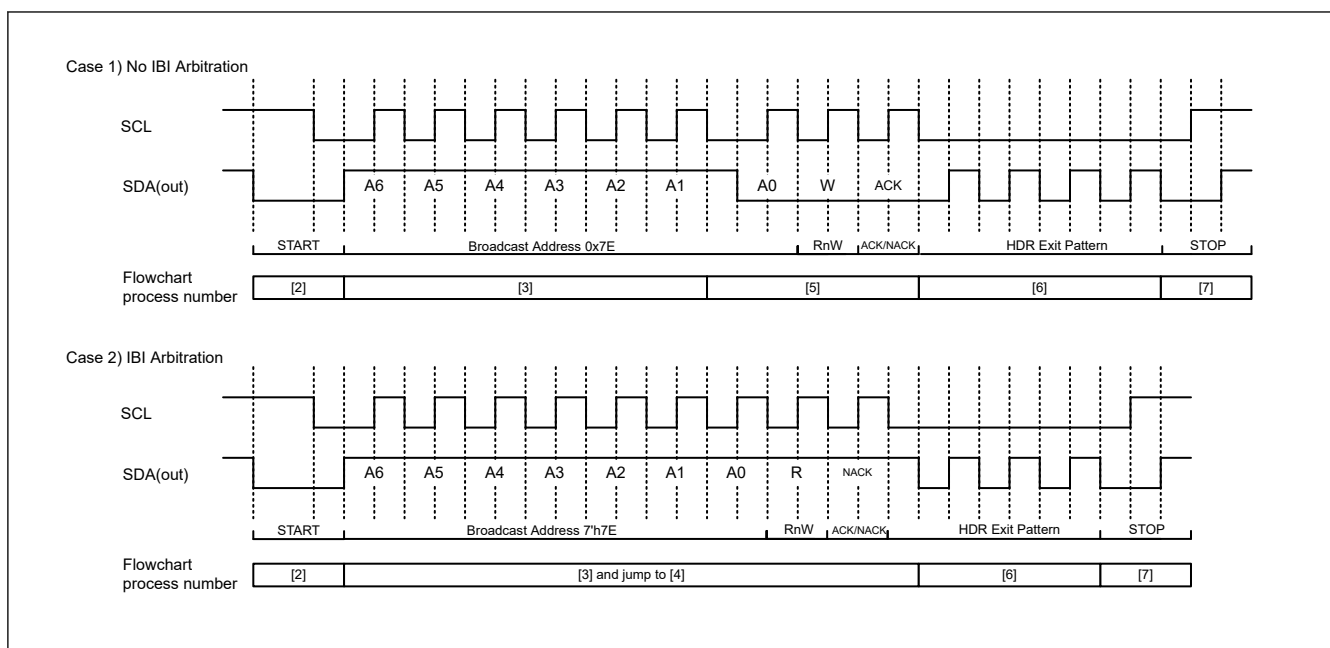


Figure 40.125 I3C Escalation Handling timing chart for I3C Master

40.3.2.5 Low Power Function

40.3.2.5.1 Wake Up Function [^I2C Mode]

I3C is equipped with the Wake-up function that causes the microcomputer to transition from low power consumption mode with system clock (ICLK) is stopped (software standby mode, etc.) to the normal operation. The Wake-up function is used to generate a Wake-up interrupt signal when the received data matches the address set to Wake-up interrupt factor also receives data in a state where the operating clock (PCLK/TCLK) is stopped (PCLK/TCLK asynchronous operation). This wake-up interrupt signal causes the microcomputer to transition to the normal operation. After Wake-up interrupt occurs, switch I3C to PCLK/TCLK synchronous operation, it will be able to continue the communication operation.

The Wake-up function has four wake-up operation modes (normal WU mode 1, normal WU mode 2, command recovery mode, and EEP response mode). The table below describes the behavior in these four wake-up operation modes.

Table 40.17 Wake-up operation mode

	ACK response timing	ACK Type responded before recovery to PCLK/TCLK synchronous operation	SCL state before recovery to PCLK/TCLK synchronous operation
Normal WU mode 1	Before recovery to PCLK/TCLK synchronous operation ^{*1}	ACK	Fixed to L
Normal WU mode 2	After recovery to PCLK/TCLK synchronous operation ^{*2}	Before recovery: no response (NACK level retained) After recovery: ACK response	Fixed to L
Command recovery mode	Before recovery to PCLK/TCLK synchronous operation ^{*1}	ACK	Open
EEP response mode	Before recovery to PCLK/TCLK synchronous operation ^{*1}	NACK	Open

Note 1. Switching timing from PCLK/TCLK asynchronous operation to PCLK/TCLK synchronous operation is the fall of the 9th clock of SCL.

Note 2. Switching timing from PCLK/TCLK asynchronous operation to PCLK/TCLK synchronous operation is the fall of the 8th clock of SCL.

The following is able to select as Wake-Up interrupt factor.

- Host address detection (valid when SVCTL.HOAE = 1)
- General call address detection (valid when SVCTL.GCAE = 1)
- Slave address 0^{*1} detection (valid when SVCTL.SVAE[0] = 1)
- Slave address 1^{*1} detection (valid when SVCTL.SVAE[1] = 1)

- Slave address 2^{*1} detection (valid when SVCTL.SVAE[2] = 1)

Note 1. 7-bit address only can be set. Set SDADLS bit to 0 in SDATBASn.

(1) Normal Wake-Up mode 1

This section describes the behavior, the timing, and a use case of normal WU mode 1.

A wake-up interrupt triggered by the match of the slave address makes the transition to the normal operation in the manner described below. Also, the detailed timing is provided in [Figure 40.128](#).

Before wake-up recovery:	ACK is sent in response to the data received with its own slave address.
During wake-up recovery:	ACK response is made at the 9th clock cycle of SCL, and the SCL is held low afterwards.*1
After wake-up recovery:	Normal operation continues.

Note 1. Between ninth clock cycle and first clock cycle during Wake-Up recovery, SCSTRCTL.RWE = 1 does not work.

If the slave address does not match, the I3C_SCL line is not held low after the fall of the 9th clock cycle of SCL, and the slave operation continues.

See [Figure 40.126](#) below for a use case.

A wake-up interrupt is not generated at the transition to the normal operation if the transition is triggered by a cause (other recovery causes (IRQ)) other than a wake-up interrupt signal generated by a slave address match. BST.WUCNDDF is not set in this case. Carry out the following processing according to [Figure 40.127](#).

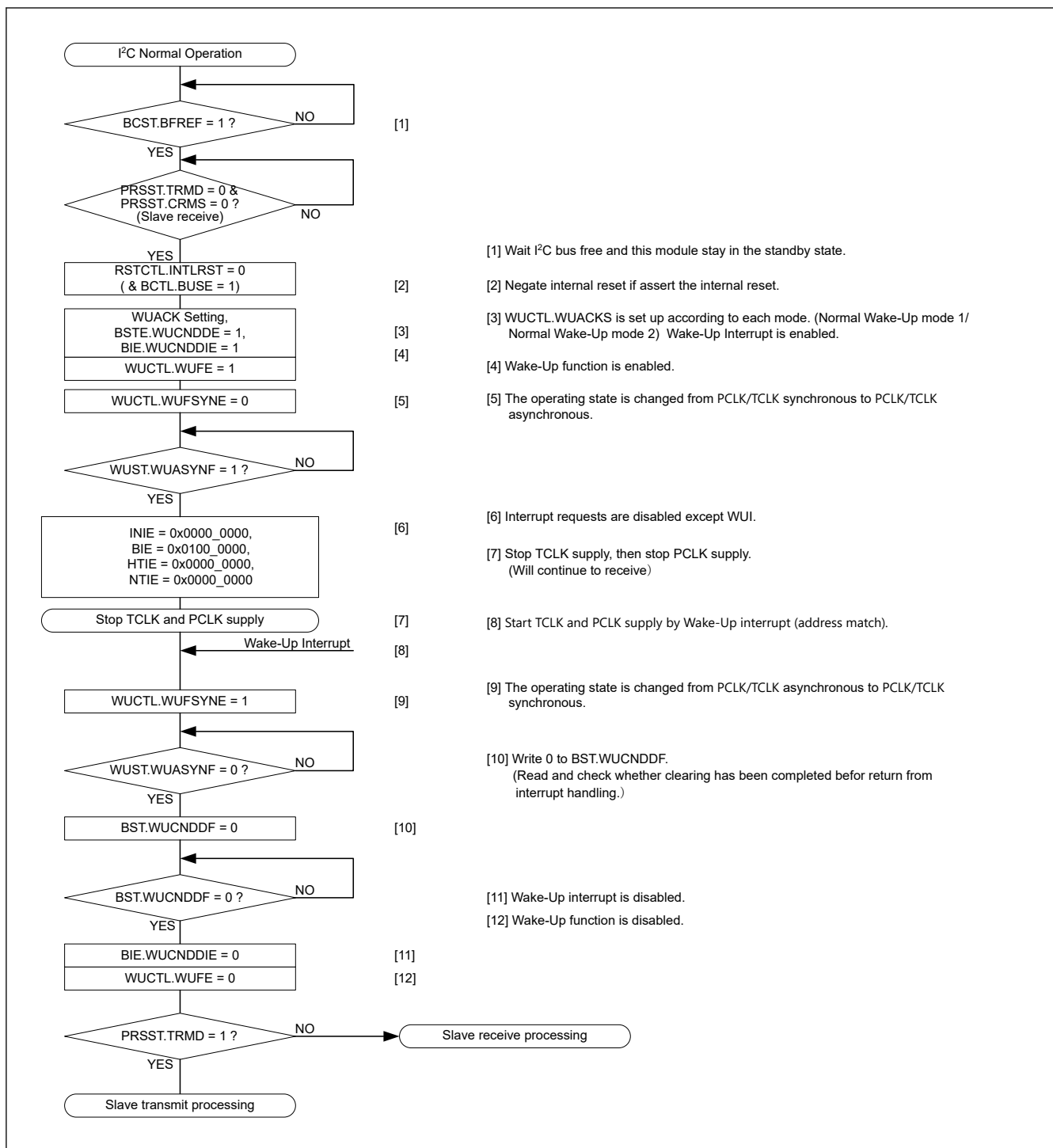


Figure 40.126 Use case of normal WU mode 1 (wake-up recovery by a wake-up interrupt triggered by the match of the slave address)

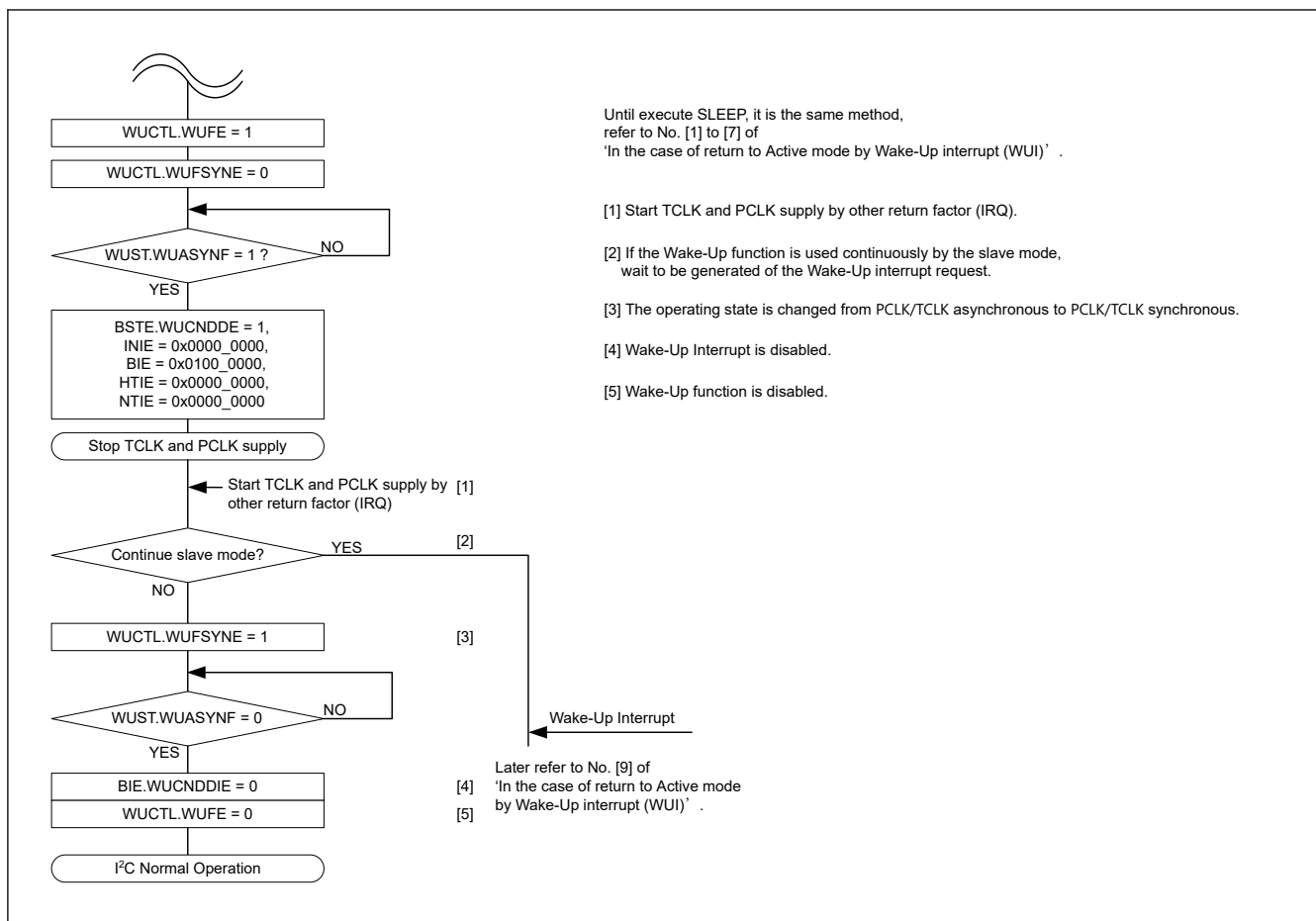


Figure 40.127 Use case of normal WU modes 1 and 2 (wake-up recovery by other recovery causes (IRQ))

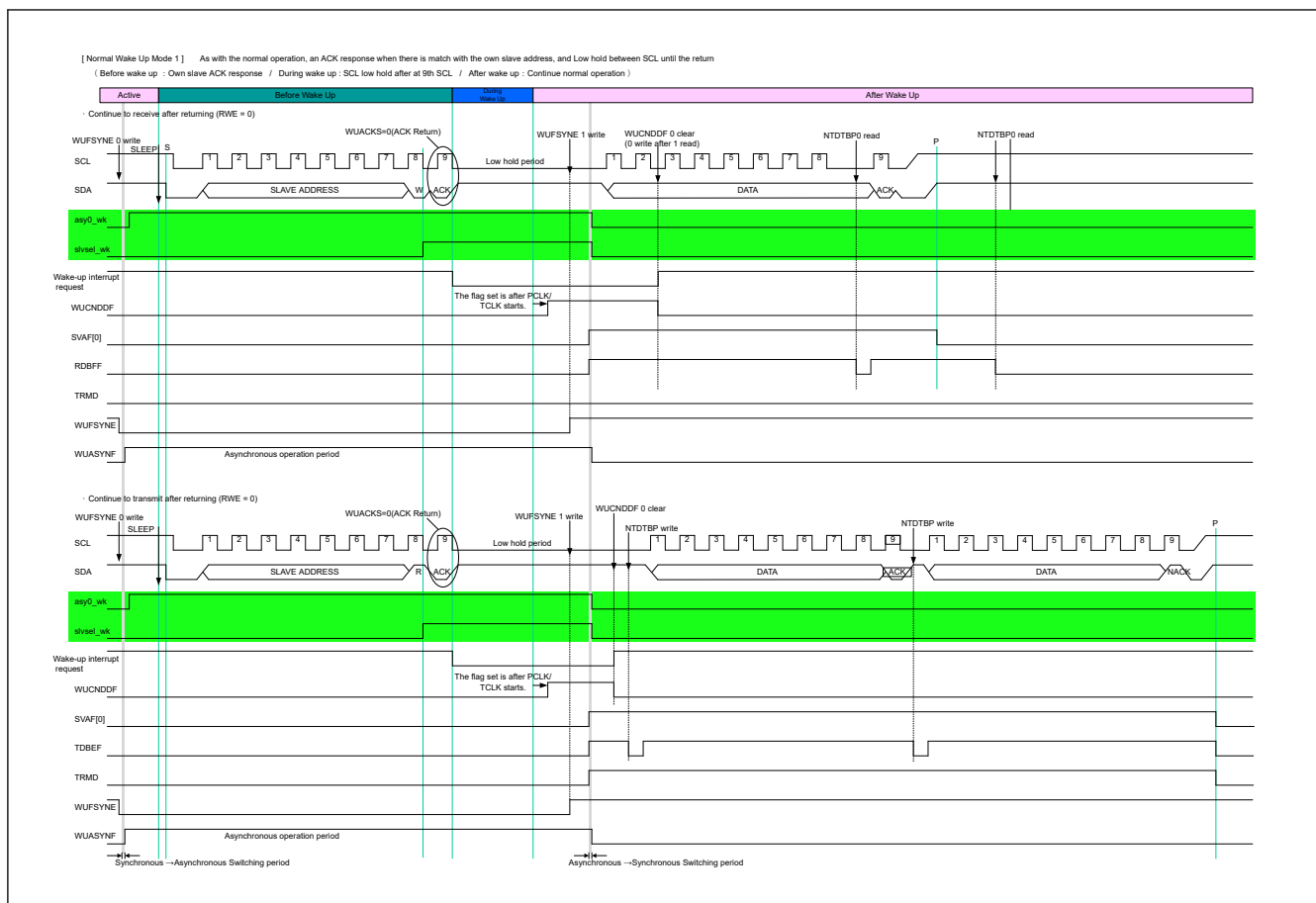


Figure 40.128 Timing of normal wake up mode 1

(2) Normal Wake Up Mode 2

This section describes the behavior, the timing, and a use case of normal WU mode 2.

A wake-up interrupt triggered by the match of the slave address makes the transition to the normal operation in the manner described below.

Also, the detailed timing is provided in [Figure 40.130](#).

Before wake-up recovery:	No response to the data received with its own slave address (until 8th SCL cycle end)
During wake-up recovery:	Holding the I3C_SCL line low during the 8th and 9th clock cycles
After wake-up recovery:	Returning ACK at the 9th clock cycle of SCL, and continuing the normal operation

If the slave address does not match, the I3C_SCL line is not held low after the fall of the 8th SCL v clock cycle. The slave operation continues.

See [Figure 40.129](#) below for a use case.

A wake-up interrupt is not generated at the transition to the normal operation if the transition is triggered by a cause (other recovery causes (IRQ)) other than a wake-up interrupt signal generated by a slave address match. BST.WUCNDDF is not set in this case. Carry out the following processing according to [Figure 40.127](#).

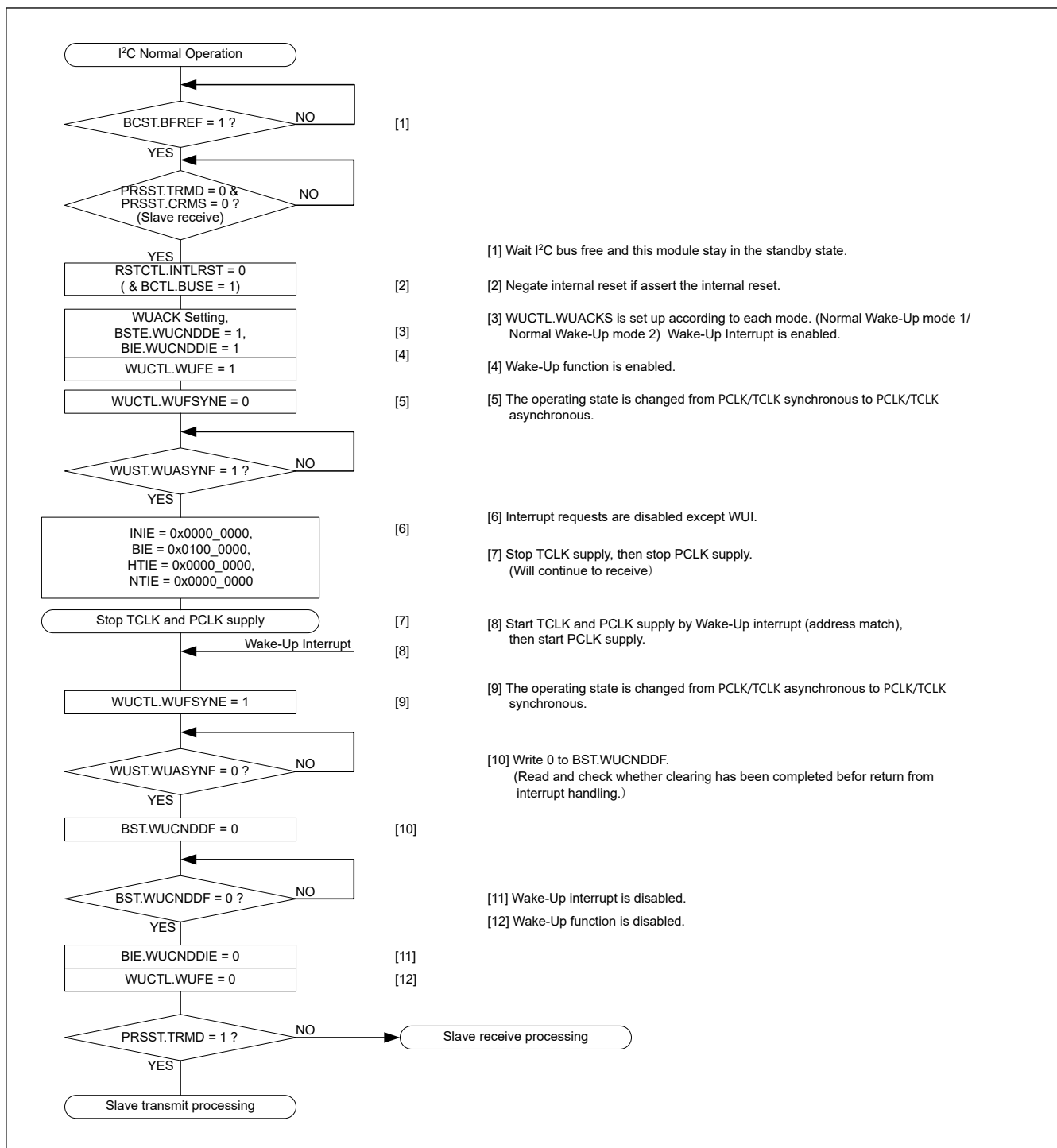


Figure 40.129 Use case of normal WU mode 2 (wake-up recovery by a wake-up interrupt triggered by the match of the slave address)

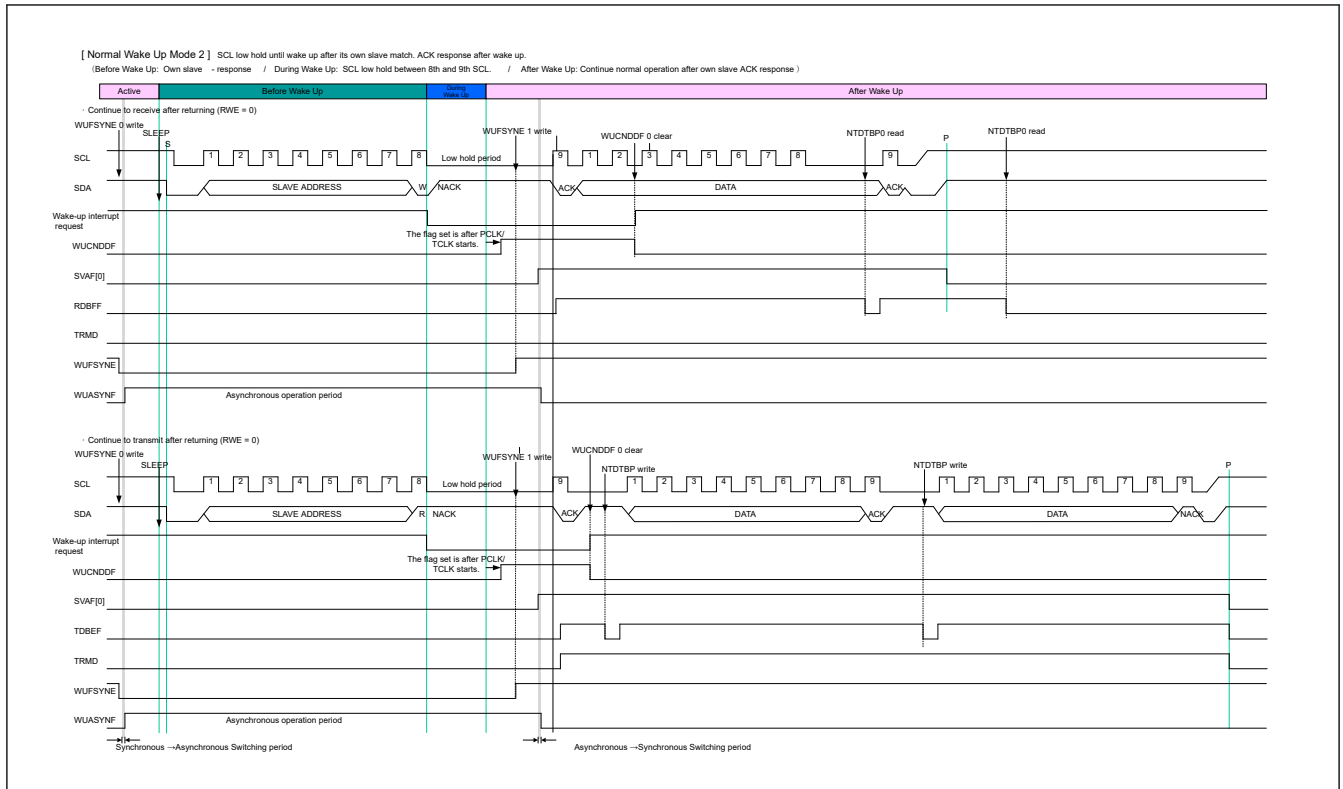


Figure 40.130 Timing of normal wake up mode 2

(3) Command recovery mode/ EEP response mode (Special Wake Up mode)

In the command recovery mode and EEP response mode, the I3C_SCL line is not held low during the wake-up recovery period (after the rise of the 9th clock cycle of SCL), so other I²C/I3C devices can use the I²C bus during this period. This section describes the behavior, the timing, and use cases of the command recovery mode and the EEP response mode.

A wake-up interrupt triggered by the match of the slave address makes the transition to the normal operation in the manner described below. Also, the detailed timing is provided in [Figure 40.133](#).

Before wake-up recovery: In response to the data received with its own slave address, ACK (command recovery mode) or NACK (EEP response mode) is returned.

During wake-up recovery: The I3C_SCL line is not held low.

After wake-up recovery: Normal operation continues after I3C initial setting.

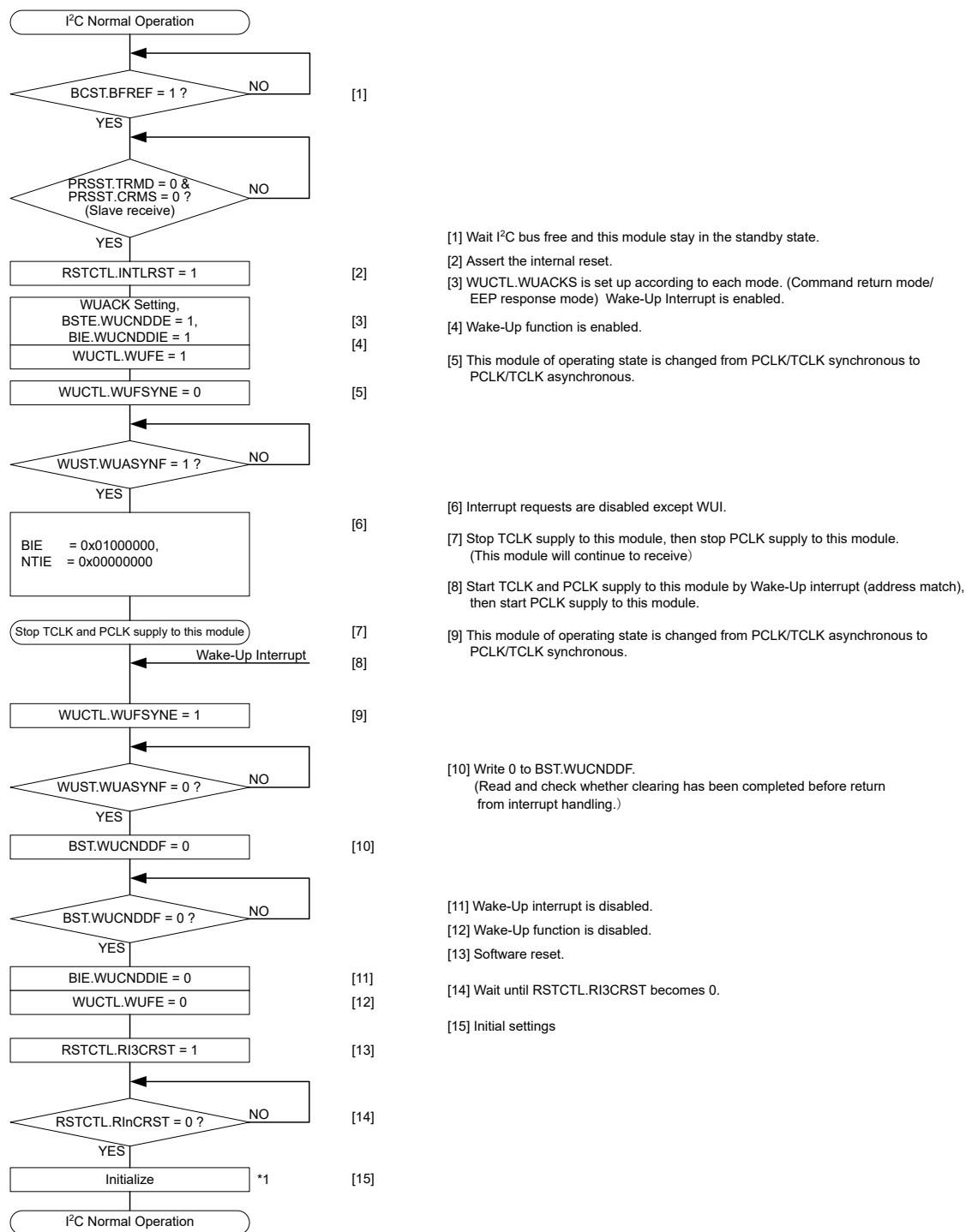
Note: Because the I3C_SCL line is not held low during wake-up recovery, the transmission/reception of the data that follows the slave address is not possible.

Note: The command recovery mode and the EEP response mode are internal reset (RSTCTL.INTLRST = 1) states. Therefore, the match of the slave address does not set the SVST flags (HOAF, GCAF, and SVAF[2:0]).

If the slave address does not match, the slave operation continues.

See [Figure 40.132](#) below for a use case.

A wake-up interrupt is not generated at the transition to the normal operation if the transition is triggered by a cause (other recovery causes (IRQ)) other than a wake-up interrupt signal generated by a slave address match. BST.WUCNDDF is not set in this case. Carry out the following processing according to [Figure 40.132](#).



Note 1. See [section 40.3.3.1. Initial Setting Flow](#) for details of initialization.

Figure 40.131 Use case of command recover mode and EEP response mode (wake-up recovery by a wake-up interrupt triggered by the match of the slave address)

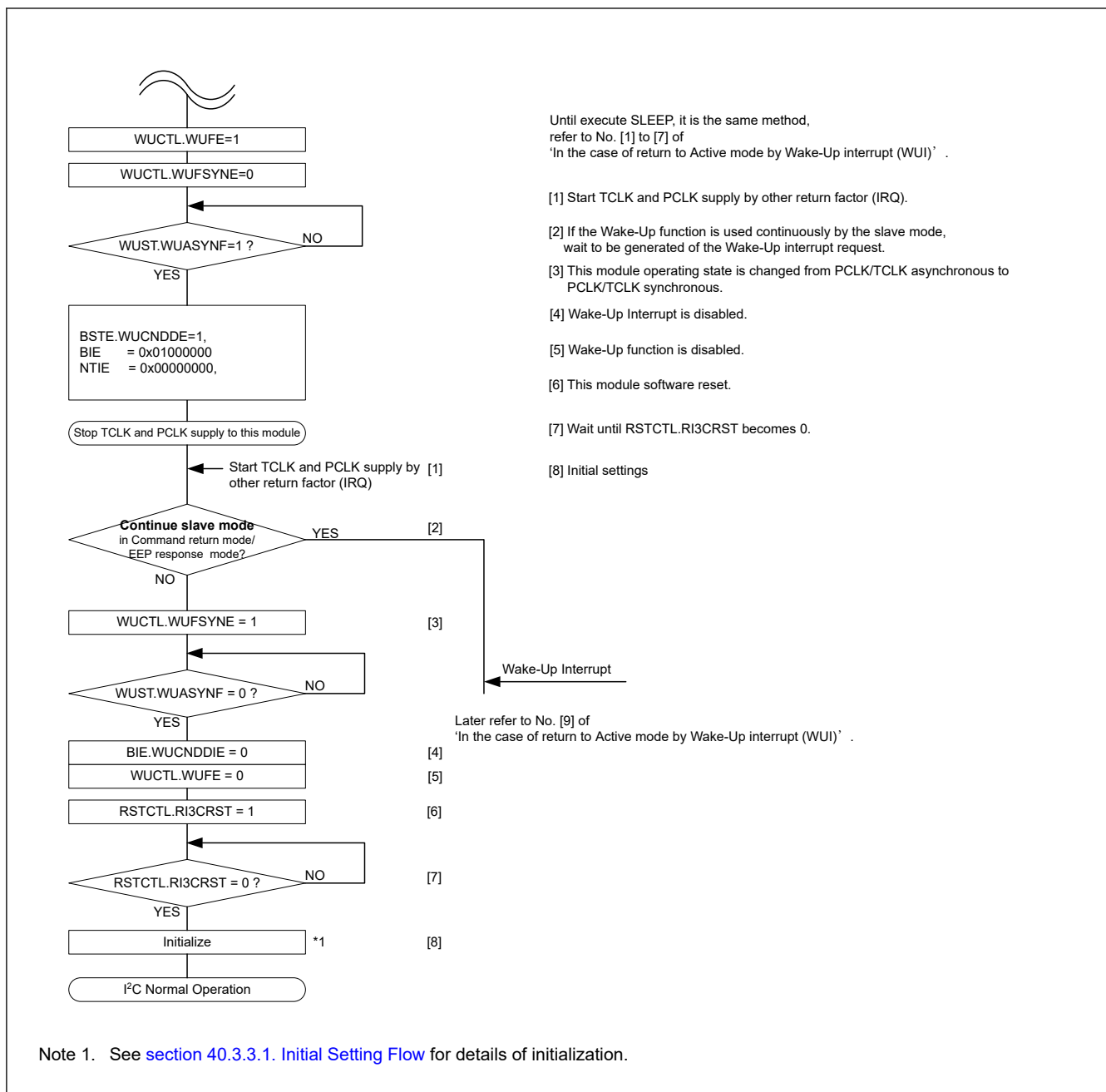


Figure 40.132 Use case of command recover mode and EEP response mode (wake-up recovery by other recovery causes (IRQ))

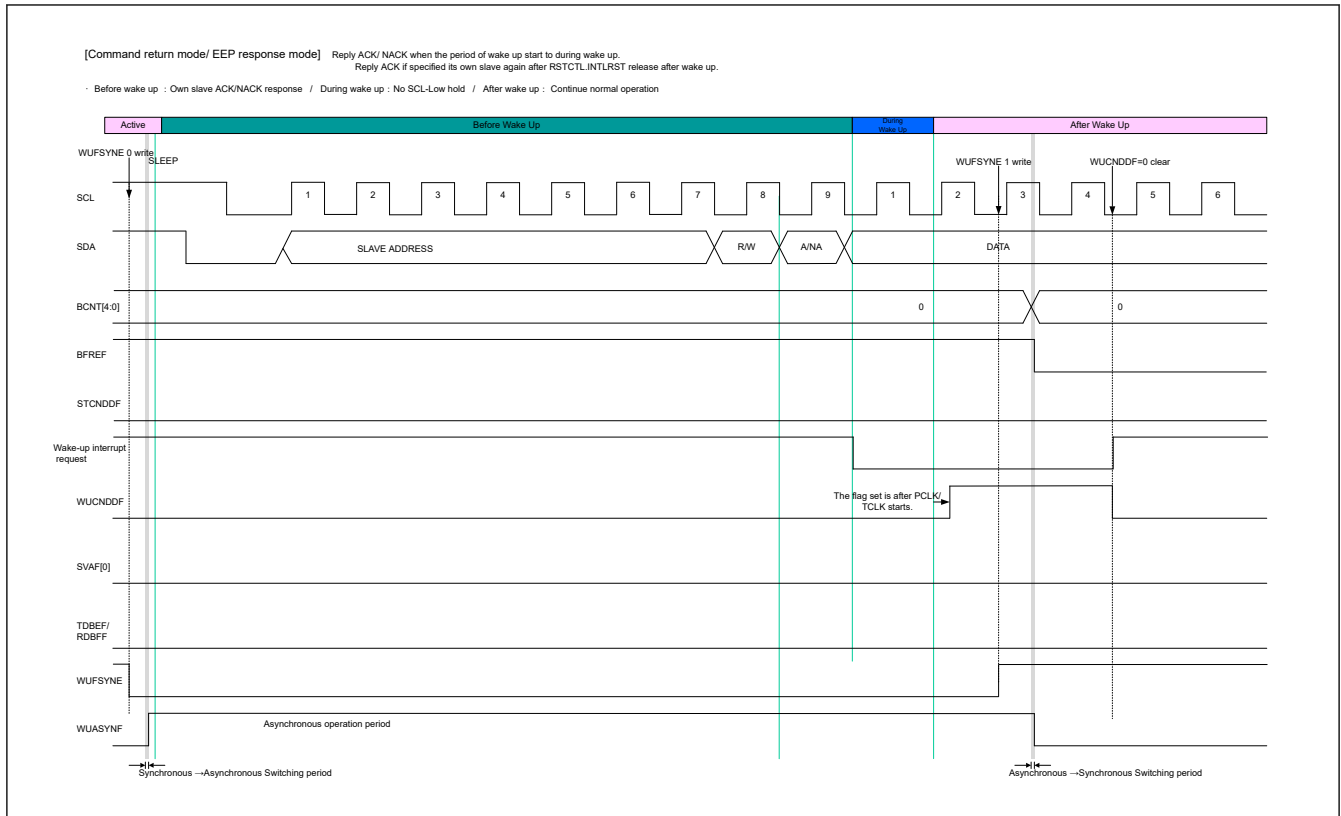


Figure 40.133 Timing of command recovery mode/EEP response mode

(4) Precautions on the use of the Wake-Up function

Precautions on the use of the Wake-up function is shown below.

- Do not change the registers in I3C except the WUCTL.WUFSYNE bit while the WUST.WUASYNF flag = 1 (while PCLK/TCLK asynchronous operation).
- Set WUCTL.WUFE = BSTE.WUCNDDDE = BIE.WUCNDDIE = 1 and PRSST.CRMS = PRSST.TRMD = 0 (slave reception mode) before switching PCLK/TCLK asynchronous mode.
- Cannot select the device ID and the 10-bit slave address for wake-up interrupt factor. Set the DVIDE bit in SVCTL and SDADLS bit in SDATBASn (n = 0 to 2) to 0.
- Sets all bits in BIE (TENDIE, NACKDIE, SPCNDDIE, STCNDDIE, ALIE, TODIE) and TDBEIE0 and RDBFIE0 bits in NTIE to 0 (Interrupt disabled) before switching the asynchronous operation.
- Do not use the timeout function while the Wake-up function is enabled (WUCTL.WUFE = 1).
- Wake-up interrupt is generated while PCLK/TCLK asynchronous operation (when WUST.WUASYNF = 1). In case of detecting slave address matching, The case of detect slave address match in PCLK/TCLK synchronous mode (WUST.WUASYNF = 0), does not occur Wake-up interrupt, and BST.WUCNDDF flag will be not set also.
- If WUCTL.WUFSYNE bit to 0 write timing and START condition of detecting a conflict, I3C might start the next reception in PCLK/TCLK synchronous operation mode. In this case, WUST.WUASYNF flag becomes 1 (switch to PCLK/TCLK asynchronous mode) when data communication is finished and detected STOP condition and starts the Wake-up event detection.
- If you want to switch from PCLK/TCLK asynchronous operation to PCLK/TCLK synchronous operation without address match detection, it will switch in the STOP condition detection. When the WUCTL.WUFSYNE bit was set to 1 in a bus free state, it is continued PCLK/TCLK asynchronous operation (Reception operation: waiting communication frame). WUST.WUASYNF flag becomes to 0 when I3C detect the STOP condition of the next communication frame, and I3C switches to PCLK/TCLK synchronous operation.
- After writing 0 to WUFSYNE bit in WUCTL, do not change I3C operation mode setting register (BFCTL, SCSTRCTL, ACKCTL, INCTL, SVCTL, SDATBASn (n = 0 to 2)) until switched to the PCLK/TCLK asynchronous operation from

PCLK/TCLK synchronous operation (while WUST.WUASYNF flag = 1). If register value changes by the interrupt processing etc. in this period, I3C might malfunction without succeeding to the setting to the asynchronous operation.

- During PCLK/TCLK asynchronous operation (WUST.WUASYNF = 1), do not refer to each flag of SVST, BST, NTST, HTST register and BCST.BFREF flag.
- Do not set ACKCTL.ACKT = 1 in order to make an ACK response in the synchronization unit when Wake-up is performed by slave address match in Normal wake-up mode 2.

40.3.2.5.2 Wake Up Function

(1) I3C Master Wake-Up

Wake-up interrupt causes of I3C master are shown below.

- SDA low detection (IBI request from I3C slave)

The operation when transitioning to active mode (normal operation) by Wake-Up interrupt of SDA Low detection is shown below.

Before wake-up recovery: SDA Low Drive is detected and the I3C_WU interrupt is asserted.

During wake-up recovery: Keep I3C_SCL Line High.

After wake-up recovery: Drive SCL Low and complete START condition.
SCL is supplied on the I3C bus and IBI from I3C slave is received.

If transition to active mode (normal operation) due to other factors, disable the Wake-up function as necessary.

After confirming PRSTDBG.SDILV = 1, set WUCTL.WUFE = 0.

Do not use the timeout function while the Wake-up function is enabled (WUCTL.WUFE = 1).

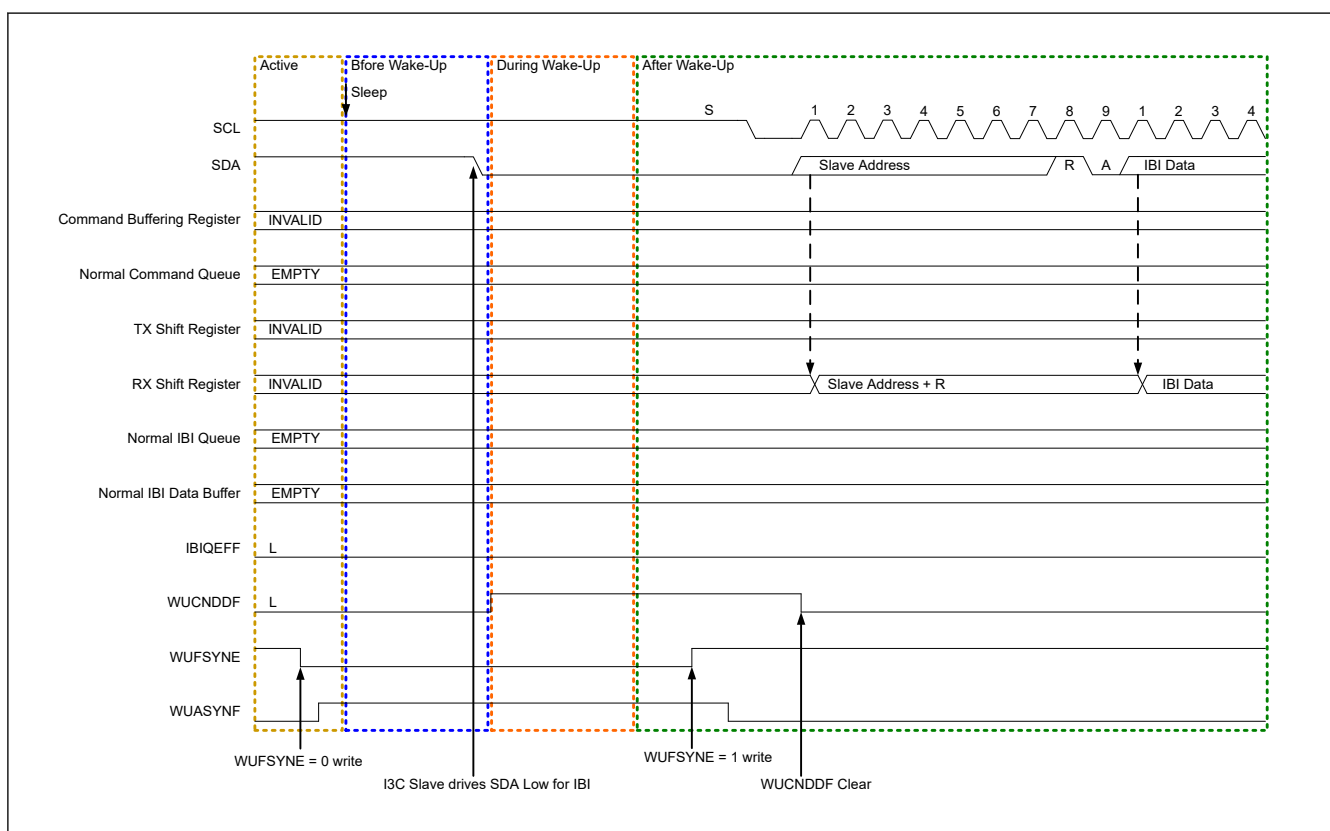


Figure 40.134 I3C master wake-up operation

(2) I3C Slave Wake-Up

Wake-up interrupt causes of I3C slave are shown below.

- Broadcast address (0x7E) and detects its own Slave address match

The operation when the Broadcast address (0x7E) and transition to active mode (normal operation) by Wake-up interrupt by detecting its own Slave address match is shown.

- Before wake-up recovery:
1. If I3C detects BA (0x7E/W) following a START (or Repeated START) Condition, then I3C shall generate ACK (after 0x7E/W).
 2. If I3C detects its own Dynamic address after a Repeated START condition following Step1, then I3C shall generate NACK (after its own Dynamic address) and then issues a I3C_WU interrupt.
- During wake-up recovery: I3C always generates NACK.
- After wake-up recovery: Normal operation continues.

If transition to active mode (normal operation) due to other factors, disable the Wake-up function as necessary. Do not use the timeout function while the Wake-up function is enabled (WUCTL.WUFE = 1).

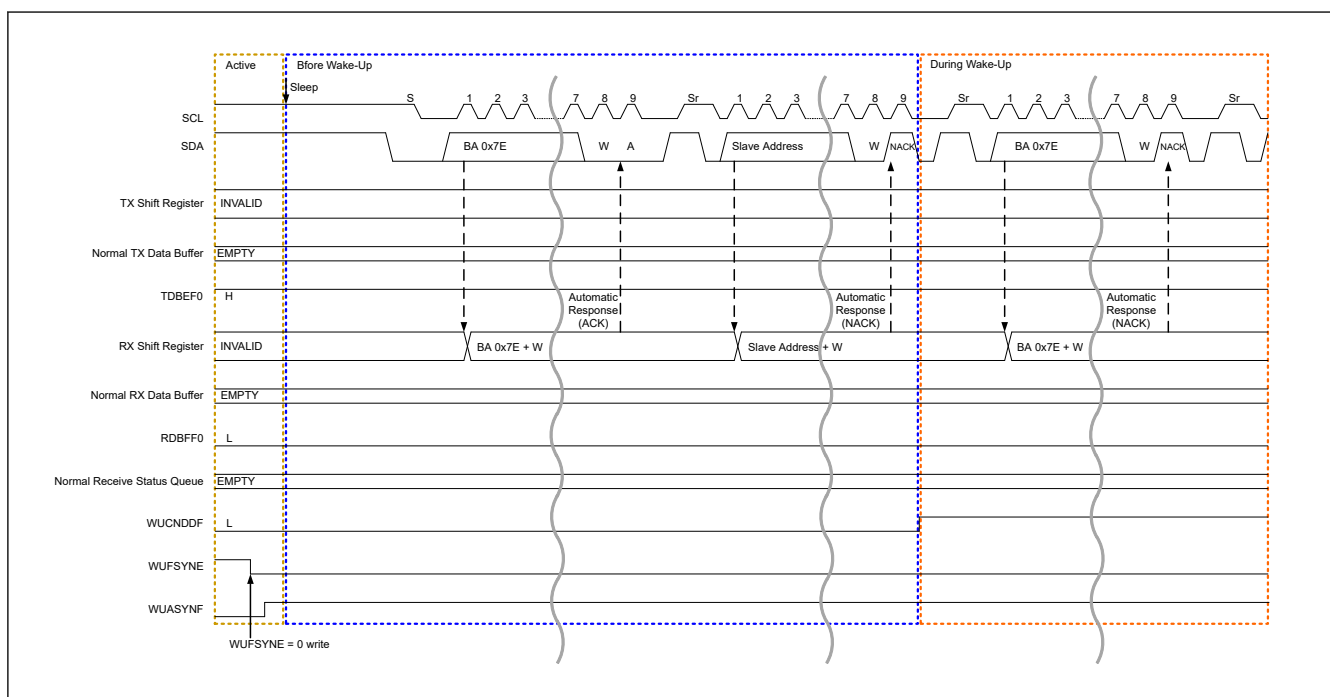


Figure 40.135 I3C slave wake-up operation (1/2)

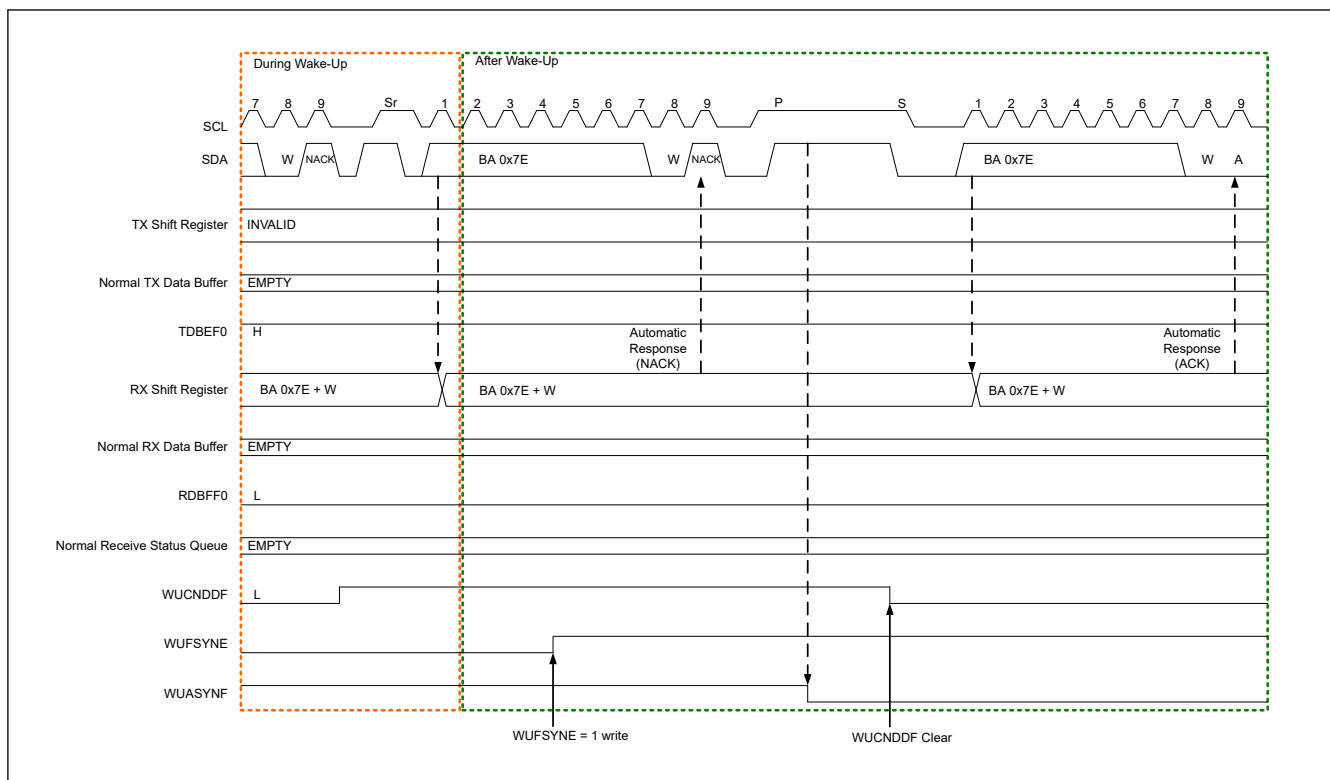


Figure 40.136 I3C slave wake-up operation (2/2)

40.3.2.6 Other

40.3.2.6.1 SCL Synchronization Circuit [I²C Mode]

This function is enabled while the PRTS.PRTMD bit is set to 1.

In generation of the SCL clock, I3C starts counting out the value for width at high level specified in STDBR.SBRHO[7:0] when it detects a rising edge on the I3C_SCL line and drives the I3C_SCL line low once counting of the width at high level is complete.

When I3C detects the falling edge of the I3C_SCL line, it starts counting out the width at low level period specified in STDBR.SBRLO[7:0], and then stops driving the I3C_SCL line (releases the line) once counting of the width at low level is complete. The SCL clock is thus generated.

If multiple master devices are connected to the I²C bus, a collision of SCL signals may arise due to contention with another master device. In such cases, the master devices have to synchronize their SCL signals. Since this synchronization of SCL signals must be bit by bit, I3C is equipped with a facility (the SCL synchronization circuit) to obtain bit-by-bit synchronization of the SCL clock signals by monitoring the I3C_SCL line while in master mode.

When I3C has detected a rising edge on the I3C_SCL line and thus started counting out the width at high level specified in STDBR.SBRHO[7:0], and the level on the I3C_SCL line falls because an SCL signal is being generated by another master device, I3C stops counting when it detects the falling edge, drives the level on the I3C_SCL line low, and starts counting out the width at low level specified in STDBR.SBRLO[7:0]. When I3C finishes counting out the width at low level, it stops driving the I3C_SCL line to the low level (releases the line). At this time, if the width at low level of the SCL clock signal from the other master device is longer than the width at low level set in this module, the width at low level of the SCL signal will be extended. Once the width at low level for the other master device has ended, the SCL signal rises because the I3C_SCL line has been released. When I3C finishes outputting the low-level period of the SCL clock, the I3C_SCL line is released and the SCL clock rises. That is, in cases of contention of SCL signals from more than one master, the width at high level of the SCL signal is synchronized with that of the clock having the narrower width, and the width at low level of the SCL signal is synchronized with that of the clock having the broader width. However, such synchronization of the SCL signal is only enabled when the SCSYNE bit in BFCTL is set to 1.

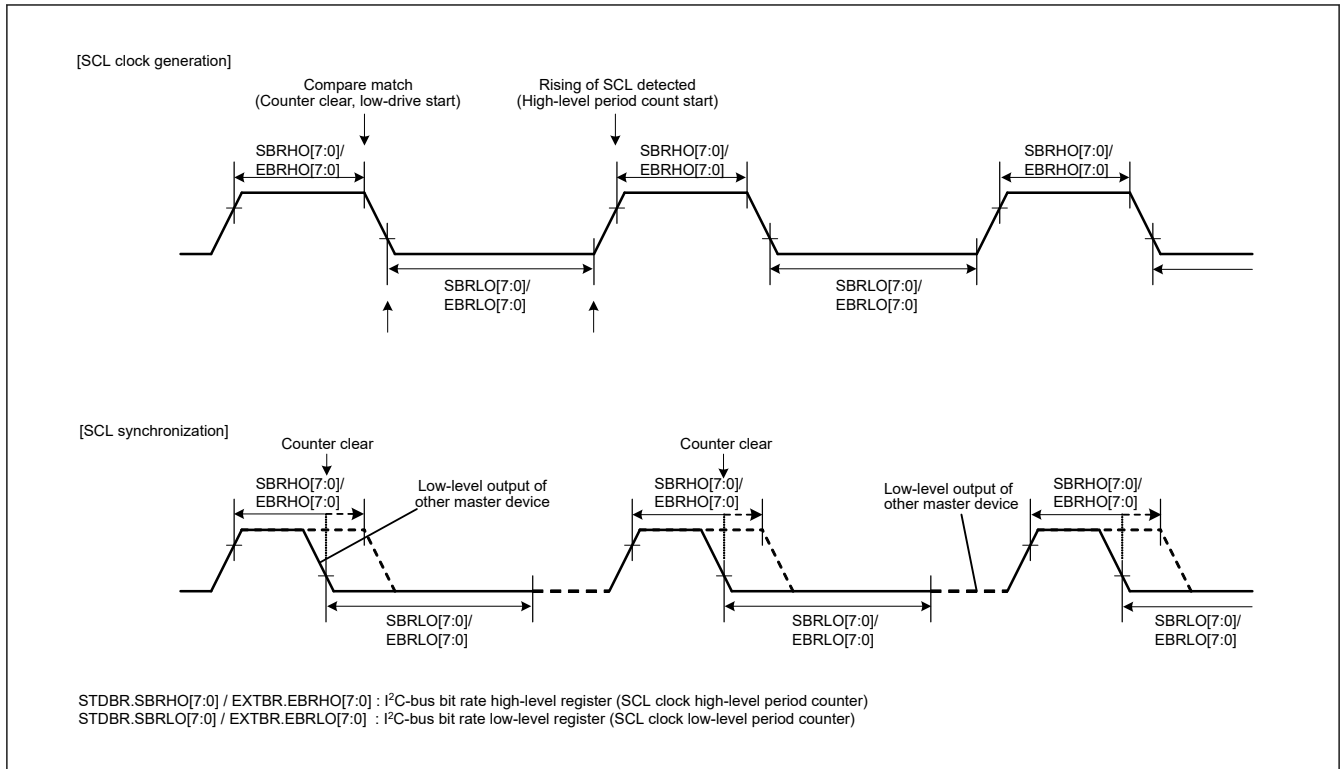


Figure 40.137 Generation and synchronization of the SCL signal

40.3.2.6.2 Facility for Delaying SDA Output [I²C Mode]

I3C module incorporates a facility for delaying output on the I3C_SDA line. The delay can be applied to all output (issuing of the START, Repeated START, and STOP conditions, data, and the ACK and NACK signals) on the I3C_SDA line.

With the SDA output delay facility, SDA output is delayed from detection of a falling edge of the SCL signal to ensure that the SDA signal is output within the interval over which the SCL clock is at the low level. Doing this leads to usage with the aim of preventing erroneous operation of communications devices, with the aim of satisfying the 300 ns (minimum) data-hold time requirement of the SMBus specification.

The output delay facility is enabled by setting the SDOD[2:0] bits in OUTCTL to any value other than 000b, and disabled by setting the same bits to 000b.

While the SDA output delay facility is enabled (while the SDOD[2:0] bits in OUTCTL are set to any value other than 000b), the SDODCS bit in OUTCTL selects the clock source for counting by the SDA output delay counter as the internal base clock (I3C ϕ) for I3C module or as a clock signal derived by dividing the frequency of the internal base clock by two (I3C ϕ /2). The counter counts the number of cycles set in the SDOD[2:0] bits in OUTCTL. After counting of the set number of cycles of delay is completed, I3C module places the required output (START, Repeated START, or STOP condition, data, or an ACK or NACK signal) on the I3C_SDA line.

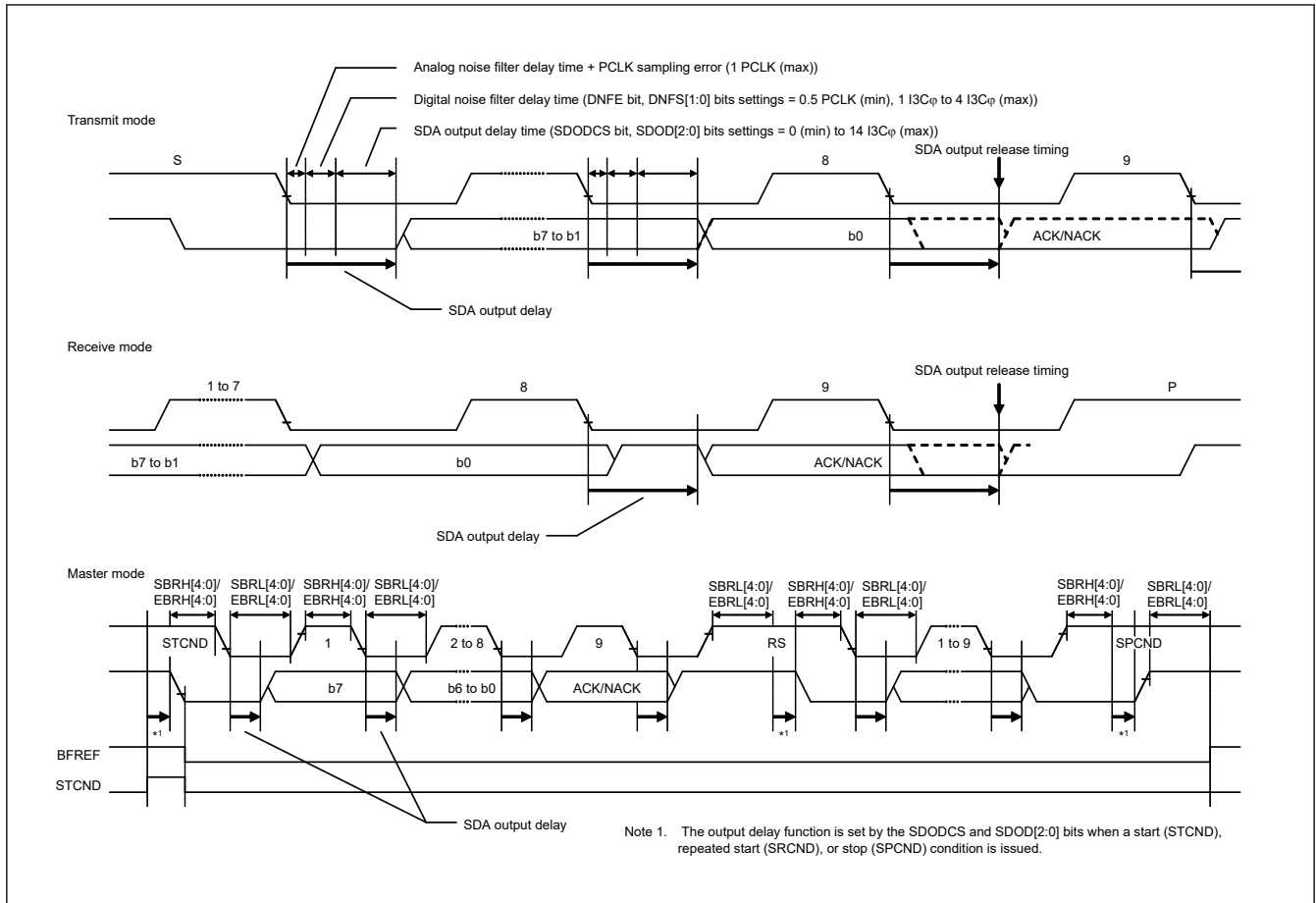


Figure 40.138 SDA output delay facility

40.3.2.6.3 Digital Noise-Filter Circuits [I²C Mode]

The states of the I3C_SCL and I3C_SDA pins are conveyed to the internal circuitry through digital noise-filter circuits. [Figure 40.139](#) is a block diagram of the digital noise-filter circuit.

The on-chip digital noise-filter circuit of I3C consists of 16 flip-flop circuit stages connected in series and a match detection circuit. When HS mode is selected, only the first four flip-flop circuit stages are enabled.

The number of effective stages in the digital noise filter is selected by the INCTL.DNFS[3:0] bits. The selected number of effective stages determines the noise-filtering capability as a period from one to sixteen I3Cφ cycles.

The input signal to the I3C_SCL pin (or I3C_SDA pin) is sampled on rising edges of the I3Cφ signal. When the input signal level matches the output level of the number of effective flip-flop circuit stages as selected by the INCTL.DNFS[3:0] bits, the signal level is conveyed to the subsequent stage. If the signal levels do not match, the previous value is retained.

If the ratio between the frequency of the internal operating clock (TCLK) and the transfer rate is small (For example, data transfer at 400 kbps with TCLK = 4 MHz), the characteristics of the digital noise filter may lead to the elimination of needed signals as noise.

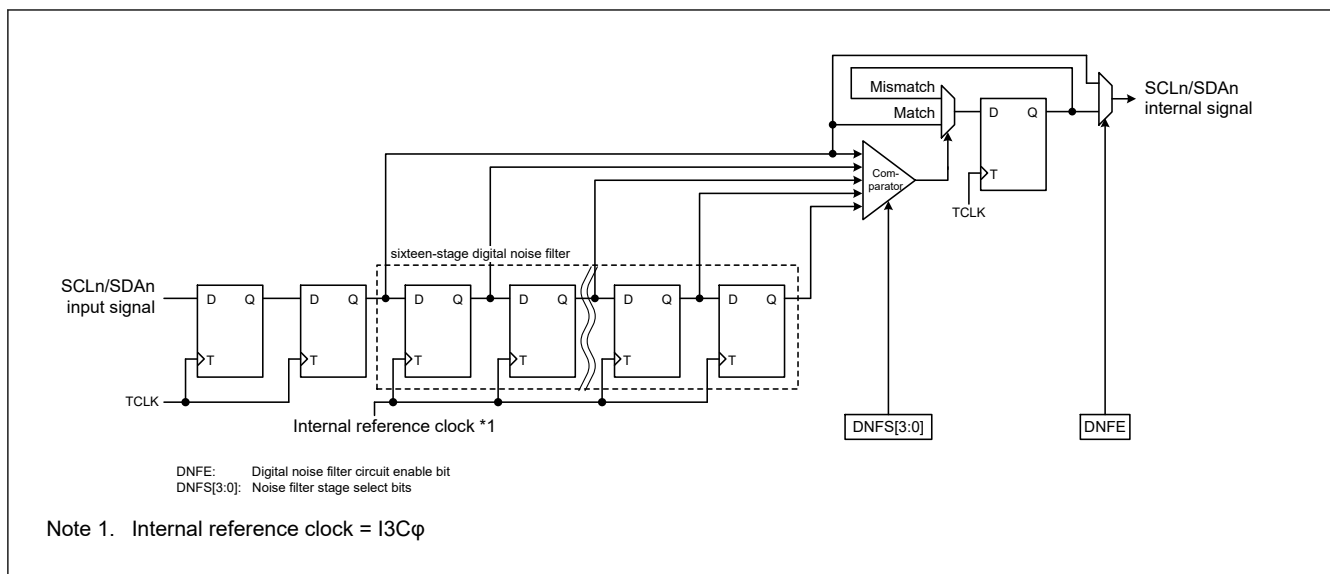


Figure 40.139 Block diagram of digital noise filter circuit

40.3.3 Operation

40.3.3.1 Initial Setting Flow

40.3.3.1.1 I²C Initial Setting Flow (Single Buffer Transfer)

Before starting data transmission and reception in I²C protocol mode, initialize I3C module.

First, set the CECTL.CLKE to 1, and set the BCTL.BUSE bit to 0 (I3C_SCL, I3C_SDA pins not driven).

Next, Set the RSTCTL.RI3CRST bit to 1. This initializes the registers and internal state of I3C module. Then, it waits for RI3CRST to become "0". For flags and registers to be initialized by this operation, see [section 40.6. Reset Descriptions](#).

After that, set each register required for operation. For detail procedure, see [Figure 40.140](#).

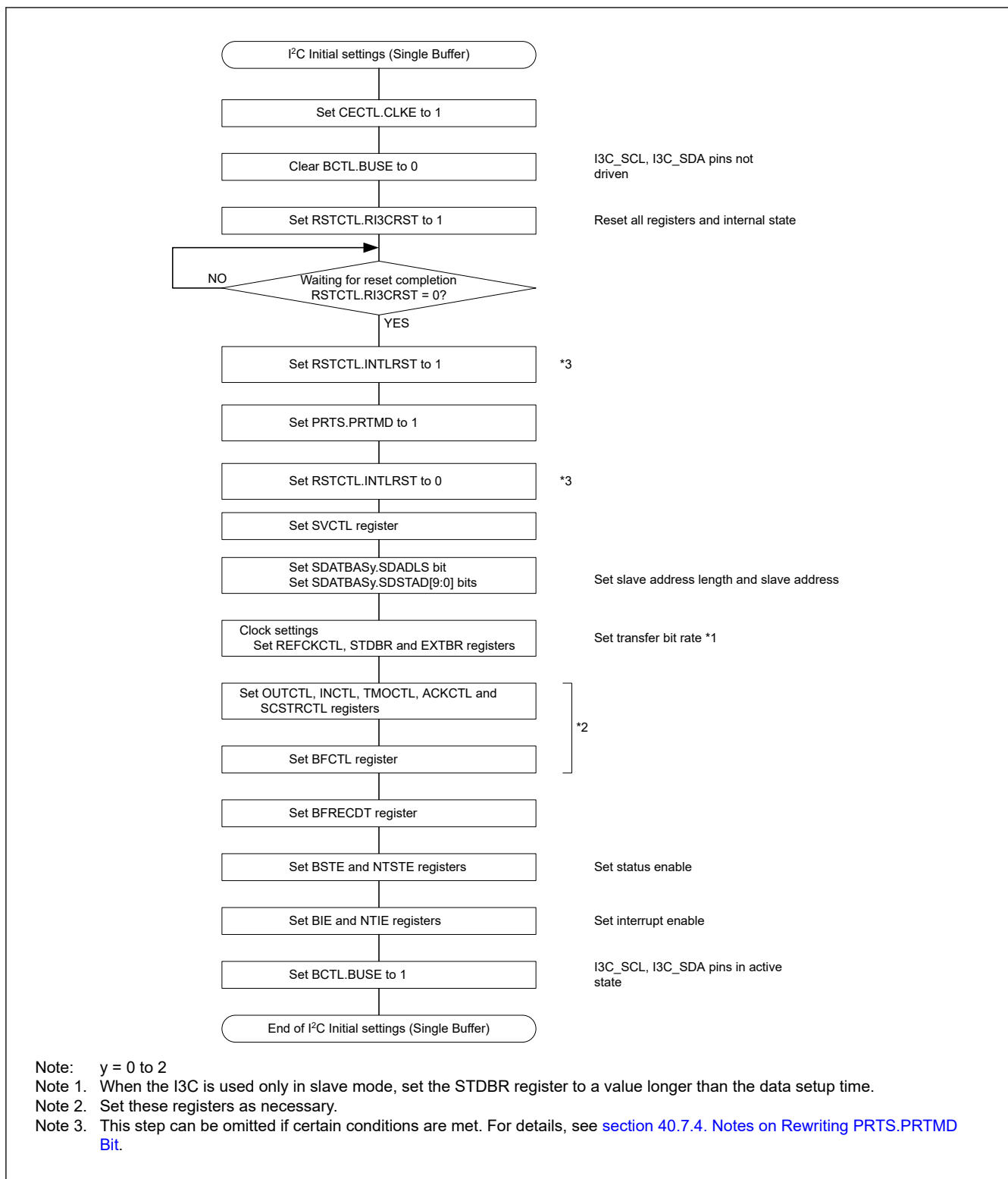


Figure 40.140 Example of I²C initialization flowchart (single buffer transfer)

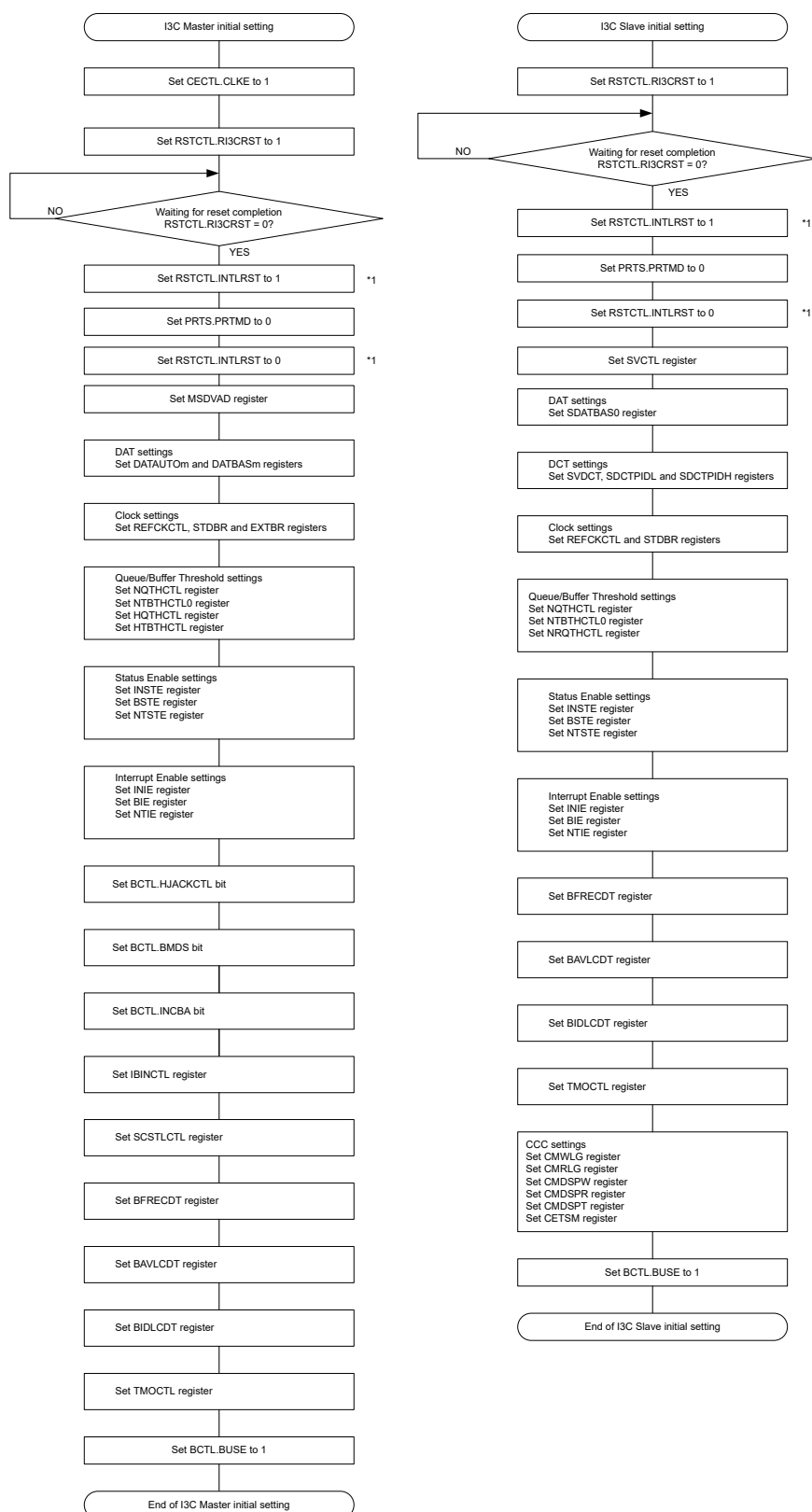
40.3.3.1.2 I3C Initial Setting Flow

Before starting data transmission and reception in I3C protocol mode, initialize I3C module.

First, set the CECTL.CLKE to 1.

Next, Set the RSTCTL.RI3CRST bit to 1. This initializes the registers and internal state of I3C module. Then, it waits for RI3CRST to become "0". For flags and registers to be initialized by this operation, see [section 40.6. Reset Descriptions](#).

After that, set each register required for operation. For detail procedure on each of the master mode and slave mode, see [Figure 40.141](#).



Note: y = 0 to 2

Note 1. This step can be omitted if certain conditions are met. For details, see [section 40.7.4. Notes on Rewriting PRTS.PRTMD Bit](#).

Figure 40.141 Example of each initialization flowchart for I3C master mode and slave mode

40.3.3.2 I3C Communication Flow

Figure 40.142 illustrates how I3C communication is initiated:

- All I3C communication occurs within a frame. The frame begins with a START, followed by one or more transfers, and a STOP.
- For the HDR modes:
 - First the dedicated Broadcast I3C address (0x7E) is issued to all Slaves on the I3C bus.
 - Then one of the EnterHDR CCCs is issued, indicating that the Master is entering an HDR mode. Each HDR mode has its own EnterHDR CCC.
 - This is followed by one or more HDR transfers.
 - HDR mode is ended by using the HDR exit pattern protocol.

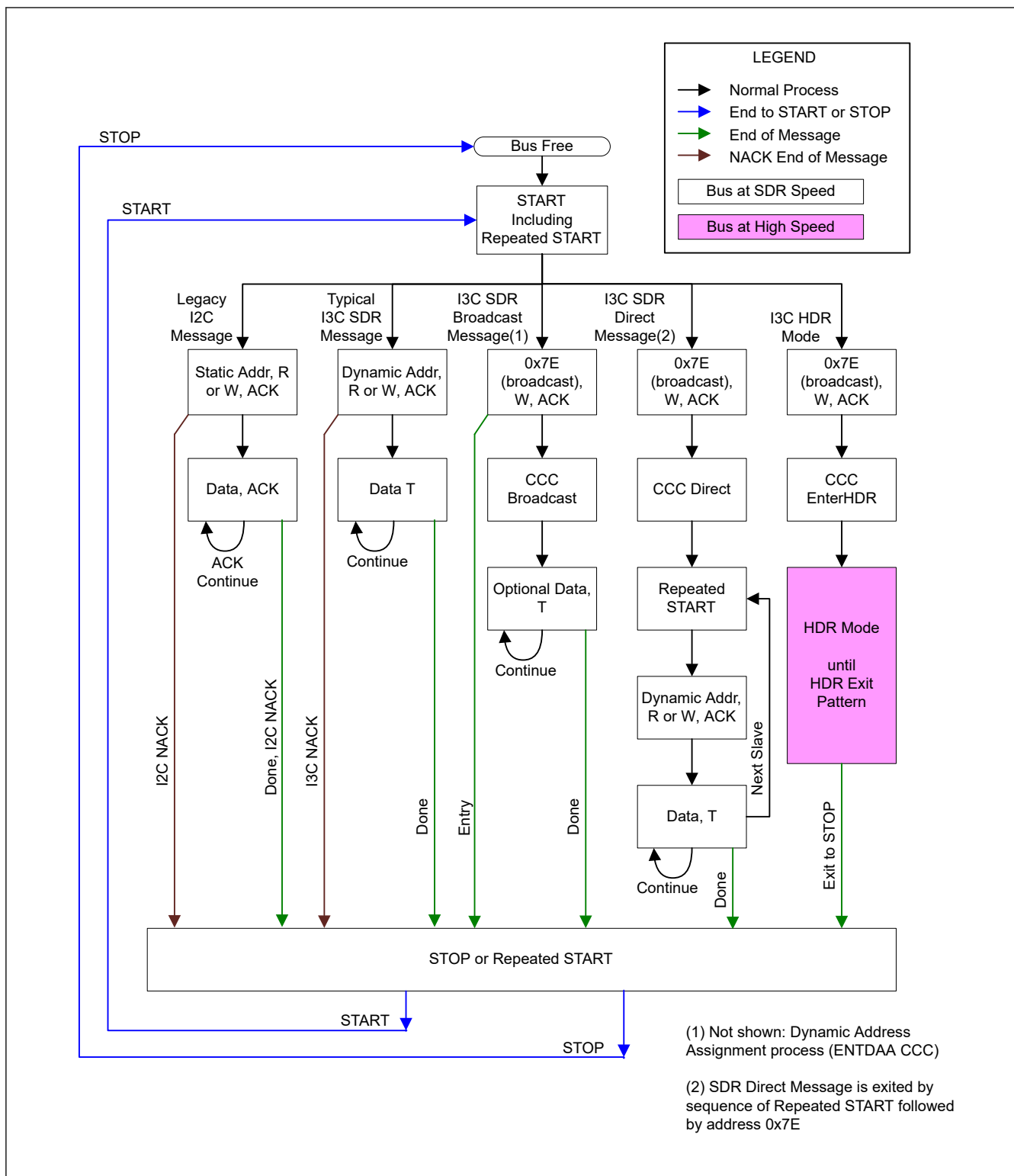


Figure 40.142 I3C communication flow

I3C is based on a frame encapsulation approach. A frame includes a data payload. The transfer protocol for the data payload is either SDR or HDR. Frames are bordered by I²C-like bus management.

The I3C frame always includes at least the START, the Header, the Data, and the STOP. The Header following a START allows for Bus Arbitration. The Master uses the Header to address Slave device (s). Slave devices (s) may use the Header Arbitration for multiple purposes: for In-Band Interrupt, and for Secondary Master functionality.

Common Command Codes (CCCs) are used to enter the High Data Rate (HDR) modes. It is important to understand that I3C bus activity for the HDR Message does not follow the Legacy I²C format.

I3C allows only one Master to have control of the I3C bus at a time. Mechanisms for handoff of the Master role from one device to another device are provided.

40.3.3.3 Master Mode Communication Flow

40.3.3.3.1 I²C Master Transmission Flow (Single Buffer Transfer)

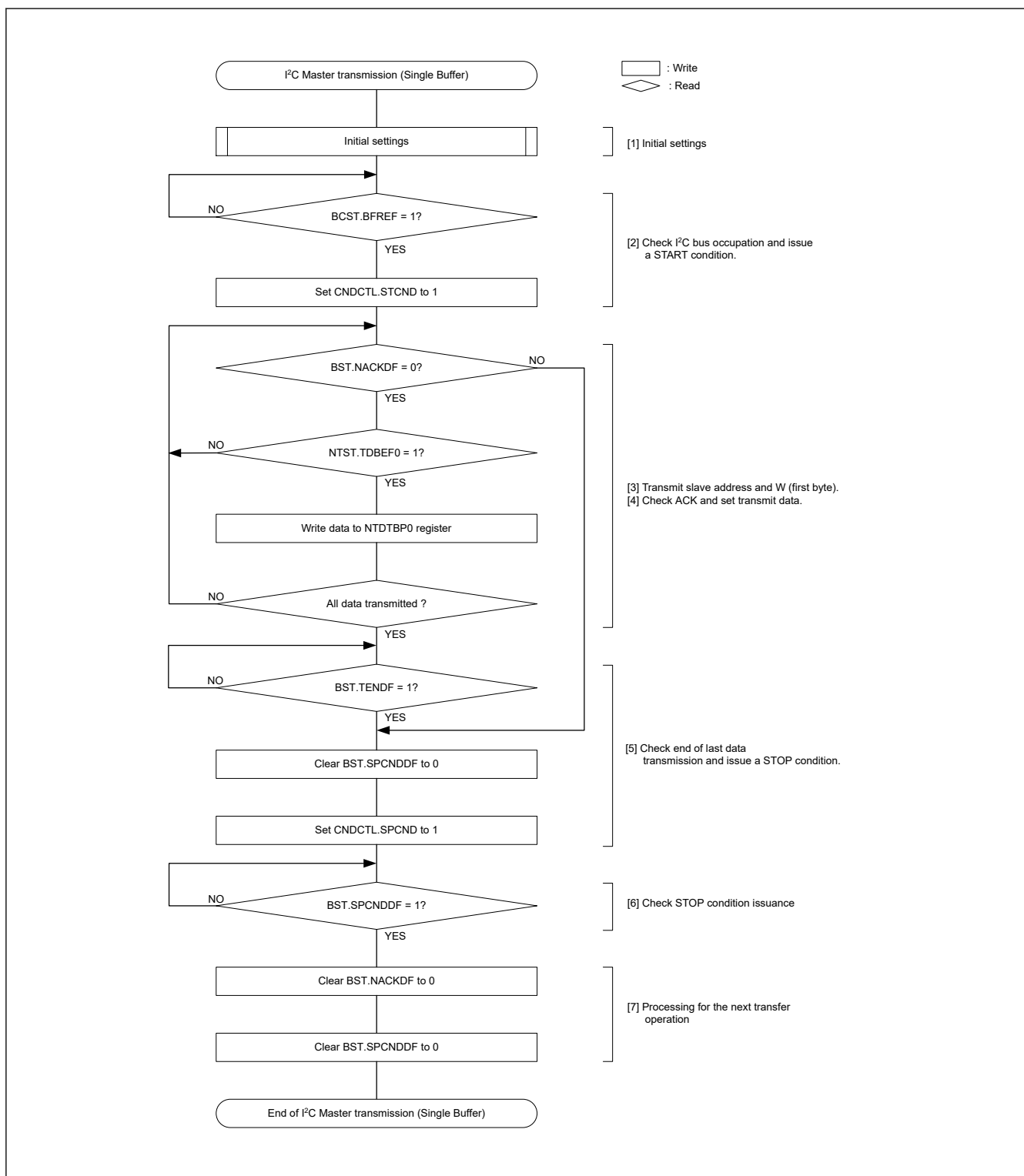
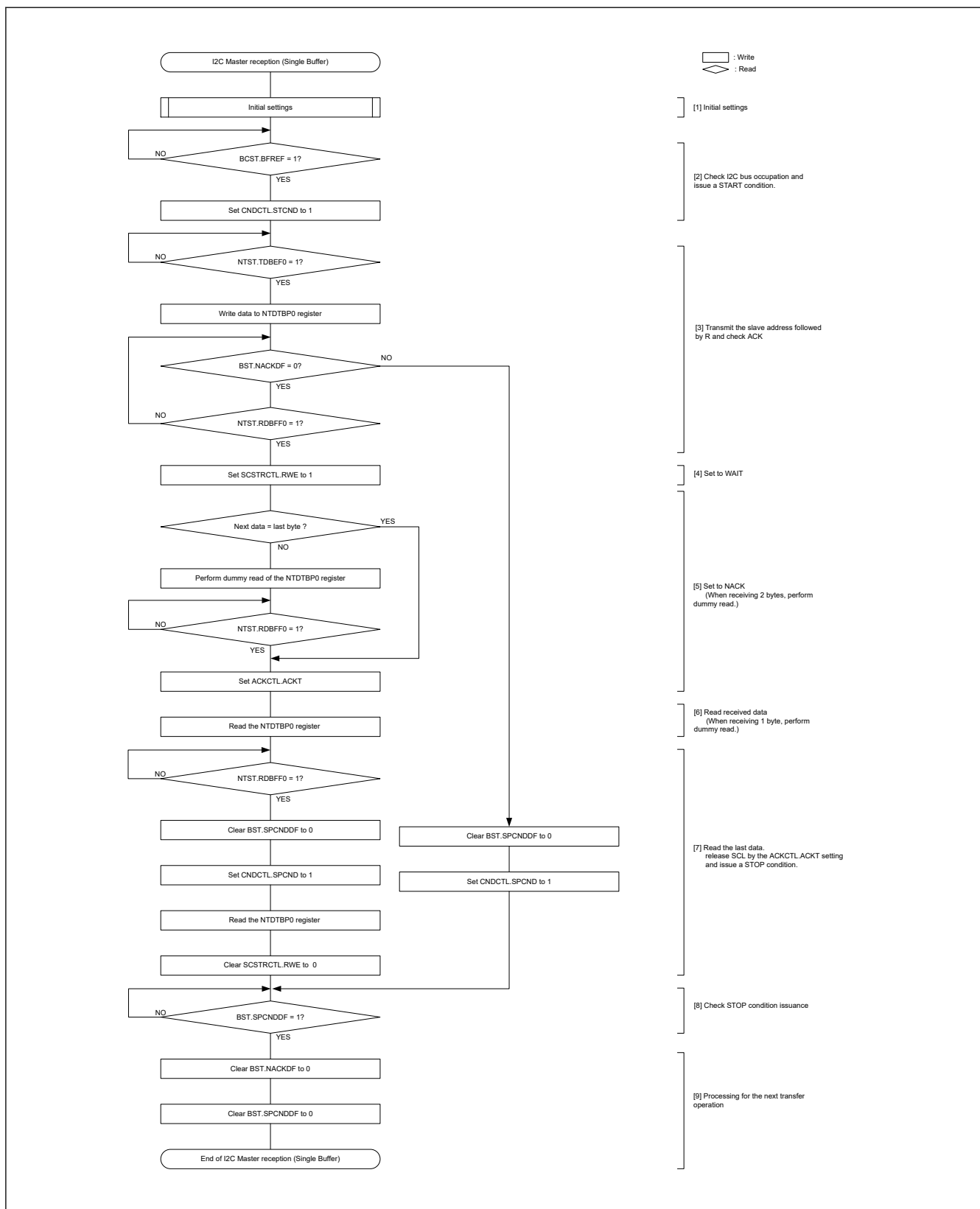
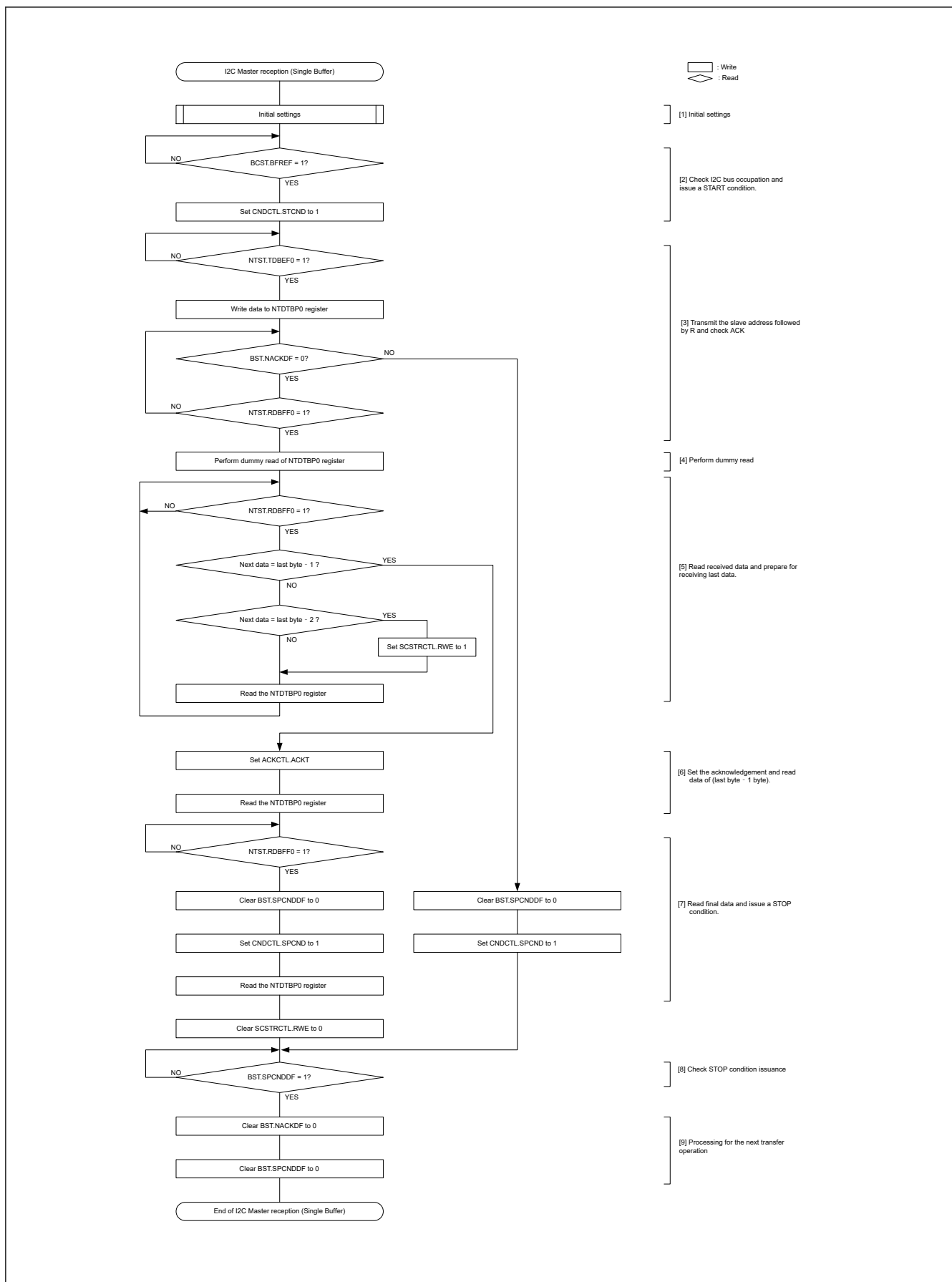


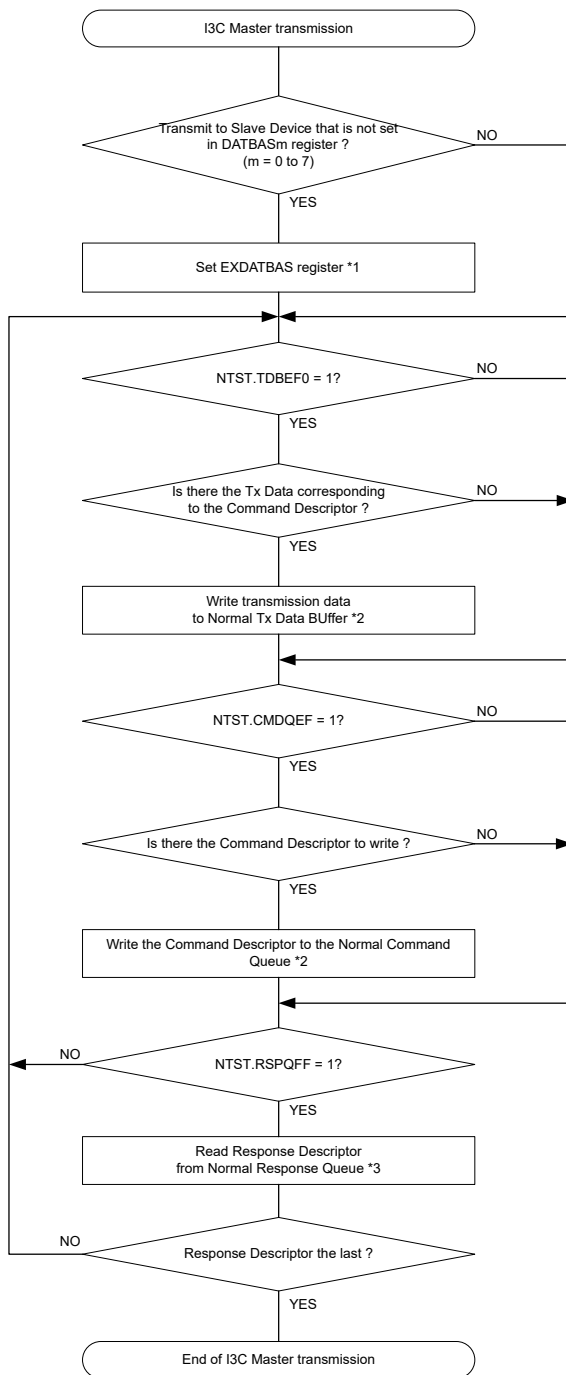
Figure 40.143 Example of I²C master transmission flowchart (single buffer transfer)

40.3.3.3.2 I²C Master Reception Flow (Single Buffer Transfer)Figure 40.144 Example of I²C master reception flowchart (7-bit address format, 1 or 2 bytes)

Figure 40.145 Example of I²C master reception flowchart (7-bit address format, 3 bytes or more)

40.3.3.3.3 I3C Master Transmission Flow (Normal FIFO Buffer Transfer)

Master transmission flow in I3C normal FIFO buffer transfer is common to Legacy I²C and SDR (Private Transfer, Broadcast CCC, Direct CCC, HDR-DDR, and HDR-TSP/TSL).



Note 1. Changing the EXDATBAS register is prohibited until all response descriptors of command for the Slave device set in the EXDATBAS register are stored.

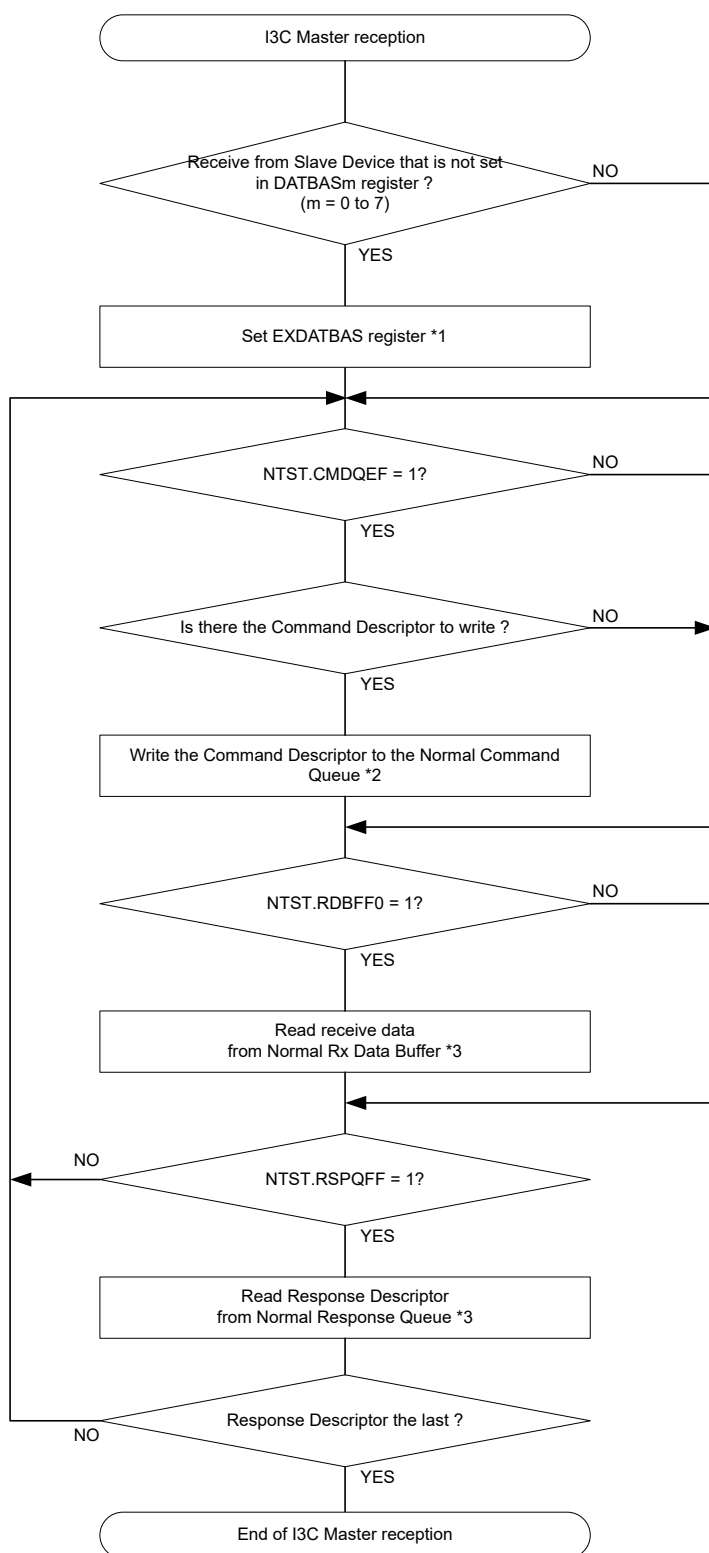
Note 2. In the case the register is write by a way other than DMAC/DTC, clear corresponding flags after write access.

Note 3. In the case the register is read by a way other than DMAC/DTC, clear corresponding flags after read access.

Figure 40.146 Example of I3C master transmission flowchart (normal FIFO buffer transfer)

40.3.3.3.4 I3C Master Reception Flow (Normal FIFO Buffer Transfer)

Master reception flow in I3C normal FIFO buffer transfer is common to Legacy I²C and SDR (Private Transfer, Broadcast CCC, Direct CCC), HDR-DDR, and HDR-TSP/TSL.



Note 1. Changing the EXDATBAS register is prohibited until all response descriptors of command for the Slave device set in the EXDATBAS register are stored.

Note 2. In the case the register is write by a way other than DMAC/DTC, clear corresponding flags after write access.

Note 3. In the case the register is read by a way other than DMAC/DTC, clear corresponding flags after read access.

Figure 40.147 Example of I3C master reception flowchart (normal FIFO buffer transfer)

40.3.3.3.5 I3C Master Transmission Flow (High Priority FIFO Buffer Transfer)

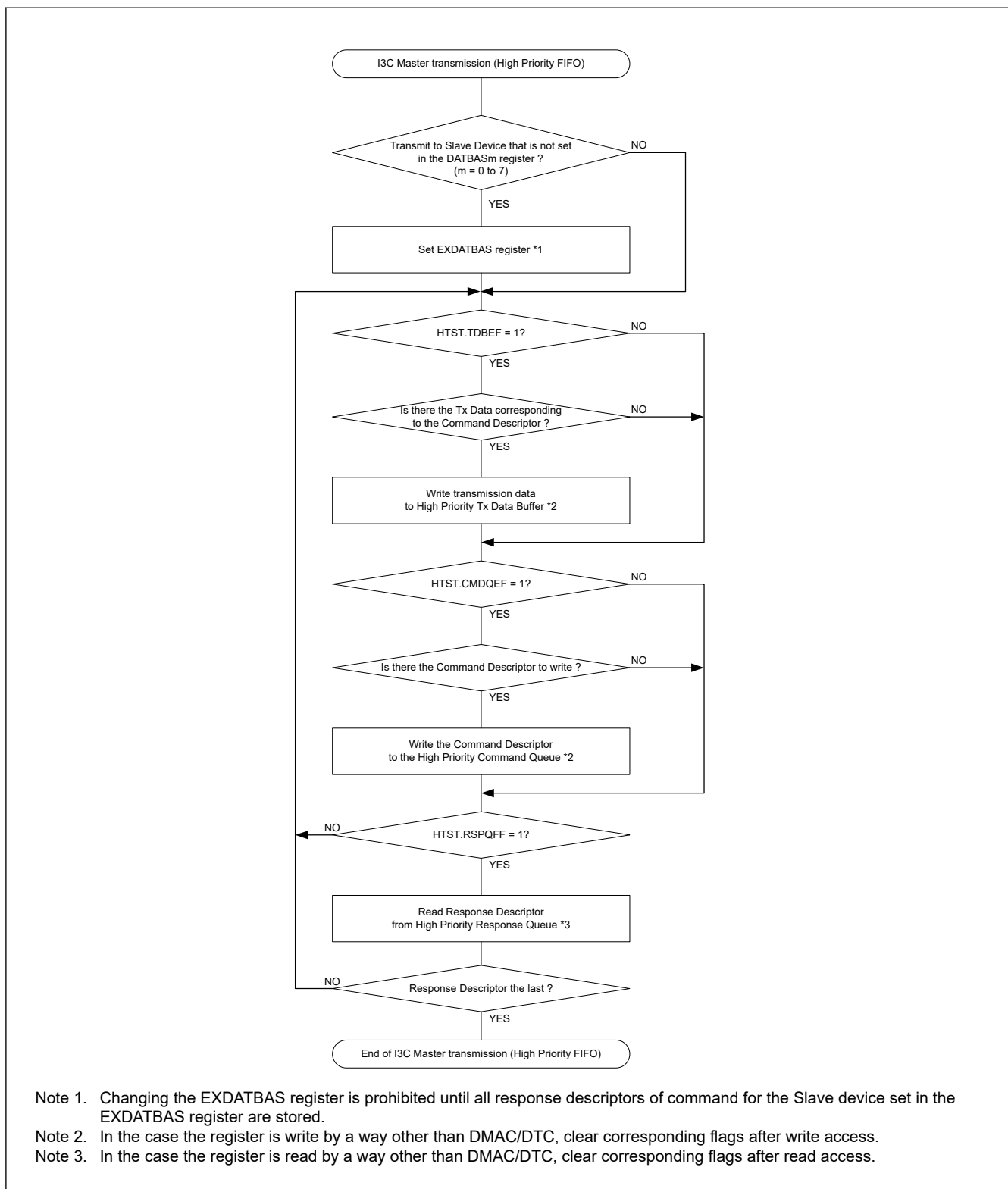
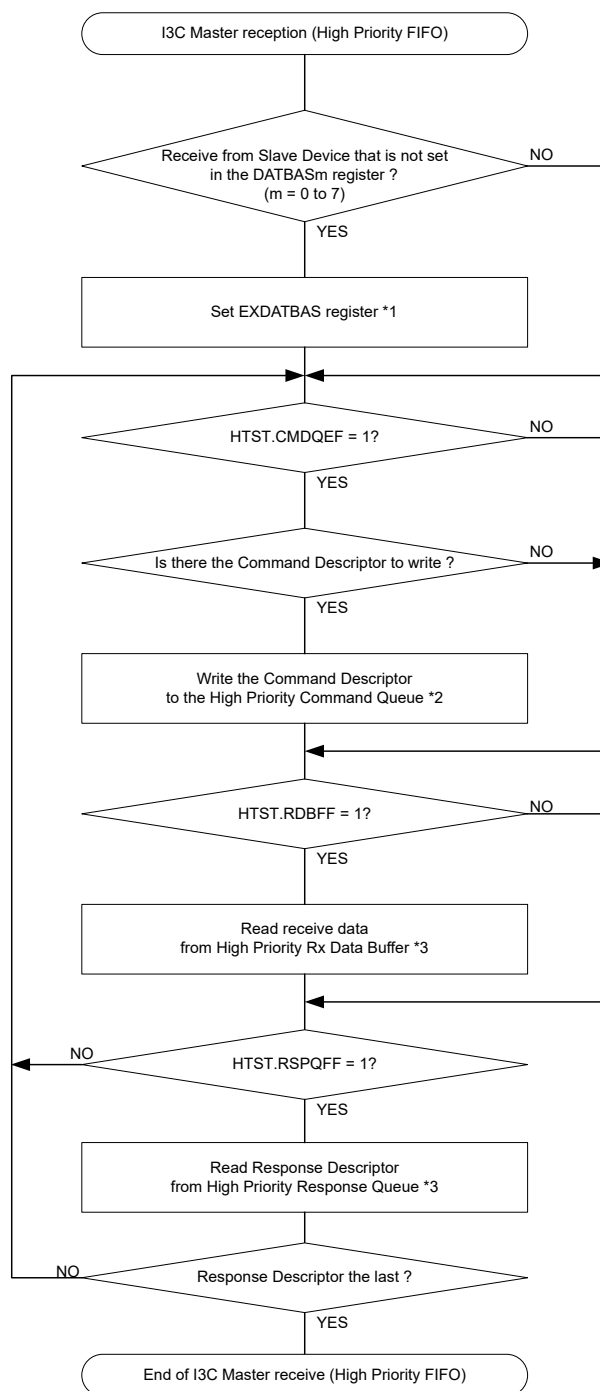


Figure 40.148 Example of I3C master transmission flowchart (high priority FIFO buffer transfer)

40.3.3.3.6 I3C Master Reception Flow (High Priority FIFO Buffer Transfer)



Note 1. Changing the EXDATBAS register is prohibited until all response descriptors of command for the Slave device set in the EXDATBAS register are stored.

Note 2. In the case the register is write by a way other than DMAC/DTC, clear corresponding flags after write access.

Note 3. In the case the register is read by a way other than DMAC/DTC, clear corresponding flags after read access.

Figure 40.149 Example of I3C master reception flowchart (high priority FIFO buffer transfer)

40.3.3.3.7 I3C Master IBI Reception Flow

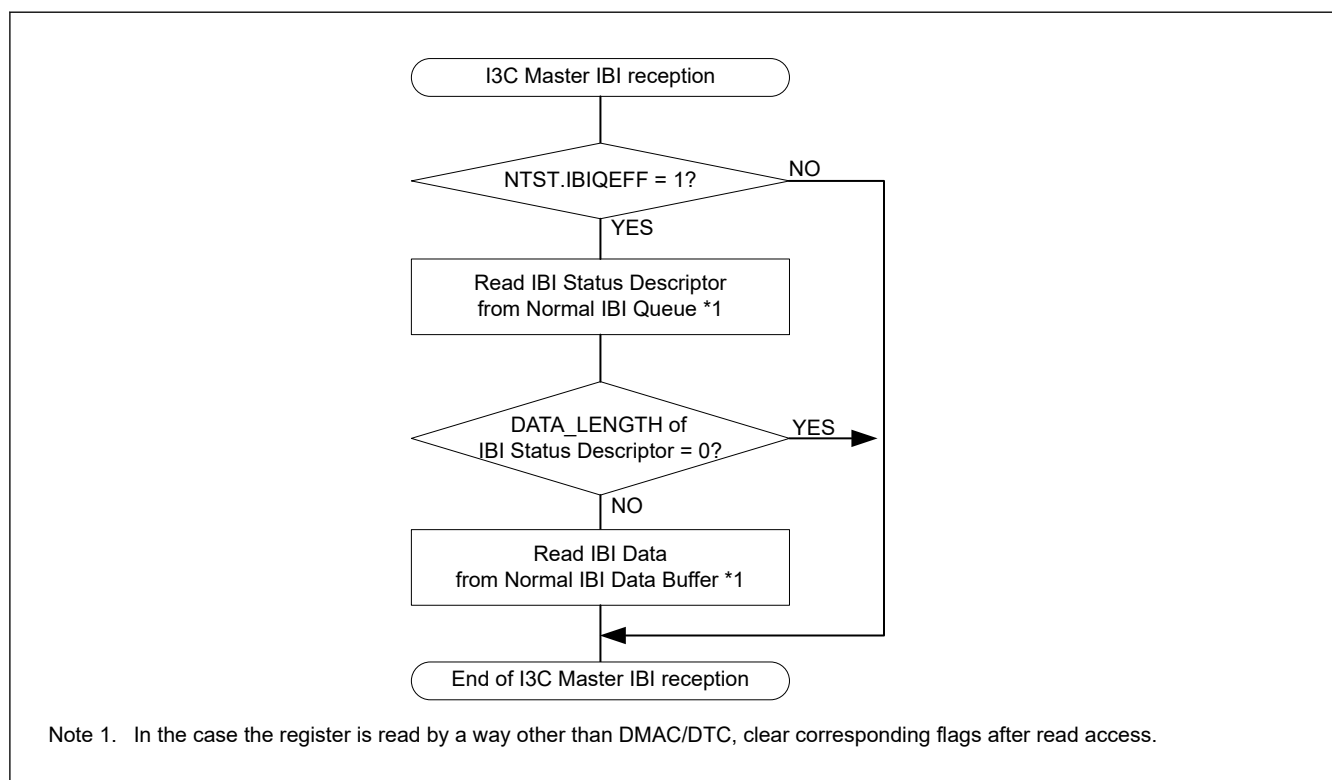


Figure 40.150 Example of I3C master IBI reception flowchart

40.3.3.3.8 I3C Master Wake-Up Flow

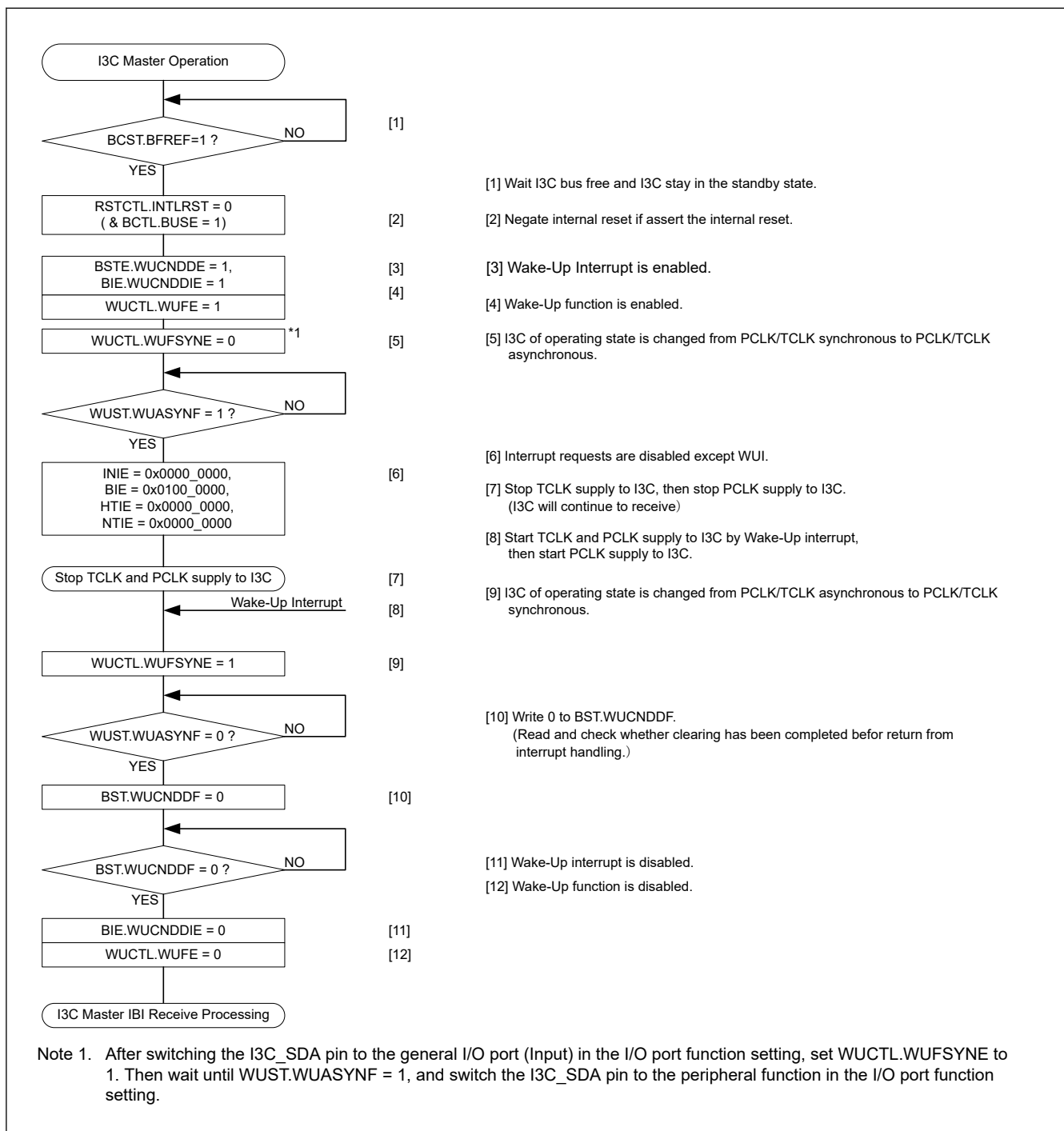
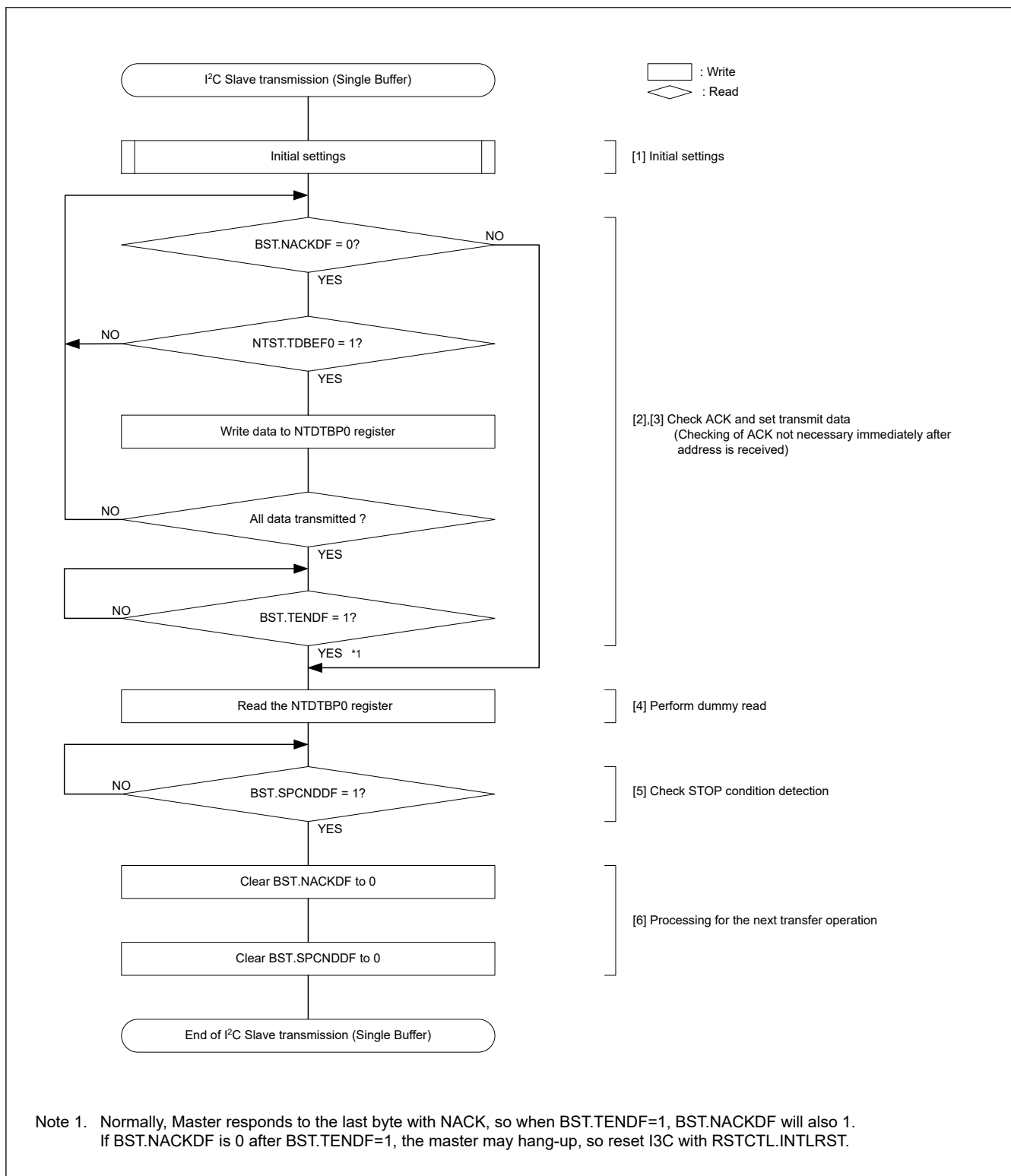


Figure 40.151 Use case of I3C master wake-up

40.3.3.4 Slave Mode Communication Flow

40.3.3.4.1 I²C Slave Transmission Flow (Single Buffer Transfer)Figure 40.152 Example of I²C slave transmission flowchart (single buffer transfer)

40.3.3.4.2 I²C Slave Reception Flow (Single Buffer Transfer)

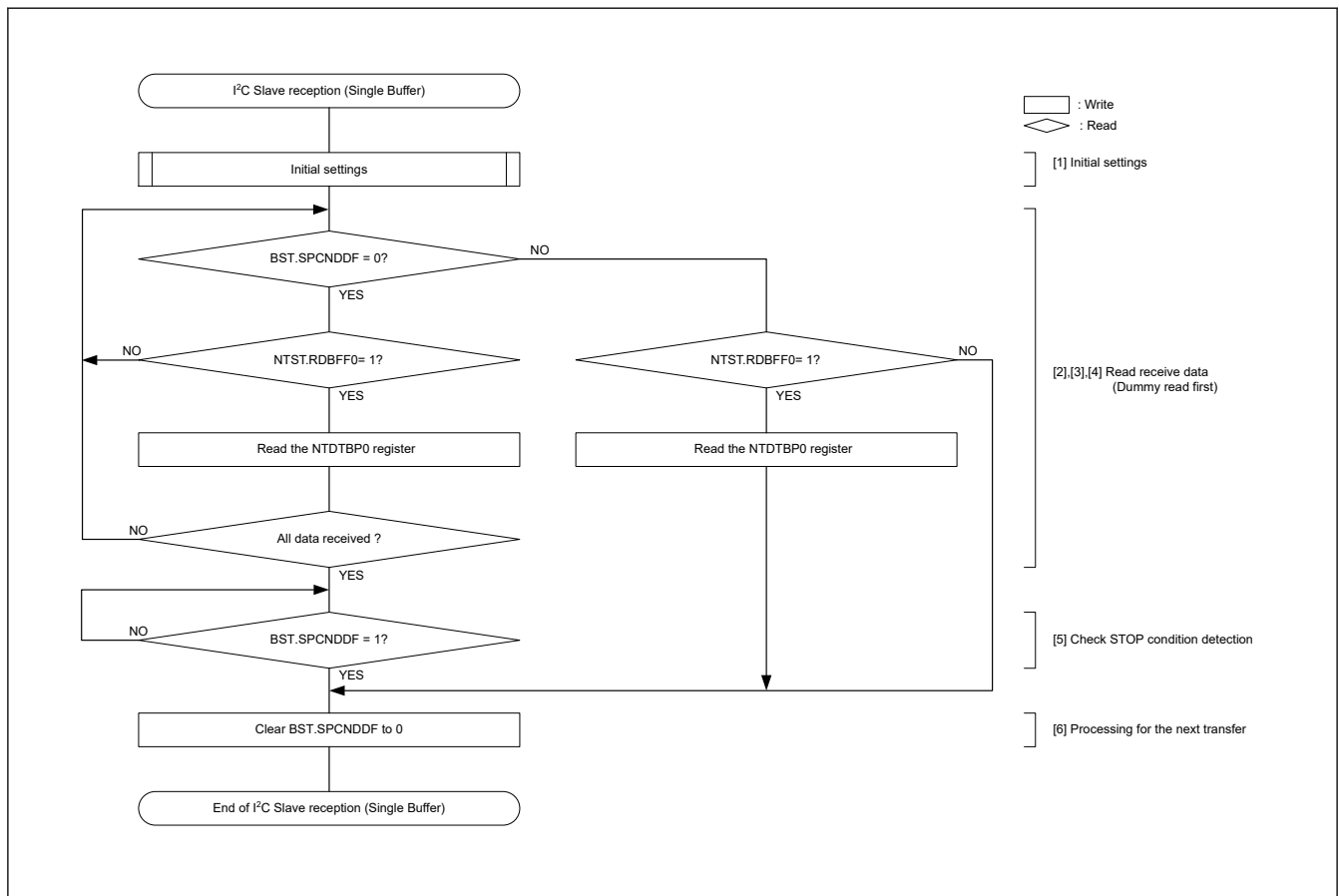


Figure 40.153 Example of I²C slave reception flowchart (single buffer transfer)

40.3.3.4.3 I3C Slave Transmission Flow (Normal FIFO Buffer Transfer)

Slave Transmission Flow in I3C normal FIFO buffer transfer is common to Legacy I²C, SDR (Private Transfer, Broadcast CCC, Direct CCC), HDR-DDR, and HDR-TSP/TSL.

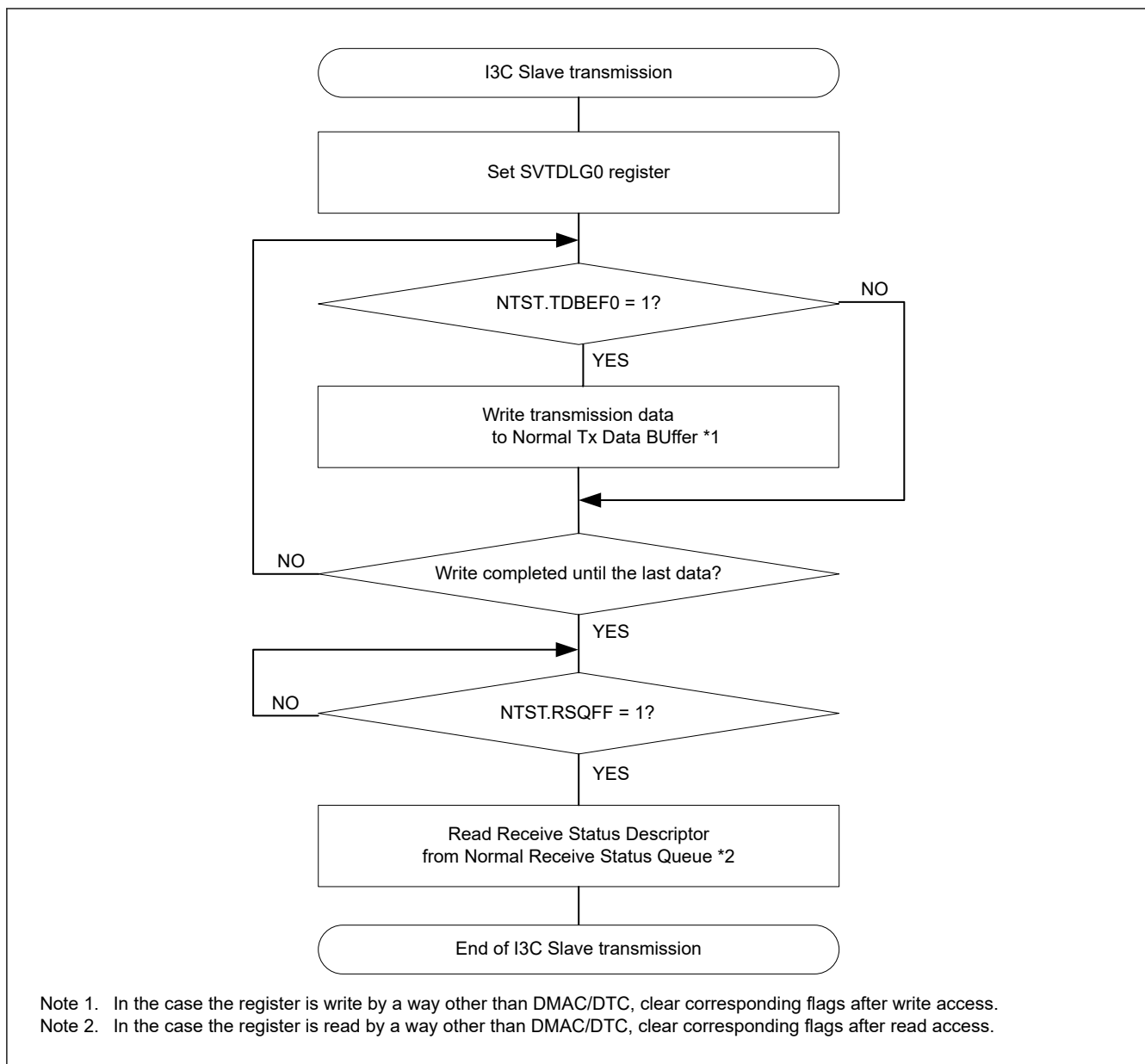
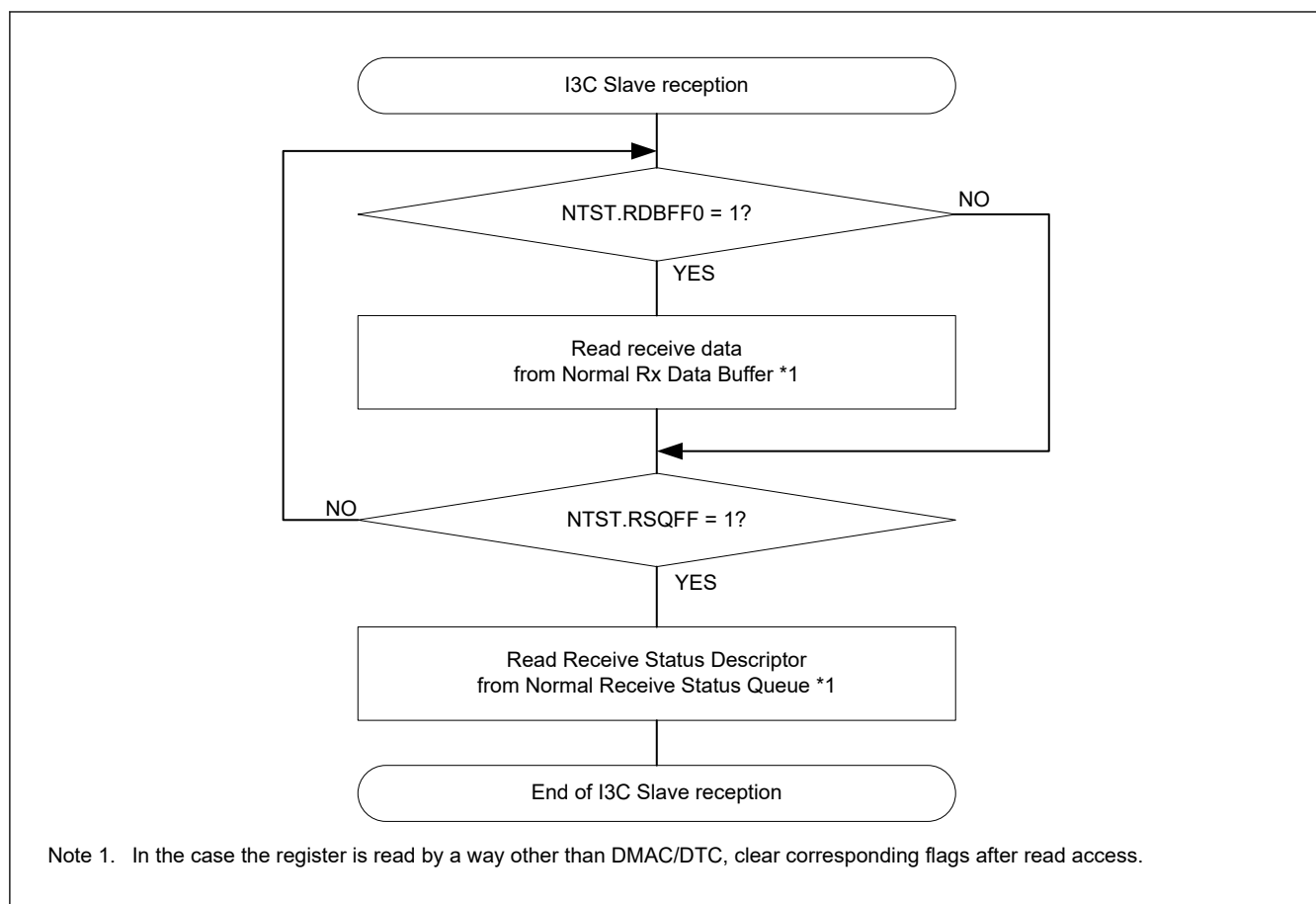


Figure 40.154 Example of I3C slave transmission flowchart (normal FIFO buffer transfer)

40.3.3.4.4 I3C Slave Reception Flow (Normal FIFO Buffer Transfer)

Slave Reception Flow in I3C normal FIFO buffer transfer is common to Legacy I²C, SDR (Private Transfer, Broadcast CCC, Direct CCC), HDR-DDR, and HDR-TSP/TSL.

**Figure 40.155** Example of I3C slave reception flowchart (normal FIFO buffer transfer)

40.3.3.4.5 I3C Slave IBI Transmission Flow

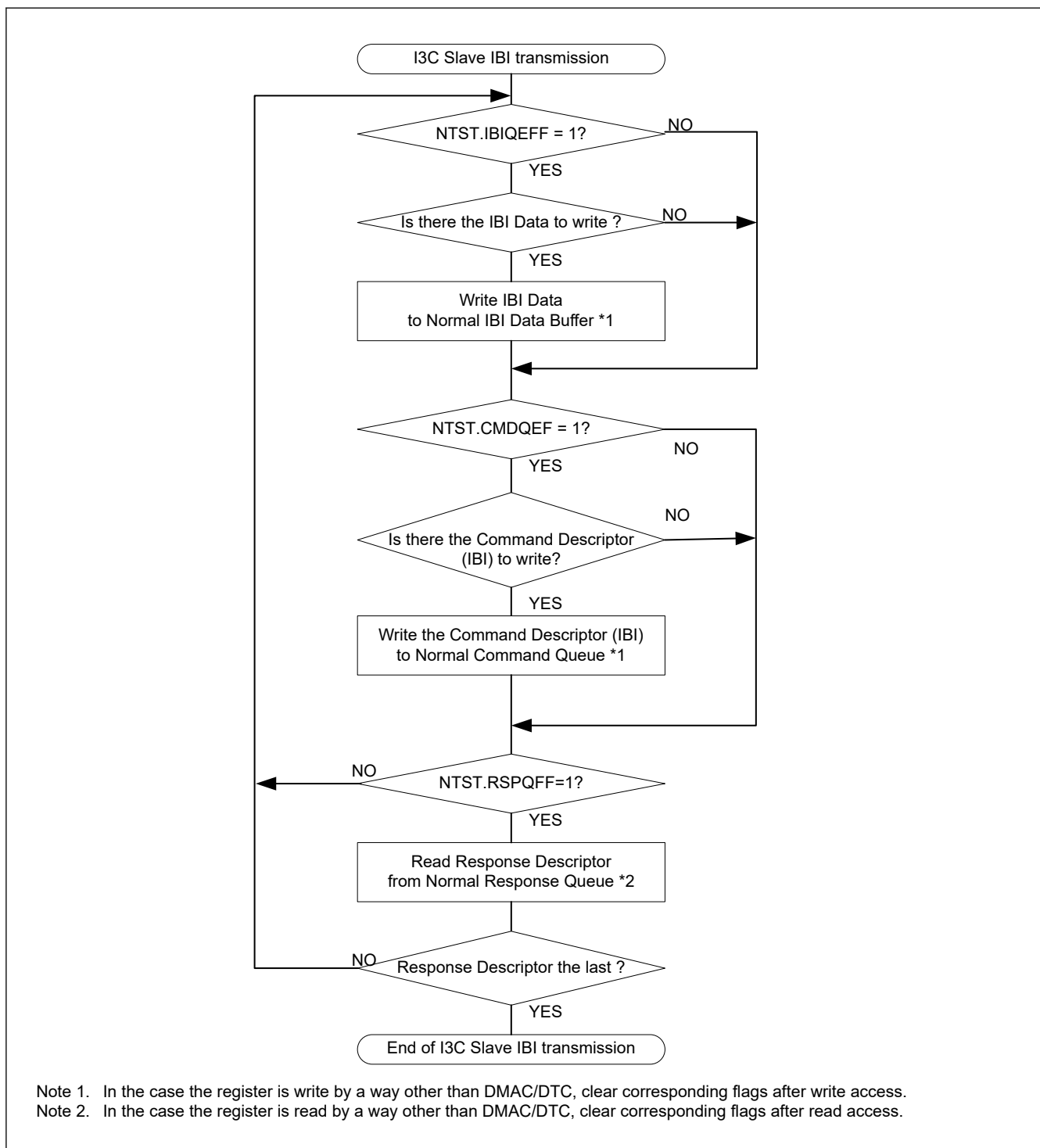


Figure 40.156 Example of I3C slave IBI transmission flowchart

40.3.3.4.6 I3C Slave Wake-Up Flow

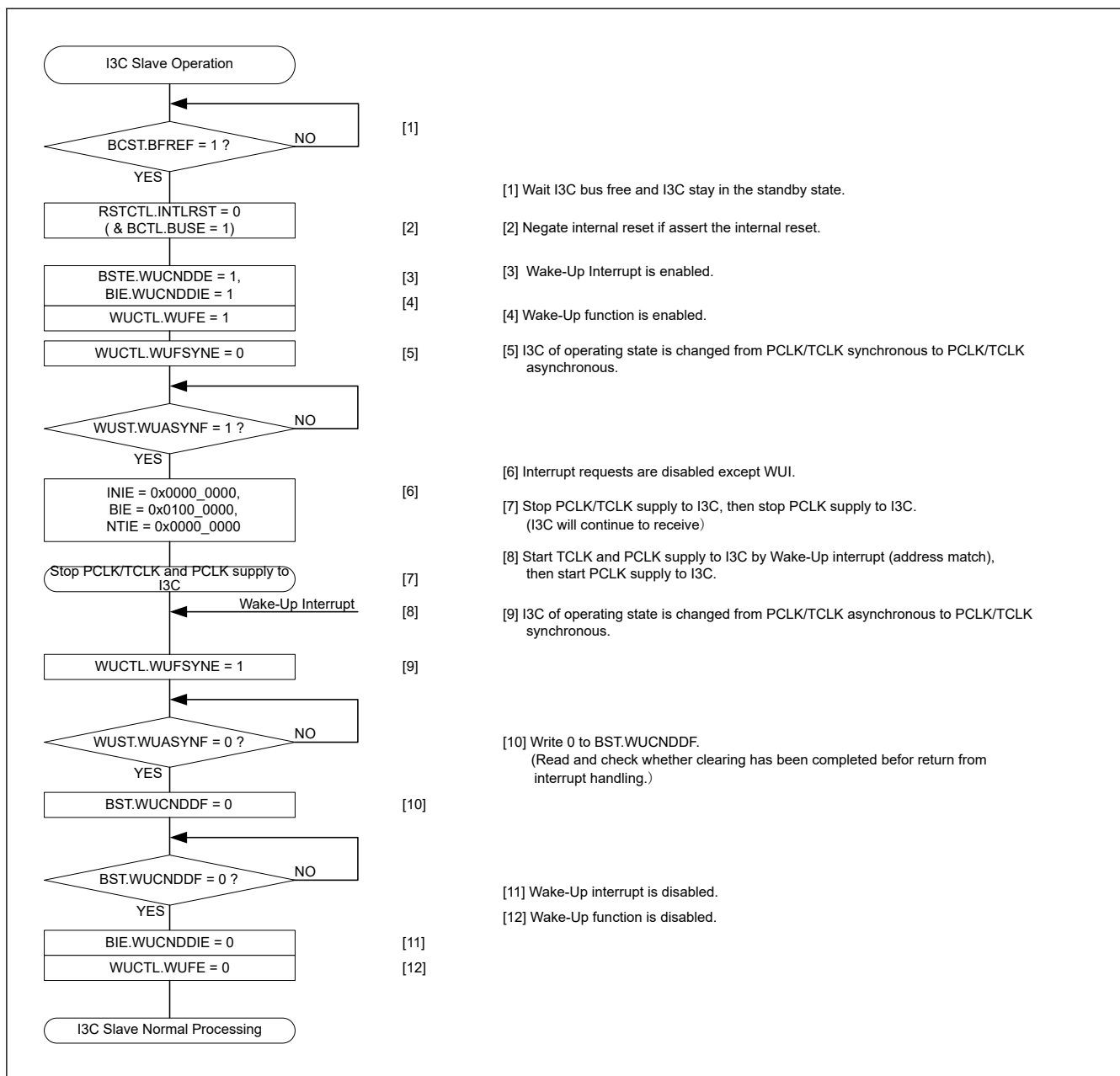


Figure 40.157 Use case of I3C slave wake-up (wake-up recovery by a wake-up interrupt triggered by the match of the slave address)

40.4 Interrupt Sources

I3C can generate the following interrupt requests:

40.4.1 Overview

The I3C has the interrupt factors shown in [Table 40.18](#).

Table 40.18 Interrupt Generation (1 of 2)

Symbol	Interrupt source	Support			
		I2C	I3CM	I3C2M	I3CS
I3C_RESP	Normal Response Queue full	—	✓	✓	✓

Table 40.18 Interrupt Generation (2 of 2)

Symbol	Interrupt source	Support			
		I2C	I3CM	I3C2M	I3CS
I3C_CMD	Normal Command queue empty	—	✓	✓	✓
I3C_IBI	Normal IBI Queue Empty/Full	—	✓	✓	✓
I3C_RX	Normal Rx Data buffer full	✓	✓	✓	✓
I3C_TX	Normal Tx Data buffer empty	✓	✓	✓	✓
I3C_RCV	Normal Receive Status queue full	—	—	✓	✓
I3C_HRESP	High Priority Response queue full	—	✓	✓	—
I3C_HCMTD	High Priority Command queue empty	—	✓	✓	—
I3C_HRX	High Priority Rx Data buffer full	—	✓	✓	—
I3C_HTX	High Priority Tx Data buffer empty	—	✓	✓	—
I3C_TEND	Transmit end	✓	—	—	—
I3C_EEI	Non-recoverable internal error	—	✓	✓	✓
	Normal Transfer Error	—	✓	✓	✓
	Normal Transfer Abort	—	✓	✓	✓
	High Priority Transfer Error	—	✓	✓	—
	High Priority Transfer Abort	—	✓	✓	—
	START condition detection	✓	✓	✓	✓
	STOP condition detection	✓	✓	✓	✓
	HDR Exit Pattern detection	—	✓	✓	✓
	NACK detection	✓	—	—	—
	Arbitration lost	✓	—	—	—
	Timeout detection	✓	✓	✓	✓
I3C_STEV	Synchronous Timing	—	✓	✓	✓
I3C_MREFOVF	MREF Counter Overflow	—	✓	✓	—
I3C_MREFCPT	MREF Capture	—	✓	✓	—
I3C_AMEV	Additional Master-initiated bus Event	—	✓	✓	—
I3C_WU	Wake-up condition detection	✓	✓	✓	✓

Note: ✓ : Support

— : Not support

Note: I²C: I²C Master/Slave (Single Buffer)

I3CM: I3C Master

I3C2M: I3C Secondary Master

I3CS: I3C Slave

Note: There is a delay time between the execution of a write instruction for a peripheral module by the CPU and actual writing to the module. Thus, when an interrupt flag has been cleared, read the relevant flag again to check whether clearing has been completed, and then return from interrupt handling. Returning from interrupt handling without checking that writing to the module has been completed creates a possibility of repeated processing of the same interrupt.

For I2C Protocol mode:

- Since I3C_TX is an edge-detected interrupt, it does not require clearing. Furthermore, the NTST.TDBEF0 flag (a condition for I3C_TX) is automatically set to 0 when data for transmission are written to NTDTBP0 or a STOP condition is detected (SPCNDDF flag = 1 in BST).

- Since I3C_RX is an edge-detected interrupt, it does not require clearing. Furthermore, the NTST.RDBFF0 flag (a condition for I3C_RX) is automatically set to 0 when data are read from NTDTBP0.

For I3C Protocol mode:

For details, refer to the detailed explanation of each flag bit.

- The I3C_CMD, I3C_TX, I3C_HCMD, I3C_HTX and I3C_IBI (I3C Slave) interrupts are cleared under the following conditions.
On completion of the last write access by DMAC/DTC.
Write 0 to this bit after 1 state is read by CPU.
- The I3C_RESP, I3C_IBI (I3C Master), I3C_RX, I3C_RCV, I3C_HRESP and I3C_HRX interrupts are cleared under the following conditions.
On completion of the last read access by DMAC/DTC.
Write 0 to this bit after 1 state is read by CPU.

40.4.2 Buffer Operation for Buffer Full/Empty Interrupts

If the conditions for generating each buffer full/empty interrupts are satisfied while the corresponding IR flag is 1, the interrupt request is not output for the ICU but retained internally (the capacity for internal retention is one request per source).

An interrupt request that was being retained within the ICU is output when the value of the IELSRn.IR flag becomes 0. Internally retained interrupt requests are automatically cleared under normal conditions of usage. Internally retained interrupt requests can also be cleared by writing 0 to the interrupt enable bit within the given peripheral module.

40.5 Event Link Output

I3C handles event output for the event link controller (ELC) corresponding to the following sources.

(1) Communication event

When a Communication event (arbitration-lost detection, detection of NACK, detection of timeout, detection of a START condition, or detection of a STOP condition) occurs, the corresponding event signal can be output for another module via the ELC.

(2) Rx Data buffer full

When a receive data register becomes full, the corresponding event signal can be output for another module via the ELC.

(3) Tx Data buffer empty

When a transmit data register becomes empty, the corresponding event signal can be output for another module via the ELC.

(4) Transmit end

On completion of transfer, the corresponding event signal can be output for another module via the ELC.

Other events are also available. For details, see [section 19, Event Link Controller \(ELC\)](#) 18.2.3 ELSRn : Event Link Setting Register n.

40.5.1 Interrupt Handling and Event Linking

I3C module produces four kinds of interrupt: communication event (arbitration-lost detection, detection of NACK, detection of timeout, detection of a START condition, or detection of a STOP condition), Rx Data buffer full, Tx Data buffer empty, and transmit end interrupts. Each of these has an enable bit to control enabling and disabling of the interrupt signal. An interrupt request signal is output for the CPU when an interrupt source condition is satisfied while the setting of the corresponding enable bit is enabled.

The corresponding event link output signals are sent to other modules as event signals via the ELC when the interrupt source conditions are satisfied, regardless of the settings of the interrupt enable bits. For details on interrupt sources, see [section 40.4.1. Overview](#).

40.6 Reset Descriptions

Table 40.19 Register states when issuing each condition (1) (1 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTLRST	CMDQrst	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQrst	HRSPQRST	HTDBRST	HRDBRST
PRTS	PRTMD	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
BCTL	BUSE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSM	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ABT	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	BMDs	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	INCBA	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
MSDVAD	MDYADV	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	MDYAD[6:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
RSTCTL	INTLRST	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HRDBRST	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HTDBRST	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HRSPQRST	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HCMDQrst	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSQRST	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	IBIQRST	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RDBRST	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TDBRST	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSPQRST	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQrst	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	R13CRST	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
PRsST	PRsSTWP	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TRMD	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CRMS*1	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
INST	INEF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.19 Register states when issuing each condition (1) (2 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R3CRST	INTLRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
INSTE	INEE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
INIE	INEIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
INSTFC	INEFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
DVCT	IDX[4:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
IBINCTL	NRSIRCTL	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	NRMRACTL	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
BFCTL	HSME	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	FMPE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SMBS	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SCSYNE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SALE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	NALE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	MALE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Note: In reset: To be reset (The FIFO corresponding to this register is cleared)

Note 1. In I3C mode, CRMS is not reset by INTLRST. In I2C mode, CRMS is reset by INTLRST

Table 40.20 Register states when issuing each condition (2) (1 of 3)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R3CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
SVCTL	SVAE[2]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SVAE[1]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SVAE[0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HOAE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DVIDE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HSMCE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	GCAE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
REFCKCTL	IREFCKS [2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
STDBR	DSBRPO	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SBRHP[5:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SBRLP[5:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SBRHO[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SBRLO[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
EXTBR	EBRHP[5:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	EBRLP[5:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	EBRHO[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	EBRLO[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
BFRECDT	FRECYC [8:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
BAVLCDT	AVLCYC[8:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
BIDLCDT	IDLCYC [17:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.20 Register states when issuing each condition (2) (2 of 3)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
OUTCTL	SDODCS	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SDOD[2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	EXCYC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SOCWP	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SCOC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SDOC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
INCTL	SDID[1:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DNFE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DNFS[3:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
TMOCTL	TOMDS[1:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TOHCTL	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TOLCTL	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TODTS[1:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
WUCTL	WUFE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	WUFSYNE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	WUANFS	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	WUACKS	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
ACKCTL	ACKTWP	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ACKT	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ACKR	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
SCSTRCTL	RWE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ACKTWE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.20 Register states when issuing each condition (2) (3 of 3)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
SCSTLCTL	ACKPE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	PARPE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	AAPE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	STLCYC [15:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
SVTDLG0	STDLG[15:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Note: In reset: To be reset (The FIFO corresponding to this register is cleared)

Table 40.21 Register states when issuing each condition (3) (1 of 3)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
STCTL	STOE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
ATCTL	CDIV[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	AMEOE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	MREFOE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ATTRGS	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
ATTRG	ATSTRG	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
ATCCNTE	ATCE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CNDCTL	SPCND	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SRCND	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	STCND	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
NCMDQP	NCMDQP [31:0]	In reset	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
NRSPQP	NRSPQP [31:0]	In reset	In reset	In reset	Saved	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
NTDTBP0	NTDTBP0 [31:0]	In reset	In reset	In reset	Saved	Saved	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved
NIBIQP	NIBIQP[31:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	In reset	Saved	Saved	Saved	Saved	Saved

Table 40.21 Register states when issuing each condition (3) (2 of 3)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R3CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
NRSQP	NRSQP [31:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	In reset	Saved	Saved	Saved	Saved
HCMDQP	HCMDQP [31:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	In reset	Saved	Saved	Saved
HRSPQP	HRSPQP [31:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	In reset	Saved	Saved
HTDTBP	HTDTBP [31:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	In reset	In reset
NQTHCTL	IBIQTH[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	IBIDSSZ[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSPQTH [7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQTH [7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
NTBTHCTL0	RXSTTH [2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TXSTTH[2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RXDBTH [2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TXDBTH [2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
NRQTHCTL	RSQTH[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
HQTHCTL	RSPQTH [7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQTH [7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
HTBTHCTL	RXSTTH [2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TXSTTH[2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RXDBTH [2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TXDBTH [2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.21 Register states when issuing each condition (3) (3 of 3)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R3CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
BST	WUCNDDF	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TODF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ALF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TENDF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	NACKDF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HDREXDF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SPCNDDF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	STCNDDF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
BSTE	WUCNDDE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TODE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ALE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TENDE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	NACKDE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HDREXDE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SPCNDDE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	STCNDDE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Note: In reset: To be reset (The FIFO corresponding to this register is cleared)

Table 40.22 Register states when issuing each condition (4) (1 of 3)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R3CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
BIE	WUCNDDIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TODIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ALIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TENDIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	NACKDIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HDREXDIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SPCNDDIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	STCNDDIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
BSTFC	WUCNDDFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TODFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ALFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TENDFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	NACKDFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HDREXDFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SPCNDDFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	STCNDDFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.22 Register states when issuing each condition (4) (2 of 3)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R3CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
NTST	RSQFF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	In reset	Saved	Saved	Saved	Saved
	TEF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TABTF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSPQFF	In reset	In reset	In reset	Saved	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQEF	In reset	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	IBIQEFF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	In reset	Saved	Saved	Saved	Saved	Saved
	RDBFF0	In reset	In reset	In reset	Saved	Saved	Saved	In reset	Saved	Saved	Saved	Saved	Saved	Saved
	TDBEF0	In reset	In reset	In reset	Saved	Saved	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved
NTSTE	RSQFE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TEE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TABTE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSPQFE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQEE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	IBIQEFE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RDBFE0	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TDBEE0	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.22 Register states when issuing each condition (4) (3 of 3)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTLRST	CMDQCRST	RSPQCRST	TDBRST	RDBRST	IBIQCRST	RSQCRST	HCMDQCRST	HRSPQCRST	HTDBRST	HRDBRST
NTIE	RSQFIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TEIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TABTIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSPQFIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQEIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	IBIQEFIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RDBFIE0	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TDBEIE0	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Note: In reset: To be reset (The FIFO corresponding to this register is cleared)

Table 40.23 Register states when issuing each condition (5) (1 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTLRST	CMDQCRST	RSPQCRST	TDBRST	RDBRST	IBIQCRST	RSQCRST	HCMDQCRST	HRSPQCRST	HTDBRST	HRDBRST
NTSTFC	RSQFFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TEFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TABTFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSPQFFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQEFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	IBIQEFFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RDBFFC0	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TDBEFC0	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.23 Register states when issuing each condition (5) (2 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R3CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
HTST	TEF	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TABTF	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSPQFF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	In reset	Saved	Saved
	CMDQEF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	In reset	Saved	Saved	Saved
	RDBFF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	In reset
	TDBEF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	In reset	Saved
HTSTE	TEE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TABTE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSPQFE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQEE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RDBFE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TDBEE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
HTIE	TEIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TABTIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSPQFIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQEIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RDBFIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TDBEIE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Note: In reset: To be reset (The FIFO corresponding to this register is cleared)

Table 40.24 Register states when issuing each condition (6) (1 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R3CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
HTSTFC	TEFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TABTFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RSPQFFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQEFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	RDBFFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TDBEFC	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
BCST	BIDLF	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	BAVLF	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	BFREF	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
SVST	SVAF[2]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SVAF[1]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SVAF[0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HOAF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DVIDF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	HSMCF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	GCAF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
WUST	WUASYNF	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
MRCCPT	MRCCPT [31:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.24 Register states when issuing each condition (6) (2 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
DATBAS _m (m = 0 to 7)	DVTYP	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DVNACK [1:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DVDYAD [7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DVIBITS	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DVMRRJ	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DVSIRRJ	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DVIBIPL	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DVADLS	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DVSTAD [9:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Note: In reset: To be reset (The FIFO corresponding to this register is cleared)

Table 40.25 Register states when issuing each condition (7) (1 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
EXDATBAS	EDTYP	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	EDNACK [1:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	EDDYAD [7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	EDADLS	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	EDSTAD [9:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
SDATBAS _n (n = 0 to 2)	SDDYAD [6:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SDIBIPL	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SDADLS	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SDSTAD [9:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
MSDCT _m (m = 0 to 7)	RBCR _n	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.25 Register states when issuing each condition (7) (2 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTLRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
SVDCT	TBCRn	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TDCR[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Note: In reset: To be reset (The FIFO corresponding to this register is cleared)

Table 40.26 Register states when issuing each condition (8) (1 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTLRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
SDCTPIDL	SDCTPIDL [31:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
SDCTPIDH	SDCTPIDH [31:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
SVDVADn (n = 0 to 2)	SDYADV	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SSTADV	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SADLG	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SVAD[9:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CSECMD	MSRQE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SVIRQE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CEACTST	ACTST[3:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CMWLG	MWLG[15:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CMRLG	IBIPSZ[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	MRLG[15:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CETSTMD	TSTMD[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CGDVST	VDRSV[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ACTMD[1:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	PRTE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	PNDINT[3:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.26 Register states when issuing each condition (8) (2 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTLRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
CMDSPW	MSWDR[2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CMDSPR	CDTTIM[2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	MSRDR[2:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CMDSPT	MRTE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	MRTTIM [23:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Note: In reset: To be reset (The FIFO corresponding to this register is cleared)

Table 40.27 Register states when issuing each condition (9) (1 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R13CRST	INTLRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
CETSM	INAC[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	FREQ[7:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SPTASYN [3:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SPTSYN	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CGHDRCAP	TSLEN	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	TSPEN	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	DDREN	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CETSS	ICOVF	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	ASYNE[3:0]	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SYNE	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
BITCNT	BCNT[4:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Table 40.27 Register states when issuing each condition (9) (2 of 2)

Register symbol	Register bit name	System reset	RSTCTL Register											
			R3CRST	INTRST	CMDQRST	RSPQRST	TDBRST	RDBRST	IBIQRST	RSQRST	HCMDQRST	HRSPQRST	HTDBRST	HRDBRST
NQSTLV	IBISCNT[4:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	In reset	Saved	Saved	Saved	Saved	Saved
	IBIQLV[7:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	In reset	Saved	Saved	Saved	Saved	Saved
	RSPQLV [7:0]	In reset	In reset	In reset	Saved	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	CMDQFLV [7:0]	In reset	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
NDBSTLV0	RDBLV[7:0]	In reset	In reset	In reset	Saved	Saved	Saved	In reset	Saved	Saved	Saved	Saved	Saved	Saved
	TDBFLV[7:0]	In reset	In reset	In reset	Saved	Saved	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved
NRSQSTLV	RSQVLV[7:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	In reset	Saved	Saved	Saved	Saved
HQSTLV	RSPQLV [7:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	In reset	Saved	Saved
	CMDQLV [7:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	In reset	Saved	Saved	Saved
HDBSTLV	RDBLV[7:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	In reset
	TDBFLV[7:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	In reset	Saved
PRSTDBG	SDOLV	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SCOLV	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SDILV	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
	SCILV	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
MSERRCNT	M2ECNT [7:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
SC1CPT	SC1C[15:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
SC2CPT	SC2C[15:0]	In reset	In reset	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved
CECTL	CLKE	In reset	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved	Saved

Note: In reset: To be reset (The FIFO corresponding to this register is cleared)

40.7 Usage Notes

40.7.1 Settings for the Operating Clock

The following relation is required between the frequencies of the bus clock (PCLK) and transfer clock(TCLK).

$$TCLK/2 \leq PCLK \leq TCLK$$

40.7.2 Settings for the SCL Frequency Clock

As described in Chapter 5.1.2.4 of MIPI I3C Specification V1.0, the SCL frequency setting is required depending on the bus configuration.

In Mixed Bus with both I3C and legacy I2C devices are present on the bus, If the Legacy I2C device has a 50ns Spike Filter, I3C Master can change the frequency by setting the SCL High period to less than 50ns (tDIG_H_MIXED).

40.7.3 Module-stop Function

I3C operation can be disabled or enabled using Module Stop Control Register B (MSTPCRB). The I3C module is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).

40.7.4 Notes on Rewriting PRTS.PRTMD Bit

When the PRTS.PRTMD bit is rewritten, the SCL terminal (and SDA terminal) may output low. In this case, the bus line is unintentionally put into a busy state, and communication cannot be started even if the communication start setting is made after that.

Set the RSTCTL.INTRST bit to 1 and issue an internal software reset before rewriting the PRTMD bit. After rewriting, set the RSTCTL.INTRST bit to 0 to release the internal software reset.

However, no countermeasure is necessary if the following conditions are applicable.

- (1) When the PRTS.PRTMD bit is not changed from its initial value of 1 (I2C protocol mode) to 0 (I3C protocol mode).
- (2) When the clock source selected by SCKSCR.CKSEL[2:0] and I3CCKCR.I3CCKSEL[3:0] are the same.
- (3) In master mode, when the timeout error detection function is enabled under the following conditions and a software reset is set in the I3C error interruption routine to release unintended low output.
 1. Set the BSTE.TODE bit to 1 to enable the timeout error detection function.
 2. Select TMOCTL.TOMDS[1:0] = 00b or 01b as the timeout operation mode and set the TMOCTL.TOLCTL bit = 1 and TMOCTL.TOHCTL bit = 1 to treat the SCL line fixed at low and fixed at high as errors.
 3. In the I3C error interruption routine, set the RSTCTL.RI3CRST bit to 1 to perform an I3C software reset, or set the RSTCTL.INTLRST bit to 1 to perform an internal software reset.
- (4) In slave mode, when the timeout error detection function is enabled under the following conditions and a software and TMOCTL.TOHCTL bit = 1 to treat the SCL line fixed at low and fixed at high as errors.
 1. Set the BSTE.TODE bit to 1 to enable the timeout error detection function.
 2. Select TMOCTL.TOMDS[1:0] = 01b as the timeout operation mode and set TMOCTL.TOLCTL = 1 and TMOCTL.TOHCTL = 1 to treat the SCL line fixed at low and fixed at high as errors.
 3. In the I3C error interruption routine, set the RSTCTL.RI3CRST bit to 1 to perform an I3C software reset, or set the RSTCTL.INTLRST bit to 1 to perform an internal software reset.